

1. When the Trunk port sends the data frame, how to deal with?

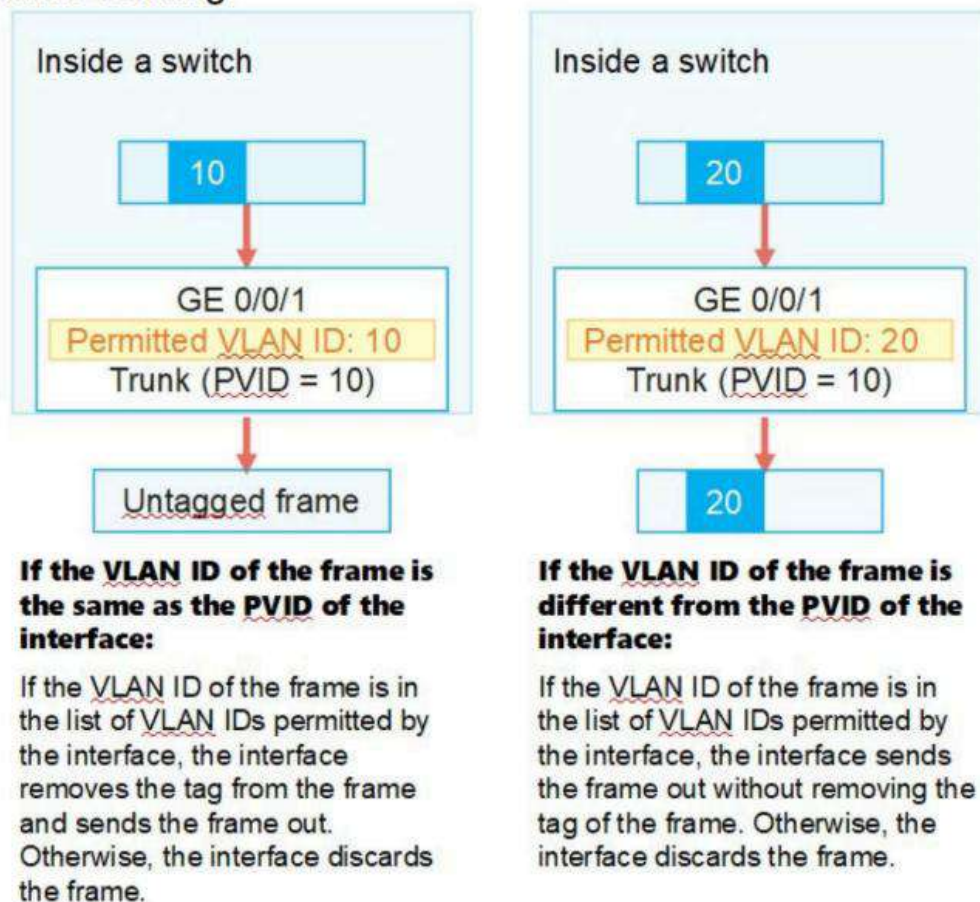
- A. When the VLAN ID is the same as the PVID of the port and is a VLAN ID allowed by the port, remove the Tag and send the message
- B. When the VLAN ID is different from the PVID of the port, the data frame is dropped
- C. When the VLAN ID is different from the PVID of the port, replace it with PVID forwarding
- D. When the VLAN ID is different from the PVID of the port, strip off the TAG and forward

Answer: A

Justification

Source: Slide Lecture # 8(VLAN Principles and Configuration), Page # 19

Frame sending



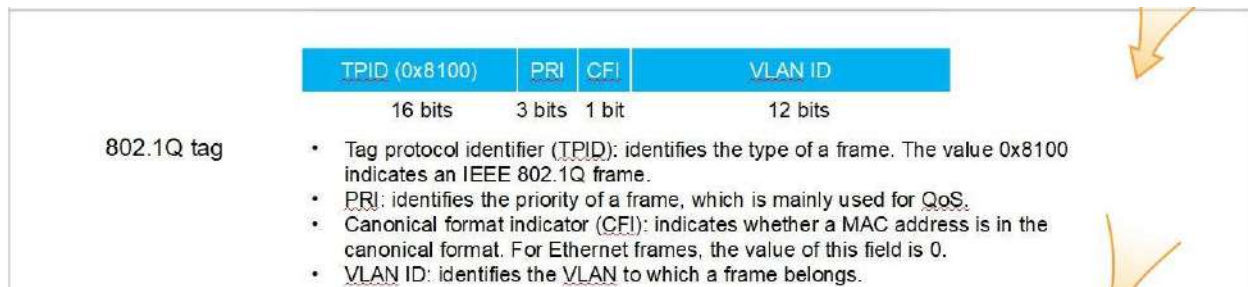
2. How many bits are there in the VLAN ID in the VLAN format defined by the IEEE 802.1Q?

- A. 12
- B. 6
- C. 10
- D. 8

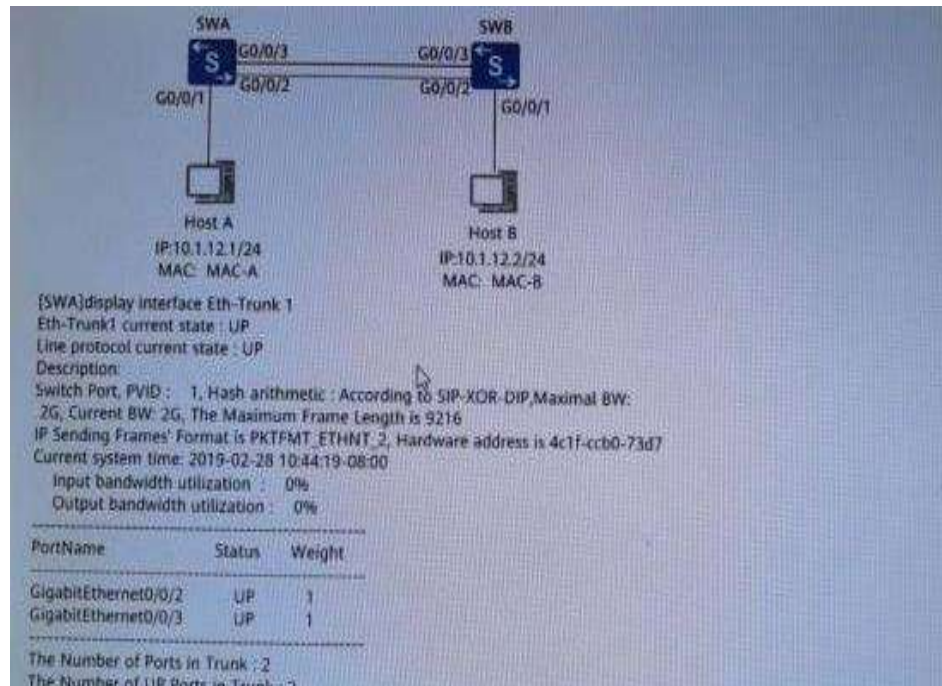
Answer: A

Justification

Source: Slide Lecture # 8(VLAN Principles and Configuration), Page # 19



3. In the network shown in the figure below, the output information of switch A is shown in the figure. Then, in the MAC address table of switch A, which interface does the MAC address of host B correspond to?



- A. GigabitEthernet 0/0/3
- B. GigabitEthernet0/0/2
- C. Eth-Trunk 1
- D. GigabitEthernet0/0/1

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000015892/e54350af/display-eth-trunk>

Format

```
display eth-trunk [ trunk-id [ interface interface-type interface-number [ verbose ] ] ]
```

4. All ports on the root-bridge switch are designated ports.

- A. True
- B. False

Answer: A

Justification

Source: (<https://support.huawei.com/enterprise/en/doc/EDOC1000017271/bd821590/stp-rstpoverview>)

Designated port: forwards BPDUs to the downstream switching device. All the ports on the root bridge are designated ports. A designated port is selected on each network segment. The device where the designated port resides is called the designated bridge on the network segment.

5. In the configuration BPDU sent by the switch, which bridge ID is unlikely to appear?

- A. 8192 00-01-02-03-04-CC
- B. 4096 00-01-02-03-04-CC
- C. 0 00-01-02-03-04-CC
- D. 2048 00-01-02-03-04-CC

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086964#:~:text=The%20default%20priority%20value%20of,elected%20as%20the%20root%20bridge.>

The default priority value of a device is 32768. The value is an integer multiple of 4096, such as 0, 4096 and 8192. A smaller value indicates a higher priority of the device. A device with a higher priority is more likely to be elected as the root bridge. Low-performance devices at lower network layers are not suitable for root bridges, so you need to set low priorities for these devices.

6. The number of VLANs that can be created using the commands "vlan batch 10 20" and "vlan batch 10 to 20" respectively is

- A. 11 and 11
- B. 11 and 2
- C. 2 and 2
- D. 2 and 11

Answer: D

Justification

Source: Huawei Slide Lecture # 8 (VLAN Principles and Configuration), Page # 31

```
[Huawei] vlan batch { vlan-id1 [ to vlan-id2 ] }
```

This command creates VLANs in a batch. In this command:

- **batch**: creates VLANs in a batch.
- *vlan-id1*: specifies a start VLAN ID.
- *vlan-id2*: specifies an end VLAN ID.

7. By default, the root path cost of the root bridge in the STP protocol must be 0.

- A. True
- B. False

Answer: A

Justification

Source: Huawei Slide Lecture # 9 (Basic STP Concepts: RSTP), page # 18

- There may be multiple paths from a non-root bridge to the root bridge. Each path has a total cost, which is the sum of all port costs on this path. A non-root bridge compares the costs of multiple paths to select the shortest path to the root bridge. The path cost of the shortest path is called the root path cost (RPC), and a loop-free tree network is generated. The RPC of the root bridge is 0.

8. The administrator wants all hosts to be unable to access the Internet between 8:00-17:00 every day, so which ACL rule needs to be bound to the traffic-filter outbound in the GE0/0/1 interface allocation?

- A. `time-range am9topm5 08:00 to working-day`
`acl number 2003`
`rule 5 deny time-range am9topjni5`
- B. `time-range am9topm5 08:00 to 17:00 off-day`
`acl number 2004`
`rule 5 permit time-range`
- C. `time-range am9topm5 08:00 to 17:00 daily`
`acl number 2001`
`rule 5 deny time-range am9topm5`
- D. `time-range am9topm5 07:00 to 18:00 daily`
`acl number 2002`

Answer: C

Justification

Source: Huawei Slide Lecture # 12 (ACL Principles and Configuration), page # 25



Basic Configuration Commands of Advanced ACLs (2)

2. Configure a rule for the advanced ACL.

You can configure advanced ACL rules according to the protocol types of IP packets. The parameters vary according to the protocol types.

- When the protocol type is **IP**, the command format is:

```
rule [ rule-id ] { deny | permit } ip [ destination { destination-address destination-wildcard | any } | source { source-address source-wildcard | any } | time-range time-name | [ dscp dscp | [ tos tos | precedence precedence ] ] ]
```

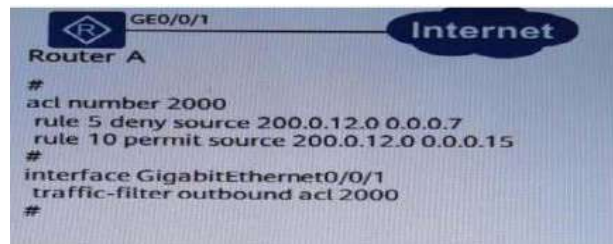
In the advanced ACL view, you can run this command to configure a rule for the advanced ACL.

- When the protocol type is **TCP**, the command format is:

```
rule [ rule-id ] { deny | permit } { protocol-number | tcp } [ destination { destination-address destination-wildcard | any } | destination-port { eq port | gt port | lt port | range port-start port-end } | source { source-address source-wildcard | any } | source-port { eq port | gt port | lt port | range port-start port-end } | tcp-flag { ack | fin | syn } * | time-range time-name ] *
```

In the advanced ACL view, you can run this command to configure a rule for the advanced ACL.

9. As shown in the figure below, the wrong statement is:



- A. The host whose source IP address is 200.0.12.4 cannot access the Internet
- B. The host whose source IP address is 200.0.12.2 cannot access the Internet
- C. The host whose source IP address is 200.0.12.8 cannot access the Internet
- D. The host whose source IP address is 200.0.12.6 cannot access the Internet

Answer: C

Justification

Explanation: The ACL rule deny source 200.0.12.0 0.0.0.7 denies access to all hosts with IP addresses in the range from 200.0.12.0 to 200.0.12.7. These hosts cannot access the internet. Therefore, the host with ip address 200.0.12.8 does not belong to this range.

10. On Huawei equipment, if you use aaa authentication for authorization. When the remote server does not respond, you can authorize from the network equipment side.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100278979/51916bb/aaa-anduser-management-configuration>

NOTE

The next authorization mode can be used only when the current authorization mode provides no response (for example, the server provides no response). If the authorization is successful or fails, the next authorization mode will not be used.

11. What percentage does the DHCP client expect to send the lease renewal message for the first time?

- A. 0.5
- B. 0.875
- C. 1
- D. 0.25

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100112351/4b2f29de/how-a-dhcpclient-renews-its-ip-address-lease>

1. When the lease reaches 50% (T1) of its validity period, the DHCP client unicasts a DHCP Request message to the DHCP server to request lease renewal. If the server renews the lease (counted from 0), it sends a DHCP ACK message to the client. If the server rejects the renewal request, it sends a DHCP NAK message to the client, which must then send a DHCP Discover message to apply for a new IP address.

12. After the administrator successfully logs in to the router through Telnet, and finds that the router's interface IP address cannot be allocated, the possible reasons are ()

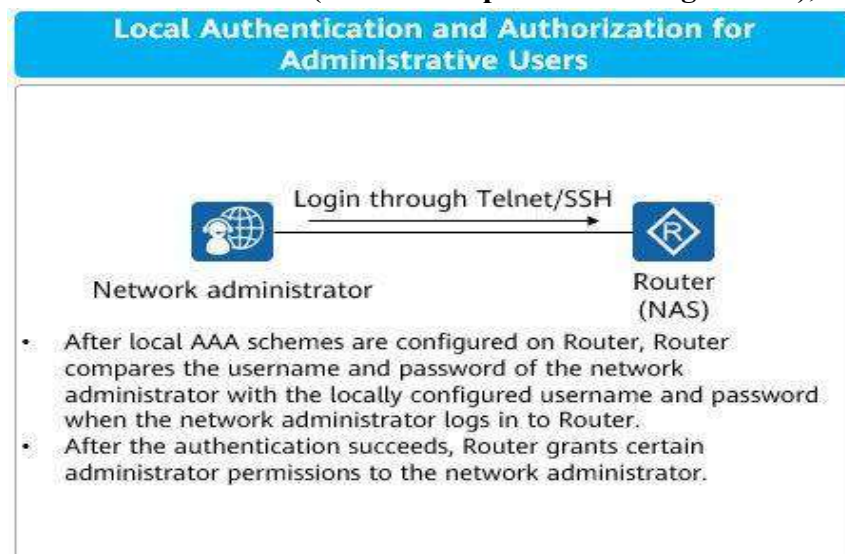
- A. The Telnet user level is incorrectly configured
- B. SNMP parameter is incorrectly configured
- C. The Telnet user authentication mode is incorrectly configured
- D. The administrator uses the telnet terminal software to prohibit the corresponding operation

Answer: A

Justification

Explanation: The reason behind this choice is that user levels in Telnet can restrict the commands a user can execute. If the Telnet user level is set too low, the administrator might not have the necessary permissions to configure or modify the interface settings, including setting IP addresses. This restriction could prevent the allocation or modification of interface IP addresses.

Source: Huawei Slide Lecture # 13 (AAA Principles and Configuration), Page # 12



13. vty configuration as shown in the figure, the user authority level is set to level

```
[Huawei]user-interface vty 0 14
```

```
[Huawei-ui-vty0-14]acl 2000 inbound
```

```
[Huawei-ui-vty0-14]user privilege level 3
```

```
[Huawei-ui-vty0-14]authentication-mode password
```

Please configure the login password (max length 16):huawei

A. False

B. True

Answer: B

Justification

Source: https://support.huawei.com/enterprise/en/doc/EDOC1000044778/d5dab65b/configuring-the-vty-user-interface#EN-US_TASK_0133019816

Procedure

1. Run:

```
system-view
```

The system view is displayed.

2. Run:

```
user-interface vty first-ui-number [ last-ui-number ]
```

The VTY user interface view is displayed.

3. Run:

```
user privilege level level
```

The user level is set.

14. When AP and AC are on different Layer 3 networks, what method is recommended for AP to discover AC? (Multiple Choice)

- A. Broadcasting method Correct
- B. DNS
- C. DHCP
- D. Manually specified on AP

Answer: B, C A

Justification

Source:

https://www.cisco.com/c/en/us/td/docs/wireless/access_point/cw916x/cw9164/installguide/b-hig-cw9164i/installation-overview.html

Ensure that the Cisco Controller Distribution System (DS) port is connected to the network. Use the procedure for CLI or GUI, as described in the release-appropriate [Cisco Catalyst 9800 Series Wireless Controller Software Configuration Guide](#).

- Enable Layer 3 connectivity between APs, Cisco Controller Management, and AP-Manager interface.
- Configure the switch to which your AP has to attach. See the [Cisco Wireless Controller Configuration Guide](#) for the release you are using, for additional information.
- **Ensure that the DHCP is enabled on the network. The AP must receive its IP address through DHCP.**



Note

An 802.11ax AP is assigned an IP address from the DHCP server only if a default router (gateway) is configured on the DHCP server (enabling the AP to receive its gateway IP address) and the gateway ARP is resolved.

15. What frequency does IEEE 802.11n support? (Multiple Choice)

- A. 2.5GHz
- B. 2.4GHz
- C. 5GHz
- D. 6GHz

Answer: BC

Justification

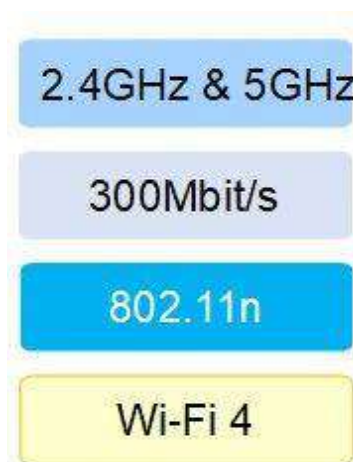
Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100081215>

Frequency Band

Original Wi-Fi systems define 2.4 GHz or 5 GHz frequency band. 802.11n supports both 2.4 GHz and 5 GHz frequency bands. There are obvious problems at the 2.4 GHz frequency band as Wi-Fi applications are increasingly used.

- Congested frequencies: A large number of non-Wi-Fi devices such as baby monitors, microwave ovens, and cordless telephones also work at the 2.4 GHz frequency band. Interferences from these devices affect Wi-Fi performance, and Wi-Fi cannot effectively solve these problems.
- Fewer frequency resources: The 2.4 GHz frequency band has only 83.5 MHz frequency resources. Fewer frequency resources indicate more frequency multiplexing and interferences. In addition, high-channel-bandwidth networking is limited, and the Wi-Fi maximum throughput cannot be fully used.

802.11ac does not support the 2.4 GHz frequency band. It prevents interferences at the 2.4 GHz frequency band and promotes popularity of terminals at the 5 GHz frequency band. In the 802.11n era, over half of terminals on the live network support only the 2.4 GHz frequency band.



16. In IPv4 networks, AP supports static and DHCP methods to obtain IP addresses.

- A. True
- B. False

Answer: A

Justification

Source:

https://www.cisco.com/c/en/us/td/docs/wireless/access_point/cw916x/cw9164/install-guide/b-hig-cw9164i/m_installing-your-access-point.html

- Configure the switch to which your AP has to attach. See the *Cisco Wireless Controller Configuration Guide* for the release you are using, for additional information.
- Ensure that the DHCP is enabled on the network. **The AP must receive its IP address through DHCP**



Note

An 802.11ax AP is assigned an IP address from the DHCP server only if a default router (gateway) is configured on the DHCP server (enabling the AP to receive its gateway IP address) and the gateway ARP is resolved.

- CAPWAP UDP ports must not be blocked in the network.
- The AP must be able to find the IP address of the controller. This can be accomplished using DHCP, DNS, or IP subnet broadcast. This guide describes the DHCP method to convey the controller IP address. For other methods, see the product documentation. See also [Configuring DHCP Option 43](#) for more information.

17. What is the maximum negotiation rate supported by the IEEE 802.11g standard?

- A. 300Mbps
- B. 150Mbps
- C. 54Mbps
- D. 1200Mbps

Answer: C

Justification

Source: Huawei Slide # 16 (WLAN Overview), Page # 8

2.4GHz 、 5GHz

54Mbit/s

802.11a 、 802.11g

Wi-Fi 3

18. Which WLAN security policies support the link authentication method of Open mode? (Multiple Choice)

- A. WPA2-802.1X
- B. WPA2-PSK
- C. WEP
- D. WPA

Answer: ACD

Justification

Source: Huawei Slide # 16 (WLAN Overview), Page # 45



- As WLAN technologies use radio signals to transmit service data, service data can be easily intercepted or tampered with by attackers when being transmitted on open wireless channels. Ensuring WLAN security is crucial to building safe and effective wireless networks.
- Common security policy:

Security Policy	Link Authentication	Access Authentication	Data Encryption	Description
WEP	Open system	N/A	No encryption or WEP	Insecure policy
	Shared-key Authentication	N/A	WEP	Insecure policy
WPA/WPA2-802.1X	Open system	802.1X (EAP)	TKIP or CCMP	A more secure policy, applicable to large enterprises
WPA/WPA2-PSK	Open system	PSK	TKIP or CCMP	More secure policy, applicable to small- and medium-sized enterprises or household users

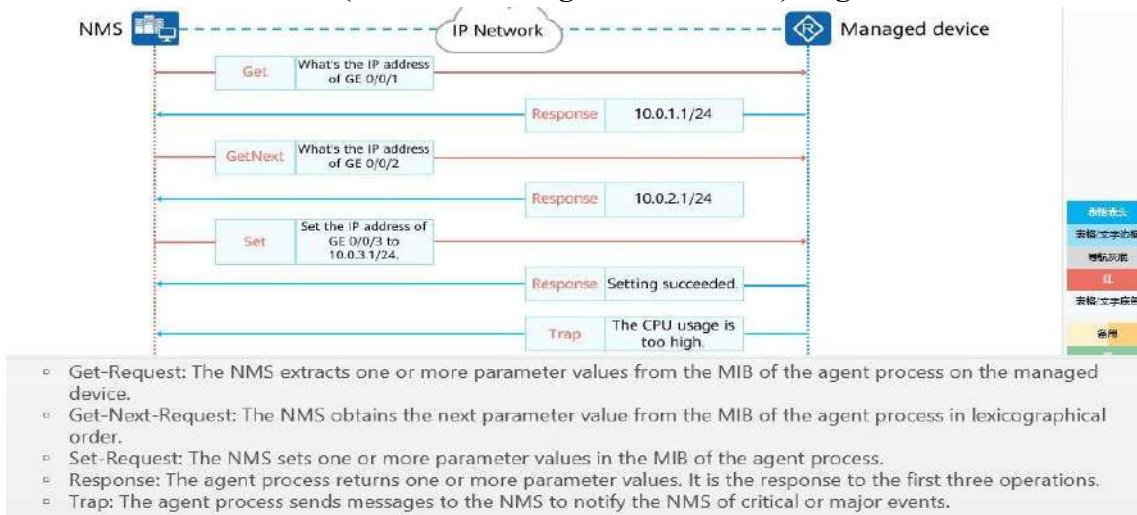
19. Which of the following SNMP packets are sent to the NMS by the Agent on the managed device?

- A. Set-Request
- B. Get-Request
- C. Get-Next-Request
- D. Response

Answer: D

Justification

Source: Huawei Slide # 18 (Network Management and OM) Page # 16

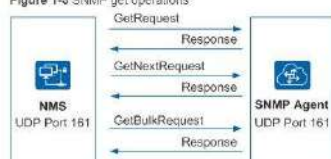


SNMP Get

The NMS can send get requests to an SNMP agent to obtain data, as shown in Figure 1-3. After receiving a get request, the SNMP agent executes the corresponding instruction in the MIB and sends the result to the NMS. SNMP get operations include Get, GetNext, and GetBulk. SNMPv1 does not support the GetBulk operation.

- Get: This operation allows the NMS to obtain one or more variables from the SNMP agent.
- GetNext: This operation allows the NMS to obtain one or more subsequent variables from the SNMP agent.
- GetBulk: This operation is equal to consecutive GetNext operations. You can set the number of GetNext operations to be included in one GetBulk operation.

Figure 1-3 SNMP get operations



20. Which field is added to the IPv6 packet header that is not added to the IPv4 packet header?

- A. Destination Address
- B. Flow Label
- C. Version
- D. Source Address

Answer: B

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178170/490ce67d/ipv6packet-format>

- Flow Label: 20 bits long. This field was added in IPv6 to differentiate traffic. A flow label and source IP address identify a data flow. Intermediate network devices can effectively differentiate data flows based on this field.

IPv4 packet header (20–60 bytes)

Version	IHL	ToS	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol	Head-Checksum		
Source Address				
Destination Address				
Options				Padding

Basic IPv6 header (40 bytes)

Version	Traffic Class	Flow Label	
Payload Length		Next Header	Hop Limit
Source Address			
Destination Address			
Deleted	Reserved	Name/Location changed	New

21. Which of the following statements about RA and RS in IPv6 are correct? (Multiple Choice)

- A. RS is used to reply address prefix information
- B. RA is used to request address prefix information
- C. RS is used to request address prefix information
- D. RA is used to reply address prefix information

Answer: CD

Justification

- Router Solicitation (RS) message
When a host is not configured with a unicast address, for example, when the host has just started, it sends an RS message. The RS message helps the host rapidly perform address autoconfiguration without waiting for the RA message that is periodically sent by an IPv6 routing device. An RS message is of ICMPv6 type 133.
- Router Advertisement (RA) message
Interfaces on each IPv6 routing device periodically send RA messages when they are enabled to do so. After receiving an RS message from an IPv6 node on the local link, a routing device responds with an RA message. The RA message is sent to the multicast address FF02::1 of all nodes or to the IPv6 unicast address of the node that sent the RS message. An RA message is of ICMPv6 type 134.
After an IPv6 host on the local link receives an RA message, it learns the updated default router list, prefix list, and other configurations from the message.

Source: Huawei Slide Lecture # 19 (IP v6 Basics), Page # 28

- DHCPv6 is used. An IPv6 client obtains a complete 128-bit IPv6 address, as well as other address parameters, such as DNS and SNTP server address parameter, from a DHCPv6 server.
- The DHCPv6 server records the allocation of the IPv6 address (this is where stateful comes).
- This method is complex and requires high performance of the DHCPv6 server.
- Stateful address configuration is mainly used to assign IP addresses to wired terminals in an enterprise, facilitating address management.
- For SLAAC, M = 0, O = 0:
 - ICMPv6 is used.
 - The router enabled with ICMPv6 RA periodically advertises the IPv6 address prefix of the link connected to a host.
 - Alternatively, the host sends an ICMPv6 RS message, and the router replies with an RA message to notify the link's IPv6 address prefix.

22. Telnetlib is a module that implements the Telnet protocol that comes with Python.

- A. True
- B. False

Answer: A

Justification

Source: Huawei Lecture # 21(Network Programmability and Automation) Page # 24



Introduction to telnetlib

- telnetlib is a module in the standard Python library. It provides the telnetlib.Telnet class for implementing the Telnet function.
- Different methods in the telnetlib.Telnet class are called to implement different functions.

Import the Telnet class of the telnetlib module. --

Create a Telnet connection to a specified server. --

Invoke the read_all() method. --

```
from telnetlib import Telnet
tn = Telnet(host=None, port=0[, timeout])
tn.read_all()

...
```

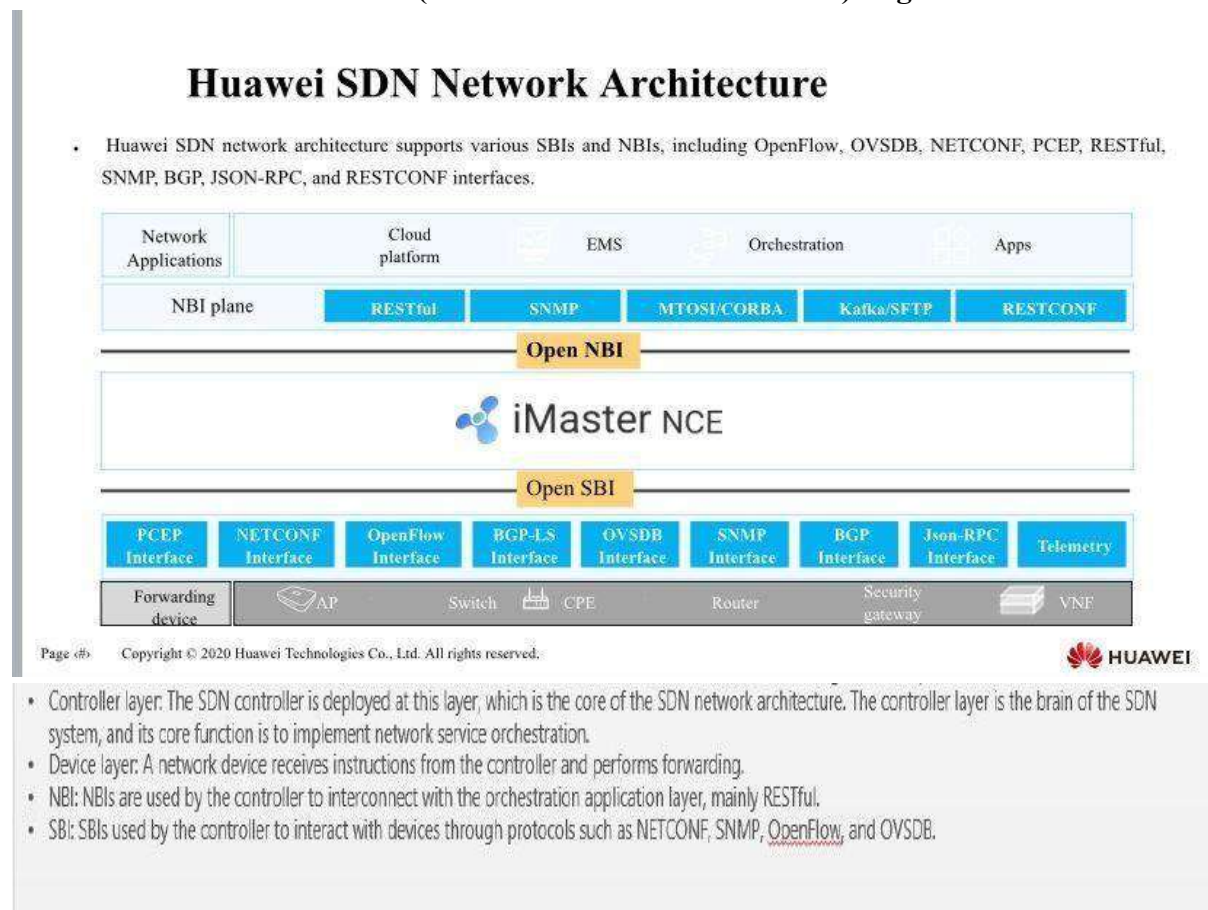
23. The controller is the core component of SDN. The controller connects to the device through the southbound interface. Which of the following belongs to the controller southbound protocol? (Multiple Choice)

- A. SNMP
- B. PCEP
- C. OpenFlow
- D. NETCONF

Answer: ACD

Justification

Source: Huawei Lecture # 20 (Introduction to SDN and NFV) Page # 19



24. In the campus network, network reliability can be improved by using link aggregation and stacking technology.

- A. True
- B. False

Answer: A

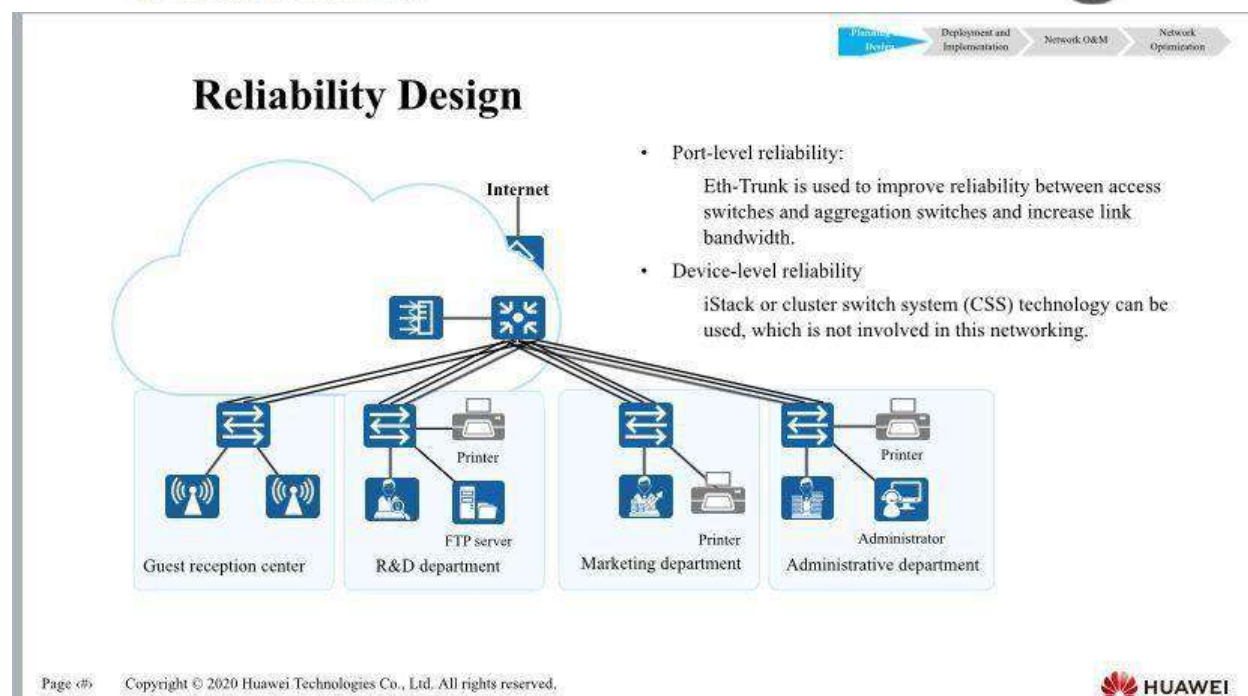
Justification

Source:

https://support.huawei.com/enterprise/en/doc/EDOC1100193624/20c00492/reliabilitydesign#EN-US_TOPIC_0276661504

On a campus network, two uplinks are usually used to improve the reliability of links between devices. In addition, for redundant links, Link Aggregation Control Protocol (LACP) is commonly used to virtualize multiple physical links into a logical Eth-Trunk. The interfaces on the Eth-Trunk are called Eth-Trunk interfaces. As shown in **Figure 2-150**, link aggregation has the following advantages:

- Increasing bandwidth: The maximum bandwidth of a link aggregation group (LAG) is the sum of bandwidth of the member interfaces in the LAG.
- Improving reliability: When an active link fails, the traffic carried over this failed link is switched to another functional active link, improving LAG reliability.
- Achieving load balancing: In a LAG, traffic is load balanced among all functional active links.



25. Using the command **mkdir test** in the VRP operating platform, the system will create a directory named test.

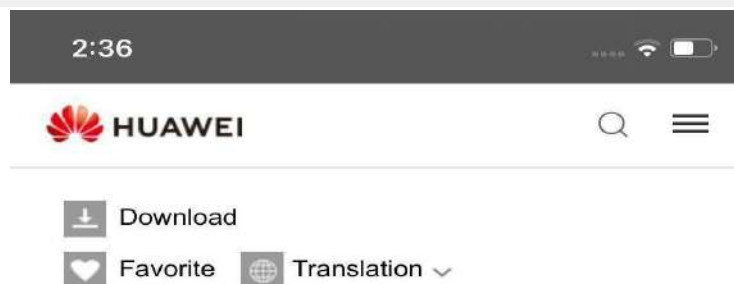
- A. True
- B. False

Answer: A

Justification

Source: Huawei Lecture # 3 (Huawei VRP) Page # 29

- VRP uses the file system to manage files and directories on a device. To manage files and directories, you often need to run basic commands to query file or directory information. Such commonly used basic commands include **pwd**, **dir [/all] [filename | directory]**, and **more [/binary] filename [offset] [all]**.
 - The **pwd** command displays the current working directory.
 - The **dir [/all] [filename | directory]** command displays information about files in the current directory.
 - The **more [/binary] filename [offset] [all]** command displays the content of a text file.
 - In this example, the **dir** command is run in the user view to display information about files in the flash memory.
- Common commands for operating directories include **cd directory**, **mkdir directory**, and **rmdir directory**.
 - The **cd directory** command changes the current working directory.
 - The **mkdir directory** command creates a directory. A directory name can contain 1 to 64 characters.



mkdir (User view)

Function

The **mkdir** command creates a directory in the current storage device.

Format

mkdir *directory*

26. As shown in the figure, after the two routers are configured with OSPF, the administrator configures the `<silent-interface s0/0/0>` command on RTA, then the correct description of the following is (). (Multiple Choice)



- A. The neighbor relationship between the two routers will be down
- B. The neighbor relationship between the two routers will not be affected
- C. RTA will no longer send OSPF packets
- D. RTA will continue to receive, analyze and process OSPF packets sent by RTB

Answer: BCD ACD

Justification

Explanation: When the silent-interface command is applied to an interface on RTA in an OSPF configuration, it causes RTA to stop sending OSPF packets, including Hello packets, which are crucial for establishing and maintaining OSPF neighbor relationships. However, RTA will still listen to and process OSPF packets received on that interface. Therefore, RTA will not form or maintain a neighbor relationship with RTB (option A), it will not send OSPF packets (option C), but it will continue to receive and process OSPF packets from RTB (option D). This confirms why B is incorrect and A, C, D are the correct responses.

Usage Scenario

To ensure that OSPF routing information cannot be obtained by the devices of a certain network and the local device does not receive routing update information advertised by other devices, you can run the **silent-interface** command to **disable an interface from receiving and sending OSPF packets**.

Disabling interfaces from receiving or sending OSPF packets is a solution to routing loops.

Configuration Impact

After an OSPF interface is set to be in the silent state, the interface can still advertise its direct routes. **Hello packets on the interface**, however, **will be blocked and no neighbor relationship can be established on the interface**. This can enhance the networking adaptability of OSPF and reduce the consumption of system resources.

27. Which of the following statements about the default route are correct? (Multiple Choice)

- A. If the destination address of the message cannot match any other routing entries in the routing table, the router will forward the message according to the default route
- B. There must be a default route in the routing table of any router
- C. The default route can only be manually configured by the administrator
- D. In the routing table, the default route appears as a route to the network 0.0.0.0 (the mask is also 0.0.0.0)

Answer: AD

Justification

Source: Huawei Lecture # 5 (IP Routing Basics), Page # 31

Default Routes

- Default routes are used only when packets to be forwarded do not match any routing entry in an IP routing table.
- In an IP routing table, a default route is the route to network 0.0.0.0 (with the mask 0.0.0.0), namely, 0.0.0.0/0.

RTA needs to forward packets to a subnet that is not directly connected to it and forwards the packets to 10.0.0.2.

```
[RTA] ip route-static 0.0.0.0 0 10.0.0.2
```

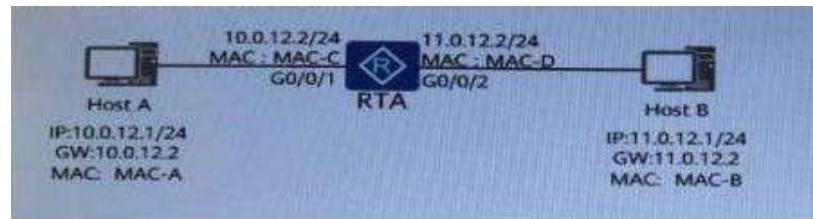
HUAWEI

Default Routes

Default routes are special routes used only when packets to be forwarded do not match any routing entry in a routing table. If the destination address of a packet does not match any entry in the routing table, the packet is sent through a default route. If no default route exists and the destination address of the packet does not match any entry in the routing table, the packet is discarded. An Internet Control Message Protocol (ICMP) packet is then sent, informing the originating host that the destination host or network is unreachable.

In a routing table, a default route is the route to network 0.0.0.0 (with the mask 0.0.0.0). You can run the **display ip routing-table** command to check whether a default route is configured. Generally, administrators can manually configure default static routes. Default routes can also be generated through dynamic routing protocols such as OSPF and IS-IS.

28. In the network shown in the figure below, the router receives a data packet with a destination IP address of 11.0.12.1 from host A. After this data packet is forwarded by the router, what are the destination MAC and destination IP?



- A. MAC-C, 11.0.12.1
 - B. MAC-D, 10.0.12.2
 - C. MAC-D, 11.0.12.1
 - D. MAC-B, 11.0.12.1
- IP never changes
MAC = next hop (changes every router)

Answer: C D, C is correct for source MAC

Justification

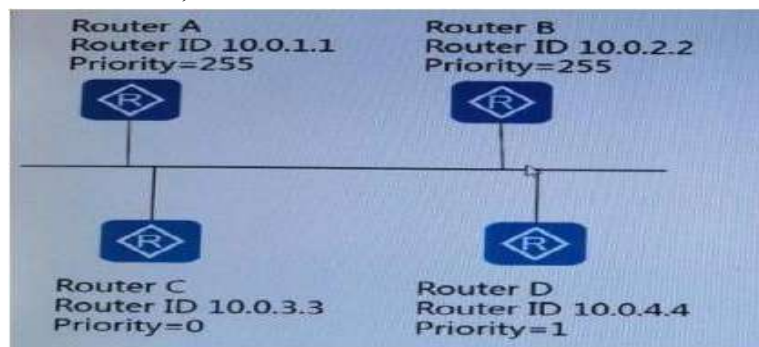
Source: Lecture # 5 (IP Routing Basics) Page # 31

Explanation: The correct answer is C: MAC-D, 11.0.12.1. This is because in a network where a router is used, when a packet needs to cross from one network to another (in this case from 10.0.12.1/24 to 11.0.12.1/24), the router's MAC address for the outbound interface (MAC-D) is used as the destination MAC in the forwarded packet.

Simple Explanation:
When Host A sends data to Host B (IP 11.0.12.1), the router receives it. The router checks where to send the packet next and uses its outgoing interface (G0/0/2, MAC D) to forward it. The **destination MAC** changes to the router's outgoing interface (MAC-D), but the **destination IP** stays the same (11.0.12.1).

When a router receives an IP packet, the router searches its routing table for the best matching route based on the destination IP address of the packet and forwards the packet through the outbound interface or next-hop IP address for the route. The routing table contains the routing entries (routes) learned by the router in various ways. The router can obtain routing entries in static or dynamic mode and maintain its own routing table.

29. In the network shown in the figure below, assuming that all routers enable the OSPF protocol at the same time, which router is the BDR in this network?



- A. Router D
- B. Router B
- C. Router C
- D. Router A

Answer: D

Justification

Explanation: For electing the DR and BDR, we consider the priority of the routers. In the above example, two routers have the highest priority, i.e., 255. Therefore, as both RTA and RTB have the same priority, we now have to consider the router ID. The router with the highest router ID will be considered as DR, which in this case is RTB. Whereas the router with second second-highest router ID will be selected as BDR, which in this case is the RTA.

30. There are multiple route entries to the same destination network in the routing table. These routes are called:

- A. Sub-optimal routing
- B. Multi-path routing
- C. Default route
- D. Equal cost routing

Answer: D

Justification

Source: Huawei Lecture # 05 (IP Routing Basics), Page # 39

Equal-Cost Route

Route Recursion Equal-Cost Route Floating Route Route Summarization

- When there are equal-cost routes in the IP routing table, a router forwards IP packets to be sent to the destination subnet through all valid outbound interfaces and next hops in the equal-cost routes, achieving load balancing.

RTA's IP routing table

Destination/Mask	Next Hop
10.0.0.0/30	20.1.1.2
	30.1.1.2

If there are multiple routes to the same destination from the same source, with the same cost, but pointing to different next hops, the routes are installed in the IP routing table as equal-cost routes. Traffic to be sent to the destination will be distributed to these equal-cost routes.

31. The function of the OSPF hello message is (). (Multiple Choice)

- A. Neighbor discovery
- B. Maintain neighbor relations
- C. Update LSA information
- D. Synchronize the LSDB of the router

Answer: AB

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100082074>

Table 1-1 Packet types

Packet Type	Function
Hello packet	Sent periodically to discover and maintain OSPF neighbor relationships.

Source: Huawei Slide # 06 (OSPF basics), Page # 18

 OSPF Packet Types

- There are five types of OSPF protocol packets and implement different functions in interaction between OSPF routers.

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.

32. The broadcast address is a special address in which the host bits in the network address are all set to 1, and it can also be used as a host address.

- A. True
- B. False

Answer: B

Justification

Source: Lecture 4(Network Layer Protocols and IP Addressing) Page.No.20

- Broadcast address: After the host part of this address is set to all 1s, the obtained result is the broadcast address used on the network where the IP address is located.
- Number of IP addresses: 2^n , where n indicates the number of host bits.
- Number of available IP addresses: $2^n - 2$, where n indicates the number of host bits.

- Broadcast address
 - The network ID is X, and each bit in the host ID is 1.
 - It cannot be assigned to a host interface.

- A network range defined by a network ID is called a network segment.

- **Network address:** identifies a network.

Example: 192.168.10.0/24

192.	168.	10.	00000000
------	------	-----	----------

- **Broadcast address:** a special address used to send data to all hosts on a network.

Example: 192.168.10.255/24

192.	168.	10.	11111111
------	------	-----	----------

- **Available addresses:** IP addresses that can be allocated to device interfaces on a network.

Example: 192.168.10.1/24

192.	168.	10.	00000001
------	------	-----	----------

Note

- Network and broadcast addresses cannot be directly used by devices or their interfaces.
- Number of available addresses on a network segment is $2^n - 2$ (n is the number of bits in the host part).

33. Which of the following cannot be the length of the IPv4 packet header?

- A. 32B
- B. 60B
- C. 20B
- D. 64B

Answer: D

Justification

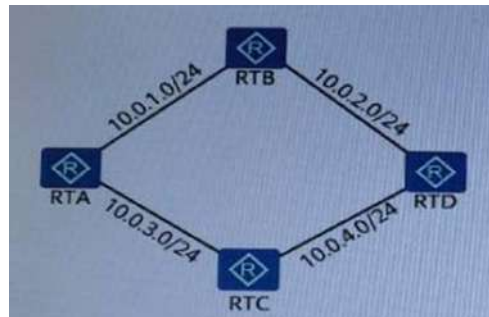
IPv4 packet header (20–60 bytes)				
Version	HLE	ToS	Total Length	
Identification			Flags	Fragment Offset
TTL	Protocol		Head-Checksum	
Source Address				
Destination Address				
Options				Padding

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Source: Lecture 4(Network Layer Protocols and IP Addressing) Page.No.8

- The IP packet header contains the following information:
 - Version: 4 bits long. Value 4 indicates IPv4. Value 6 indicates IPv6.
 - Header Length: 4 bits long, indicating the size of a header. If the Option field is not carried, the length is 20 bytes. The maximum length is 60 bytes.
 - Type of Service: 8 bits long, indicating a service type. This field takes effect only when the QoS differentiated service (DiffServ) is required.
 - Total Length: 16 bits long. It indicates the total length of an IP data packet.

34. In the network shown in the figure, all links are Ethernet links, and all interfaces of all routers run OSPF. How many DRS are elected in the entire network?



- A. 4
- B. 1
- C. 3
- D. 2

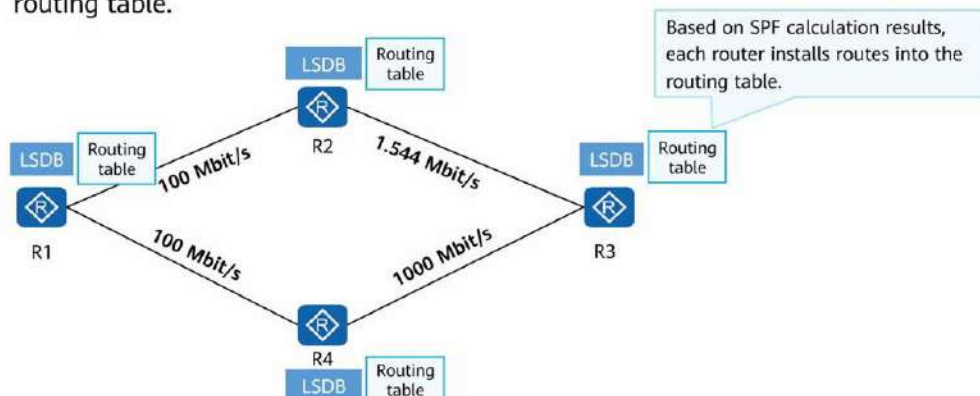
Answer: A

Justification

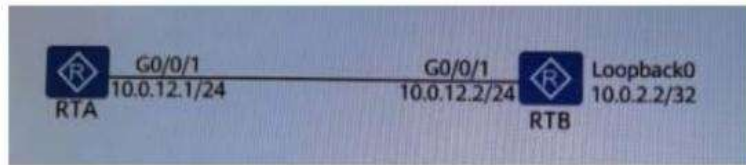
Explanation: In the OSPF (Open Shortest Path First) protocol, a Designated Router (DR) is elected on each multiaccess network segment, such as Ethernet. Since the network diagram shows that each link between the routers (RTA, RTB, RTC, RTD) represents a separate Ethernet link and each is considered a multi-access network, a DR will be elected on each of these links. Therefore, there are four separate segments, implying that each segment will elect its own DR. Hence, the total number of DRS elected across the network is 4.

Link-State Routing Protocol - Routing Table Generation

- Ultimately, the router installs routes for the calculated preferred paths into its routing table.



35. In the network shown in the figure below, if Router A and Router B's loopback O communicate through static routing, which command needs to be entered on Router A?



- A. `ip route-static 10.0.2.2 O GigabitEthernet O/O/O`
- B. `ip route-static 10.0.2.2 32 GigabitEthernet O/O/O`
- C. `ip route-static 10.0.2.2 255.255.255.255 10.0.12.1`
- D. `ip route-static 10.0.2.2 255.255.255.255 10.0.12.2`

Answer: D

Justification

Source: Lecture 5(IP Routing Basics) Page.No.29

Static Route Configuration

- Specify a next-hop IP address for a static route.

[Huawei] `ip route-static ip-address { mask } mask-length { nexthop-address`

Reason: In this case, the destination ip address is that of Loopback0 which is 10.0.2.2 with its mask of /32 with hop address of 10.0.12.2 of RTB to create a static route.

36. What is the priority value of the direct route entry?

- A. 0
- B. 30
- C. 10
- D. 20

Answer: A

Justification

Source: Lecture 5(IP Routing Basics) Page.No.14

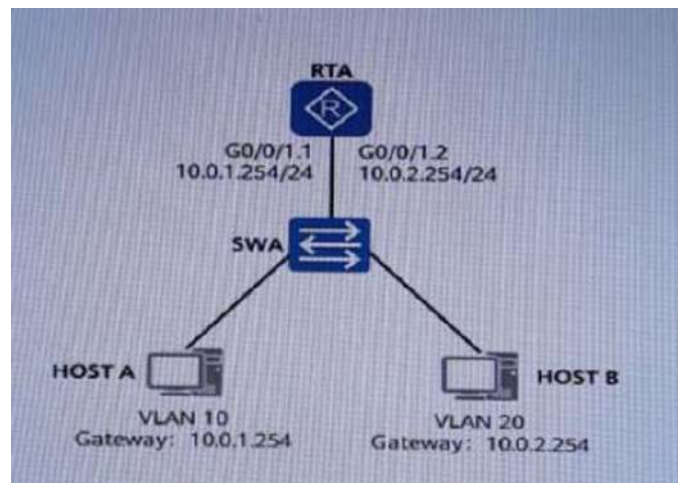
<Quidway> display ip routing-table
Route Flags: R - relay, D - download to fib

Routing Tables: Public
Destinations : 6 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.1/32	Static	60	0	D	0.0.0.0	NULL0
2.2.2.2/32	Static	60	0	D	100.0.0.2	Vlanif100
100.0.0.0/24	Direct	0	0	D	100.0.0.1	Vlanif100
100.0.0.1/32	Direct	0	0	D	127.0.0.1	Vlanif100
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Destination/Mask Protocol Route preference Cost (Metric) Flag Next-hop address Outbound interface

37. AS shown in the figure, if host A and host B want to achieve inter-VLAN communication through single-arm routing, which configuration should be done under the `GE0/0/1.1` interface of RTA?



- A. `dot1q termination vid 30`
- B. `dot1q termination vid 20`
- C. `dot1q termination vid 10`
- D. `dot1q termination vid 40`

Answer: C

Justification

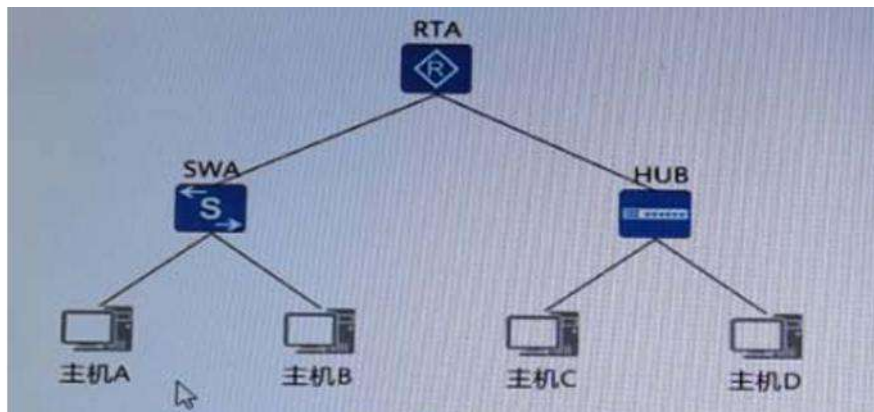
Source: Lecture 10(Inter VLAN Communication) Page.No.11 Reason:

- The **`dot1q termination vid`** command enables Dot1q VLAN tag termination for single-tagged packets on a sub-interface. By default, Dot1q VLAN tag termination for single-tagged packets is not enabled on sub-interfaces. The **`arp broadcast enable`** command enables ARP broadcast on a VLAN tag termination sub-interface. By default, ARP broadcast is not enabled on VLAN tag termination sub-interfaces. VLAN tag termination sub-interfaces cannot forward broadcast packets and automatically discard received ones. To allow a VLAN tag termination sub-interface to forward broadcast packets, run the **`arp broadcast enable`** command.

Contents	
1.	Background
2.	Using Routers' Physical Interfaces or Sub-interfaces to Implement Inter-VLAN Communication
3.	Using VLANIF Interfaces to Implement Inter-VLAN Communication
4.	Layer 3 Communication Process

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38. As shown in the figure, the correct description of this network is ().



- A. There are 6 collision domains in this network
- B. There are 2 broadcast domains in this network
- C. There are 6 broadcast domains in this network
- D. There are 12 conflict domains in this network

Answer: B

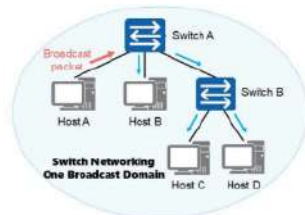
Hub = 1 collision domain for all ports; Switch = 1 collision domain per interface.
And for broadcast domains:
Switch breaks collision domains; Router breaks broadcast domains.

Justification

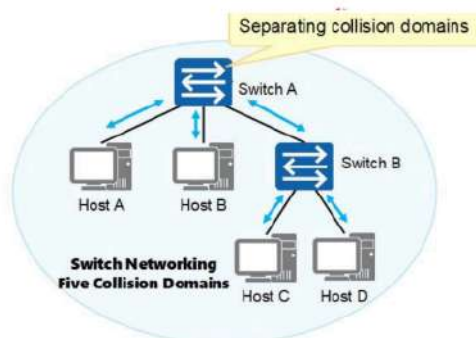
Reason:

The router separates these groups of devices into two broadcast domains because routers do not forward broadcasts between their interfaces. There are 2 broadcast domains in this scenario: □ One for the switch and its connected hosts (all ports on the switch are in the same VLAN by default).

- One for the hub and its connected hosts.



A switch forwards broadcast packets to all interfaces. Therefore, the nodes connected to all interfaces of the switch belong to the same broadcast domain.



39. The network administrator created VLAN 10 on the Layer 3 switch and configured an IP address under the virtual interface of the VLAN. When the command "display ip interface brief" is used to check the port status, it is found that the VLANIF 10 interface is in the down state. What should be done to restore the VLANIF 10 interface to normal?

- A. Add an interface that must be Up and must be of type Trunk into VLAN 10
- B. Assign any physical interface to VLAN 10
- C. Assign a physical interface that must be up to VLAN 10
- D. Use the command "undo shutdown" under the VLANIF 10 interface

Answer: D

Justification

Explanation: When we run the shutdown command on the VLANIF 10 interface, its state is changed to down. In order to recover it, run the undo shutdown command in the VLANIF interface view to start the VLANIF interface, and after that its state will be converted to UP and working.

40. The output information of the sub-interface of a router aggregate port 1 is shown in the figure. Based on this information, which of the following statements are correct? (Multiple Choice)

```
[Router A]display Interface Eth-Trunk 1.100
Eth-Trunk1.100 current state : UP
Line protocol current state : UP
Last line protocol up time : 2019-03-04 10:22:40 UTC -08 :00
Description: HUAWEI, AR Series, Eth-Trunk1.100 Interface
Route Port, Hash arithmetic : According to SIP-XOR-DIP, Maximal BW: 2G, Current BW: 2G, The
Maximum Transmit Unit is 1500
Internet Address is 10.0.12.2/24
IP Sending Frames' Format is PKTFIT ETENT 2.Hardware address 1s 00e0-fc3b-2015
Current system time: 2019-03-04 10:24:29 -08:00
Last 300 seconds input rate 0 bits/sec, 0 packets/sec
Last 300 seconds output rate 0 bits/sec, 0 packets/sec
Realtime 0 seconds input rate 0 bits/sec, 0 packets/sec
Realtime 0 seconds output rate 0 bits/sec, 0 packets/sec
Input: 6 packets, 574 bytes,
Output: 7 packets, 638 bytes,
Input bandwidth utilization : 0%
Output bandwidth utilization : 0%
```

PortName	Status	Weight
GigabitEthernet0/0/1	UP	1
GigabitEthernet0/0/2	UP	1

```
The Number of Ports in Trunk : 2
The Number of UP Ports in Trunk : 2
```

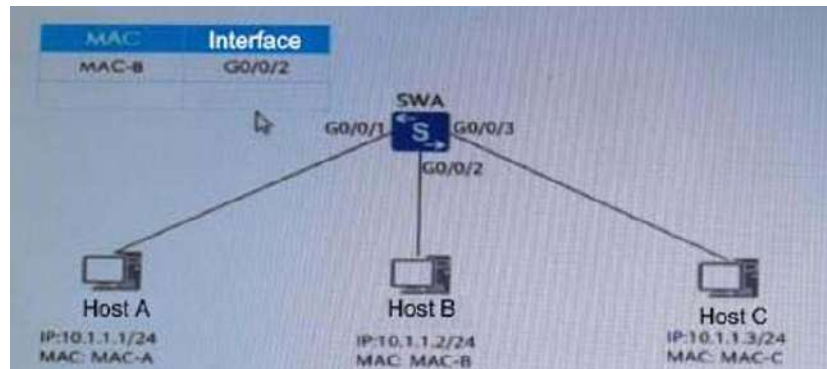
- A. Two links exist on the aggregate port
- B. The IP address of the sub-interface of the aggregate port is 10.0.12.2/24
- C. The sub-interface number of the aggregate port is 100
- D. The sub-interface of the aggregate port carries VLAN TAG 100 when forwarding data frames

Answer: ABC

Justification

Explanation: The output lists two active (GigabitEthernet0/0/1 and GigabitEthernet0/0/2) under the aggregate port, indicating two links are present. The output explicitly states that the IP address for EthTrunk1. is 10.0.12.2/24. The sub-interface number of the aggregate port is 100 - The interface name 'Eth-Trunk1.10 indicates the sub-interface number is 100.

41. As shown in the figure below, suppose the MAC address table of SWA is as follows. Now host A sends a data frame with the destination MAC address MAC-B. Which of the following statements is correct?



- A. Forward this data frame from port G0/0/2
- B. Forward this data frame port G0/0/3
- C. STA discards the data frame
- D. Flood this data frame

Answer: A

Justification

Explanation: When Host A sends a data frame to Host B, the switch (SWA) uses its MAC address table to determine the outgoing port. Since the MAC address table of SWA shows that the MAC address of Host B (MAC-B) is associated with port G0/0/2, the switch will specifically forward the data frame to this port.



MAC Address Table

Each device maintains a MAC address table. A MAC address table records the MAC address, VLAN ID and outbound interfaces learned from other devices. When forwarding a data frame, the device searches the MAC table for the outbound interface according to the destination MAC address and VLAN ID in the frame. This helps the device reduce broadcasting.

Packet Forwarding Based on the MAC Address Table

The device forwards packets based on the MAC address table in either of the following modes:

- Unicast mode: If the destination MAC address of a packet can be found in the MAC address table, the device forwards the packet through the outbound interface specified in the matching entry.
- Broadcast mode: If a packet is a broadcast or multicast packet or its destination MAC address cannot be found in the MAC address table, the device broadcasts the packet to all the interfaces in the VLAN except the inbound interface.

42. Which of the following is not included in AAA?

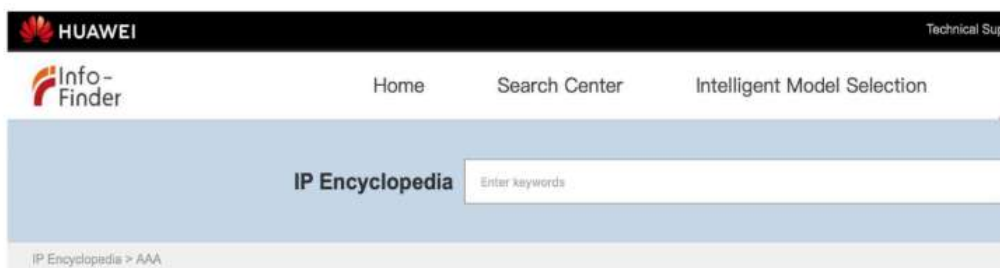
- A. Authentication
- B. Accounting
- C. Authorization
- D. Audit

Answer: D

Justification

Source: Slide Lecture # 13 (AAA Principles and Configuration) Page # 2

- Authentication, authorization, and accounting (AAA) is a management framework that provides a security mechanism for authorizing some users to access specified resources and recording the operations of these users. AAA is widely used because of its good scalability and easy implementation of centralized management of user information. AAA can be implemented through multiple protocols. In actual applications, the Remote Authentication Dial-In User Service (RADIUS) protocol is the most commonly used to implement AAA.
- This course describes the basic concepts, implementation, basic configurations, and typical application scenarios of AAA.



What Is AAA?

Authentication, Authorization, and Accounting (AAA) is a security management framework for network access control. It determines which users can access the network and which resources or services are available to authorized users. This document introduces the three elements of AAA and its implementation, used protocols, as well as applications.

43. On Huawei ARG3 series routers, what authentication modes does AAA support?

- A. 802.1x
- B. Local
- C. None
- D. HWTACACS

Answer: BCD

Justification

Source: Huawei Lecture # 13 (AAA Principles and Configuration), Page # 8

- The AAA authorization function grants users the permission to access specific networks or devices. AAA supports the following authorization modes:
 - Non-authorization: Authenticated users have unrestricted access rights on a network.
 - Local authorization: Users are authorized based on the domain configuration on the NAS.
 - Remote authorization: The RADIUS or HWTACACS server authorizes users.
 - In HWTACACS authorization, all users can be authorized by the HWTACACS server.
 - RADIUS authorization applies only to the users authenticated by the RADIUS server. RADIUS integrates authentication and authorization. Therefore, RADIUS authorization cannot be performed singly.
- When remote authorization is used, users can obtain authorization information from both the authorization server and NAS. The priority of the authorization information configured on the NAS is lower than that delivered by the authorization server.

Device

- Configuring Local Authentication and Authorization
- Configuring RADIUS AAA
- Configuring HWTACACS AAA
- Maintaining AAA
- Configuration Examples

NAC Configuration

ACL Configuration

DHCP Snooping Configuration

Local Attack Defense Configuration

IPSG Configuration

ARP Security Configuration

MFF Configuration

Traffic Suppression and Storm Control Configuration

AAA Features Supported by the Device

The device supports RADIUS or HWTACACS authentication, authorization, and accounting, and local authentication and authorization.

The device supports the combination of local, Remote Authentication Dial In User Service (RADIUS), and Huawei Terminal Access Controller Access Control System (HWTACACS) authentication, authorization, and accounting. For example, the device provides local authentication, local authorization, and RADIUS accounting. In practice, the following schemes are used separately:

- Local authentication and authorization

If users need to be authenticated or authorized but no RADIUS server or HWTACACS server is deployed on the network, use local authentication and authorization. Local authentication and authorization feature fast processing and low operation cost, whereas the amount of information that can be stored is limited by the device hardware capacity.

Local authentication and authorization are often used for administrators.
- RADIUS authentication and accounting

RADIUS protects a network from unauthorized access, which is often used on the networks demanding high security and remote user access control.
- HWTACACS authentication, authorization, and accounting

HWTACACS protects a network from unauthorized access and supports command-line authorization. Compared with RADIUS, HWTACACS is more reliable in transmission and encryption, and is more suitable for security control.

Multiple authentication or authorization modes can be used in a scheme. For example, local authentication is used as a backup of RADIUS authentication and HWTACACS authentication, and local authorization is used as a backup of HWTACACS authorization.

44. Which of the following protocols consists of the PPP protocol? (Multiple Choice)

- A. LCP
 - B. PPP0E
 - C. NCP
 - D. Authentication protocol
- PAP (Password Authentication Protocol)
CHAP (Challenge Handshake Authentication Protocol)

Answer: ACD

Justification

Source: <https://forum.huawei.com/enterprise/intl/en/thread/configuring-pppoe-users-connected-to-lac-headquarters-over-l2tp/667218205573267457?blogId=667218205573267457>

PPP has the following advantages:

- Supports both synchronous and asynchronous transmission.
- Good expansibility.
- Provides LCP for negotiation of various link layer parameters.
- Various NCPs are provided for the negotiation of network layer parameters.
- Provides authentication protocols, such as CHAP and PAP, to ensure network security.
- No retransmission mechanism, low network overhead, and high speed

45. Which of the following statements but Prefix is wrong?

- A. Prefix Segment spreads to network elements through the IGP, which is globally visible and effective
- B. The Prefix Segment needs to be configured manually.
- C. Prefix Segment is used to identify a destination address prefix in the network (prefix).
- D. Prefix Segment can be automatically allocated by the IGP

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000173015>

Label	Generation Mode	Function
Prefix Segment	Manually configured.	Identifies the prefix of a destination address. An IGP floods the prefix segment to the other NEs. The prefix segment is visible globally and takes effect globally. The prefix segment is identified by the prefix segment ID (SID). A prefix SID is an offset within the SRGB range and advertised by a source node. The receive end uses the local SRGB to compute label values and generate forwarding entries.

46. The output information of a certain device is shown in the figure. Which of the following statements is correct? (Multiple Choice)

```
<Huawei>display interface G1gabitEthernet 0/0/0
G1gabi tEthernet0/0/0 current state : UP
Line protocol current state : UP
Route Port, The Maximum Transmit Unit is 1000
Internet Address is 10.0.12.2/24
IP Sending Frames' Format is PRTFMT ETENT_2.Hardware address is 00e0- fc22- 4fff
Port Mode: FORCE COPPER
Speed : 100.Loopback: NONE
Duplex: FULL,Negotiation: DISABLE
Md1 : AUTO
Last 300 seconds input rate 24 bits/sec, 0 packets/sec
Last 300 seconds output rate 24 bits/sec, 0 packets/sec
```

- A. The IP address of the interface is 10.0.12.2/24
- B. The MAC address of the interface is 00e0-fc22-4fff
- C. The working rate of this interface is 1 Gbps
- D. The MTU value of this interface is 1000

Answer: ABD

Justification

Explanation: The IP address is explicitly list under the interface confirming the IP configuration. The MAC address is directly given as 'OeO-fc22-4fff," indicating the MAC address Of the interface. The Maximum Transmit Unit (MTU) specified for the interface is 10, clearly stated in the output, Option C is incorrect as the output mentions "Speed: 100," suggesting the interface is operating at Mbps, not 1 Gbps.

47. In the OSI reference complete end-to-end error detection and control?

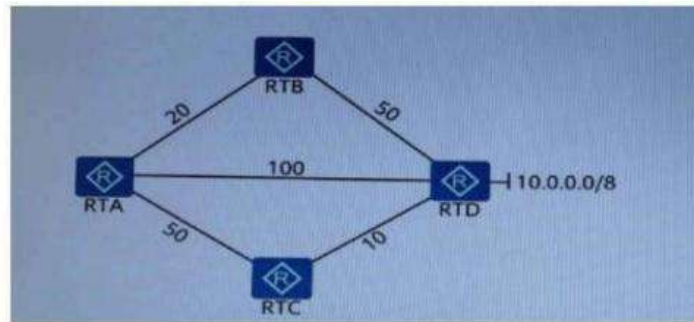
- A. Physical layer
- B. Data Enk layer perform error detection, but only at a local level
- C. Network layer
- D. Transport layer

Answer: D

Justification

- The OSI reference model is also called the seven-layer model. The seven layers from bottom to top are as follows:
 - Physical layer: transmits bit flows between devices and defines physical specifications such as electrical levels, speeds, and cable pins.
 - Data link layer: encapsulates bits into octets and octets into frames, uses MAC addresses to access media, and implements error checking.
 - Network layer: defines logical addresses for routers to determine paths and transmits data from source networks to destination networks.
 - Transport layer: implements connection-oriented and non-connection-oriented data transmission, as well as error checking before retransmission.

48. In the network shown in the figure below, all routers run OSPF protocol. Above the link is the value of the Cost value, then the path for RTA to reach the network 10.0.0.0/8 is:



- A. A-B-D
- B. RTA cannot reach 10.0.0.0/8
- C. A-C-D
- D. A-D

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100313446>

OSPF Route Selection Rules

OSPF supports multi-process. Multiple OSPF processes can run independently on the same router and do not affect one another. Route selection between multiple processes complies with the following rules:

1. When selecting the optimal route, the protocol preference of the two processes are checked first. The route with the highest protocol priority is selected as the optimal route.
2. If the protocol priorities of the two processes are the same, the cost values of the two routes are compared. The route with a smaller cost value is selected as the optimal route.

Reason: Since the path A-C-D has the lowest cost of 60, therefore the path for RTA to reach the network will be of A-C-D.

49. Which of the following statements about static routing on Huawei equipment is wrong?

- A. The cost value (Cost) of the static route cannot be modified.
- B. The default value of the static route is 60
- C. The range of the static priority value is 1-255
- D. When the priority of the static route is 0, the route will definitely be preferred.

Answer. C

Justification

preference <i>preference</i>	Specifies the preference of a static route. A smaller value indicates a higher preference.	The value is an integer that ranges from 1 to 255. The default value is 60.
--	--	---

50. Which of the following statements is correct?

- A. The router works at the physical layer
- B. The switch works at the physical layer
- C. The switch works at the network layer
- D. The router works at the network layer

Answer: D

Justification

Source: Lecture 2 (Network Referencing Model) Page.No.8

- The OSI reference model is also called the seven-layer model. The seven layers from bottom to top are as follows:
 - Physical layer: transmits bit flows between devices and defines physical specifications such as electrical levels, speeds, and cable pins.
 - Data link layer: encapsulates bits into octets and octets into frames, uses MAC addresses to access media, and implements error checking.
 - Network layer: defines logical addresses for routers to determine paths and transmits data from source networks to destination networks.

51. On interfaces that use PPP as the data link layer protocol, you can configure static routes by specifying the next hop address or outbound interface.

- A. True
- B. False

Answer. A

Justification

Source: Huawei Lecture # 5 (IP Routing Basics) Page # 29

Static Route Configuration

- Specify a next-hop IP address for a static route.

```
[Huawei] ip route-static ip-address { mask | mask-length } nexthop-address
```

- Specify an outbound interface for a static route.

```
[Huawei] ip route-static ip-address { mask / mask-length } interface-type interface-number
```

- Specify both the outbound interface and next hop for a static route.

```
[Huawei] ip route-static ip-address { mask / mask-length } interface-type interface-number [ nexthop-address ]
```

When creating a static route, you can specify both the outbound interface and next hop. Alternatively, you can specify either the outbound interface or next hop, depending on the interface type:

For a point-to-point interface (such as a serial interface), you must specify the outbound interface.

For a broadcast interface (for example, an Ethernet interface) or a virtual template (VT) interface, you must specify the next hop.

incorrect?

52. Which of the following description about the Layer 2 Ethernet switch is correct?

- A. It is necessary to make certain modifications to the layer 3 header of the forwarded message, and then forward it
- B. Able to learn MAC address
- C. Forward according to the Layer 2 header information of the Ethernet frame
- D. Layer 2 Ethernet switches work at the data link layer

Answer: A

Justification

Source: Lecture 7 (Ethernet Switching Basics) Page.No.23

- Layer 2 switches work at the data link layer and forward frames based on MAC addresses. Switch interfaces used to send and receive data are independent of each other. Each interface belongs to a different collision domain, which effectively isolates collision domains on the network.
- Layer 2 switches maintain the mapping between MAC addresses and interfaces by learning the source MAC addresses of Ethernet frames. The table that stores the mapping between MAC addresses and interfaces is called a MAC address table. Layer 2 switches look up the MAC address table to determine the interface to which frames are forwarded based on the destination MAC address.

53. As shown in the information in the figure, when learning this switch needs to forward the frame with the destination MAC address of 5489-98ec-f011, which of the following description is correct?

```
<Huawei>display mac-address
```

MAC Address	VLAN/VSI	Learned-From	Type
5489-98ec-f018	1/-	GE0/0/13	dynamic

Total items displayed = 1

- A. The switch will flood the frame on all ports except the port that received the frame
- B. The switch will send a message that the destination is unreachable to the source device
- C. The switch does not find a matching entry in the MAC address table, so it will discard the frame
- D. The switch needs to send a request to discover the device with the MAC address of 5489 98ec-f011

Answer: A

Justification

Source: Lecture 7 (Ethernet Switching Basics) Page.No.24

- A MAC address table records the mapping between MAC addresses and interfaces of other devices learned by a switch. When forwarding a frame, the switch looks up the MAC address table based on the destination MAC address of the frame. If the MAC address table contains the entry corresponding to the destination MAC address of the frame, the frame is directly forwarded through the outbound interface in the entry. If the MAC address table does not contain the entry corresponding to the destination MAC address of the frame, the switch floods the frame on all interfaces except the interface that receives the frame.

54. What is the range of VLAN ID that users can use?

- A. 1-4096
- B. 1-4094
- C. 0-4096
- D. 0-4095

Answer: B

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100278261/55c108dd/crea?ngand-dele?ng-a-vlan>

Create a VLAN and enter the VLAN view. If the VLAN already exists, this command will directly display the VLAN view. The VLAN ID ranges from 1 to 4094 (0 and 4095 are reserved).

```
vlan vlan-id
```

55. By default, in the SNMP protocol, which does the agent process use to send alarm messages to NS?

- A. 164
- B. 163
- C. 162
- D. 161

Answer: C

Justification

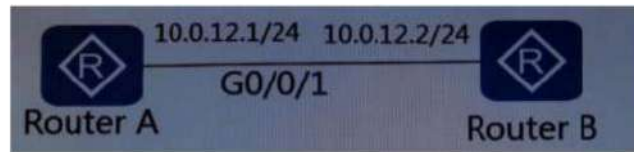
Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000060766/e35b53a6/snmpport-numbers>

- Port 162: This port number is used by the SNMP agent to send traps or inform messages to the NMS.

This port number is configurable. Ensure that the port number used by the SNMP agent to send traps or inform messages is the same as that used by the NMS to listen to traps or inform messages.

56. As shown in the figure below, router A has Telnet router B through IP address 10.0.12.2. In the current interface, which of the following operations will cause the Telnet session of Router A and Router B to be interrupted? (Multiple Choice)



- A. Modify the IP address G0/0/1 interface
- B. Close G0/0/1 interface
- C. Configure static routing
- D. Enable OSPF protocol under G0/0/1 interface

Answer: AB

Justification

Explanation: Modifying the IP address of the G0/0/1 interface (Option A) or closing it (Option B) directly disrupts the network path used by the Telnet session Router A and Router B. Changing the IP address alters the routing path, making the current connection invalid, while closing the interface terminates all data transmission, effectively severing the connection immediately. Both actions result in an immediate loss of connectivity for the Telnet session.

Source: Huawei Lecture # 6 (OSPF basics), Page # 17

Basic OSPF Concepts: Cost Value

Area
Router-ID
Cost Value

- OSPF uses costs as route metric values. Each OSPF-enabled interface maintains a cost value. Default cost value = $\frac{100 \text{ Mbit/s}}{\text{Interface bandwidth}}$, where, 100 Mbit/s is the default reference value specified by OSPF and is configurable.
- Generally, the cost of an OSPF route is the sum of costs of all inbound interfaces along a path from the destination network segment to the local router.

Cost Value of an OSPF Interface

Each OSPF interface has a specific cost because of the particular bandwidth value.

Accumulated Costs on an OSPF Path

In the routing table of R3, the cost of the OSPF route to 1.1.1.0/24 is 75 (10 + 1 + 64).

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57. What is the port number used by the FTP protocol control plane?

- A. 23
- B. 24
- C. 21
- D. 22

Answer: C

Justification

Source: Lecture 15 (Network Services and Applications) Page.No.7

- In active mode, the FTP client uses a random port (with the number greater than 1024) to send a connection request to port 21 of the FTP server. After receiving the request, the FTP server sets up a control connection with the FTP client to transmit control messages. In the meantime, the FTP client starts to listen on port P (another random port with the number greater than 1024) and uses the PORT command to notify the FTP server. When data needs to be transmitted, the FTP server sends a connection request from port 20 to port P of the FTP client to establish a TCP connection for data transmission.

58. There is no channel overlap in the 14 available frequency bands in the 2.4 GHz frequency band.

- A. True
- B. False

Answer: B

Justification

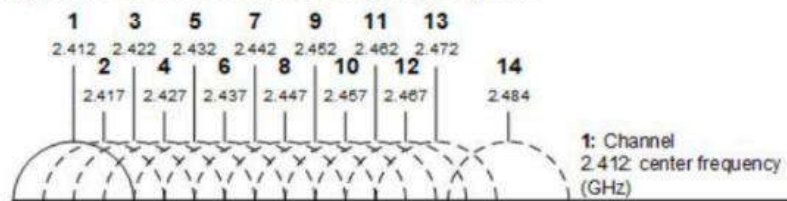
Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000113315/2d2a4a3c/frequency-bandand-channel>

2.4 GHz Frequency Band

The 2.4 GHz frequency band can be divided into fourteen 20 MHz channels, with the channel ID ranging from 1 to 14 and adjacent channels overlapping, as shown in Figure 8-7.

Figure 8-7 Channels on the 2.4 GHz frequency band



59. After the link authentication is completed, the STA will continue to initiate link negotiation. Which of the following messages is used to achieve the specific negotiation?

- A. Join
- B. Handshake
- C. Association
- D. Negotiate

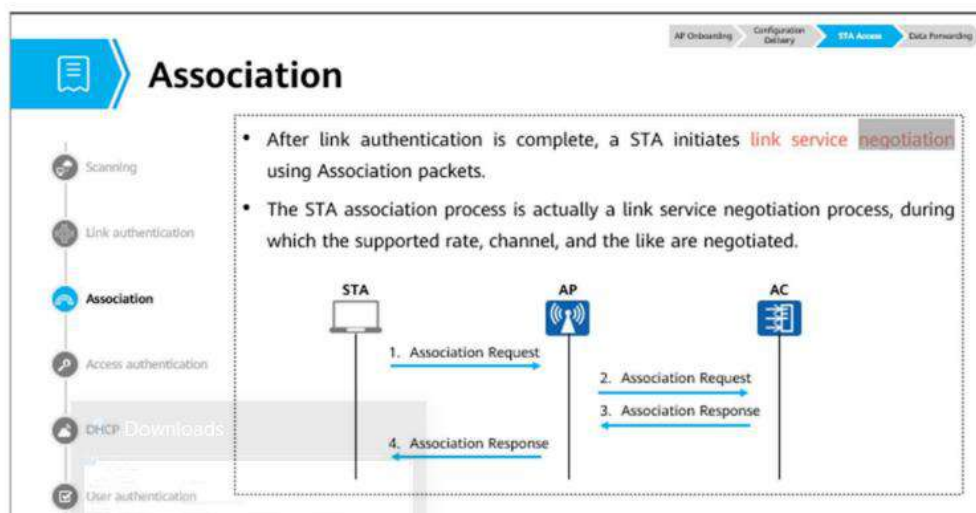
Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100033726/e88c00c0/sta-access>

Association

STA association is also known as link negotiation. After link authentication is complete, a STA initiates link negotiation using Association packets. Figure 4-15 shows the association process in the Fit AP architecture.



60. The Layer 2 networking of WLAN means that the STA uses the AP as its gateway.

- A. True uses the Access Controller (AC) as its gateway, not the Access Point (AP)
- B. False

Answer: A B

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000069520/82ee2e5b/standalone-ac-solution-core-switches-and-ac-function-as-the-gateways-for-wired-and-wireless-users-respectively>

Networking Requirements

Core switches set up a CSS that functions as the core of the entire campus network to implement high network reliability and for network.

Aggregation switches set up stacks to implement device-level backup and increase the interface density and forwarding bandwidth.

In this example, core switches and standalone ACs function as the gateways for wired and wireless users on the entire network.

Figure 2-13 Core switches and standalone ACs functioning as the gateways for wired and wireless users respectively

61. What is the IEEE802.11 standard corresponding to Wi-Fi 6?

- A. 802.11au
- B. 802.11ax
- C. 802.11at
- D. 802.11ac

Answer: B

Justification

What Is Wi-Fi 6?

Wi-Fi 6 is short for the next-generation 802.11ax standard. With the evolution of Wi-Fi standards, Wi-Fi Alliance (WFA) renames Wi-Fi standards using sequence numbers to help Wi-Fi users and device vendors easily learn about their connected and supported Wi-Fi device models. Additionally, the new naming convention is to better highlight the significant progress of Wi-Fi technology. The latest standard supports a large number of new functions to provide higher throughput, faster speed, and more concurrent connections. **Table 1-1** lists the new naming conventions for 802.11 standards announced by WFA.

**62. Which of the following working modes are supported by Huawei Enterprise AP?
(Multiple Choice)**

- A. Local
- B. Cloud
- C. FIT
- D. FAT

Answer: BCD

Justification

Source: Lecture 16 (WLAN Overview) Page.No.11

- The AP can switch flexibly among the Fat, Fit, and cloud modes based on the network plan.
- Fat AP: applies to home WLANs. A Fat AP works independently and requires separate configurations. It provides only simple functions and is cost-effective. The Fat AP independently implements functions such as user access, authentication, data security, service forwarding, and QoS.
- Fit AP: applies to medium- and large-sized enterprises. Fit APs are managed and configured by the AC in a unified manner, provide various functions, and have high requirements on network maintenance personnel's skills. Fit APs must work with an AC for user access, AP going-online, authentication, routing, AP management, security, and QoS.
- Cloud AP: applies to small- and medium-sized enterprises. Cloud APs are managed and configured by a cloud management platform in a unified manner, provide various functions, support plug-and-play, and have low requirements on network maintenance personnel's skills.

63. Which of the following statements Adjacency Segment is wrong?

- A. Adjacency Segment is identified by Adjacency Segment ID (SID)
- B. Adjacency Segment spreads to other network elements through IGP protocol, which is visible globally and effective locally
- C. The adjacency Segment must be configured manually
- D. Adjacency Segment is used to identify the adjacency of a node in the network

Answer: C

Justification

Adjacency segment	Dynamically allocated by the source node through a protocol or manually configured	Identifies an adjacency on a network. Adjacency segments are propagated to other devices through an IGP. They are visible to all the devices but effective only on the local device. Each adjacency segment is identified by an adjacency SID, which is a local SID out of the SRGB.
-------------------	--	--

64. If during the PPP authentication process, the authenticator sends the wrong name to the authenticator, what type of message will the authenticator send to the authenticator?

- A. Authenticate-Reject
- B. Authenticate-Ack
- C. Authenticate-Nak
- D. Authenticate-Reply

Answer: C

Justification

Source: Lecture 17 (WAN Technologies) Page.No.19

- The peer sends the configured username and password to the authenticator in clear text through an Authenticate-Request packet.
- After receiving the username and password from the peer, the authenticator checks whether the username and password match those in the locally configured database. If they match, the authenticator returns an Authenticate-Ack packet, indicating that the authentication is successful. If they do not match, the authenticator returns an Authenticate-Nak packet, indicating that the authentication is unsuccessful.

Code	Name	Content
0x01	Configure-Request	Configuration request packet.
0x02	Configure-Ack	Configuration success packet.
0x03	Configure-Nak	Configuration parameters need to be negotiated.
0x04	Configure-Reject	Configuration parameters cannot be identified.

65. Which version of the SNMP protocol below supports encryption features?

- A. SNMPv3
- B. SNMPv2c
- C. SNMPv1
- D. SNMPv2

Answer: A

Justification

Table 9-3 Comparison of security in different SNMP versions

Protocol Version	User Checksum	Encryption
v1	No. Uses a community name.	No
v2c	No. Uses a community name.	No
v3	Yes. User-name-based verification.	Yes

66. SNMPv1 defines 5 protocol operations.

- A. True
- B. False

Answer: A

Justification

Source: Lecture 18 (Network Management and OM) Page.No.16

- SNMPv1 defines five protocol operations.
 - Get-Request: The NMS extracts one or more parameter values from the MIB of the agent process on the managed device.
 - Get-Next-Request: The NMS obtains the next parameter value from the MIB of the agent process in lexicographical order.
 - Set-Request: The NMS sets one or more parameter values in the MIB of the agent process.
 - Response: The agent process returns one or more parameter values. It is the response to the first three operations.
 - Trap: The agent process sends messages to the NMS to notify the NMS of critical or major events.

67. The IPv6 address ABEF: 224E: FFE2: BCCO: COO: DDBE: 8058 cannot be abbreviated.

- A. True
- B. False

Answer: A

Justification

Reason: There are no sequences of consecutive zero sections (e.g., 0000:0000) that could be replaced with '::'. Therefore, the only possible abbreviation is the removal of a leading zero from any sections that have them. In this case, the sections either don't have leading zeros to remove or, like CD00, the zero is not leading but a part of the significant value. Hence, it cannot be abbreviated.

- For convenience, IPv6 can be abbreviated according to the following rules.

Abbreviation Specifications	Abbreviation Examples
2001 : 0DB8 : 0000 : 0000 : 0008 : 0800 : 200C : 417A The leading 0s in each 16-bit segment can be omitted. However, if all bits in a 16-bit segment are 0s, at least one 0 must be reserved. The trailing 0s cannot be omitted.	Before 0000:0000:0000:0000:0000:0000:0000:0001 After ::1
2001 : DB8 : 0 : 0 : 8 : 800 : 200C : 417A If one or more consecutive 16-bit segments contain only 0s, a double colon (::) can be used to represent them, but only one :: is allowed in an entire IPv6 address.	Before 2001:0DB8:0000:0000:FB00:1400:5000:45FF After 2001:DB8:FB00:1400:5000:45FF
2001 : DB8 : :: : 8 : 800 : 200C : 417A If an abbreviated IPv6 address contains two double colons (::), the IPv6 address cannot be restored to the original one.	Before 2001:0DB8:0000:0000:0000:2A2A:0000:0001 After 2001:DB8::2A2A:0:1 Before 2001:0DB8:0000:1234:FB00:0000:5000:45FF After 2001:DB8::1234:FB00:0:5000:45FF or 2001:DB8:0:1234:FB00::5000:45FF

68. The DHCPv6 server supports providing other configuration information as a DNS server address, for the host.

- A. True
- B. False

Answer: A

Justification

- DHCPv6 stateful autoconfiguration: DHCPv6 servers automatically provide IPv6 addresses, PD prefixes, and other network configuration parameters, such as addresses of the DNS, NIS, and SNTP servers.
- DHCPv6 stateless autoconfiguration: IPv6 addresses are generated based on RA packets. A DHCPv6 server does not provide IPv6 addresses but provides other configuration parameters about the DNS, NIS, and SNTP servers.

69. Which planes of the following does a network device have? (Multiple choice)

- A. Control plane
- B. Business plane
- C. Data plane
- D. Management plane

Answer: ACD

Justification

Source: Lecture 20 (Introduction to SDN and NFV) Page.no.7

- The switch is used as an example to describe the forwarding plane, control plane, and management plane.
- Forwarding plane: provides high-speed, non-blocking data channels for service switching between service modules. The basic task of a switch is to process and forward various types of data on its interfaces. Specific data processing and forwarding, such as Layer 2, Layer 3, ACL, QoS, multicast, and security protection, occur on the forwarding plane.
- Control plane: provides functions such as protocol processing, service processing, route calculation, forwarding control, service scheduling, traffic statistics collection, and system security. The control plane of a switch is used to control and manage the running of all network protocols. The control plane provides various network information and forwarding query entries required for data processing and forwarding on the data plane.
- Management plane: provides functions such as system monitoring, environment monitoring, log and alarm processing, system software loading, and system upgrade. The management plane of a switch provides network management personnel with Telnet, web, SSH, SNMP, and RMON to manage devices, and supports, parses, and executes the commands for setting network protocols. On the management plane, parameters related to various protocols on the control plane must be pre-configured, and the running of the control plane can be intervened if necessary.
- Some Huawei series products are divided into the data plane, management plane, and monitoring plane.

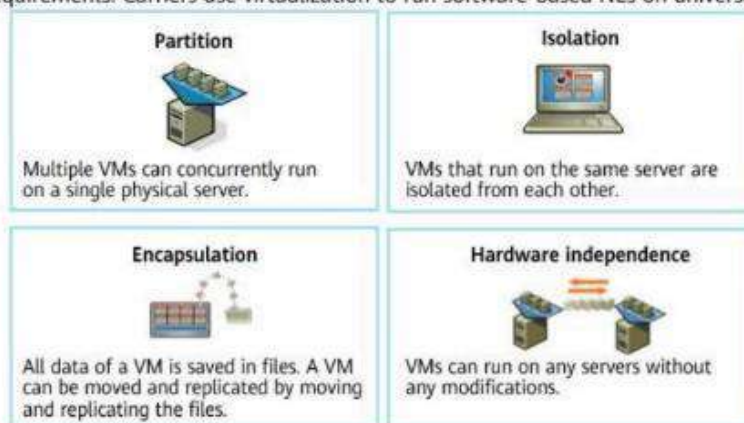
70. Virtualization is the basis for the realization of NEV. Which of the following statements about the characteristics of virtualization is correct? (Multiple choice)

- A. Hardware independence: virtual machine and hardware support virtual machine servers.
- B. Encapsulation: All virtual machines share the operating system, and encapsulation is through name spaces.
- C. Isolation: The virtual machines on the same physical server are isolated from each other.
- D. Partition: You can run multiple virtual machines on a single physical server at the same time.

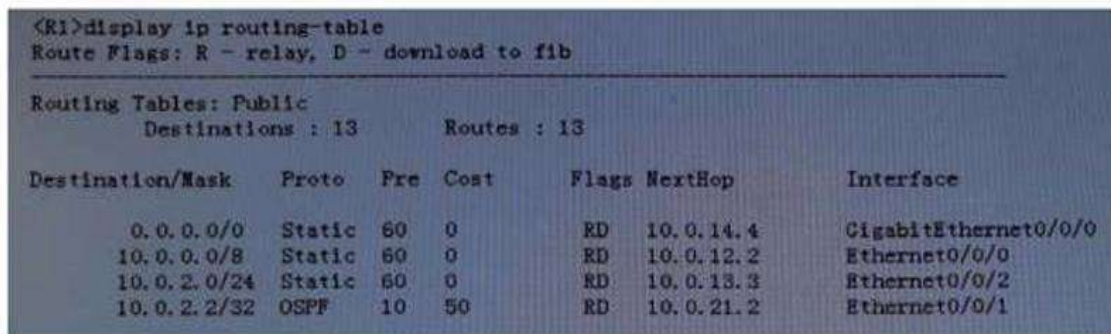
Answer: ACD

Justification

- On traditional telecom networks, each NE is implemented by dedicated hardware, resulting in high costs and difficult O&M. Virtualization features partition, isolation, encapsulation, and independence from hardware, which can meet NFV requirements. Carriers use virtualization to run software-based NEs on universal infrastructures.



71. The output information of the router R1 routing table is shown in the figure. Which of the following statements is correct? (Multiple Choice)



```
<R1>display ip routing-table
Route Flags: R - relay, D - download to fib

Routing Tables: Public
Destinations : 13      Routes : 13

Destination/Mask    Proto    Pre    Cost    Flags NextHop         Interface
0.0.0.0/0           Static   60     0        RD    10.0.14.4      GigabitEthernet0/0/0
10.0.0.0/8           Static   60     0        RD    10.0.12.2      Ethernet0/0/0
10.0.2.0/24          Static   60     0        RD    10.0.13.3      Ethernet0/0/2
10.0.2.2/32          OSPF     10     50       RD    10.0.21.2      Ethernet0/0/1
```

- A. The router will discard data packets whose destination network is 11.0.0.0/8
- B. The router will forward data packets whose destination network is 12.0.0.0/8
- C. The data packet whose destination network is 11.0.0.0/8 will be forwarded from the router's GigabitEthernet0/0/0 interface
- D. The data packet whose destination network is 12.0.0.0/8 will be forwarded from the router's Ethernet0/0/0 interface

Answer: BC

Justification

Explanation:

- **Option B: The router will forward data packets whose destination network is 12.0.0.0/8.** This is assumed correct based on typical routing behavior where packets are forwarded unless explicitly denied or unmatched.
- **Option C: The data packet whose destination network is 11.0.0.0/8 will be forwarded from the router's GigabitEthernet0/0/0 interface.** This is directly confirmed by the routing table showing an entry for 11.0.0.0/8 with the next hop via GigabitEthernet0/0/0.

**72. Which of the following descriptions about the Router ID of OSPF is incorrect?
(Multiple Choice)**

- A. Router ID must be the IP address of an Interface on the router
- B. The prerequisite for the normal operation of the OSPF protocol is that the router has a Router ID (Its correct not incorrect)
- C. The Router ID must be manually configured
- D. The Router ID must be the same in the same area, and the Router ID in different areas can be different

Answer: ABCD

Justification

You can configure OSPF to look up Domain Naming System (DNS) names for use in all OSPF show EXEC command displays. You can use this feature to more easily identify a router, because the router is displayed by name rather than by its **router ID** or neighbor ID.

OSPF uses the largest IP address configured on the interfaces as its **router ID**. If the interface associated with this IP address is ever brought down, or if the address is removed, the OSPF process must recalculate a new **router ID** and resend all its routing information out its interfaces.

If a loopback interface is configured with an IP address, the Cisco IOS software will use this IP address as its **router ID**, even if other interfaces have larger IP addresses. Because loopback interfaces never go down, greater stability in the routing table is achieved.

Router ID

A router requires a router ID if it is to run OSPF. A router ID is a 32-bit unsigned integer, uniquely identifying a router in an AS.

A router ID can be manually configured or automatically selected by a router:

- It is recommended that you manually configure a router ID to ensure OSPF stability.
- If no router ID is manually configured for a router, the router selects an interface IP address as its router ID. A router selects its router ID based on the following rules:
 1. The router preferentially selects the largest IP address among loopback interface addresses as its router ID.
 2. If no loopback interface is configured, the router selects the largest IP address among other interface addresses as its router ID.

73. Which of the following statements about direct routing is correct?

- A. Direct routing requires the administrator to manually configure the destination network and the next-hop address
- B. The direct route has lower priority than the static route
- C. Direct route priority is lower than dynamic route
- D. Direct routing has the highest priority.

Answer: D

Justification

- Pre (Preference): indicates the routing protocol preference of the route. There may be multiple routes to the same destination, which have different next hops and outbound interfaces. These routes may be discovered by different routing protocols or be manually configured. A router selects the route with the highest preference (with the lowest preference value) as the optimal route.

Table 2-1 Routing protocols and their default external preferences

Routing Protocol or Route Type	Default External Preference
Direct	0
OSPF	10
IS-IS	15
Static	60
RIP	100
OSPF ASE	150
OSPF NSSA	150
IBGP	255
EBGP	255

74. What are the advantages of the OSPF protocol? (Multiple Choice)

- A. Division of support areas
- B. OSPF supports load sharing for equal-cost routes
- C. OSPF supports message authentication
- D. OSPF supports type-less inter-domain routing (CIDR)

Answer: ABCD

Justification

Why Do We Need OSPF?

Before the emergence of OSPF, the Routing Information Protocol (RIP) was the most widely used IGP.

RIP is a distance-vector routing protocol which is gradually being replaced by OSPF, due to the former's slow convergence, tendency to form routing loops, and poor scalability.

OSPF is a link-state routing protocol featuring:

- **Multicast** packet transmission, reducing load on the routers that do not run OSPF
- Support for Classless Inter-Domain Routing (CIDR)
- Load balancing among equal-cost routes
- Packet authentication

With the preceding advantages, OSPF is widely accepted and used as an IGP.

75. The route entry 10.0.0.24/29 may be aggregated from which of the following subnet routes? (Multiple Choice)

- A. 10.0.0.26/30
- B. 10.0.0.23/30
- C. 10.0.0.24/30
- D. 10.0.0.28/30

Answer: ACD

Justification

Explanation:

- The route entry 10.0.0.24/29 covers the IP range from 10.0.0.24 to 10.0.0.31.
- Analyzing the options subnets 10.0.0.26/30, 10.0.0.24/30, and 10.0.0.28/30 fit entirely within this range.
- Therefore, the correct options are **A, C, D**.

76. When OSPF routers form a 2-WAY relationship, the LSDB has been synchronized.

neighbor relationship formed, not full adjacency

- A. True
- B. False

Answer: B

Justification

- After the neighbor status changes to Loading, R1 sends an LSR to R2 to request the LSAs that are discovered through DD packets in the Exchange state but do not exist in the local LSDB.
- After receiving the LSU, R2 sends an LSU to R1. The LSU contains detailed information about the requested LSAs.
- After R1 receives the LSU, R1 replies with an LSAck to R2.
- During this process, R2 also sends an LSA request to R1. When the LSDBs on both ends are the same, the neighbor status changes to Full, indicating that the adjacency has been established successfully.

77. If the Protocol of a certain route in the routing table is Direct, the priority of this route is set to 0.

- A. True
- B. False

Answer: A

Justification

Source: Lecture 5 (IP Routing Basics) Page.No.14

<Quidway> display ip routing-table
Route Flags: R - relay, D - download to fib

Routing Tables: Public
Destinations : 6 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
1.1.1.1/32	Static	60	0	D	0.0.0.0	NULL0
2.2.2.2/32	Static	60	0	D	100.0.0.2	Vlanif100
100.0.0.0/24	Direct	0	0	D	100.0.0.1	Vlanif100
100.0.0.1/32	Direct	0	0	D	127.0.0.1	Vlanif100
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

Destination/Mask Protocol Route preference Cost (Metric) Flag Next-hop address Outbound interface

78. What are the network types supported by the OSPF protocol? (Multiple Choice)

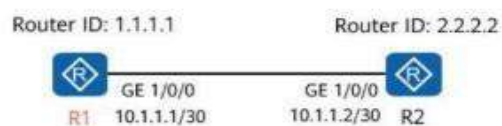
- A. Point-to-Point
- B. Non-Broadcast Multi-Access
- C. Point-to-Multi-point
- D. Broadcast

Answer: ABCD

Justification

Source: Lecture 6(OSPF BASICS) Page.No.29

- The default OSPF network type of an interface depends on the data link layer encapsulation used by the interface.
- As shown in the figure, OSPF has four network types: broadcast, NBMA, P2MP, and P2P.



[R1-GigabitEthernet1/0/0] <code>ospf network-type ?</code>	
<code>broadcast</code>	Specify OSPF broadcast network
<code>nbma</code>	Specify OSPF NBMA network
<code>p2mp</code>	Specify OSPF point-to-multipoint network
<code>p2p</code>	Specify OSPF point-to-point network

79. The election of the root switch in STP only compares the switch priority, while in RSTP, the switch priority and MAC address are compared at the same time.

- A. True
- B. False

Answer: B

Justification

Source: Lecture 9 (STP Principles and Configuration) Page.No.19

Root Bridge

- One of the main functions of STP is to calculate a loop-free STP tree on the entire switching network.
- The root bridge is the root of an STP network.
- After STP starts to work, it elects a root bridge on the switching network. The root bridge is the key for topology calculation of the spanning tree and is the root of the loop-free topology calculated by STP.
- On an STP network, the device with the smallest BID acts as the root bridge.

During BID comparison, devices first compare bridge priorities. A smaller priority value indicates a higher priority of a device. The switch with the smallest priority value becomes the root bridge. If priority values are the same, the switch with the smallest MAC address becomes the root bridge.

80. Which of the following is not the reason for RSTP improving the convergence speed?

- A. Introduction of edge ports
- B. Fast switching of the root port
- C. P/A mechanism
- D. Forward Delay canceled

Answer: D

Justification

In this manner, RST BPDUs are flooded on the network.

RSTP provides rapid convergence using the Proposal/Agreement Mechanism, Fast Switchover of the Root Port, and Edge Port.

81. The IP addresses of the VLANIF interfaces of the same switch cannot be the same.

- A. True
- B. False

Answer: A

Justification

Each IP address configured on a VLAN interface must be unique in the switch it cannot be used by a VLAN interface or another loopback interface.

82. The MAC address table of the existing switch is shown in the figure. Which of the following statements are correct?

```
<Huawei>display mac-address
MAC address table of slot 0:
```

MAC Address	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR-ID MAC-Tunnel
5489-9811-0b49	1	-	-	Eth0/0/3	static	-

Total matching items on slot 0 displayed = 1

```
MAC address table of slot 0:
```

MAC Address	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR-ID MAC-Tunnel
5489-989d-1d30	1	-	-	Eth0/0/1	dynamic	0/-
5489-9885-18a8	1	-	-	Eth0/0/2	dynamic	0/-

Total matching items on slot 0 displayed = 2

- A. When the switch restarts, the MAC address learned by port Eth0/2 does not need to be relearned.
- B. When the switch restarts, the MAC address learned by port Eth0/3 needs to be learned again.
- C. The data frame with the source MAC address of 5489-9811-bd49 and the destination MAC address of 5489-989d-1d30 received from the port is forwarded from the Eth0/2 port.
- D. The data frame with the source MAC address of 5489-9885-18a8 and the destination MAC address of 5489-989d-1d30 received from the port is forwarded from the Eth0/1 port.

Answer: D

Justification

Explanation: This statement would be true as the MAC address table shows that the destination MAC address 5489-989d-1d30 is associated with Eth0/1. If the switch knows that the destination MAC is located on Eth0/1, it will forward the frame through that port.

83. The MAC address table of a certain switch is shown in the figure. If the switch receives a data frame with a destination on MIAC of 5489-9885-18a8 from port Eth0/0/2, which of the following statements is correct?

```
<Huawei>display mac-address
MAC address table of slot 0:
```

MAC Address	VLAN/ VSI/SI	FEVLAN	CEVLAN	Port	Type	LSP/LSR-ID MAC-Tunnel
5489-9885-18a8	1	-	-	Eth0/0/2	dynamic	0/-
5489-989d-1d30	1	-	-	Eth0/0/1	dynamic	0/-

Total matching items on slot 0 displayed = 2

- A. Forward this data frame from Eth0/0/1 port
- B. Flood this data frame
- C. Discard this data frame
- D. Forward this data frame from the Eth0/0/2 port

Answer: D

Justification

Explanation: This statement would be true as the MAC address table shows that the destination MAC address 5489 9885-18a8 is associated with Eth0/0/2. If the switch knows that the destination on the MAC is located on Eth0/0/2, it will forward the frame through that port.

84. The switch G0/0/1 port configuration information is shown in the figure. Which VLAN data frame does the switch forward without a VLAN tag?

```
#  
interface GigabitEthernet0/0/1  
port link-type trunk  
port trunk pvid vlan 10  
port trunk allow-pass vlan 10 20 30 3040  
#
```

- A. 10
- B. 30
- C. 20
- D. 40

Answer: A

Justification

Explanation: The correct answer is **10**, because VLAN 10 is explicitly configured as the native VLAN on this trunk port, and thus, it is the VLAN whose traffic is forwarded without a VLAN tag.

85. When the Access type port sends a message, it will ().

- A. Send messages with a tag
- B. Strip the VLAN information of the message, and then send it out
- C. Type the PVID information of this port, and then send it out.
- D. Add the ZLAN message to the message, and then send it out

Answer: B

Justification

- Access port

As shown in **Figure 8-22**, the access port on a switch connects to the port on a host. The access port can only connect to an access link. Only the VLAN whose ID is the same as the default VLAN ID is allowed on the access port. Ethernet frames sent from the access port are untagged frames.

86. The configuration information of a certain port of the switch is shown in the figure. This port carries a VLAN tag when sending data frames carrying which VLAN?

```
#  
interface GigabitEthernet0/0/1  
port hybrid tagged vlan 2 to 3 100  
port hybrid untagged vlan 4 6  
#
```

- A. 1, 2, 3, 100
- B. 2, 3, 100
- C. 1, 2, 3, 4, 6, 100
- D. 2, 3, 4, 6, 100

Answer: B

Justification

Explanation: The correct answer is **B. 2, 3, 100**, which aligns with the configuration specifying that VLANs 2, 3, and 100 are tagged. This means that data frames associated with these VLANs will include the VLAN tag when sent from this port. VLANs 4 and 6, however, will send untagged frames as per the configuration.

87. What is the value of the field in the RSTP configuration BPDU message?

- A. 0x01
- B. 0x00
- C. 0x02
- D. 0x03

Answer: C

Justification

Version	Type	Name
0	0x00	Configuration BPDU
0	0x80	TCN BPDU
2	0x02	RST BPDU
3	0x02	MST BPDU

88. The Priority field in the VLAN tag can identify the priority of the data frame. What is the range of this priority?

- A. 0-15
- B. 0-63
- C. 0-3
- D. 0-7

Answer: D

Justification

Field	Length	Description	Value
TPID	2 bytes	Tag Protocol Identifier (TPID), indicating the frame type.	The value 0x8100 indicates an IEEE 802.1Q frame. An 802.1Q-incapable device discards 802.1Q frames. Device vendors can define their own TPID values, and users can then change the value to realize interconnection of devices from different vendors.
PRI	3 bits	Priority (PRI), indicating the 802.1p priority of a frame.	The value is in the range from 0 to 7. A larger value indicates a higher priority. If congestion occurs, the switch sends packets with the highest priority first.
CFI	1 bit	Canonical Format Indicator (CFI), indicating whether a MAC address is encapsulated in canonical format over different transmission media. CFI is used to ensure compatibility between Ethernet and token ring networks.	The value 0 indicates that the MAC address is encapsulated in canonical format, and the value 1 indicates that the MAC address is encapsulated in non-canonical format. The CFI field has a fixed value of 0 on Ethernet networks.
VID	12 bits	VLAN ID (VID), indicating the VLAN to which a frame belongs.	The VLAN ID is in the range from 0 to 4095. The values 0 and 4095 are reserved, and therefore available VLAN IDs are in the range from 1 to 4094.

89. What elements can NAT convert?

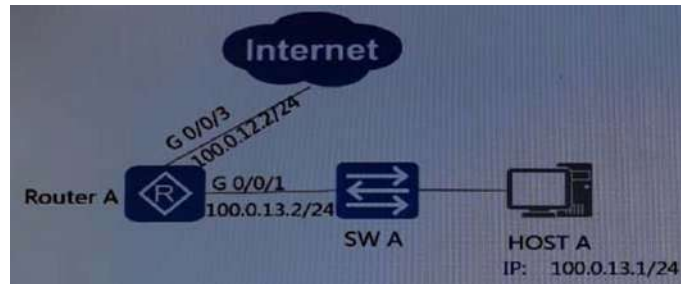
- A. Only the MAC address
- B. IP address + port number
- C. MAC address + port number
- D. Only IP address

Answer: B

Justification

After receiving a response packet from the server on the public network, the AP queries the reverse NAT entry based on the packet's destination IP address and port number. The AP translates the packet's destination IP address and port number to the private IP address and port number of the host on the private network based on the reverse NAT entry, and sends the packet to the host. For example, after the translation is performed on the packet sent from the server to Host A, the packet's destination IP address is 10.1.1.100, and its destination port number is 1025.

90. In the network shown in the figure below, from a security perspective, router A refuses to receive OSPF packets and ICMP packets received from the GigabitEthernet0/0/1 interface. Which of the following commands can achieve this requirement? (Multiple Choice)



- A.

```
acl number 3000
rule 5 deny ospf
rule 10 deny icmp
#
interface GigabitEthernet0/0/1
traffic-filter inbound acl 3000
#
```
- B.

```
acl number 2000
rule 5 deny 89
Rule 10 deny 1
#
interface GigabitEthernet0/0/1
traffic-filter inbound acl 2000
#
```
- C.

```
acl number 3000
rule 5 deny 89
rule 10 deny icmp
#
interface GigabitEthernet0/0/1
traffic-filter inbound acl 3000
#
```
- D.

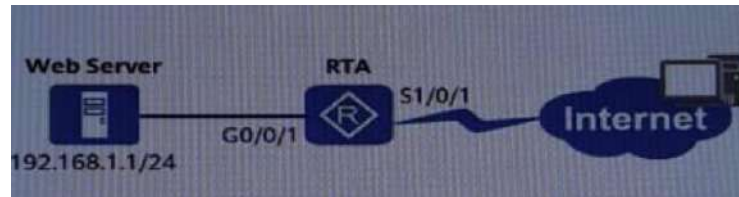
```
acl number 3000
rule 5 deny 89
rule 10 deny 1
interface GigabitEthernet0/0/1
traffic-filter inbound acl 3000
```

Answer: ACD

Justification

Explanation: Options A, C, and D provide correct configurations that meet the requirement to deny OSPF and ICMP packets. They include specific rules for denying these protocols and correctly applying these ACLs to the interface GigabitEthernet0/0/1. Each option utilizes either protocol names or numbers correctly and ensures the ACL is applied to the interface from where OSPF and ICMP packets should be blocked.

91. As shown in the figure, there is a web server in the private network that needs to provide HTTP services to public network users. Therefore, the network administrator needs to configure NAT on the gateway router RTA to achieve the requirements. Which of the following configurations can meet the requirement?



- A. RTA-GigabitEthernet0/0/1]nat server protocol tcp global 192.168.1.1 www inside 202.10.10.18080
- B. [RTA-Serial1/0/1]nat server protocol tcp global 192.168.1.1 www inside 202.10.10.1 8080
- C. [RTA-GigabitEthernet0/0/1]nat server protocol tcp global 202.10.10.1 www inside 192.168.1.18080
- D. [RTA-Serial 1/0/1]nat server protocol tcp global 202.10.10.1 www inside 192.168.1.1 8080

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064352/e7aab215/nat-server>

Format

```
nat server protocol { tcp | udp } global { global-address | current-interface | interface interface-type interface-number } global-port [ global-port2 ] inside host-address [ host-address2 ] [ host-port ] [ acl acl-number ] [ description description ]

nat server [ protocol { protocol-number | icmp | tcp | udp } ] global { global-address | current-interface | interface interface-type interface-number } inside host-address [ acl acl-number ] [ description description ]
```

92. A network engineer receives the error message "Error: Unrecognized command found at '^' position" when entering a command. Which of the following correctly describes the issue?

- A. The input command is incomplete
- B. Wrong parameter type
- C. No keyword found
- D. Input command is not clear

Answer: C

Justification

Source: Huawei Lecture # 3, Page # 24

Interpreting Command Line Error Messages

- If a command passes the syntax check, the system executes it. Otherwise, the system reports an error message.

```
[Huawei] sysname
      ^
Error:Incomplete command found at '^' position.    #A supplement needs to be made at the position pointed by the arrow.

[Huawei] router if 1.1.1.1
      ^
Error: Unrecognized command found at '^' position.    #An identification failure occurs at the position pointed by the arrow.
Check whether the command is correct.

[Huawei] a
      ^
Error: Ambiguous command found at '^' position.    #More than one command matches the keyword at the position pointed by
the arrow. In this example, it indicates that there are multiple keywords starting with a.

[Huawei-GigabitEthernet0/0/0]ospf cost 800000
                        ^
Error: Wrong parameter found at '^' position.    #The parameter value at the position pointed by the arrow is invalid.
```

93. <huawei>system-view

[huawei]command-privilege level 3 view user save

The correct statement about the above configuration command is:

- A. Modify the user's authority level to 3, and save the configuration
- B. Modify the permission level of the user view command to 3, and save the configuration
- C. Modify the authority level of the save command used by a user to 3
- D. Modify the permission level of the save command in the user view to 3

Answer: D

Justification

Source: [command-privilege level - CX91x Series Switch Modules V100R001C00 Command Reference 15 - Huawei](#)

2. Set the command privilege level.

- Run **command-privilege level level view** view-name command-key

The command privilege level is set in the specified view.

94. Which of the following descriptions about the switch is wrong?

- A. The switch can complete the exchange of data frames
- B. The switch can provide network access services for terminal equipment (PC, Server)
- C. Switches generally work at the data link layer
- D. Switches are generally used as network export equipment

Answer: D

Justification

Explanation:

Option A is correct about the switch.

On Ethernet networks, frames contain a source MAC address and a destination MAC address. Switches receive a frame from the source device and quickly forward it toward the destination device.

Source: Lecture # 1, Page # 12

Option B is correct about the switch.



Option C is also correct about the switch.

- On a campus network, a switch is the device closest to end users and is used to connect terminals to the campus network. Switches at the access layer are usually Layer 2 switches and are also called Ethernet switches. Layer 2 refers to the data link layer of the TCP/IP reference model.

95. Which of the following descriptions about firewalls is wrong?

- A. Firewall can realize user identity authentication
- B. Firewall can realize access control between different networks
- C. Firewall can isolate networks with different security levels
- D. The firewall cannot achieve network address translation

Answer: D

Justification

Source: Huawei Lecture # 1, Page # 14



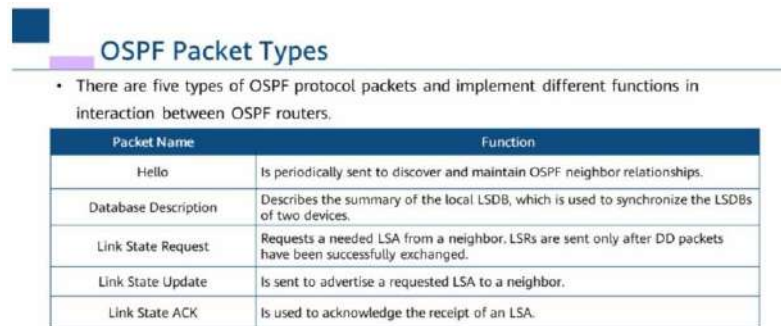
96. What kind of message does the OSPF protocol use to discover and maintain neighbor relationship?

- A. DD
- B. LSR
- C. LSU
- D. HELLO

Answer: D

Justification

Source: Huawei Lecture # 6, Page # 17

The slide is titled "OSPF Packet Types" and features a blue header bar. Below the title, a bullet point states: "There are five types of OSPF protocol packets and implement different functions in interaction between OSPF routers." A table with two columns, "Packet Name" and "Function", lists the five packet types and their purposes.

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.

97. On the VRP platform, the default protocol priority of routes in the direct routing, static routing, RIP, OSPF area is:

- A. Direct routing, OSPF, Static routing, RIP
- B. Direct routing, Static routing, RIP, OSPF
- C. Direct routing, RIP, Static routing, OSPF
- D. Direct routing, OSPF, RIP, Static routing

Answer: A

Justification

Source: support.huawei.com/enterprise/en/doc/EDOC1000047418/e3a3303d/routing-protocol-preference

Table 2-1 Routing protocols and their default external preferences

Routing Protocol or Route Type	Default External Preference
Direct	0
OSPF	10
IS-IS	15
Static	60
RIP	100

98. In the network shown in the figure below, the host has an ARP cache, and host A sends a packet to host B. What are the destination MAC and destination IP of this packet?



- A. MAC-B, 11.0.12.1
- B. MAC-D, 11.0.12.1
- C. MAC-C, 10.0.12.2
- D. MAC-A, 11.0.12.1

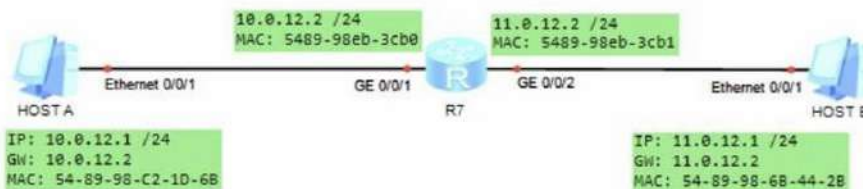
MAC-C, 11.0.12.1 as send by Host A

Answer: A

Justification

Source: Huawei eNSP

For the same ping the packet forwarding was captured on both interfaces of router (G0/0/1 and G0/0/2)



The image shows two Wireshark capture windows and a console window. The left Wireshark window shows a capture on interface 10.0.12.1, displaying ICMP echo request and reply packets. The right Wireshark window shows a capture on interface 11.0.12.1, also displaying ICMP echo request and reply packets. The console window shows the output of a ping command from Host A to Host B, indicating successful connectivity.

99. On Huawei routers, what is the default value of the OSPF protocol internal routing priority?

- A. 20
- B. 10
- C. 30
- D. 0

Answer: B

Justification

Source: Huawei Lecture # 5, Page # 18

Protocol	Route Type	Default Preference
Direct	Direct route	0
Static	Static route	60
Dynamic routing protocol	OSPF internal route	10
	OSPF external route	150

100. Now there are the following four segments: 10.24.0.0/24, 10.24.1.0/24, 10.24.2.0/24, 10.24.3.0/24, which route can point to these four network segments at the same time? (Multiple Choice)

- A. 10.24.1.0/23
- B. 0.0.0.0/0
- C. 10.24.0.0/22
- D. 10.24.0.0/21

Answer: BCD

Justification

JUSTIFICATION:

Option A: 10.24.1.0/23 (Incorrect)

The /23 subnet mask covers two /24 subnets.

This route does **not** cover all four given subnets (10.24.2.0/24 and 10.24.3.0/24 are missing).

Option B: 0.0.0.0/0 (Correct)

This **is** a **default** route that matches all IPv4 addresses.

It covers any **and** all network segments.

Option C: 10.24.0.0/22 (Correct)

The /22 subnet mask covers four /24 subnets.

10.24.0.0/22 includes:

10.24.0.0/24 10.24.1.0/24 10.24.2.0/24

10.24.3.0/24

This route perfectly matches all four given subnets.

Option D: 10.24.0.0/21 (Correct)

The /21 subnet mask covers eight /24 subnets.

10.24.0.0/21 includes:

10.24.0.0/24 through 10.24.7.0/24

This route includes the four given subnets **and** more.

101. Which fields in the IPv4 header are related to fragmentation? (Multiple Choice)

- A. TTL
- B. Fragment Offset
- C. Identification
- D. Flags

Answer: BCD

Justification

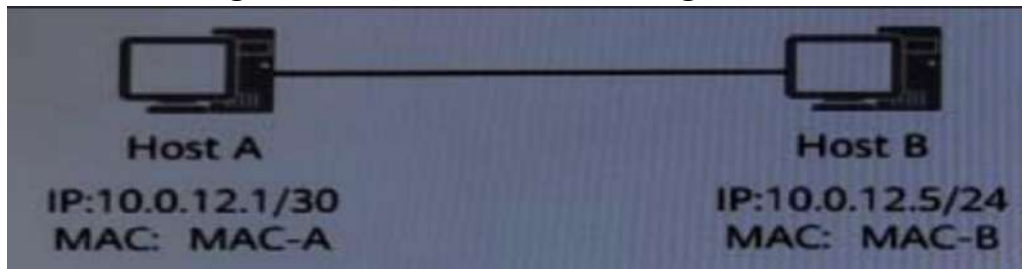
Source: Huawei Lecture # 4, Page # 9

- Identification: 16 bits long. This field carries a value assigned by a sender host and is used for fragment reassembly.
- Flags: 3 bits long.
 - Reserved Fragment: 0 (reserved).
 - Don't Fragment: Value 1 indicates that fragmentation is not allowed, and value 0 indicates that fragmentation is allowed.
 - More Fragment: Value 1 indicates that there are more segments following the segment, and value 0 indicates that the segment is the last data segment.
- Fragment Offset: 12 bits long. This field is used for fragment reassembly. This field indicates the relative position of a fragment in an original packet that is fragmented. This field is used together with the More Fragment bit to help the receiver assemble the fragments.

Source: support.huawei.com/enterprise/en/doc/EDOC1100112351/dd76eaf/ipv4-packet-format

Identification	16 bits	IPv4 software maintains a counter in the storage device to record the number of IP datagrams. The counter value increases by 1 every time a datagram is sent, and is filled in the identification field.
Flags	3 bits	Only the rightmost two bits are valid. The rightmost bit indicates whether the datagram is not the last data fragment. The value 1 indicates the last fragment, and the value 0 indicates non-last fragment. The middle bit is the fragmentation flag. The value 1 indicates that the datagram cannot be fragmented, and the value 0 indicates that the datagram can be fragmented.
Fragment Offset	13 bits	Specifies the location of a fragment in a packet.
Time to Live (TTL)	8 bits	Specifies the life span of a datagram on a network. TTL is measured by the number of hops.

102. As shown in the figure below, which of the following statements are correct?



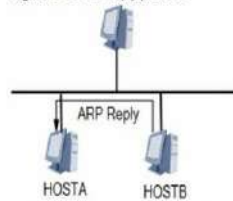
- A. Host A can ping host B
- B. Host A and Host B cannot obtain each other's MAC address
- C. The following entry 10.0.12.5 MAC-B exists in the ARP cache of host A
- D. The broadcast address of host A and host B are the same

Answer: B

Justification

Source: Huawei Lecture # 2, Page # 29

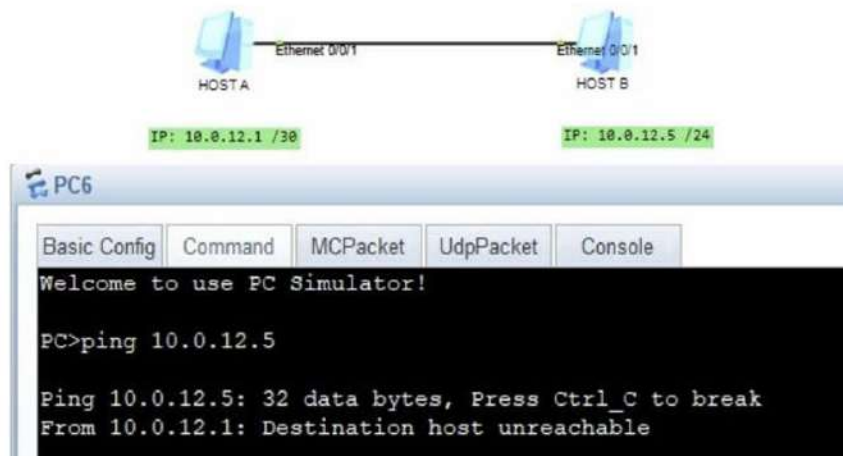
Figure 8-60 ARP reply process



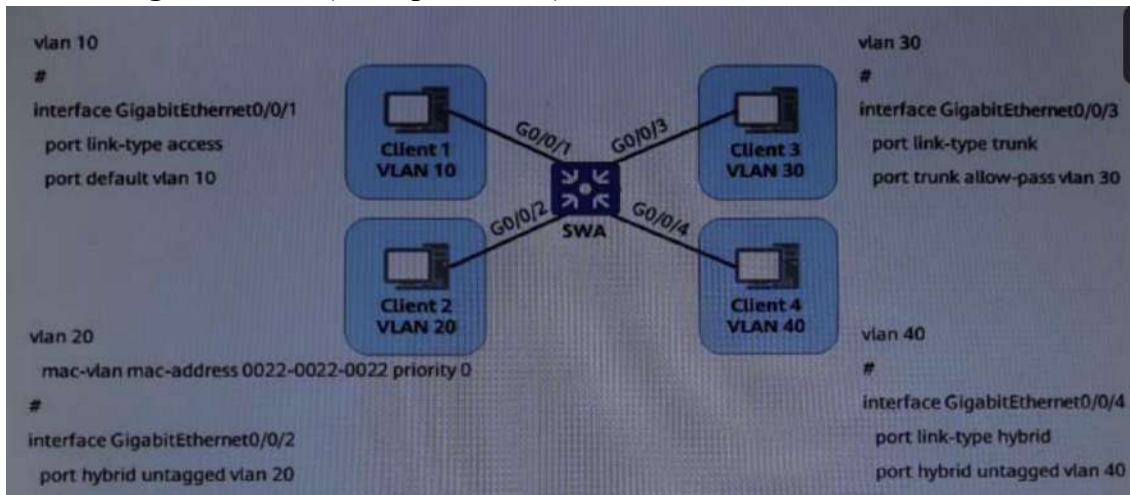
HOSTB compares its IP address with the destination IP address in the ARP Request packet. If HOSTB finds that its IP address is the same as the destination IP address, HOSTB adds the IP address and MAC address of the sender (HOSTA) to the local ARP table. Then HOSTB unicasts an ARP Reply packet, which contains its MAC address, to HOSTA, as shown in Figure 8-60.

After receiving the ARP Reply packet, HOSTA adds HOSTB's MAC address into the local ARP table. Meanwhile, HOSTA encapsulates the IP packets and forwards them to HOSTB.

Source: Verified in eNSP:



103. As the figure shown: (Multiple Choice)



- A. Client3 belongs to VLAN30, and the command for dividing VLAN on the switch is correct
- B. Client4 belongs to VLAN40, and the command for dividing VLAN on the switch is correct
- C. Client2 belongs to VLAN20, and the switch based on the MAC address division command is correct
- D. Client1 belongs to VLAN10, and the command for dividing VLAN on the switch is correct

Answer: ABD

Justification

Source: Huawei Lecture # 4, Page # 9

Run **vlan** **vlan-id**

Run the **interface** **interface-type** **interface-number** command to enter the view of the Ethernet interface that wants to join the VLAN.

Run the **port link-type { access | hybrid | trunk | dot1q-tunnel }** command to configure the link type of the Ethernet interface.

- Access or QinQ interface

Run the **port default vlan** **vlan-id** command to add the interface to the specified VLAN.

Run the **port interface-type { interface-number1 [to interface-number2] } &<1-10>** command in the VLAN view to add one interface or a group of interfaces to a VLAN.

- Trunk interface

- Run the **port trunk allow-pass vlan { [vlan-id1 [to vlan-id2] } &<1-40> | all }** command to add the interface to the specified VLAN.

- Hybrid interface

- Run the following command as required.

- Run the **port hybrid untagged vlan { [vlan-id1 [to vlan-id2] } &<1-10> | all }** command to add a hybrid interface to a VLAN in untagged mode.

In untagged mode, an interface removes tags from outgoing packets. This mode applies to the scenario where an Ethernet interface directly connects to a terminal.

- Run the **port hybrid tagged vlan { [vlan-id1 [to vlan-id2] } &<1-10> | all }** command to add a hybrid interface to a VLAN in tagged mode.

In tagged mode, an interface does not remove tags from outgoing packets. This mode applies to the scenario where an Ethernet interface connects to an interface of another switch.

Dividing a LAN into VLANs Based on MAC Addresses:

Run **vlan** **vlan-id**

Run **mac-vlan** **mac-address** **mac-address** [**priority** **priority**]

Run the **interface** **interface-type** **interface-number** command to enter the view of the interface.

Run the **port hybrid untagged vlan { [vlan-id1 [to vlan-id2] } &<1-10> | all }** command to configure the hybrid interface to allow frames with a specified VLAN ID to pass through.

Run **mac-vlan enable**

104. The load sharing mode at both ends of the Eth-Trunk can be inconsistent.

- A. True
- B. False

Answer: B

Justification

Source: [Configuring and Troubleshooting Eth-Trunks for S5700 Switches - Huawei](#)

2. Check whether the interface configurations on both ends of an Eth-Trunk are consistent.

- Devices at both ends of the Eth-Trunk must use the same number of physical interfaces, duplex mode, and flow control mode.
- Devices at both ends of an Eth-Trunk must use the same link aggregation mode.

105. When Huawei switches allow STP, the priority of the switch is by default:

- A. 32768
- B. 8192
- C. 16384
- D. 4096

Answer: A

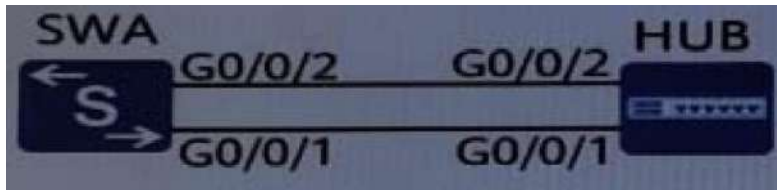
Justification

Source: Huawei Lecture # 9, Page # 13

09 STP Principles and Configuration

- In STP, each switch has a bridge ID (BID), which consists of a 16-bit bridge priority and a 48-bit MAC address. On an STP network, the bridge priority is configurable and ranges from 0 to 65535. The default bridge priority is 32768. The bridge priority can be changed but must be a multiple of 4096. The device with the highest priority (a smaller value indicates a higher priority) is selected as the root bridge. If the priorities are the same, devices compare MAC addresses. A smaller MAC address indicates a higher priority.

106. As shown in the figure, the switch runs STP with default parameters, which of the following ports will be elected as the designated port?



- A. G0/0/1 port of switch A
- B. Port G0/0/2 of switch A
- C. HUB's G0/0/1 port
- D. HUB's G0/0/2 port

Answer: A

Justification

Source: Huawei Lecture 9, page # 15,16

- Each port on a switch has a cost in STP. By default, a higher port bandwidth indicates a smaller port cost.

Port Rate	Port Mode	Recommended STP Cost		
		IEEE 802.1d-1998	IEEE 802.1t	Huawei Legacy Standard
100 Mbit/s	Half-duplex	19	200,000	200
	Full-duplex	18	199,999	199
	Aggregated link: two ports	15	100,000	180
1000 Mbit/s	Full-duplex	4	20,000	20
	Aggregated link: two ports	3	10,000	18
10 Gbit/s	Full-duplex	2	2000	2
	Aggregated link: two ports	1	1000	1
40 Gbit/s	Full-duplex	1	500	1
	Aggregated link: two ports	1	250	1
100 Gbit/s	Full-duplex	1	200	1
	Aggregated link: two ports	1	100	1

Explanation: Since both ports are on the same switch, the bridge ID is the same, therefore the selection falls to the port ID. Port IDs are numerically assigned based on the interface number. G0/0/1 has a lower numerical value than G0/0/2.

107. What is the Solicited-node multicast address corresponding to IPv6 address 2019::8:AB?

- A. FF02::1::FF08:AB
- B. FF02::FF08:AB
- C. FF02::1::FF20:19
- D. FF02::1::FE08:AB

Answer: A

Justification

Source: [Understanding IPv6 - Wireless Access Controller \(AC and Fit AP\) V200R020C00 CLI-based Configuration Guide - Huawei](#)

The solicited-node multicast address consists of the prefix FF02::1:FF00:0/104 and the last 24 bits of the corresponding unicast address.

108. For the description of the PPP link establishment process, which of the following statement is wrong?

- A. In the Establish phase, the PPP link performs LCP parameter negotiation. The negotiation content includes the maximum receiving unit MRU, authentication method, magic word and other options.
- B. After the NCP negotiation is successful, the PPP link will maintain the communication state and enter the Terminate phase.
- C. The Dead phase is also called the physical layer unavailable phase. When the two ends of the communication detect that the physical line is activated, they will migrate from the Dead phase to the Establish phase, which is the link establishment phase.
- D. In the Network phase, the PPP link performs NCP negotiation. Select and configure a network layer protocol through NCP negotiation, and negotiate network layer parameters E. PPP link supports half-duplex and full-duplex modes

Answer: B

Justification

Source: Huawei Lecture # 17, Page # 10

- A PPP link can be set up after going through the link establishment, authentication, and network layer negotiation phases. The details are as follows:
 1. Two communicating devices enter the Establish phase when starting to set up a PPP connection.
 2. In the Establish phase, they perform LCP negotiation to negotiate an MRU, authentication mode, magic number, and other options. If the negotiation is successful, the devices enter the Opened state, indicating that the lower-layer link has been established.
 3. If authentication is configured, the devices enter the Authenticate phase. Otherwise, the devices directly enter the Network phase.
 4. In the Authenticate phase, link authentication is performed based on the authentication mode negotiated in the link establishment phase. Two authentication modes are available: PAP and CHAP. If the authentication succeeds, the devices enter the Network phase. Otherwise, the devices enter the Terminate phase, tear down the link, and set the LCP status to Down.
 5. In the Network phase, NCP negotiation is performed on the PPP link to select and configure a network layer protocol and to negotiate network layer parameters. The most common NCP protocol is IPCP, which is used to negotiate IP parameters.

PPP LINK ESTABLISHMENT PROCESS: (JUNIPER NETWORKS)

4. If a network layer protocol is configured, the PPP link enters the Network-Layer Protocol phase for NCP negotiation, such as IPCP negotiation and IPv6CP negotiation.
 - If the NCP negotiation succeeds, the link goes up and becomes ready to carry negotiated network-layer protocol packets.
 - If the NCP negotiation fails, NCP reports a Down event and enters the Link Termination phase.

If the interface is configured with an IP address, the IPCP negotiation is performed. IPCP configuration options include IP addresses and DNS server IP addresses. After the IPCP negotiation succeeds, the link can carry IP packets.
5. After the NCP negotiation is performed, the PPP link remains active until either of the following events occurs:
 - Explicit LCP or NCP frames close the link.
 - Some external events take place (for example, the intervention of a user).

109. Which statement about SNMP protocol is correct?

- A. SNMP uses UDP as the transport layer protocol
- B. SNMP uses ICMP as the network layer protocol
- C. SNMP protocol uses multicast to send management messages
- D. The SNMP protocol only supports the sending of flash management messages on the Ethernet link

Answer: A

Justification

Source: [SNMP-based Network Management - HUAWEI Bar 500, Box 500, Box 700, Box 900, Board 20.0.0 MIB Reference - Huawei](#)

- SNMP is the carrier for the communication between the NMS and the agent. Data is encapsulated in the protocol data unit (PDU) defined by SNMP. SNMP does not transmit data. Instead, data exchange tasks are implemented by the transport layer protocols such as UDP.

110. Which of the following commands can be used to view the router's CPU usage?

- A. display cpu-state
- B. display interface
- C. display memory
- D. display cpu-usage

Answer: D

Justification

Source: [Displaying the CPU Usage - S1720, S2700, S5700, and S6720 V200R011C10 Configuration Guide - Device Management - Huawei](#)

Procedure

- Run the **display cpu-usage** [slave | slot slot-id] [vcpu vcpu] command to view CPU usage statistics.
- Run the **display cpu-usage history** [1hour | 24hour | 72hour] [slave | slot slot-id] [vcpu vcpu] command to view CPU usage statistics within a period.
- Run the **display cpu-usage configuration** [slave | slot slot-id] command to view CPU usage configurations.

111. Which of the following messages are used for IPv6 stateless address autoconfiguration? (Multiple Choice)

- A. NS
- B. RA
- C. NA
- D. RS

Answer: BD

Justification

Source: Huawei Lecture # 19, Page # 23

SLAAC is a highlight of IPv6. It enables IPv6 hosts to be easily connected to IPv6 networks, without the need to manually configure IPv6 addresses and to deploy application servers (such as DHCP servers) to assign addresses to hosts. SLAAC uses ICMPv6 RS and RA messages.

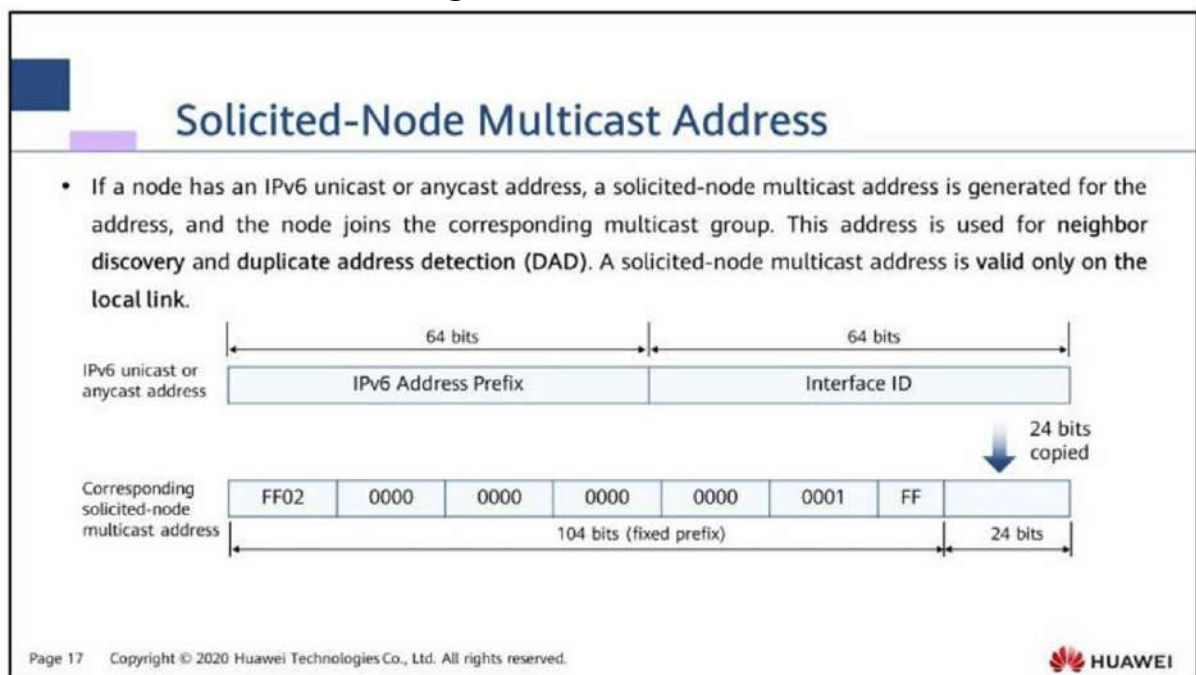
112. Which of the following IPv6 addresses is a multicast address?

- A. FE80::2E0:FCFF:FEEF:FEC
- B. FC00::2E0:FCFF:FEEF:FEC
- C. FF02::2E0:FCFF:FEEF:FEC
- D. 2000::2E0:FCFF:FEEF:FEC

Answer: C

Justification

Source: Huawei Lecture # 19, Page # 17



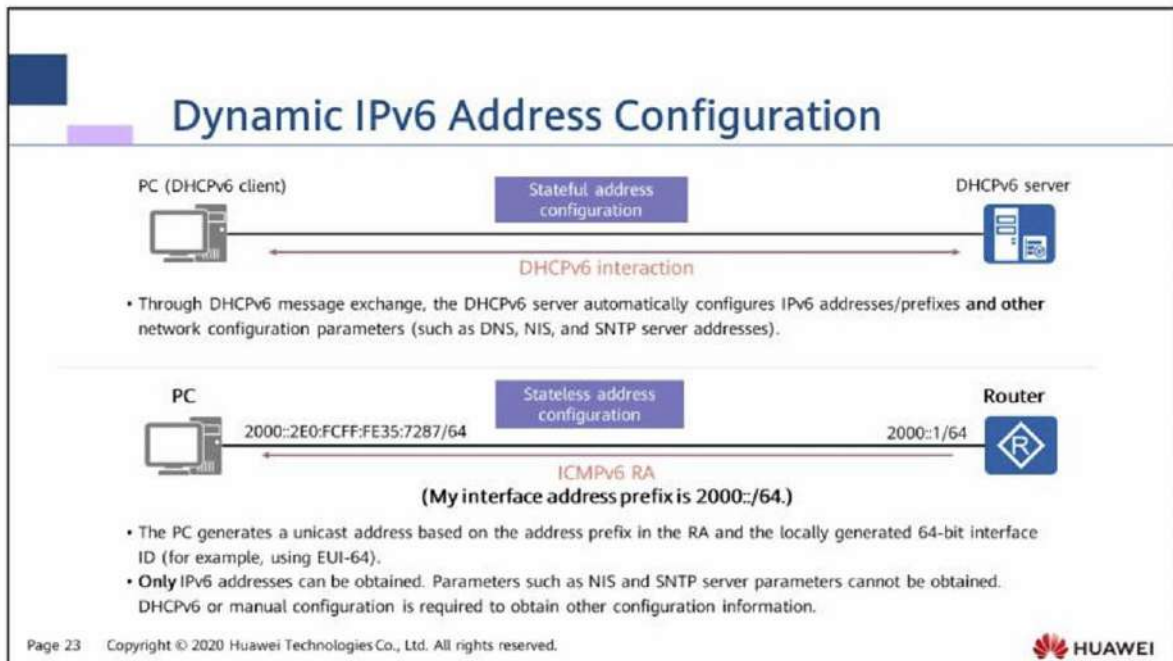
113. Which of the following statements about IPv6 address configuration are correct?
(Multiple Choice)

- A. IPv6 supports stateless automatic configuration
- B. IPv6 address supports automatic configuration in multiple ways
- C. IPv6 supports DHCPv6 for address configuration
- D. IPv6 address can only be configured manually

Answer: ABC

Justification

Source: Huawei Lecture # 19, Page # 23



114. A device has been configured to complete the Telnet configuration. The device login address is 10.1.1.10. The Telnet user name is admin and the password is Huawei@123. Which of the following is the correct way to log in to this device using telnetlib?

- A. `telnetlib.Telnet(10.1.1.10,23,admin,Huawei@123)`
- B. `telnetlib.Telnet(10.1.1.1.0)`
- C. `telnetlib.Telnet(10.1.1.1.0,23,Huawei@123)`
- D. `telnetlib.Telnet(10.1.1.1.0,admin)`

Answer: A

Justification

Even though the username and password are not directly used as parameters, they are needed for login. Therefore, option A is correct. One example of login using telnetlib is as follows:

Source: [Python - Telnet to Huawei routers with script - Huawei Enterprise Support Community](#)

```
import telnetlib
import socket
ipfile = "129.9.0.104"
user = "root"
password = "Test1234."
outputfile = ((ipfile)+".output")
f = open(outputfile, 'w')
f.write("")
f.close()

with open(ipfile) as ips:
    all_ips = [x.rstrip() for x in ips] # get all ips in a list and strip newline

for ip in all_ips:

    try:
        tn = telnetlib.Telnet(ip,23,2)
        tn.read_until(b"Username: ")
        tn.write(user.encode('ascii') + b"\n")
        if password:
            tn.read_until(b"Password: ")
            tn.write(password.encode('ascii') + b"\n")
            tn.write(b"screen 0 tem\n")
            tn.write(b"display version\n")          # print(tn.read_all().decode('ascii'))

            with open(outputfile,"ab") as f:      #write to a file
                f.write(tn.read_all())
    except socket.timeout:
        print((ip)+" timeout")
```


115. The classic network forwarding method is based on router routing table forwarding. The forwarding method of OpenFlow switches is based on flow table forwarding. Which of these two forwarding methods are correct? (Multiple Choice)

- A. The routing table is of fixed length. A device can only have one public routing table
- B. The flow table is variable length. A network device can only have one flow table
- C. The matching method of the flow table is to match the MAC address and IP address of the traffic at the same time
- D. The matching method of the address table is to match the route of the destination network segment with the longest mask

Answer: ACD

Justification

Source: [IP Routing Basics - Huawei](#)

Matching with FIB Table

After selecting an optimal route from the routing table, a router sends it to the FIB table. When receives a packet, the router compares it against the FIB table to find the optimal route to forward the packet.

Each entry in the FIB table contains the physical or logical interface through which a packet is sent to a network segment or host to reach the next router. An entry can also indicate whether the packet can be sent to a destination host in a directly connected network.

The router performs the "AND" operation on the destination address in the packet and the network mask of each entry in the FIB table. The router then compares the result of the "AND" operation with the entries in the FIB table to find a match and chooses the optimal route to forward packets according to the longest match rule.

Explanation:

- A. The routing table is of fixed length. A device can only have one public routing table True. Traditional routing tables in routers are typically fixed in terms of the structure of the entries, where each entry consists of a destination network, the associated mask, the next-hop IP address, and possibly some other fixed parameters. While the size of the routing table (number of entries) can grow, the structure is fixed. In traditional routing architectures, there is generally one main routing table that the device uses to make forwarding decisions.
- B. The flow table is variable length. A network device can only have one flow table False. In OpenFlow switches, flow tables can be of variable length, meaning the number of entries can vary based on the flows configured. However, a single network device can have multiple flow tables, allowing for complex packet processing pipelines where packets can be matched against multiple tables in sequence.
- C. The matching method of the flow table is to match the MAC address and IP address of the traffic at the same time True. OpenFlow flow tables allow for matching on multiple fields within packet headers, including both MAC addresses and IP addresses, as well as other parameters like VLAN tags, IP protocols, TCP/UDP ports, etc. This flexibility allows OpenFlow switches to perform detailed packet classification and forwarding based on multiple header fields simultaneously.
- D. The matching method of the address table is to match the route of the destination network segment with the longest mask True.

116. The router running OSPF can reach the FULL state after LSDB synchronization is completed.

- A. True Correct
- B. False

Answer: B A

Justification

Source: Huawei Lecture # 6, Page # 23

06 OSPF Basics

Relationships Between OSPF Routers

- There are two important concepts about the relationship between OSPF routers: neighbor relationship and adjacency.
- On a simple network, two routers are directly connected. OSPF is enabled on interconnected interfaces. The routers start to send and listen to Hello packets. After the two routers discover each other through Hello packets, they establish a neighbor relationship.
- The establishment of a neighbor relationship is just the beginning. A series of packets, such as DD, LSR, LSU, and LSAck packets, will be exchanged later. When LSDB synchronization between two routers is complete and the two routers start to calculate routes independently, the two routers establish an adjacency.

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117. In the OSPF broadcast network, which routers will a Dother router exchange link state information with? (Multiple Choice)

- A. BDR
- B. All OSPF neighbors
- C. DR Other
- D. DR

Answer: AD

Justification

Source: [Fundamentals of OSPF - S1720, S2700, S5700, and S6720 V200R011C10 Configuration Guide - IP Unicast Routing - Huawei](#)

OSPF Network Types

Four OSPF network types are classified based on link layer protocols.

- **Broadcast**

Networks using Ethernet or Fiber Distributed Data Interface (FDDI) at the link layer are broadcast networks by default.

On broadcast networks:

- Hello packets, LSU packets, and LSAck packets are sent in multicast mode. 224.0.0.5 is the IP multicast address reserved for OSPF DR others. 224.0.0.6 is the IP multicast address reserved for OSPF DRs or BDRs.
- DD and LSR packets are sent in unicast mode.

Reason for DR/BDR Election

On broadcast and non-broadcast multiple access (NBMA) networks, any two routers need to exchange routing information. On the network in **Figure 6-8**, there are n routers that need to establish $n \times (n - 1) / 2$ adjacencies. A route change on any router is transmitted to the other routers, which wastes bandwidth resources.

To solve this problem, the concept of DR is defined in OSPF. After a DR is elected, all the other routers send routing information only to the DR, and the DR broadcasts LSAs.

To prevent service interruption caused by DR re-election when the DR fails, a BDR is also elected during DR election. The routers excluding DR and BDR are called DR others. The DR others do not establish adjacencies or exchange any routing information with each other. This reduces the number of adjacencies established between routers on broadcast and NBMA networks.

118. Which of the following IPv4 addresses the host uses cannot directly access the Internet?

- A. 100.1.1.1
- B. 50.1.1.1
- C. 10.1.1.1
- D. 200.1.1.1

Answer: C

Justification

Source: [What is IPv4 - Huawei](#)

Some main network addresses are used as private addresses and therefore cannot be used on the Internet.			
Category	Scope	Private Address	Scale
Class A address	IP address whose first octet starts with 0, that is, the first octet ranges from 00000001 to 01111111 (binary) or from 1 to 127 (decimal). IP addresses starting with 01111111 (127) are used only for loopback tests.	All addresses on the 10.0.0.0 network segment are private addresses.	There are 125 ($2^7 - 3$) network segments. Each network segment can contain 16,777,214 ($2^{24} - 2$) hosts. Class A addresses are applicable to super-large networks.
Class B address	IP address whose first octet starts with 10, that is, the first octet ranges from 10000000 to 10111111 (binary) or from 128 to 191 (decimal).	All addresses in the range of 172.16.0.0 to 172.31.255.255 are private addresses.	There are 16,384 ($2^{14} - 16$) network segments. Each network segment can contain 65,534 ($2^{16} - 2$) hosts. Class B addresses are applicable to large networks.
Class C address	IP address whose first octet starts with 110, that is, the first octet ranges from 11000000 to 11011111 (binary) or from 192 to 223 (decimal).	All addresses in the range of 192.168.1.0 to 192.168.255.255 are private addresses.	There are 2,096,896 ($2^{21} - 256$) network segments. Each network segment can contain 254 ($2^8 - 2$) hosts. Class C addresses are applicable to small networks.

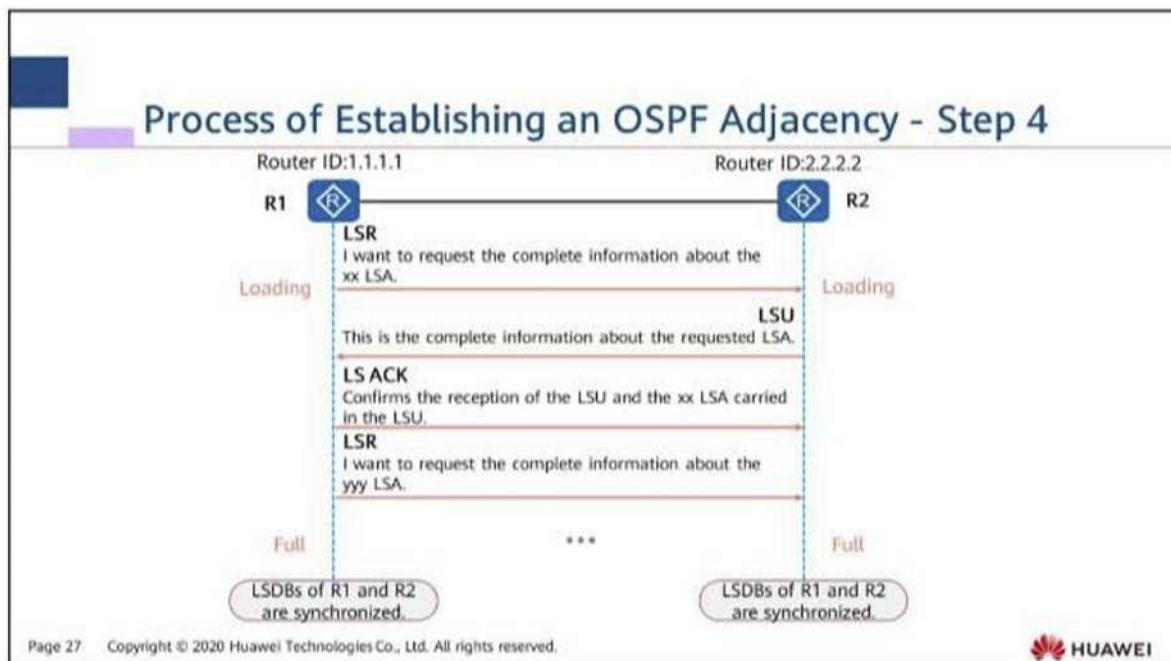
119. What kind of message does the OSPF protocol use to confirm the received LSU message?

- A. LSU
- B. LSR
- C. LSACK
- D. LSA

Answer: C

Justification

Source: Huawei Lecture # 6, Page # 27



120. The link aggregation interface can only be used as a Layer 2 interface.

- A. TRUE
- B. FALSE

Answer: B

Justification

Source: [Creating an Eth-Trunk Interface and Configure a Link Aggregation Mode for It - NetEngine AR5700, AR6700, and AR8000 V600R023C00 Configuration Guide - Ethernet Switching Configuration - Huawei](#)

- An Eth-Trunk interface can either work in Layer 2 or Layer 3 mode. To add an Eth-Trunk interface to a VLAN or configure it to transmit Layer 2 data packets, you need to configure the Eth-Trunk interface to work in Layer 2 mode. In this case, all Layer 3 functions of the Eth-Trunk interface are disabled, and the Eth-Trunk interface uses the system MAC address. To use an Eth-Trunk interface to transmit Layer 3 data packets, configure the Eth-Trunk interface to work in Layer 3 mode. In this case, you can configure the IP address, MAC address, and MTU for the Eth-Trunk interface.
- Physical interfaces can be added to an Eth-Trunk in either Layer 2 or Layer 3 mode.

121. The RSTP protocol uses the P/A mechanism to speed up the speed of the upstream port to the Forwarding state, but what is the reason why there is no temporary loop?

- A. Speed up port role election
- B. Through the synchronization mechanism to ensure that no temporary loop occurs
- C. Edge ports are introduced
- D. Shorten the time of Forward Delay

Answer: B

Justification

Explanation: In Rapid Spanning Tree Protocol (RSTP), synchronization ensures that no temporary loops occur during rapid transitions. When a switch receives a proposal for a new root port, it blocks all its designated ports before acknowledging the proposal. This process ensures that no loops are formed during topology changes. The synchronization mechanism allows RSTP to achieve rapid convergence while maintaining a loop-free topology.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100378012/b4e804a4/understanding-stp-rstp>

THE P/A PROCESS CAN PROTECT TO DOWNSTREAM DEVICES.

Although STP can select designated ports quickly, to prevent loops, all ports must wait at least one interval of the Forward Delay timer before forwarding traffic. To overcome this bottleneck, RSTP blocks non-root ports to prevent loops, and uses the P/A mechanism to shorten the time that an upstream port waits before transitioning to Forwarding state.

NOTE

The P/A mechanism requires a P2P full-duplex link between two devices. If P/A fails, a designated port is elected after two intervals of the Forward Delay timer. This is the same as designated port election in STP.

Enhanced P/A mechanism

In **Figure 6-39**, the enhanced P/A mechanism works as follows:

1. At the beginning of negotiation, all devices consider themselves as the root bridge, on which all ports are designated ports in Discarding state. When the synced variable is set to 1, the Proposal and Agreement fields are also set to 1. The upstream device sends a proposal BPDU to the downstream device, requesting that the port connected to the downstream device rapidly enters the Forwarding state. After receiving this BPDU, the downstream device sets its port connected to the upstream device as the root port and blocks all non-edge ports.
2. The upstream device sends an agreement BPDU. After receiving this BPDU, the root port on the downstream enters the Forwarding state.
3. The downstream device replies with an agreement BPDU. After receiving this BPDU, the upstream device sets its port connected to the downstream device as the designated port, and the port then enters the Forwarding state.

122. Which of the following statements about router-on-a-stick are correct?

- A. Use only one physical link connection between the switch and the router
- B. Need to create a sub-interface on the router
- C. One physical connection per VLAN
- D. On the switch, configure the port connected to the router as a Trunk port, and allow frames of the relevant VLAN to pass

Answer: ABD

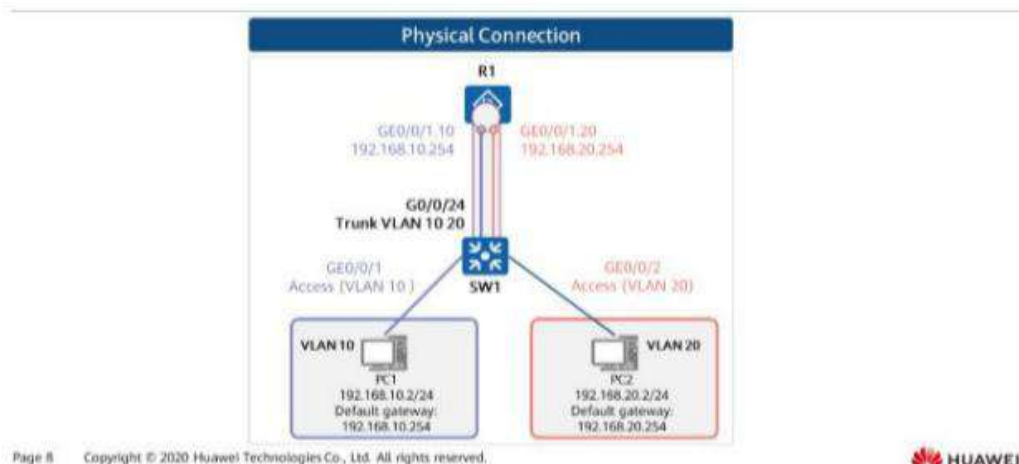
Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000138063/516eb78d/example-for-configuring-sub-interfaces-to-implement-inter-vlan-communication>

SOURCE: Topic 10, SLIDE 7

In this example, it is clear that one physical connection is needed between a switch and a router so option A is correct and subinterfaces are created so option B is also correct. Using the one physical connection, virtual interfaces for vlans are created so option C is incorrect and switch configures Trunk port on the port that connects it to the router so option D is also correct.



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- R1 connects to SW1 through a physical interface (GE 0/0/1). Two sub-interfaces (GE 0/0/1.10 and GE 0/0/1.20) are created on the physical interface and used as the default gateways of VLAN 10 and VLAN 20, respectively.

123. After the system restarts the static MAC address table, the saved entries will not be lost.

- A. TRUE
- B. FALSE

Answer: A

Justification

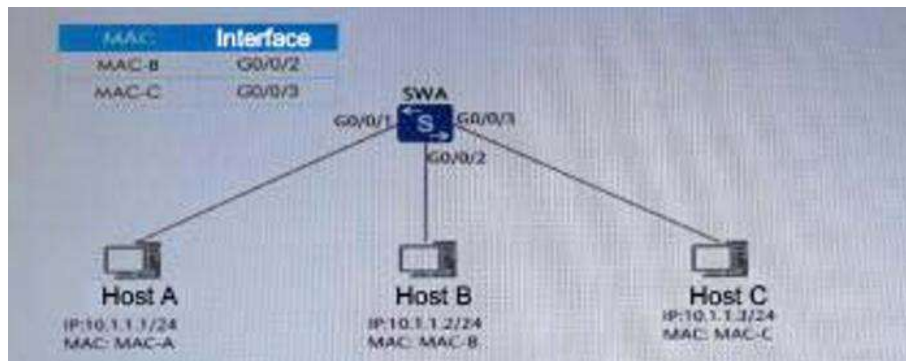
Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100277053/9ef704bd/configuring-a-sta?c-macaddress-entry>

Static MAC address entries have the following characteristics:

- A static MAC address entry will not be aged out. After being saved, a static MAC address entry will not be lost after a system restart, and can only be deleted manually.

124. As shown in the figure below, the MAC address table of SWA is as follows. The switch can't learn the MAC address of HOST A all the time. Which of the following is impossible?



- A. The G0/0/1 port of the switch has disabled MAC address learning
- B. SWA turns off the MAC address learning function of the VLAN to which host A belongs
- C. HOST A did not send any data frames
- D. The G0/0/1 port of the switch is set to ACCESS mode

Answer: D

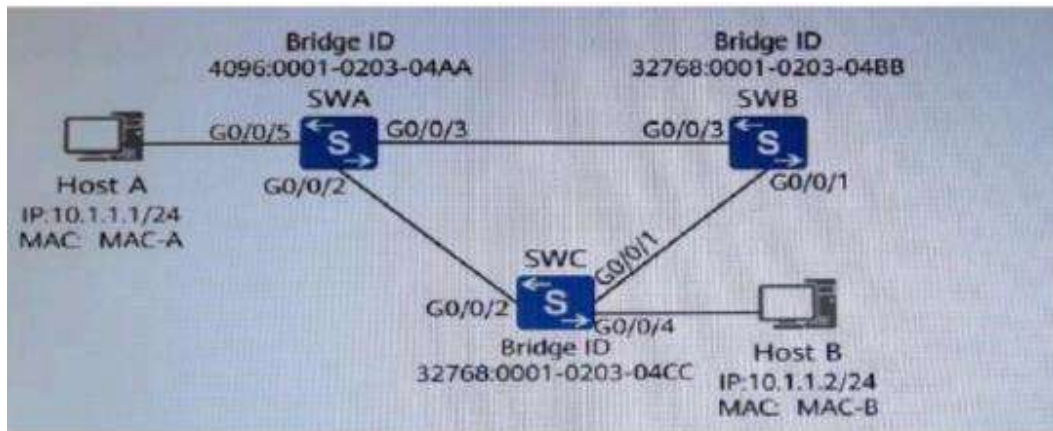
Justification

Explanation:

- A.** If MAC learning is disabled on the port, the switch cannot learn MAC addresses → **Possible**.
- B.** If MAC learning is disabled on the VLAN, MAC addresses won't be learned → **Possible**.
- C.** If Host A sends no frames, SWA can't learn its MAC address → **Possible**.
- D.** **Access mode** just means the port belongs to a single VLAN; it **does NOT stop** MAC address learning → **Impossible** (MAC learning still works in access mode).

The switch (SWA) cannot learn the MAC address of Host A. This suggests that SWA is not receiving frames from Host A. If the G0/0/1 port were configured as a trunk mode port, SWA could potentially receive frames from Host A even if Host A belonged to a different VLAN. However, since the port is in access mode, SWA will only forward frames for the VLAN assigned to that port and will not learn the MAC addresses of devices on other VLANs.

125. As shown in the network below, all switches enable STP protocol. Disable SWA's G0/0/2 port configuration BPDU sending function, SWC G0/0/1 re-converges to become the root port, about this process, which of the following statements are correct?



- A. SWC sends TCN BPDU message to SWB
- B. SWB forwards TCN BPDU to SWA
- C. SWB sends configuration BPDU with TCA set to SWC
- D. SWA sends configuration BPDU with TC set

Answer: ABCD

Config BPDU: Sent every 2 seconds by the root bridge (and relayed by other switches) to maintain the STP topology.

TCN BPDU: Sent by a non-root switch whenever it detects a topology change to notify the root bridge.

Justification

Explanation:

- **SWC sends TCN BPDU to SWB:** When something changes in the network (like a link going down), SWC notices it and sends a special message (TCN BPDU) to SWB to tell it about the change. (Correct.)
- **SWB forwards TCN BPDU to SWA:** SWB gets the TCN BPDU from SWC, and it passes it on to SWA (the main/root switch) to continue informing the network. (Correct)
- **SWB sends Configuration BPDU with TCA set to SWC:** After SWB gets the TCN BPDU from SWC, it sends a confirmation (TCA set) back to SWC to say, "I got your message." (Correct)
- **SWA sends Configuration BPDU with TC set:** Once the root switch (SWA) knows about the change, it tells everyone else in the network by setting the TC flag in the BPDUs it sends out. (Correct)

126. In standard STP mode, which of the following non-root switches will forward the TC set BPDU generated by the root switch?

- A. Designated port
- B. Root port
- C. Backup port
- D. Preparation

Answer: A

Justification

Explanation: Non-root switches receive it via their Root Port and forward it out their Designated Ports to inform downstream switches.

SOURCE

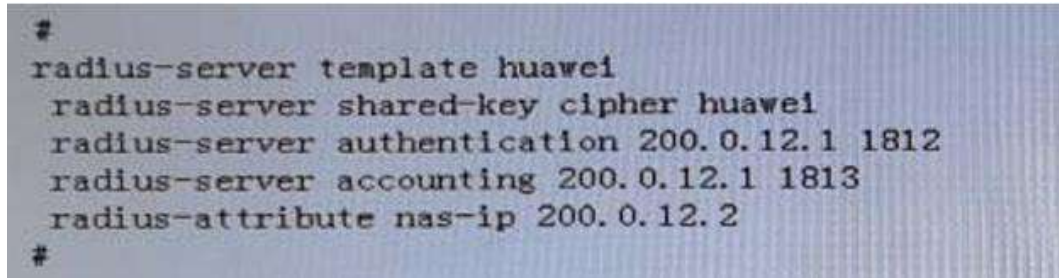
Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086964>

Designated port: The designated port is located on a designated bridge and forwards configuration BPDUs to the device or LAN

SOURCE: Topic 9, SLIDE 26

- Therefore, a designated port is selected for each link to send BPDUs along the link.
- Note: Generally, the root bridge has only designated ports.

127. The router's Radius information configuration is shown in the figure. Which of the following statements is correct?



```
#
radius-server template huawei
radius-server shared-key cipher huawei
radius-server authentication 200.0.12.1 1812
radius-server accounting 200.0.12.1 1813
radius-attribute nas-ip 200.0.12.2
#
```

- A. The source IP address of the Radius packet sent by the router is 200.0.12.2
- B. The IP address of the accounting server is 200.0.12.1
- C. The IP address of the authorization server is 200.0.12.1
- D. The IP address of the authentication server is 200.0.12.1

Answer: BD

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100112357/142fad68/example-forconfiguring-the-primary-and-secondary-radius-servers>

Explanation:

Options B and D are correct as they are both clearly configured in the image.

B. The IP address of the accounting server is 200.0.12.1

- The command:
radius-server accounting 200.0.12.1 1813
Clearly shows the **accounting server IP** is **200.0.12.1** (with port 1813).

D. The IP address of the authentication server is 200.0.12.1

- The command:
radius-server authentication 200.0.12.1 1812
Confirms the **authentication server IP** is also **200.0.12.1** (with port 1812).

128. If a Windows host is started for the first time, if the IP address is obtained by DHCP, then what is the source IP address of the first packet sent by this host?

- A. 127.0.0.1
- B. 169.254.2.33
- C. 0.0.0.0
- D. 255.255.255.255

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100276747/225eec10/dhcp-messages>

ciaddr	4 bytes	Client IP Address: indicates the IP address of a DHCP client. The IP address is either the existing IP address of the client or an IP address allocated by a DHCP server to the client. During the process of a client acquiring an IP address, the value of this field is 0.0.0.0. 0.0.0.0 is an invalid destination address and is used by a DHCP-enabled device to communicate only temporarily with other devices during startup.
--------	---------	--

The DHCP initialization process works like this: A client sends a DHCPDISCOVER message with a source address of 0.0.0.0 and destination address of 255.255.255.255 (broadcast).

129. In order to implement the FIT AP online, the FIT AP first needs to obtain the IP address of the AC. Which of the following is not included in the method of obtaining the IP address of the AC by the FIT AP?

- A. Broadcasting method
- B. Static designation method on AP
- C. Multicast mode
- D. DHCP option mode

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100096325/78331d51/ap-online-process>

An AP can discover an AC in static or dynamic mode.

▪ **Static mode**

An AC IP address list is preconfigured on the AP. When the AP goes online, the AP unicasts a Discovery Request packet to each AC whose IP address is specified in the preconfigured AC IP address list. After receiving the Discovery Request packet, the ACs send Discovery Response packets to the AP. The AP then selects an AC to establish a CAPWAP tunnel according to the received Discovery Response packets.

▪ **Dynamic mode**

An AP can dynamically discover an AC in DHCP, DNS, or broadcast mode. Details on each of the modes are as follows:

- **DHCP mode:** An AP obtains the AC IP address through DHCP (by configuring a DHCP response packet to carry Option 43 for IPv4 packets or Option 52 for IPv6 packets containing the AC IP address list on the DHCP server), and sends a Discovery Request unicast packet to the AC. The AC then sends a Discovery Response packet to the AP.
- **DNS mode:** An AP obtains the AC domain name and DNS server IP address through the DHCP service (by configuring a DHCP response packet to carry Option 15 for IPv4 packets or Option 24 for IPv6 packets containing the AC domain name on the DHCP server), and sends a request to the DNS server to obtain the IP address corresponding to the AC domain name. After obtaining the AC IP address, the AP unicasts a Discovery Request packet to the AC. The AC then sends a Discovery Response packet to the AP.

After receiving the DHCP Response packet, the AP obtains the AC domain name carried in Option 15 or Option 24. The AP then automatically adds the prefix **huawei-wlan-controller** to the obtained domain name and sends it to the DNS server to obtain the IP address corresponding to the AC domain name. For example, after obtaining the AC domain name **ac.test.com** configured on the DHCP server, the AP adds the prefix **huawei-wlan-controller** to **ac.test.com** and sends the **huawei-wlan-controller.ac.test.com** to the DNS server for resolution. The IP address corresponding to **huawei-wlan-controller.ac.test.com** must be configured on the DNS server.

- **Broadcast mode:** The AP broadcasts a Discovery Request packet to automatically discover ACs in the same network segment. The AP then selects an AC to establish a CAPWAP tunnel based on the Discovery Response packets received from available ACs. The broadcast mode is used in the following cases:
 - When the AP fails to obtain the IP address of an AC, it broadcasts packets. If no AC responds to the broadcast packet for two times, a failure to discover an AC is determined.

130. How to add AP on AC?

- A. Add online
- B. Automatically discover AP
- C. Import AP offline
- D. Manually confirm APs in the uncertified list.

Answer: BCD

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100096325/1e602044/adding-aps>

You can add APs in any of the following modes:

- Importing APs offline: The APs' MAC addresses and serial numbers (SNs) are configured on an AC before APs go online. The AC starts to set up connections with the APs if the MAC addresses or SNs of the APs match the configured ones.
- Configuring the AC to automatically discover an AP: The AP authentication mode is set to no authentication; alternatively, the AP authentication mode is set to MAC or SN authentication and the AP whitelist is configured on the AC. When an AP in the whitelist connects to the AC, the AC discovers the AP, and the AP goes online.
- Manually confirming APs added to the list of unauthorized APs: The AP authentication mode is set to MAC or SN authentication, and the AP whitelist is configured on the AC. When an AP out of the whitelist connects to the AC, the AC adds the AP to the list of unauthorized APs. After the AP identity is confirmed, the AP can go online.

131. Which of the following messages is used by AP to request the software version from AC?

- A. VRP System Request
- B. Software Request
- C. Image Data Request
- D. Image package Request

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000060368/48c453a5/ap-joinprocess#sec?on5941142153214>

After receiving the Join Response message, the AP checks whether the current system software version is the same as that expected by the AC. If not, the AP enters the **Image data** state and starts to download the upgrade file and upgrade its software version in AC, FTP, or SFTP mode. After the upgrade is complete, the AP restarts and repeats the preceding phases. If the upgrade fails, the AP restarts and repeats the previous phases. Therefore, if the AP upgrade configuration is incorrect, the AP may continuously repeat the preceding phases until the AP software version is upgraded correctly.

132. The IEEE 802.11 ac standard only supports the 5 GHz frequency band.

- A. TRUE
- B. FALSE

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100081215>

No.	Improvement	Description	Benefit
1	Channel bandwidth	▪ Adds 80M channel bandwidth. ▪ Adds 160M channel bandwidth. ▪ Integrates two non-adjacent 80M channel bandwidths into a 160M channel bandwidth.	Improved maximum throughput
2	Working frequency	Works in the frequency band less than 6 GHz, excluding 2.4 GHz. The frequency band of 5 GHz is mainly used.	▪ Abundant spectrum resources ▪ Less interference

133. To query the configuration information of the device's OSPF protocol, which of the following commands can be used?

- A. display ospf peer
- B. Enter the command "display this" in the OSPF protocol view
- C. display current-configuration
- D. display ip routing-table

Answer: BC

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064353/32b784cf/display-currentconfiguration>

Source: https://support.huawei.com/enterprise/en/doc/EDOC1100015130/b3431a1f/ospf-2ospfnbrstatechange_ac?ve

Display all configurations that include **vlan**.

```
<Huawei> display current-configuration | include vlan
```

```
vlan batch 10 77 88  
port trunk allow-pass vlan 10
```

Run the **display this** command in both the interface view and OSPF view of the local and remote devices to check whether the versions of OSPF configured at both ends are the same.

134. Which of the following does not include in the VRP system login method?

- A. Telnet
- B. Netstream
- C. SSH
- D. web

Answer: B

Justification

SOURCE: Topic 3, Page 10

Device Management

There are two commonly used device management modes: CLI and web system.

Web System	CLI
• The web system provides a graphical user interface (GUI) for easy device management and maintenance. This method, however, can be used to manage and maintain only some, not all, device functions. • The web system supports the HTTP and HTTPS login modes.	• The CLI requires users to use commands provided by a device to manage and maintain the device. This mode implements refined device management but requires users to be familiar with the commands. • The CLI supports the console port, Telnet, and SSH login modes.

SOURCE: Topic 3, Page 13, 15

Login to the Web System

Take the web system for a Huawei AR router as an example. Start a browser on a PC, enter **https://192.168.1.1** in the address bar, and press **Enter**. Then, the web system login page is displayed.

CLI - Local Login (2)

PuTTY is a connection software for login through Telnet, SSH, serial interfaces, and so on.

135. In which view can the administrator modify the device name for the router?

- A. User-view
- B. Protocol-view
- C. System-view
- D. Interface-view

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100262557/bf7a8ec5/setting-the-device-name>

Setting the Device Name

Context

To differentiate devices on the network, you can set a unique device name for each device.

Procedure

1. Enter the system view.

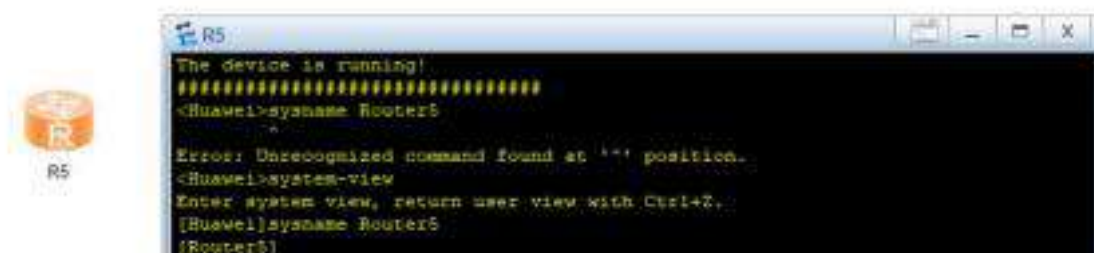
```
system-view
```

2. Set the device name.

```
sysname host-name
```

By default, the device name is HUAWEI.

You can run the **undo sysname** command to restore the default host name.



136. If the transport layer protocol is UDP, the value of the network layer Protocol field is 6.

- A. True
- B. False

Answer: B

Justification

Source: https://en.wikipedia.org/wiki/List_of_IP_protocol_numbers

Hex	Protocol Number	Keyword	Protocol	References/RFC
0x11	17	UDP	User Datagram Protocol	RFC 768 ↗

Source: <https://networklessons.com/cisco/ccna-routing-switching-icnd1-100-105/ipv4-packet-header>

Protocol: this 8 bit field tells us which protocol is encapsulated in the IP packet, for example TCP has value 6 and UDP has value 17.

137. If the application layer protocol is Telnet, what is the value of the Protocol field in the IPv4 header?

- A. 23
- B. 17
- C. 6
- D. 67

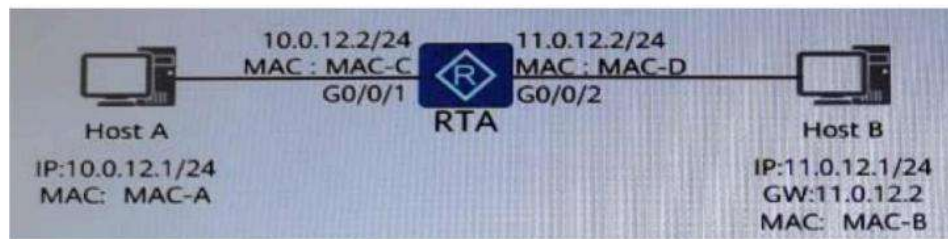
Answer: C

Justification

Source: <https://networklessons.com/cisco/ccna-routing-switching-icnd1-100-105/ipv4-packet-header>

Protocol: this 8 bit field tells us which protocol is encapsulated in the IP packet, for example TCP has value 6 and UDP has value 17.

138. In the network shown in the figure below, host A is not configured with a gateway, and host B has the ARP cache of the gateway. use the command ping 11.0.12.1 on host A. Which of the following statements are correct?

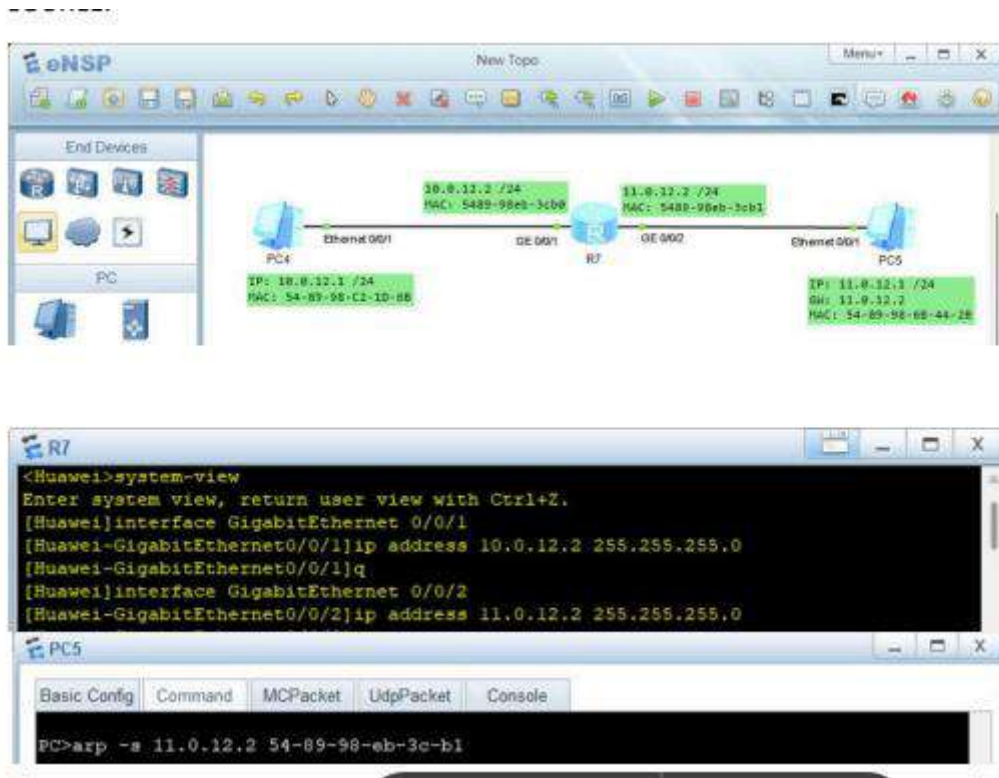


- A. No data packets will be sent from host A
- B. The IP address of the data packet sent by host A is 11.0.12.1
- C. The destination MAC address of the data frame sent by host A is MAC-C
- D. The destination MAC address of the data frame sent by host A is MAC-B

Answer: A

Justification

Source: Host A doesn't know where to send packets (no gateway + IP outside its local network), so it won't send anything.



139. Which layer of the OSI reference model does the router work at?

- A. Data link layer
- B. Transport layer
- C. Application layer
- D. Network layer

Answer: D

Justification

SOURCE: Topic 1, Page 12

- Routers:
 - Routers work at the network layer of the TCP/IP reference model.

140. VRP operating platform, which of the following commands can only view static routes?

- A. display ip routing-table statistics
- B. display ip routing-table verbose
- C. display ip routing-table
- D. display ip routing-table protocol static

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100333923/a884a514/display-ip-routing-table-protocol>

Function

The **display ip routing-table protocol** command displays routing information about a specified routing protocol.

Parameter	Description
protocol	Displays routing information about the specified routing protocol: • direct: displays direct routes. • static: displays static routes. • unr: displays user network routes.

141. Which OSPF version below is suitable for IPv6?

- A. OSPFv1
- B. OSPFv2
- C. OSPFv3
- D. OSPFv4

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100082074>

At present, OSPF Version 2, defined in RFC 2328, is intended for IPv4, and OSPF Version 3, defined in RFC 2740, is intended for IPv6.

142. `ip route-static 10.0.2.2 255.255.255.255 10.0.12.2 preference 20`. Which is correct about this command?

- A. The NextHop of this route is 10.0.12.2
- B. The priority of this route is 20
- C. This route can guide the forwarding of packets with the destination IP address 10.0.2.2.
- D. This route can guide the forwarding of packets with the destination IP address 10.0.12.2

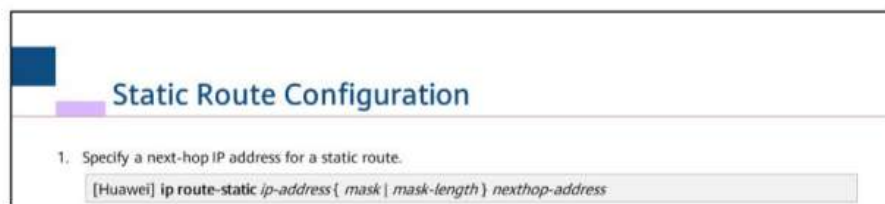
Answer: ABC

Justification

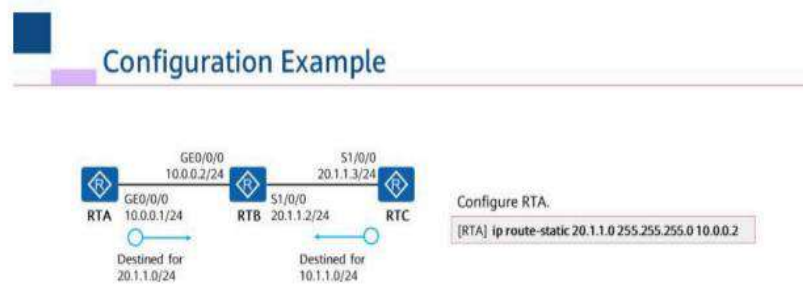
Explanation:

- A. The NextHop is **10.0.12.2** → **Correct** (clearly specified in the command).
- B. The priority (preference) is **20** → **Correct** (given after preference keyword).
- C. The route matches destination **10.0.2.2/32** → **Correct** (only packets to 10.0.2.2).
- D. **Wrong** → 10.0.12.2 is **NextHop**, not the destination.

SOURCE: Topic 5, Page 28



SOURCE: Topic 5, Page 29



SOURCE: Topic 5, Page 39

```
[RTA] ip route-static 20.0.0.0 30 10.1.1.2
```

SOURCE: Topic 5, Page 14

- Pre (Preference): indicates the routing protocol preference of the route. There may be multiple routes to the same destination, which have different next hops and outbound interfaces. These routes may be discovered by different routing protocols or be manually configured. A router selects the route with the highest preference (with the lowest preference value) as the optimal route.

143. What are the following characteristics of the default VLAN in the Layer 2 Ethernet switch?

- A. The default VLAN cannot be deleted manually
- B. By default, all ports of the switch are member ports of the default VLAN
- C. A default VLAN must be created before it can be assigned a port
- D. The IP address configured on the switch will only be applied to the member ports of the default VLAN

Answer: ABD

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000178168/6907bedb/default-vlan>

Option B is correct

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000178168/6907bedb/default-vlan>

Each interface has a default VLAN. By default, the default VLAN ID of all interfaces is VLAN 1. You can change the default VLAN ID as required.

Option A is correct, and Option C is incorrect

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100278261/55c108dd/crea?ng-and-dele?ng-avlan>

As VLAN 1 is the default VLAN, it cannot be deleted and does not need to be created.

The default VLAN ID of an interface is called the port default VLAN ID (PVID). Frames processed in a switch all carry VLAN tags. When the switch receives an untagged frame, it adds a VLAN tag to the frame according to the default VLAN of the interface that receives the frame. The PVID is used in the following scenarios:

- When an interface receives an untagged frame, the interface adds a tag with the PVID to the frame and sends the frame to the switch for processing. When an interface receives a tagged frame, the switch does not add a tag with the PVID to the frame.
- When an interface sends a frame in which the VLAN ID is the same as the PVID, the switch removes the tag from the frame before sending it out from the interface.

Each interface has a default VLAN. By default, the default VLAN ID of all interfaces is VLAN 1. You can change the default VLAN ID as required:

- The default VLAN of an access interface is the VLAN allowed by the access interface. To change the default VLAN of an access interface, change the allowed VLAN.
- Trunk and hybrid interfaces allow multiple VLANs but have only one default VLAN. Changing the allowed VLANs will not change the default VLAN.

144. Switch stacking supports more than two devices. Through stacking, multiple switches can be grouped into one logical device.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100102781>

How Many Member Switches Can a Stack Contain?

A stack can contain a maximum of nine member switches.

After stacking is enabled, multiple devices at the aggregation layer are combined into a single logical device (stack), and access devices are directly connected to the stack. This stack networking does not require MSTP or VRRP, simplifying network configurations. If a stack member is faulty, inter-device link aggregation speeds up network convergence and improves network reliability.

145. On which layer of the OSI reference model does the Ethernet switch work?

- A. Network layer
- B. Data link layer
- C. Physical layer
- D. Transport layer

Answer: B

Justification

SOURCE: Topic 1, Page 13

- Switches:
 - On a campus network, a switch is the device closest to end users and is used to connect terminals to the campus network. Switches at the access layer are usually Layer 2 switches and are also called Ethernet switches. Layer 2 refers to the data link layer of the TCP/IP reference model.

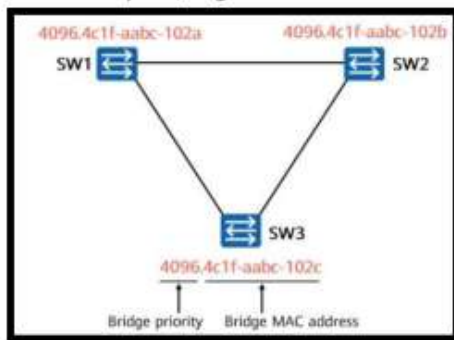
146. The bridge ID of the STP protocol in the switching network is as follows. Which of the following bridge ID switches will become the root bridge?

- A. A.4096 00-01-02-03-04-DD
- B. B.32768 00-01-02-03-04-AA
- C. 32768 00-01-02-03-04-cc
- D. D.32768 00-01-02-03-04-BB

Answer: A

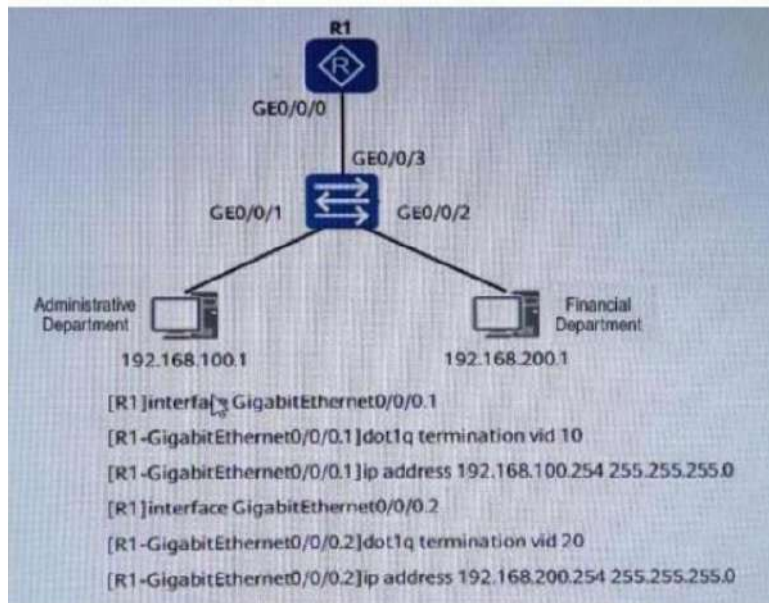
Justification

SOURCE: Topic 9, Page 13



- In STP, each switch has a bridge ID (BID), which consists of a 16-bit bridge priority and a 48-bit MAC address. On an STP network, the bridge priority is configurable and ranges from 0 to 65535. The default bridge priority is 32768. The bridge priority can be changed but must be a multiple of 4096. The device with the highest priority (a smaller value indicates a higher priority) is selected as the root bridge. If the priorities are the same, devices compare MAC addresses. A smaller MAC address indicates a higher priority.

147. Refer to the route-in-a-stick configuration in the figure below, it can be judged that the administrative department and the financial department can communicate with each other even if the ARP proxy is not enabled on the RI sub-interface.



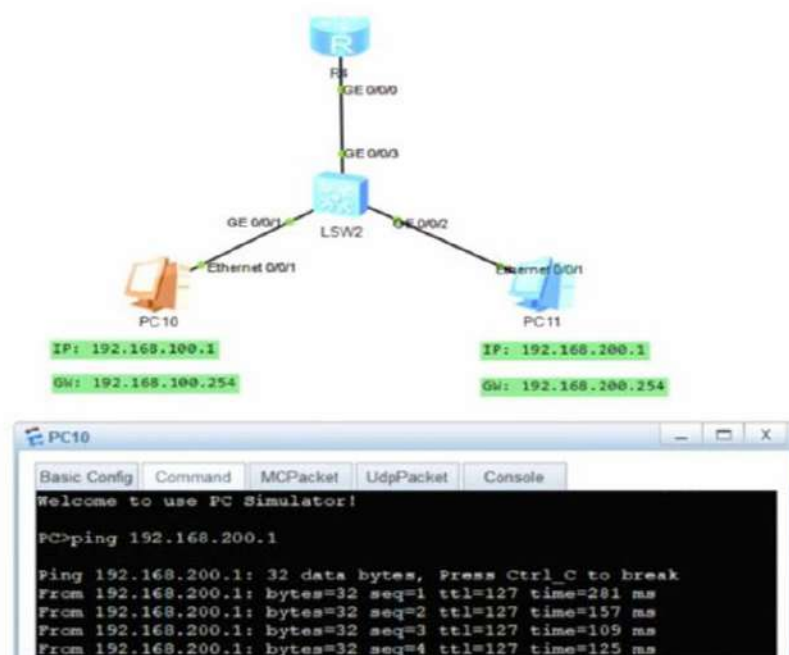
- A. TRUE
B. FALSE

ARP Proxy is primarily required when hosts are in the same subnet but different broadcast domains

Answer: A

Justification

Source: Huawei eNSP



148. Link aggregation is a common technology in enterprise networks. Which of the following descriptions are the advantages of link aggregation technology?

- A. Improve safety
- B. Realize load sharing
- C. Increase bandwidth
- D. Improve reliability

Answer: BCD

Justification

SOURCE: Topic 11, Page 10

Eth-Trunk

- Ethernet link aggregation, also called Eth-Trunk, bundles multiple physical links into a logical link to increase link bandwidth, without having to upgrade hardware.

Load Balancing Mode

- An Eth-Trunk can load balance traffic based on IP addresses or MAC addresses of packets. You can configure different load balancing modes (valid locally only for outgoing packets) to distribute data flows to different member interfaces.

Link Reliability

The diagram illustrates the concept of link reliability using Spanning Tree Protocol (STP). It shows two network configurations. In the first configuration, there is a single link between an Aggregation switch and an Access switch. In the second configuration, a second link is added between the same two switches. A callout box explains that this new link is blocked by STP and functions as a backup link to improve link reliability.

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HUAWEI

149. The network administrator configures the Huawei router RTA as shown in the figure. If a user needs to be authenticated in the authentication domain "huawei", the correct description below is ().

```
[RTA] aaa
[RTA-aaa] domain Huawei
[RTA-aaa-domain-huawei] authentication-scheme au1
[RTA-aaa-domain-huawei] authentication-scheme au2
```

- A. "authentication-scheme au1" authentication will be used, if "au1" is deleted, "au2" authentication will be used
- B. Will use "au2" authentication and will use "authentication-scheme au1" authentication
- C. Will use "authentication-scheme au2" authentication
- D. "authentication-scheme au2" authentication will be used, if "au2" is registered, "au1" authentication will be used

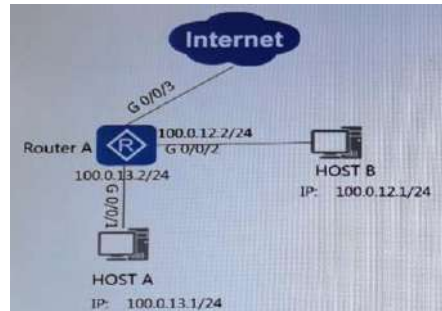
Answer: C

Justification

Explanation: The configuration [RTA-aaa-domain-huawei]authentication-scheme au2 explicitly configures the authentication scheme named "au2" within the domain "Huawei." This means that any authentication requests within the "Huawei" domain will use the "au2" authentication scheme. The configuration does not mention "au1" or imply any conditional use based on deletion or registration of authentication schemes. Therefore, option C accurately describes the configuration provided in the Huawei router.

The au2 authentication scheme is configured as the primary authentication scheme for the huawei authentication domain. This means that the router will always try to authenticate users using the au2 authentication scheme first. If the au2 authentication scheme fails, the router will not try to authenticate the user using the au1 authentication scheme.

150. In the network shown in the figure, which of the following configurations can realize that host A can access the Internet, but host B cannot? (Multiple Choice)



- A. acl number 2000
rule 5 deny source 100.0.12.0 0.0.0.255
rule 10 permit source 100.0.13.0 0.0.0.255
interface GigabitEthernet0/0/3
traffic-filter outbound acl 2000
- B. acl number 2000
rule 5 deny source 100.0.12.0 0.0.0.255
rule 10 permit
interface GigabitEthernet0/0/3
traffic-filter outbound acl 2000
- C. acl number 2000
rule 5 permit source 100.0.12.0 0.0.0.255
rule 10 deny source 100.0.13.0 0.0.0.255
interface GigabitEthernet0/0/3
traffic-filter outbound act 2000
- D. acl number 2000
rule 5 permit source 100.0.12.0 0.0.0.255
rule 10 deny
interface GigabitEthernet0/0/3
traffic-filter outbound act 2000

Answer: AB

Justification

SOURCE: Topic 12, Slide 18

TAKING THIS EXAMPLE, OPTION A AND B ARE THE ONLY CONFIGURATIONS WHERE NETWORK 100.0.12.0 IS DENIED ACCESS TO INTERNET AND THE NETWORK OF 100.0.13.0 IS PERMITTED.

Case: Use a Basic ACL to Filter Data Traffic

Requirements:
To prevent the user host on the network segment 192.168.1.0/24 from accessing the network where the server resides, configure a basic ACL on the router. After the configuration is complete, the ACL filters out the data packets whose source IP addresses are on the network segment 192.168.1.0/24 and permits other data packets.

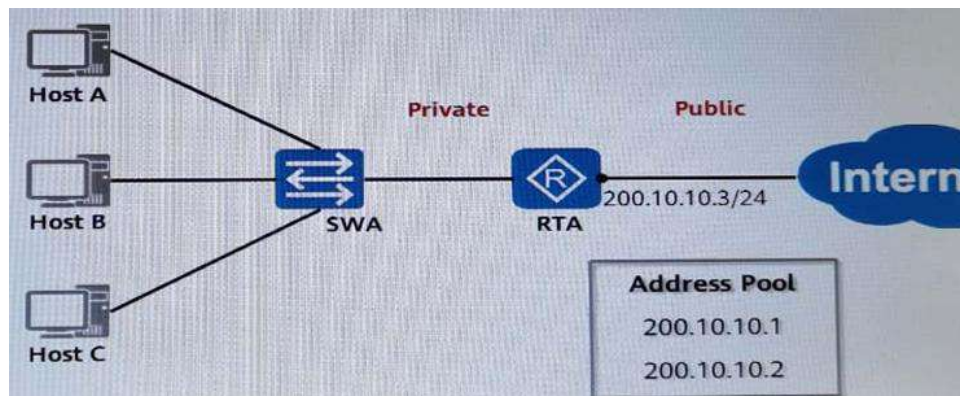
1. Configure IP addresses and routes on the router.
2. Create a basic ACL on the router to prevent the network segment 192.168.1.0/24 from accessing the network where the server resides.

```
[Router] acl 2000
[Router-acl-basic-2000] rule deny source 192.168.1.0 0.0.0.255
[Router-acl-basic-2000] rule permit source any
```
3. Configure traffic filtering in the inbound direction of GE 0/0/1.

```
[Router] interface GigabitEthernet 0/0/1
[Router-GigabitEthernet0/0/1] traffic-filter inbound acl 2000
[Router-GigabitEthernet0/0/1] quit
```

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151. As shown in the figure, RTA uses NAT technology and realizes multiple-to-multiple NAPT address translation by defining an address pool, so that hosts in the private network can access the public network. Suppose that there are only two public IP addresses in the address pool, and they have been assigned to hosts A and B for address conversion, and if host C also wants to access the public network, which of the following description is correct?



- A. Host C cannot be assigned a public network address and cannot access the public network
- B. All hosts can access the public network by using the public network address turn
- C. RTA converts the source port of host C, and host C can access the public network
- D. RTA assigns the interface address (200.10.10.3) to host C, host C can access the public network

Answer: C

Justification

Explanation: Limited Public IP Addresses: The scenario describes a situation where there are only two public IP addresses available (assigned to Host A and B) but multiple devices (including Host C) need access to the public network. NAT allows routers to translate private IP addresses on a private network to public IP addresses for internet access.

NAPT (Network Address Port Translation): The fact that the router uses NAPT indicates it translates not only the IP addresses but also the port numbers. This allows multiple devices behind the router to share the same public IP address. Port Translation: When Host C initiates a connection to the internet, the RTA (Router) translates the private IP address and source port of Host C to the public IP address (shared by other devices) and a unique, unused port number. This allows the communication to be routed to the internet and responses back to be directed to Host C based on the translated port number.

Incorrect Options:

- A. While there are only two public IP addresses, NAPT allows sharing a single public IP with multiple devices using port translation.
- B. NAPT doesn't require taking turns. Each device can access the internet simultaneously using unique port numbers.
- D. The interface address (200.10.10.3) is likely the IP address of the router itself, not something that can be directly assigned to Host C.

152. The correct description of the information as shown in the figure is ().

```
[R1] display interface G0/0/0  
current state: Administratively DOWN  
Line protocol current state: DOWN
```

- A. The GigabitEthernet 0/0/0 interface was manually closed by the administrator
- B. GigabitEthernet 0/0/10 interface does not enable dynamic routing protocol
- C. The GigabitEthernet 0/0/0 interface is not configured with an IP address
- D. GigabitEthernet 0/0/0 interface is connected to a wrong cable

Answer: A

Justification

- If the physical state of an interface is **Administratively DOWN**, the administrator has manually shut down the interface.

153. As shown in the figure below, all switches enable the STP protocol and keep other configurations in their default state. When the network is stable, which of the following statements are correct? (Multiple choice) No Figure Attached

- A. Both ports of SWB are in the forwarding state
- B. SWB is the root bridge in this network
- C. SWA is the root bridge in this network
- D. Both ports of SWC are in the forwarding state

Answer: AC

Justification

Forwarding	A port in Forwarding state can forward user traffic and process BPDUs. Only the root port and designated port can enter the Forwarding state.
------------	---

Explanation: Since SWB has both a root port and a designated port, it must be in the forwarding state. While one port of SWC is in block state, option D is incorrect, while option A is correct

Source: Huawei Lecture # 9, Page # 13

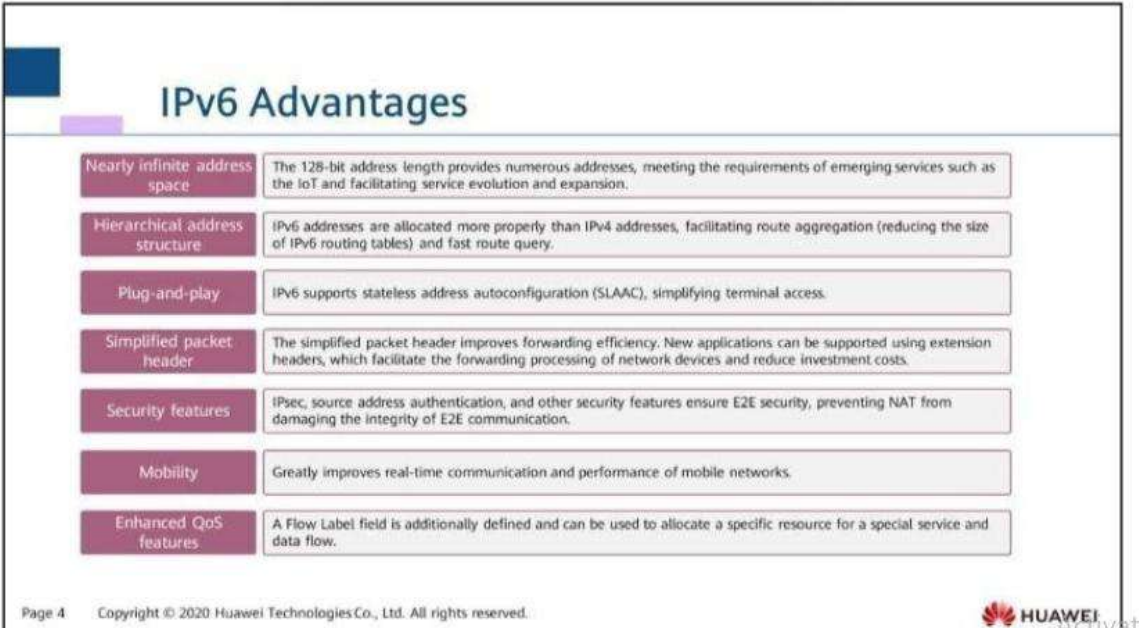
- In STP, each switch has a bridge ID (BID), which consists of a 16-bit bridge priority and a 48-bit MAC address. On an STP network, the bridge priority is configurable and ranges from 0 to 65535. The default bridge priority is 32768. The bridge priority can be changed but must be a multiple of 4096. The device with the highest priority (a smaller value indicates a higher priority) is selected as the root bridge. If the priorities are the same, devices compare MAC addresses. A smaller MAC address indicates a higher priority.

154. What is the total length of the IPv6 address in bits?

- A. 96
- B. 64
- C. 32
- D. 128

Answer: D

Justification



The slide titled "IPv6 Advantages" lists seven key features of IPv6 in a two-column table. The first column contains feature names in purple boxes, and the second column contains descriptions in white boxes. The features are: Nearly infinite address space, Hierarchical address structure, Plug-and-play, Simplified packet header, Security features, Mobility, and Enhanced QoS features. The footer includes page information, copyright notice, and the Huawei logo.

IPv6 Advantages	
Nearly infinite address space	The 128-bit address length provides numerous addresses, meeting the requirements of emerging services such as the IoT and facilitating service evolution and expansion.
Hierarchical address structure	IPv6 addresses are allocated more properly than IPv4 addresses, facilitating route aggregation (reducing the size of IPv6 routing tables) and fast route query.
Plug-and-play	IPv6 supports stateless address autoconfiguration (SLAAC), simplifying terminal access.
Simplified packet header	The simplified packet header improves forwarding efficiency. New applications can be supported using extension headers, which facilitate the forwarding processing of network devices and reduce investment costs.
Security features	IPsec, source address authentication, and other security features ensure E2E security, preventing NAT from damaging the integrity of E2E communication.
Mobility	Greatly improves real-time communication and performance of mobile networks.
Enhanced QoS features	A Flow Label field is additionally defined and can be used to allocate a specific resource for a special service and data flow.

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155. The IPv4 address assigned by the network administrator to a host in the network is 192.168.1.1/28. How many hosts can be added to the network where this host is located?

- A. 12
- B. 15
- C. 13
- D. 14

Answer: C

Justification

Explanation:

Formula to calculate the number of usable IP addresses in a subnet is $2^{\text{number of host bits}} - 2$.

For a /28 subnet:

Number of host bits = $32 - 28 = 4 \text{ bits}$

So, the number of usable IP addresses =

$$2^4 - 2 = 16 - 2 = 14$$

Since one host has already been assigned, 13 more hosts can be added to this network.

156. All ports on the root bridge switch are designated ports.

- A. True
- B. False

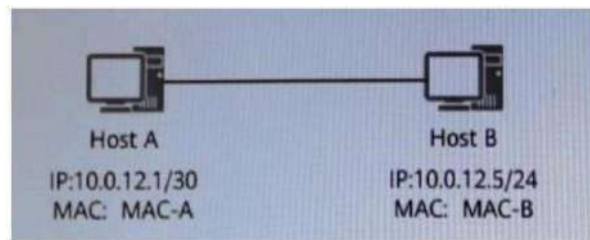
Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000017271/bd821590/stp-rstp-overview>

Designated port: forwards BPDUs to the downstream switching device. All the ports on the root bridge are designated ports.

157. As shown in the figure below, host A and host B cannot communicate.

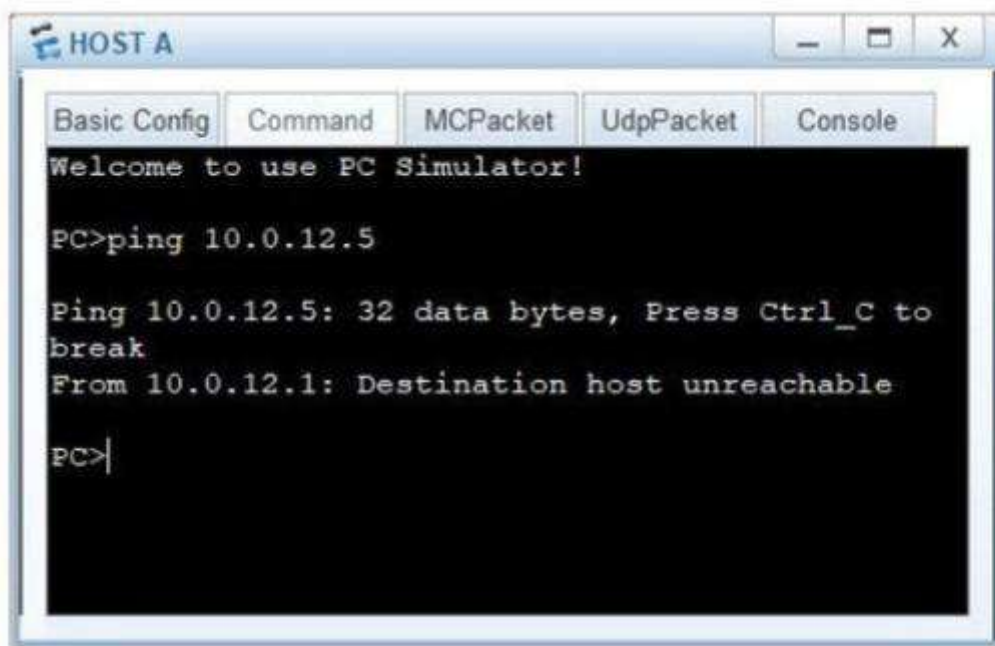


- A. True
- B. False

Answer: A

Justification

Source: Huawei eNSP



158. Which of the following statements about IPv6 multicast addresses are correct?

- A. Provide redundant functions for services
- B. Realize load sharing of services
- C. The packet whose destination address is a multicast address will be sent to the nearest network interface in the sense of routing
- D. Multicast addresses and unicast addresses use the same address space

Answer: ABC

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100365387/2db43c79/mul?cast-ipv6-loadspliBng>

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100278983/84632e62/understanding-ipv6>

159. What is the number range of advanced ACL?

- A. 3000-3999
- B. 2000-2999
- C. 4000-4999
- D. 6000-6031

Answer: A

Justification

Source: Huawei Lecture # 6, Page # 10

• ACL classification based on ACL rule definition methods		
Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
Advanced ACL	3000 to 3999	Defines rules based on source and destination IPv4 addresses, IPv4 protocol types, ICMP types, TCP source/destination port numbers, UDP source/destination port numbers, and effective time ranges.
Layer 2 ACL	4000 to 4999	Defines rules based on information in Ethernet frame headers of packets, such as source and destination MAC addresses and Layer 2 protocol types.
User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

160. The basic header length of IPv6 packets is a fixed value.

- A. True
- B. False

Answer: A

Justification

Source: Huawei Lecture # 19, Page # 35

Comparison	IPv6	IPv4
Address length	128 bits	32 bits
Packet format	A fixed 40-byte basic packet header+variable-length extension headers	A basic header containing the Options field to support extended features

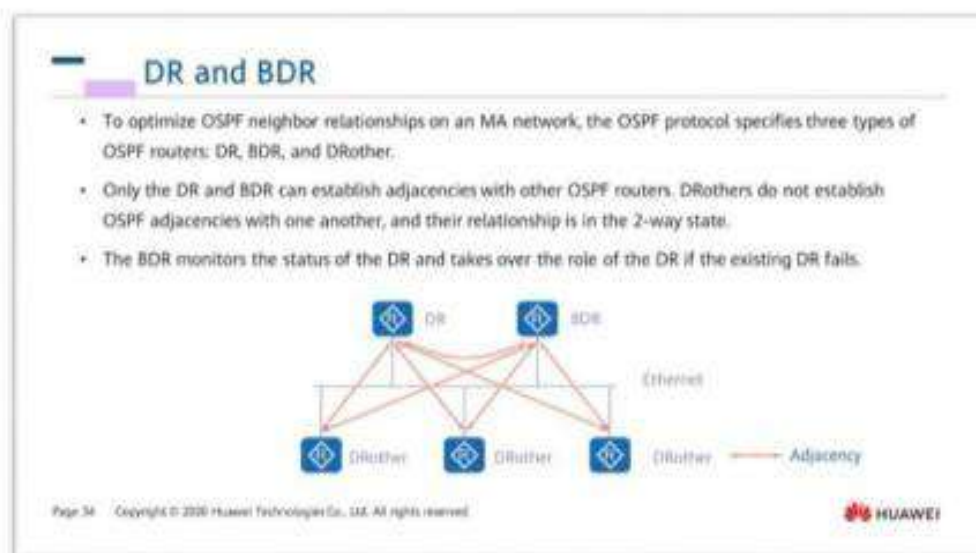
161. In which of the following network types does the OSPF protocol need to elect DR and BDR?

- A. Point to multipoint
- B. NBMA
- C. Point-to-point type
- D. Broadcast type

Answer: BD

Justification

Source: Huawei Lecture # 6, Page # 34



162. By default, how long is the lease period of the IP address assigned by the DHCP server?

- A. 12h
- B. 1h
- C. 24h
- D. 18h

Answer: C

Justification

Source: Huawei Lecture # 15, Page # 27

5. Configure the lease of IP addresses in the interface address pool of the DHCP server.

```
[Huawei-GigabitEthernet0/0/0]dhcp server lease { day day [ hour hour [ minute minute ] ] | unlimited }
```

By default, the IP address lease is one day.

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163. When deploying ACL on Huawei devices, which of the following descriptions is correct?

- A. ACL can match the TCP/UDP port number of the message, and can specify the range of port numbers
- B. The same ACL can be called under multiple interfaces
- C. When ACL defines rules, it can only proceed in the order of 10, 20, and 30
- D. ACL cannot be used to filter OSPF traffic, because OSPF traffic does not use UDP protocol encapsulation
- E. When calling ACL under the interface, it can only be applied to the outbound direction

Answer: AB

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086647>

Matching option: ACLs support many matching conditions, including Layer 2 Ethernet frame header information (source MAC, destination MAC, and Ethernet protocol type), Layer 3 packet information (destination address and protocol type), and Layer 4 packet information (TCP/UDP port number). For details about ACL matching conditions, see Matching Conditions.

An advanced ACL can filter packets based on protocol types, such as ICMP (protocol number 1), TCP (protocol number 6), UDP (protocol number 17), GRE (protocol number 47), IGMP (protocol number 2), IP (any IP layer protocol), IPinIP (protocol number 4), and OSPF (protocol number 89). The protocol number ranges from 1 to 255.

To ensure financial data security, access to the financial server is allowed only from the president office; access from the R&D department to the financial server is blocked. The implementation method is as follows:
Configure an ACL in the inbound direction of interface 1 to block the packets from the R&D department to the financial server. The ACL does not need to be configured on Interface 2, so the packets from the president office to the financial server are allowed.

Category	IP Version	Rule Definition Description	Number Range
Basic ACL	IPv4	Defines rules based on source IP addresses, fragmentation information, and time ranges.	2000-2999

Format of source port number: **source-port** { **eq** port | **gt** port | **lt** port | **range** port-start port-end }

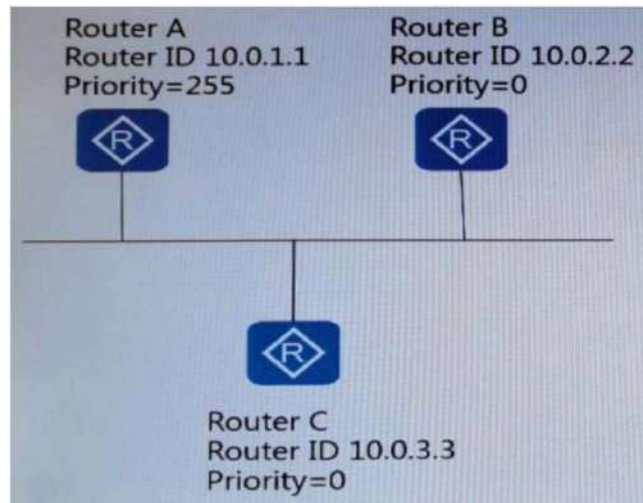
Format of destination port number: **destination-port** { **eq** port | **gt** port | **lt** port | **range** port-start port-end }

When the protocol type of an advanced ACL is specified as TCP or UDP, the device can filter packets based on TCP or UDP source/destination port numbers.

Source: <https://community.cisco.com/t5/networking-knowledge-base/cisco-access-control-lists-acl/tap/4182349>

Cisco ACLs are single or multiple permit/deny statements that filter inbound or outbound packets on a network interface. Only two ACLs are permitted per network interface per protocol. For example, a single IP ACL applied inbound and a single IP ACL applied outbound. There are a variety of ACL types available that are configured based on security requirements.

164. In the network shown in the figure below, when the OSPF neighbor status is stable, what is the neighbor status of Router B and Router C?



- A. Attempt
- B. 2-way Priority = 0 → never becomes DR or BDR (always DROther)
- C. Full For Full-State, 1 must be DR / BDR.
- D. Down

Answer: B

Justification

Source: <https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/13685-13.html>

2-Way

This state designates that bi-directional communication has been established between two routers. Bi-directional means that each router sees the hello packet from the other router. This state is attained when the router receiving the hello packet sees its own Router ID within the received hello packet neighbor field. At this state, a router decides whether to become adjacent with this neighbor. On broadcast media and non-broadcast multi-access networks, a router becomes **full** only with the designated router (DR) and the backup designated router (BDR); it stays in the 2-way state with all other neighbors. On Point-to-point and Point-to-multipoint networks, a router becomes full with all connected routers.

At the end of this stage, the DR and BDR for broadcast and non-broadcast multi-access networks are elected. For more information on the DR election process, refer to [DR Election](#).

165. Which parameter is used in the spanning tree protocol for root bridge election?

- A. Port ID
- B. Bridge priority
- C. Root path cost
- D. Bridge ID

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086964>

In addition, a switching device has a bridge ID (BID) and a port on the switching device has a port ID (PID). A PID is composed of a port priority and a port number. A BID is composed of a bridge priority and a bridge MAC address. On an STP network, the device with the smallest BID acts as the root bridge.

166. MIB (Management Information Base) is a virtual database that exists only in NMS.

- A. True
- B. False

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100111019/7f52c07b/snmp-based-networkmanagement>

- MIB

The management information base (MIB) is a set of managed objects (MOs). It acts as a communication bridge between the NMS and an agent. Each agent maintains a MIB whose object values can be read or set by the NMS.

167. Now it is necessary to implement a Python automation script Telnet to the device to view the device running configuration. Which of the following statements is wrong?

- A. Telnetlib can achieve this function.
- B. Use telnet.Telnet (host) to connect to the Telnet server.
- C. can use telnet.write(b"display current-configuration \n") to input the command to view the current configuration of the device.
- D. telnet.close () is used after a command is entered everytime; the function is that the user waits for the switch to return information.

Answer: D

Justification

Source: <https://docs.python.org/3/library/telnetlib.html#telnetlib.Telnet.close>

A screenshot of a terminal window with a dark background. The first line shows the command `telnet.close()` in a light-colored font. The second line shows the output `Close the connection.` in a slightly lighter color.

168. The network administrator of a certain company wants to uniformly divide the users who often access the company network from different switches into VLAN 10 when they frequently change office locations. Which of the following methods should be used to divide VLANs?

- A. Divide VLAN based on protocol
- B. Divide VLAN based on MAC address
- C. Divide VLAN based on the port
- D. Divide VLAN based on the subnet best for structured IP management with no mobility

Answer: D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000017271/a6bce928/dividing-a-lan-into-vlansbased-on-ip-subnets>

IP subnet-based and protocol-based VLAN division are called network layer-based VLAN division, which reduces manual VLAN configuration workload and allows users to easily join a VLAN, transfer from one VLAN to another, and exit from a VLAN. IP subnet-based VLAN division is applicable to networks that have traveling users and require simple management.

169. The picture shows the display startup information of the AR2200 router. Regarding this information, which of the following statements is wrong?

```
<Huawei>display startup
MainBoard:
  Startup system software:          sdi:/ar2220-v200r003c00spc200.cc
  Next startup system software:     sdi:/ar2220-v200r003c00spc200.cc
  Backup system software for next startup:  null
  Startup saved-configuration file:  null
  Next startup saved-configuration file:  null
  Startup license file:             null
  Next startup license file:        null
  Startup patch package:            null
  Next startup patch package:       null
  Startup voice-files:              null
  Next startup voice-files:         null
```

- A. next system startup file system can use the command "startup system software<startup-software-name>" to modify
- B. The running configuration file is not saved
- C. The system file used by the device this time is ar2220-v200r003c00spc00.cc
- D. The system file cannot be modified when the device is started next time

Answer: D

Justification

Explanation:

- A. next system startup file system can use the command "startup system software <startup- softwarename>" to modify - This statement is confirmed by the text on the display.
- B. The running configuration file is not saved - The text says "Startup saved-configuration file: null" and "Next startup saved-configuration file: null null" which indicates the configuration file is not saved.
- C. The system file used by the device this time is ar2220-v200r003c00spc200.cc - The text says "Startup system software: sdi:/ar2220-200r003c00spc200.cc".

Static floating achieved by assigning it a higher administrative distance (AD) (also called protocol priority in Huawei)

170. The administrator plans to implement route backup by configuring static floating routes. What is the correct way to achieve this?

- A. The administrator needs to configure different metric values for the main static route and the standby static route
- B. The administrator needs to configure different TAGs for the main static route and the standby static route
- C. The administrator only needs to configure two static routes
- D. The administrator needs to configure different protocol priority values for the main static route and the standby static route

Answer: D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100278527/488a24b0/understanding-sta?croutes>

Different static routes can be configured with different priorities so that routing management policies can be flexibly applied. Route backup can be implemented by specifying different priorities for multiple routes to the same destination.

In **Figure 9-17**, there are two static routes from Device A to Device C. In most cases, the only Active route is the static route with Device B as the next hop in the routing table because it has a higher priority. The other static route with Device D as the next hop functions as a backup route. The backup route is only activated to forward traffic if the primary link fails. After the primary link recovers, the static route with Device B as the next hop becomes Active to take over the traffic. Therefore, the backup route is also called a floating static route. The floating static route becomes ineffective if a fault occurs on the link between Device B and Device C.

171. There are hosts in a private network that need to access the Internet. In order to achieve this requirement, which of the following configurations should the administrator do on the edge router of the network?

- A. Default route
- B. STP
- C. DHCP
- D. NAT

Answer: AD

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100034071/e96c0933/example-for-configuringnat>

Configuration Roadmap

The configuration roadmap is as follows:

- Configure IP addresses for interfaces.
- Configure a default route.
- Configure outbound NAT and static NAT in Easy IP mode on the LAN-side interface of the router to ensure that the intranet host can use a public IP address to access the intranet server.
- Configure outbound NAT and static NAT in Easy IP mode on the WAN-side interface of the router to ensure that the intranet host can access the Internet and the extranet host can use a public IP address to access the intranet server.

172. Which of the following descriptions about the Trunk port and Access port is correct?

- A. Trunk port can only send untagged frames
- B. Access port can only send untagged frames
- C. Access port can only send tagged frames
- D. Trunk port can only send tagged frames

Answer: B

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100096323/819ade70/understanding-vlans>

- Access port

As shown in [Figure 8-22](#), the access port on a switch connects to the port on a host. The access port can only connect to an access link. Only the VLAN whose ID is the same as the default VLAN ID is allowed on the access port. Ethernet frames sent from the access port are untagged frames.

SOURCE: Topic 8 Slide 14

- Access interface

- An access interface often connects to a terminal (such as a PC or server) that cannot identify VLAN tags, or is used when VLANs do not need to be differentiated.

- Trunk interface

- A trunk interface often connects to a switch, router, AP, or voice terminal that can receive and send both tagged and untagged frames.

- Frame sending through an access interface:

- After receiving a tagged frame sent from another interface on the same switch, the access interface checks whether the VID in the tag of the frame is the same as the PVID.
 - If they are the same, the interface removes the tag from the frame and sends the untagged frame out.
 - Otherwise, the interface directly discards the tagged frame.

173. Which of the following statements about the division of the OSPF protocol area is wrong?

- A. Area 0 is the backbone area, and other areas must be connected to this area
- B. The LSDBs in the routers in the same OSPF area are completely consistent
- C. Dividing OSPF areas can reduce the LSDB scale of some routers
- D. Only ABR can be used as ASBR

Answer: D

Justification

Source: https://support.huawei.com/enterprise/en/doc/EDOC1100082074#EN-US_TOPIC_0169203192

AS Boundary Router (ASBR)	An ASBR exchanges routing information with another AS. An ASBR does not necessarily reside on the border of an AS. It can be an internal router or an ABR. An OSPF device that has imported external routing information will become an ASBR.
---------------------------	--

174. NFV (Network Function Virtualization), network function virtualization realizes the deployment of network applications in a software-based way.

- A. True
- B. False

Answer: A

Justification

Source: <https://info.support.huawei.com/info-finder/encyclopedia/en/NFV.html>

NFV Architecture

The NFV architecture is proposed by the European Telecommunications Standards Institute (ETSI) and defines standards for NFV implementation. Different from traditional physical devices with tightly coupled software and hardware, NFV aims to implement various network functions on a general-purpose hardware device using standardized software. For this purpose, NFV calls for a standard. The NFV architecture consists of:

- Network functions virtualization infrastructure (NFVI)
Similar to a mobile phone operating system, NFVI provides infrastructure components to support software and container management platforms that are needed to run network applications on hardware.
- Virtual Network Function (VNF)
Similar to applications on mobile phones, VNFs are software applications that deliver network functions, such as forwarding services and IP configurations. Based on the standardized architecture of NFVI, VNFs can run independently of hardware.
- Management and Orchestration (MANO)
MANO provides the framework for managing NFVI and VNFs, facilitating service orchestration and device management of O&M personnel.

175. UDP cannot guarantee the reliability of data transmission, and does not provide functions such as message sequencing and flow control. It is suitable for flows with low transmission reliability requirements but high transmission speed and delay requirements.

- A. True
- B. False

Answer: A

Justification

SOURCE: Topic 2, Page 9, Slide 11

- User Datagram Protocol (UDP): provides connectionless communication and does not guarantee the reliability of packet transmission. The reliability can be ensured by the application layer.

176. The administrator cannot log in to the Huawei router through Telnet, but other administrators can log in normally, so which of the following are the possible reasons?

- A. The authority level of the administrator user account has been modified to O
- B. The administrator user account has been disabled
- C. The Telnet service of the AR2200 router has been disabled
- D. The administrator user account has been deleted

Answer: BD

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100192869/9daeadebe/users-other-than-the-system-administrator-fail-to-log-in-to-the-o-m-plane>

Possible Causes

- The user is disabled by the administrator.
- The number of logins has reached the maximum.
- The account validity period is reached.
- The number of consecutive offline days of the user has reached the upper limit.
- The number of consecutive days that the account has not been used reaches the limit of the account suspension policy.

A user cannot log in when the maximum number of online sessions of the user is reached.

The current login time of the user is not within the range specified in the login time policy bound to the user.

- The IP address used for the login is out of the range specified in the client IP address policy bound to the user.
- When a multi-NIC client logs in, the IP addresses of some NICs are not added to the client IP address policy bound to the user.

A user is locked because the number of consecutive login failures of the user due to incorrect password input reaches the maximum, and the account unlocking mode is set to automatic.

A user is locked because the number of consecutive login failures of the user due to incorrect password input reaches the maximum, and the account unlocking mode is set to manual.

The IP address is locked because the number of consecutive login failures of the user due to incorrect password input reaches the maximum.

The system administrator sets the system login mode to the single-user mode.

The user password has expired, and the security administrator disables the user from changing the password.

- The user has been deleted.
- The username you entered is incorrect.
- The password you entered is incorrect.
- Remote authentication is enabled and local user authentication is not supported. The local user fails to be authenticated.

177. VRP operating system commands are divided into 4 levels: access level, Monitoring level, Configuration level, and Management level. Which level can run various business configuration commands, but cannot operate the file system?

- A. Configuration level
- B. Monitoring level
- C. Access level
- D. Management level

Answer: A

Justification

SOURCE: Topic 3, Page 12

VRP User Levels			
VRP provides basic permission control functions. It defines the levels of commands that each level of users can execute to restrict the operations of users at different levels.			
User Level	Command Level	Name	Available Command
0	0	Visit level	Network diagnosis commands (such as ping and tracert), commands for accessing external devices from the local device (such as Telnet client commands), and some display commands
1	0 and 1	Monitoring level	System maintenance commands, including display commands
2	0,1 and 2	Configuration level	Service configuration commands, including routing commands and IP configuration commands, to directly provide users with network services
3-15	0,1,2 and 3	Management level	Commands for controlling basic system operations and providing support for services, including the file system, FTP, TFTP download, user management, and command level commands, as well as debugging commands for fault diagnosis

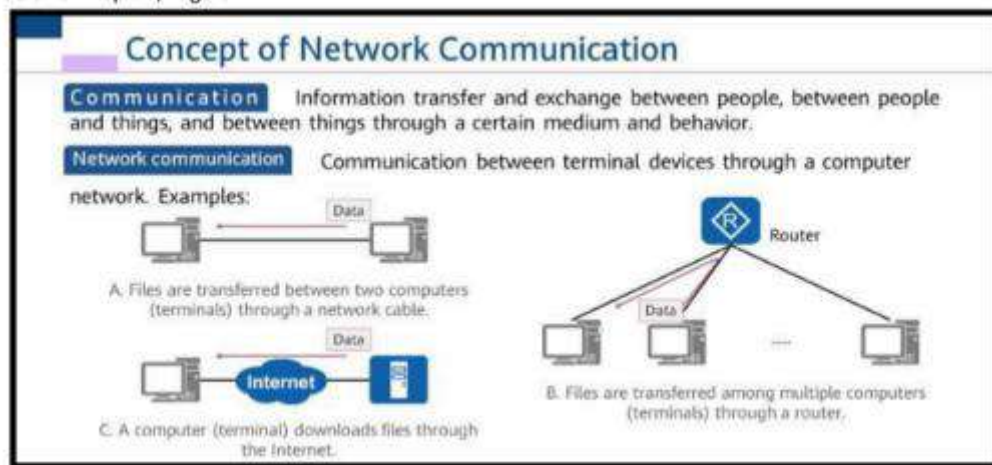
178. Using a traditional landline to make calls is a way of network communication.

- A. True
- B. False

Answer: A

Justification

SOURCE: Topic 1, Page 6



179. In the VRP interface, use the command delete vrpcfg.zip to delete a file, which must be emptied in the recycle bin to completely delete the file.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/knowledge/EKB1000114024>

When deleting a file by this command:

```
<Huawei>delete ar1220-v200r005sph006.pat
```

Actually this file will be moved into recycle bin instead of erasing from Flash.

If we run command:

```
<Huawei> dir /all
```

We will find that file still exist in Flash, but marked with "[]", which shows it is in Recycle bin.

```
<Huawei>dir /all
```

Directory of flash:/

Idx	Attr	Size(Byte)	Date	Time(LMT)	FileName
0	-rw-	105,425,408	Dec 04 2014	11:12:14	AR1220-V200R005C20SPC200.cc

.....

8	-rw-	231,168	Jan 06 2016	00:59:50	[ar1220-v200r005sph006.pat]
---	------	---------	-------------	----------	-----------------------------

248,320 KB total (25,024 KB free)

To eraser them from Flash totally, there are 2 ways:

1. when delete the file, use parameter /unreserved , so that the file will not be moved to recycle bin.

```
<Huawei>delete /unreserved ar1220-v200r005sph006.pat
```

2. Use this command to remove the files in recycle bin:

```
<Huawei>reset recycle-bin ar1220-v200r005sph006.pat
```

Then the files will be erased from Flash and the space will be released.

180. In what state does the OSPF protocol determine the master-slave relationship of DD packets?

- A. ExStart
- B. Full
- C. 2-way
- D. Exchange

Answer: A

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100278569/c1083833/understanding-ospf>

Exstart	In the Exstart state, devices establish a master/slave relationship to ensure that DD packets are sequentially exchanged.
---------	---

181. The following two configuration commands can realize the router RTA to the same destination 10.1.1.0 routing master and backup.

【RTA】 ip route-static 10.1.1.0 24 12.1.1.1 permanent

【RTA】 ip route-static 10.2.2.0 24 13.1.1.1.

- A. True
- B. False

Answer: B

Justification#1:

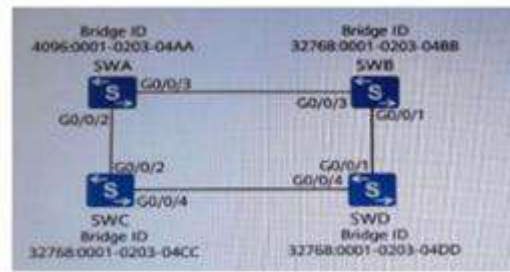
Since the destination on networks are different (10.1.1.0/24 and 10.2.2.0/24), these commands do not achieve the intended goal of routing to the same destination on with a master and backup route.

Justification#2:

Source: <https://forum.huawei.com/enterprise/intl/en/thread/STQTIC-ROUTE/667280744621424641?blogId=667280744621424641>

The permanent keyword is used in the first command, which means that the route with the next-hop 12.1.1.1 is set to be permanent, and it will not be removed even if the associated interface goes down.

182. In the network shown in the figure below, the bridge ID of the switch has been marked. Enter the command `stp root secondary` on the SWD switch, and which of the following switch will become the root bridge of this network?



- A. SWC
- B. SWA
- C. SWB
- D. SWD

Answer: B

Justification

Source: Slide: 09 STP Principle and Configuration on page Number: 24

SWA has the smallest bridge ID (4096 is lower than 28672), so SWA will remain the root bridge even after applying the `stp root secondary` command on SWD.

The switch with the smallest bridge ID becomes the root bridge.

183. If the Length/Type of an Ethernet data frame = 0x8100, which of the following statements is correct?

- A. There must be an IP header in the upper layer of this data frame
- B. This data frame must carry a VLAN TAG
- C. There must be a UDP header in the upper layer of this data frame
- D. There must be a TCP header in the upper layer of this data frame

Answer: B

Justification

Source: Slide: 08 VLAN Principle and Configuration on page Number: 10

Value

The value 0x8100 indicates an IEEE 802.1Q frame. An 802.1Q-incapable device discards 802.1Q frames.

VLAN Configuration
VLAN Assignment
VLAN Frame Forwarding

VLAN Frame

Original Ethernet frame (untagged frame)

Destination MAC address	Source MAC address	Length/Type	Data	FCS
-------------------------	--------------------	-------------	------	-----

802.1Q tag inserted between the two fields

802.1Q tag

TPID (0x8100)	PRI	CFI	VLAN ID
16 bits	3 bits	1 bit	12 bits

- Tag protocol identifier (TPID): identifies the type of a frame. The value 0x8100 indicates an IEEE 802.1Q frame.
- PRI: identifies the priority of a frame, which is mainly used for QoS.
- Canonical format indicator (CFI): indicates whether a MAC address is in the canonical format. For Ethernet frames, the value of this field is 0.
- VLAN ID: identifies the VLAN to which a frame belongs.

802.1Q frame (tagged frame)

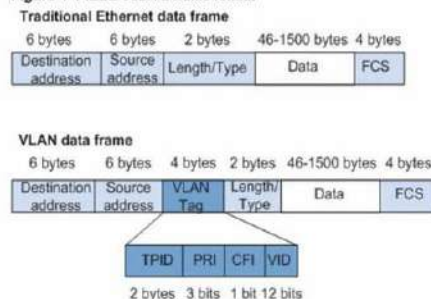
Destination MAC address	Source MAC address	Tag	Length/Type	Data	FCS
-------------------------	--------------------	-----	-------------	------	-----

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VLAN Frame Format

A switch identifies packets from different VLANs according to the information contained in its VLAN tags. IEEE 802.1Q adds a 4-byte VLAN tag between the Source/Destination MAC address and Length/Type fields of an Ethernet frame to identify the VLAN to which the frame belongs. Figure 1-1 shows the position of a VLAN tag in a VLAN data frame.

Figure 1-1 IEEE 802.1Q frame format



The following are the two types of Ethernet frames in a VLAN:

- Tagged frame: frame with a 4-byte VLAN tag
- Untagged frame: frame without a 4-byte VLAN tag

184. FAT AP can independently complete wireless user access, wireless user authentication, and business data forwarding without AC.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000060368/9e00e376/understanding-modes-supported-by-aps>

Table 22-24 AP mode description

AP Mode	Description	Application Scenario
Fat	A Fat AP can provide Wi-Fi coverage independently without the need of an additional management and control device. Since Fat APs work independently, each Fat AP must be separately configured. Deploying multiple Fat APs to provide Wi-Fi coverage in a large area leads to high configuration and maintenance costs. In addition, STAs cannot roam between Fat APs because the Fat APs control network access independently. Therefore, Fat APs are usually deployed to provide small-scale Wi-Fi coverage. In enterprise scenarios, the AC + Fit AP and cloud management platform + cloud AP networking modes are more commonly used.	Small-scale Wi-Fi coverage

conversion as well as wireless signal receiving and transmitting functions. This allows them to transmit the wireless signals of mobile devices such as mobile phones and laptops. Fat APs can implement such functions independently, whereas Fit APs must interwork with the WLAN Access Controller (AC). On the flip side, Fat APs need to be deployed and configured one by one, which is

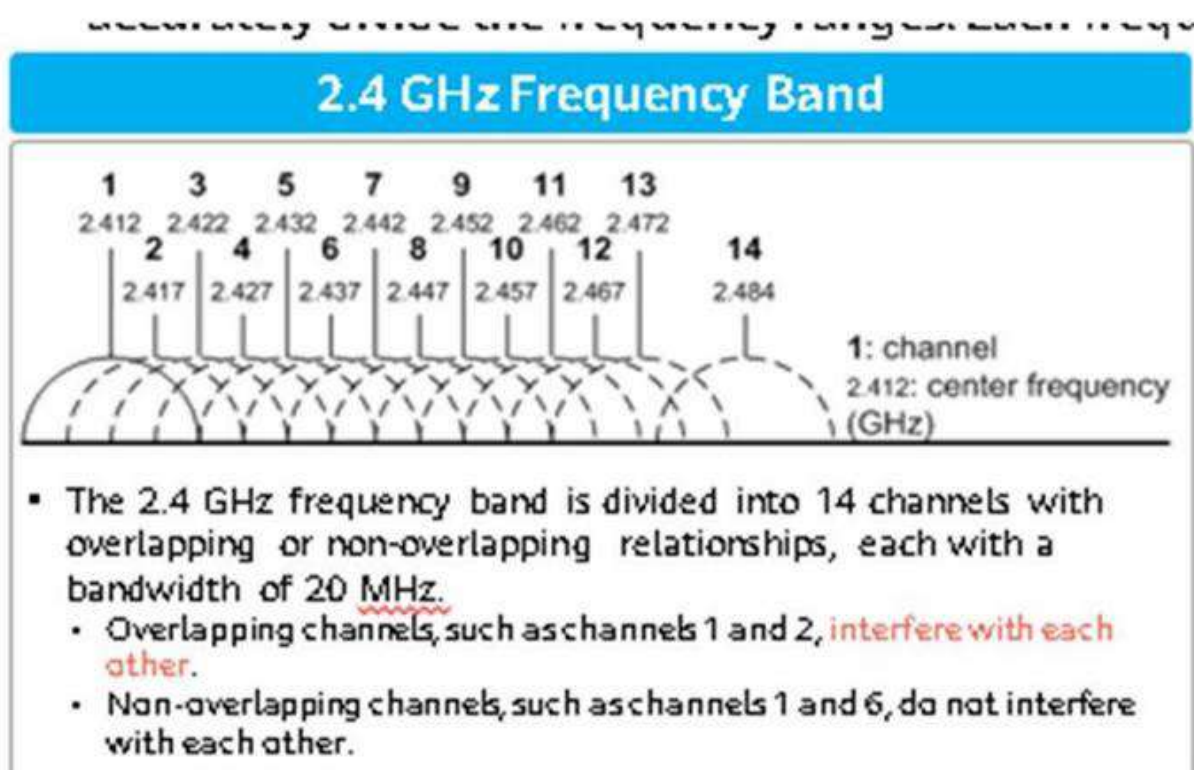
185. There is no channel overlap in the 14 available frame segments in the 185.2.4 GHz frame segment.

- A. True
- B. False

Answer: B

Justification

Source: Slide: 16 WLAN Overview Page Number: 21



186. In the WLAN network architecture, what is terminal equipment that supports the 802.11 standard called?

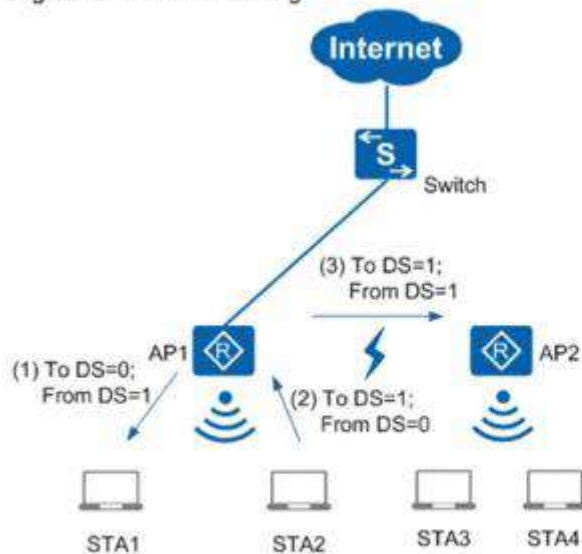
- A. Client
- B. AC
- C. STA
- D. AP

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100112363/b1db415/80211-standards>

Figure 2-4 WLAN networking



187. Which of the following characteristics does the OSPF protocol have? (Multiple Choice)

- A. Easy to produce routing loops
- B. Calculate the shortest path by number
- C. Support the division of areas
- D. Trigger an update

Answer: CD

Justification

Source: <https://www.geeksforgeeks.org/open-shortest-path-first-ospf-protocol-fundamentals/>

Option C: Support the division of areas

- To handle routing efficiently and on time, this protocol divides an autonomous system into areas.
- Area is a collection of routers, hosts, networks all contained within an autonomous system.
- The network is divided and it can be structured, or subdivided, into routing areas to simplify administration and optimize traffic and resource utilization.
- Among the areas, there is an area called ‘backbone’ to which all the other areas are connected and serve as the primary area.
- Two different areas can’t be connected directly but they need to be connected through the backbone network and it is compulsory.

Option D: Trigger an update

- Also when there is a slight change in the router configuration (routers added/removed), the routing table is updated very fast.
- If one of the links fails the protocol finds the other shortest path to reach the destination hence it possesses faster convergence.

188. Which of the following statement about static routing is wrong?

- A. Cannot automatically adapt to changes in network topology
- B. Manual configuration by the network administrator
- C. Low system performance requirements
- D. Routers need to exchange routing information

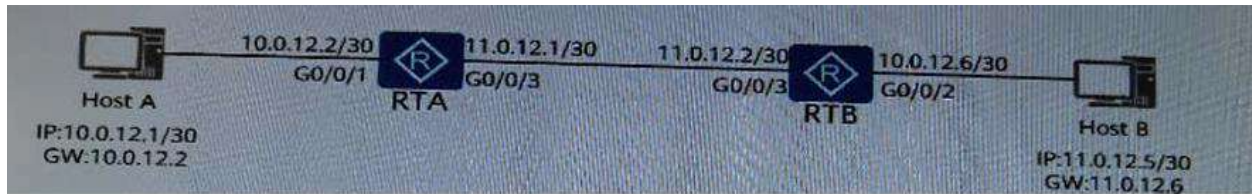
Answer: D

Justification

Source: Slide: 05 IP Routing Basics page Number: 28

The disadvantage of static routes is that they cannot automatically adapt to network topology changes and so require manual intervention.

189. In the network shown in the figure below, which of the following commands can be entered on the router to enable host A to ping host B? (Multiple choice)



- A. RTA: ip route-static 10.0.12.5 255.255.255.252 11.0.12.2
RTB: ip route-static 10.0.12.1 255.255.255.252 11.0.12.1
- B. RTA: ip route-static 10.0.12.5 255.255.255.252 11.0.12.1
RTB: ip route-static 10.0.12.1 255.255.255.252 11.0.12.2
- C. RTA: ip route-static 0.0.0.0 0.0.0.0 11.0.12.2
RTB: ip route-static 0.0.0.0 0.0.0.0 11.0.12.1
- D. RTA: ip route-static 0.0.0.0 0.0.0.0 11.0.12.1
RTB: ip route-static 0.0.0.0 0.0.0.0 11.0.12.2

Answer: AC

Justification

Source: Slide: 03 Network Layer Protocols and IP Addressing Page Number: 2

Specify a next-hop IP address for a static route.

[Huawei] **ip route-static** *ip-address* { *mask* | *mask-length* } *nexthop-address*



Default Routes

- Default routes are used only when packets to be forwarded do not match any routing entry in an IP routing table.
- In an IP routing table, a default route is the route to network 0.0.0.0 (with the mask 0.0.0.0), namely, 0.0.0.0/0.

RTA needs to forward packets to a subnet that is

190. Which of the following IPv4 addresses can be used by a host to directly access the Internet?

- A. 192.168.1.1/24
- B. 172.32.1.1/24
- C. 172.16.255.254/24
- D. 10.255.255.254/24

Answer: B

Justification

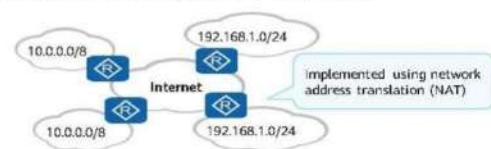
Source: Slide: 03 Network Layer Protocols and IP Addressing Page Number: 22

[Concepts](#) > [Address Classification](#) > [Address Calculation](#) > [Special Addresses](#) > [IPv4 vs. IPv6](#)



Private IP Addresses

- **Public IP address:** An IP address is assigned by the Internet Assigned Numbers Authority (IANA), and this address allocation mode ensures that each IP address is unique on the Internet. Such an IP address is a public IP address.
- **Private IP address:** In practice, some networks do not need to connect to the Internet. For example, on a network of a lab in a college, IP addresses of devices need to avoid conflicting with each other only within the same network. In the IP address space, some IP addresses of class A, B, and C addresses are reserved for the preceding situations. These IP addresses are called private IP addresses.
 - Class A: 10.0.0.0–10.255.255.255
 - Class B: 172.16.0.0–172.31.255.255
 - Class C: 192.168.0.0–192.168.255.255



Connecting a private network to the Internet

191. The switch becomes a logical switch after stacking and clustering, and Eth- Trunk across physical devices can be deployed to improve network reliability.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000069520/5e798e3e/typical-stack-configuration-of-fixedswitches>

Fixed switches are often deployed at the aggregation layer and access layer. Unlike modular switches, fixed switches are continuous; the number of ports provided by a single fixed switch may be insufficient to meet the network requirement one logical switch. You can set up a stack to improve network scalability and device reliability.

192. What is the value of TPID defined by IEEE802.1Q

- A. 0x8200
- B. 0x7200
- C. 0x8100
- D. 0x9100

Answer: C

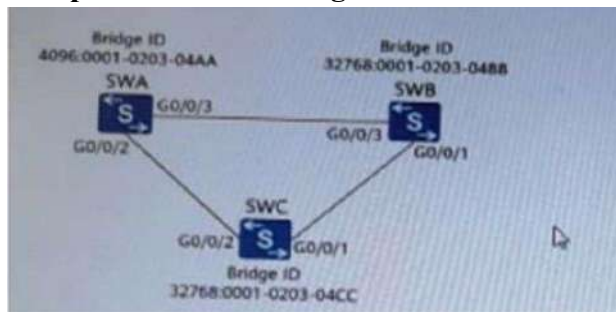
Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100277028/3e01a4f2/tpid>

TPID :

The Tag Protocol Identifier (TPID) specifies the protocol type of a VLAN tag. The TPID value defined in IEEE 802.1Q is 0x8100.

193. As shown in the figure below, assuming the other configurations of the switch are kept in the default state, which port of the following switch will become the designated port?



- A. SWC's G0/0/1
- B. SWA's G0/0/3
- C. SWC's G0/0/2
- D. SWB's G0/0/1

Answer: BD

Justification

Source: Slide: 09 STP Principles and Configurations on Page Number: 22

BPDUs Comparison Rules

Field	
Protocol Identifier	The core of STP is to calculate a loop-free topology on a switching network. During topology calculation, the comparison of configuration BPDUs is important. The Root Identifier, Root Path Cost, Bridge Identifier, and Port Identifier fields are the main fields of a configuration BPDU. STP-enabled switches compare the four fields.
Protocol Version Identifier	
BPDU Type	
Flags	
Root Identifier	STP selects the optimal configuration BPDU in the following sequence:
Root Path Cost	1. Smallest BID of the root bridge
Bridge Identifier	2. Smallest RPC
Port Identifier	3. Smallest BID of the network bridge
Message Age	4. Smallest PID
Max Age	Among the four rules (each rule corresponds to a field in a configuration BPDU), the first rule is used to elect the root bridge on the network, and the following rules are used to elect the root port and designated port.
Hello Time	
Forward Delay	

In this scenario, SWA has the lowest bridge ID therefore it is the root bridge making all its ports designated by default including port G0/0/3. For non-root bridges like SWB and SWC, we compare the path costs to the root bridge. SWB's port G0/0/1 is designated since it has a direct link to the root bridge, offering the lowest path cost. Thus, the correct designated ports are B. SWA's G0/0/3 and D. SWB's G0/0/1.

194. The management of users by NAS devices is based on domains. Each user belongs to a domain, and a domain is a group of users who belong to the same domain.

- A. True
- B. False

Answer: A

Justification

Source: [https://support.huawei.com/enterprise/en/doc/EDOC1100112357/1e2a4c59/do main-based-user-management](https://support.huawei.com/enterprise/en/doc/EDOC1100112357/1e2a4c59/do-main-based-user-management)

Domain-based User Management:

An NAS performs domain-based user management. A domain is a group of users, and each user belongs to a domain. A user uses only AAA configuration information in the domain to which the user belongs

195. Which of the following fields does the MPLS header include? (Multiple Choice)

- A. EXP
- B. TTL
- C. Label
- D. Tos

Answer: ABC

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100334522/e1707d39/understanding-mpls>

Figure 12-3 MPLS packet header structure



The MPLS header contains the following fields:

- Label: a 20-bit field that identifies a label value.
- Exp: a 3-bit I field used for extension. This field is used to implement the class of service (CoS) function, which is similar to Ethernet 802.1p.
- S: a 1-bit field that identifies the bottom of a label stack. MPLS supports multiple labels that may be stacked. If the S field value is set to 1, the label is at the bottom of the label stack.
- TTL: an 8-bit field indicating a time to live (TTL) value. This field is the same as the TTL field in IP packets.

196. Huawei ARG3 series routers have all SNMP versions (SNMPv1, SNMPv2c and SNMPv3) by default.

- A. True
- B. False

Answer: A

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000069520/dfaf8717/typical-snmp-configuration>

SNMP Overview

The Simple Network Management Protocol (SNMP) is a standard network management protocol widely used on TCP/IP networks. SNMP manages network elements by using a central computer (a network management station) running network management software.

SNMP has the following advantages:

- Automatic configuration management:

A network administrator can use SNMP to quickly query information, modify data, locate faults, and so forth on any SNMP agent. SNMP greatly improves work efficiency of network administrators.

- Multi-vendor management:

SNMP shields the physical differences between devices of different vendors. SNMP provides only a basic function set, so the managed tasks are separated from the managed physical features and lower-layer interoperation technologies. Therefore, SNMP can uniformly manage devices of multiple vendors simultaneously.

SNMP is available in three versions. SNMPv1 is the initial version of SNMP. It provides authentication based on community names. SNMPv1 has low security, and can return only a few error codes. SNMPv2c issued by IETF is the second release of SNMP. SNMPv2c has enhancements to standard error codes, data types (Counter 64 and Counter 32), and operations including GetBulk and Inform. SNMPv2c does not improve the security, so IETF issued SNMPv3 that provides encryption and authentication based on the user-based security model (USM) and access control based on the view-based access control model (VACM).

1. Configure SNMPv3 on the switch so that the NMS running SNMPv3 can manage the switch.

```
<HUAWEI> system-view
[HUAWEI] sysname Switch
[Switch] snmp-agent sys-info version v3 //By default, SNMPv3 is supported. If SNMPv3 is not disabled, skip this command.
```

197. IPv6 address 2001: ABEF: 2240E: FFE2: BCC0: CD0:DDBE:8D58 cannot be abbreviated.

- A. True
- B. False

Answer: A

Justification

Source: Slide: 19 IPv6 Basics Page Number: 14

IPv6 Address Abbreviation Specifications

- For convenience, IPv6 can be abbreviated according to the following rules.

Abbreviation Specifications	Abbreviation Examples
2001 : 0DB8 : 0000 : 0000 : 0008 : 0800 : 200C : 417A The leading 0s in each 16-bit segment can be omitted. However, if all bits in a 16-bit segment are 0s, at least one 0 must be reserved. The trailing 0s cannot be omitted. 2001 : DB8 : 0 : 0 : 8 : 800 : 200C : 417A If one or more consecutive 16-bit segments contain only 0s, a double colon (::) can be used to represent them, but only one :: is allowed in an entire IPv6 address. 2001 : DB8 : : 8 : 800 : 200C : 417A If an abbreviated IPv6 address contains two double colons (::), the IPv6 address cannot be restored to the original one.	Before 0000:0000:0000:0000:0000:0000:0000:0001 After ::1 Before 2001:0DB8:0000:0000:FB00:1400:5000:45FF After 2001:DB8:FB00:1400:5000:45FF Before 2001:0DB8:0000:0000:0000:2A2A:0000:0001 After 2001:DB8::2A2A:0:1 Before 2001:0DB8:0000:1234:FB00:0000:5000:45FF After 2001:DB8::1234:FB00:0:5000:45FF or 2001:DB8:0:1234:FB00::5000:45FF

198. Which state does the OSPF protocol use to indicate that the adjacency relationship has been established?

- A. Full
- B. 2-way
- C. Attempt
- D. Down

Answer: A

Justification

Source: <https://www.ciscopress.com/articles/article.asp?p=2294214#:~:text=State%3A%20A%20Full%20state%20represents,with%20the%20router%20ID%201.1.1.>

State:

A Full state represents the final stage of OSPF neighbor establishment process and denotes that the local router has established full neighbor adjacency with the remote OSPF neighbor. DR means that DR/BDR election process has been completed and that the remote router with the router ID 1.1.1.1 has been elected as the designated router (DR).

Full	In this state, a device establishes adjacencies with its OSPF neighbors and all LSDBs have been synchronized.
------	--

199. The TRACERT diagnostic tool in Huawei routers is used to track the data forwarding path.

- A. True
- B. False

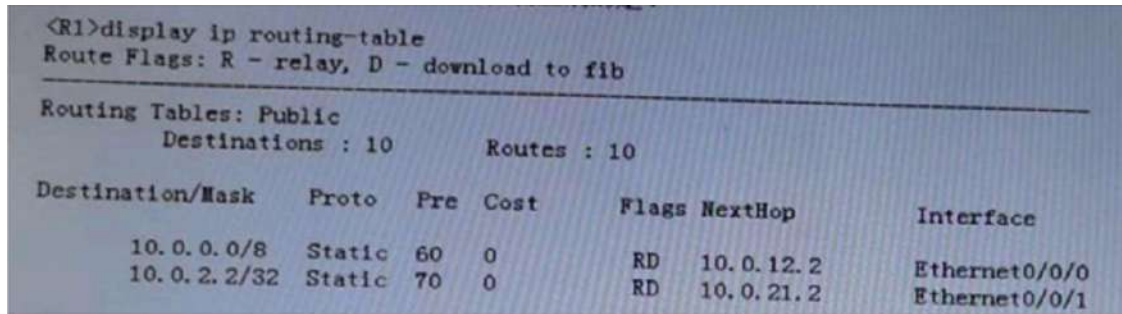
Answer: A

Justification

Source: <https://support.microsoft.com/en-gb/topic/how-to-use-tracert-to-troubleshoot-ip-problems-in-windows-e643d72b-2f4f-cdd6-09a0-fd2989c7ca8e>

This article describes TRACERT (Trace Route), a command-line utility that you can use to trace the path that an Internet Protocol (IP) packet takes to its destination.

200. The routing table output information of a router is shown in the figure. Which of the following statements is correct? (Multiple choice)



```
<R1>display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public
Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.0.0/8	Static	60	0	RD	10.0.12.2	Ethernet0/0/0
10.0.2.2/32	Static	70	0	RD	10.0.21.2	Ethernet0/0/1

- A. The local router to the NextHop of 10.0.2.2 is 10.0.21.2
- B. The local router to the NextHop of 10.0.2.1 is 10.0.21.2
- C. The local router to the NextHop of 10.0.2.2 is 10.0.12.2
- D. The local router to the NextHop of 10.0.0.1 is 10.0.12.2

Answer: AD

Justification

Explanation:

For the destination 10.0.2.2/32, the next hop is 10.0.21.2.

For the destination 10.0.0.0/8, the next hop is 10.0.12.2

201. Which of the following OSPF protocol packets can ensure the reliability of LSA updates?

- A. LSACK
- B. DD
- C. LSU
- D. LSR

Answer: A

Justification

Source: <https://forum.huawei.com/enterprise/en/OSPF/thread/667267213926744064-667213852955258880>

OSPF Packet

The following table lists the protocol packets used by OSPF. These OSPF packets are encapsulated in IP packets and transmitted in multicast or unicast mode.

Packet Type	Description
Hello	Hello packets are sent periodically to discover and maintain OSPF neighbor relationships.
Database Description (DD)	DD packets carry brief information about the local Link-State Database (LSDB) and are used to synchronize the LSDBs between OSPF neighbors.
Link State Request (LSR)	LSR packets are used to request the required LSAs from neighbors. LSR packets are sent only after DD packets are exchanged successfully between OSPF neighbors.
Link State Update (LSU)	LSU packets are used to send the required LSAs to peers.
Link State Acknowledgment (LSAck)	LSAck packets are used to acknowledge the received LSAs.

202. The last option field of IPv4 is variable-length optional information. What is the maximum length of this field?

- A. 10B
- B. 40B
- C. 60B
- D. 20B

Answer: B

Justification

Source: <https://forum.huawei.com/enterprise/en/use-of-the-op%C6%9Fons-field-ipv4header/thread/667267215742877696-667213852955258880>

The IP datagram can contain zero, one or more options, which makes the total length of the Options field in the IPv4 header variable. Each of the options can be a single byte or several bytes long, depending on the amount of information the option needs to forward (from 0 to 40 bytes maximum). When more than one option is included, they are simply concatenated and placed in the Options field as a whole. All datagrams may not contain this field. This field is optional.

203. Which of the following advantages does router-on-a-stick achieve inter-VLAN routing intercommunication?

- A. Reduce the number of equipment
- B. Reduce the use of IP addresses
- C. Reduce the number of link connections
- D. Reduce routing table entries

Answer: C

Justification

Source:

<https://community.cisco.com/t5/networking-knowledge-base/pros-and-cons-of-router-on-a-stickmethodinnetworking/tap/4951799#:~:text=Advantages%20of%20Router%20on%20a%20St ick%3A&text=Simplified%20Network%20Management%3A%20RoAS%20simplifies,routing% 20protocols%20and%20security%20policie s.>

When it comes to designing and configuring networks, the *Router on a Stick* (RoAS) method is a commonly used approach. RoAS is a technique where a single router interfaces with multiple VLANs (Virtual Local Area Networks) through a single physical connection. While this method offers several advantages, it also comes with its fair share of disadvantages. I will briefly talk about both in this article.

Advantages of Router on a Stick:

1. **Cost-Efficiency:** RoAS is cost-effective because it uses a single physical interface on the router to connect to multiple VLANs. This reduces the need for additional hardware, such as multiple routers, which can save on equipment costs.
 2. **Simplified Network Management:** RoAS simplifies network management since all inter-VLAN routing is centralized through a single router. This makes it easier to configure and manage routing protocols and security policies.
 3. **Scalability:** The Router on a Stick method is highly scalable. As your network grows and more VLANs are required, you can easily add them without significant changes to the physical infrastructure.
 4. **Traffic Isolation:** RoAS ensures that traffic between VLANs is isolated. This isolation is essential for maintaining network security, preventing broadcast storms, and controlling the flow of data between different segments of the network.
 5. **Efficient Use of Router Resources:** Since a single router handles inter-VLAN routing, it can efficiently use its resources for routing functions, without unnecessary redundancy. This can lead to better performance and resource allocation.
-

204. If STP is not turned on for a network composed of switches, there must be a Layer 2 loop.

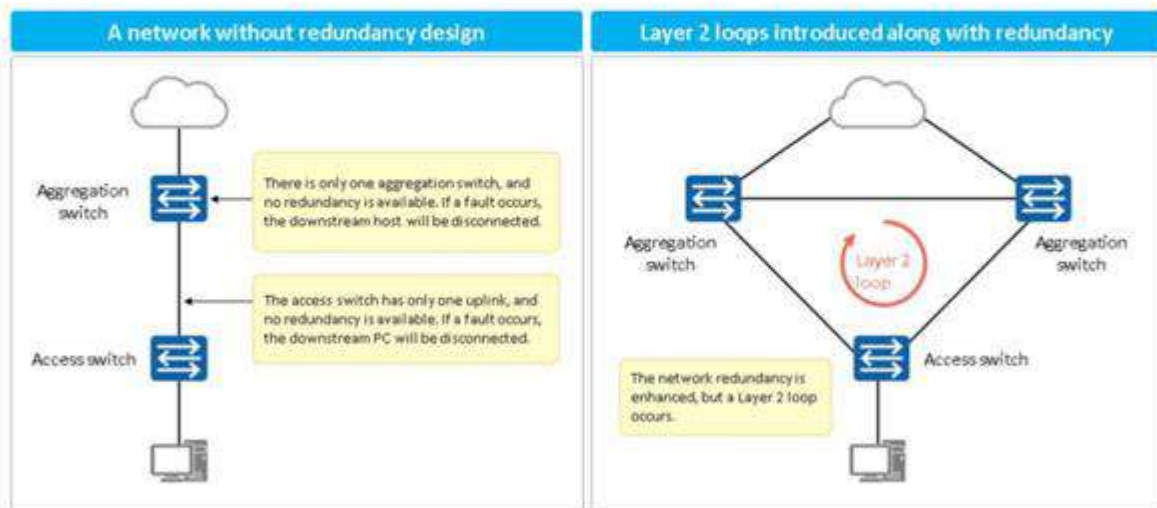
- A. True
- B. False

Answer: B

Justification

Source: Slide 09 STP Principle and Configuration

If a specific network setup does not have redundant paths (e.g., only one connection between each pair of switches), then a Layer 2 loop would not occur even without STP.



205. The switch receives a unicast data frame. If the destination MAC of the data frame can be found in the MAC table, the data frame will be forwarded from the corresponding MAC port.

- A. True
- B. False

Answer: A

Justification

Source: Slide#07, Page Number: 23

Slide: 07 Ethernet Switching Basics

Page Number: 23

After receiving a frame, the switch learns the source MAC address of the frame, searches the **MAC address table** for the destination MAC address (MAC2: 0050-5600-0002 in this example) of the frame, and **forwards** the frame through the corresponding interface.

206. The output information of the link aggregation port of a certain switch is as follows. If you want to delete Eth-Trunk 1, which of the following commands is correct?

```
<SW1>display interface Eth-Trunk 1
Eth-Trunk1 current state : UP
Line protocol current state : UP
Description:
Switch Port, PVID : 1, Hash arithmetic : According to SIP-XOR-DIP,Maximal BW:
2G, Current BW: 2G, The Maxiuma Frame Length is 9216
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 4c1f-ccb0-73d7
Current system time: 2019-02-28 10:28:09-08:00
Input bandwidth utilization : 0%
Output bandwidth utilization : 0%
```

PortName	Status	Weight
GigabitEthernet0/0/1	UP	1
GigabitEthernet0/0/2	UP	1

```
The Number of Ports in Trunk : 2
The Number of UP Ports in Trunk : 2
```

- A. undo interface Eth-trunk 1
- B. inte GigabitEthernet 0./0/2
Undo eth-trunk
Quit
Undo interface eth-trunk 1
- C. inte GigabitEthernet 0./0/1
Undo eth-trunk
Quit
inte GigabitEthernet 0./0/2
Undo eth-trunk
Quit
- D. inte GigabitEthernet 0./0/1
Undo eth-trunk
Quit
Undo interface eth-trunk 1

Answer: A

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000057410?section=j01o&topicName=deleting-an-eth-trunk>

Deleting an Eth-Trunk

Prerequisites

All member interfaces have been deleted from an Eth-Trunk. See **Deleting a Specified Member Interface from an Eth-Trunk**.

Procedure

Run the **undo interface eth-trunk trunk-id** command in the system view.

```
<HUAWEI> system-view
[HUAWEI] undo interface eth-trunk 10
```

207. Which information of the data frame does the VLANIF interface use to determine whether to perform Layer 2 or Layer 3 forwarding?

- A. Destination MAC
- B. Source MAC
- C. Destination port
- D. Source IP

Answer: A

Justification

Source: <https://forum.huawei.com/enterprise/en/tc-while/thread/667234961951965185-667213879488425984>

The switch determines whether to perform Layer 2 or Layer 3 forwarding based on the destination MAC address of the received unicast frame. When the destination MAC address of the received packet is the MAC address of the VLANIF interface, the packet is forwarded at Layer 3. Otherwise, Layer 2 forwarding is used

208. If the value of the Type/Length field of an Ethernet data frame is 0x0800, the length of the upper layer message header carried by this data frame is in the range of 20-60B.

- A. True
- B. False

Answer: A

Justification

Source: Slide#07, Page Number: 10

The Type/Length field in an Ethernet frame can indicate either the type of protocol encapsulated in the frame or the length of the payload. When the field value is 0x0800, it indicates that the payload is an IPv4 packet. The length of the IPv4 header is variable, but it is typically in the range of 20 to 60 bytes. Therefore, the statement is true.



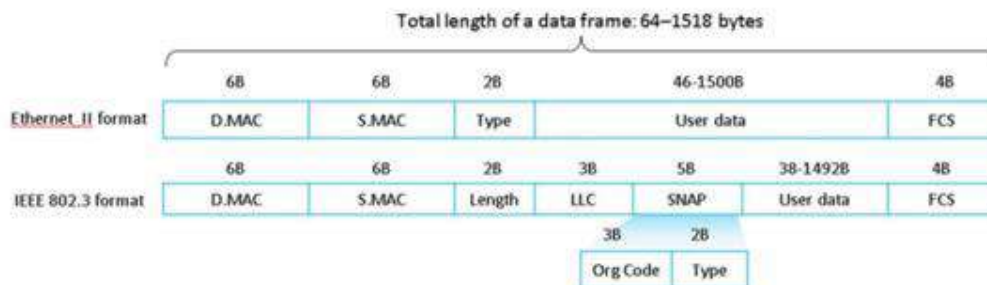
Ethernet Frame Format

Ethernet Frame Format

MAC address

Ethernet frame

- The frames used by Ethernet technology are referred to as Ethernet frames.
- Ethernet frames are in two formats: Ethernet II and IEEE 802.3.



209. When using FTP for file transfer, how many TCP connections will be established?

- A. 2
- B. 1
- C. 3
- D. 4

Answer: A

Justification

Source: <https://www.sciencedirect.com/topics/computer-science/file-transfer-protocol>

The File Transfer Protocol (FTP) sets up two connections: one for the commands, the other for the data exchange

**210. What are the user authentication methods that can be deployed in WLAN?
(Multiple choice)**

- A. Portal authentication
- B. MAC authentication
- C. Radius certification
- D. 802.1X authentication (Correct)

Answer: AB D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000178178/60b102f2/configuring-a-userauthen%C6%9Fca%C6%9Fon-mode>

** NOTE**

When configuring multi-mode authentication, pay attention to the following points:

- **An authentication profile can be bounded to an 802.1X access profile, a MAC access profile and a Portal access profile at most**
- After multi-mode authentication is configured, the device by default allows users to use multiple authentication modes. For example, if a user passes MAC address authentication, the user will not be redirected to the Portal authentication page when accessing a web page. However, if the user directly enters the Portal authentication website in the browser, Portal authentication can be performed. After the authentication succeeds, the users can obtain network access rights for Portal authentication users. To authenticate users using only one authentication mode, run the **authentication single-access** command to configure the device to allow users to pass only one access authentication.
- MAC address authentication and Portal authentication cannot be performed after 802.1X authentication succeeds.
- 802.1X + MAC address authentication is mainly applied to scenarios where dumb terminals exist. When a gateway functions as an authentication device, 802.1X + MAC address authentication is not recommended because ARP packets sent by terminals trigger MAC address authentication first. This degrades the performance of 802.1X authentication and ARP attacks may occur.

In a scenario where dumb terminals exist and a gateway functions as an authentication device, you are advised to use the following configuration mode:

- a. Ensure that dumb terminals use fixed IP addresses. You can manually configure IP addresses or bind IP addresses statically using DHCP snooping.
 - b. Do not configure multi-mode authentication on the gateway. Configure 802.1X authentication for users who do not use dumb terminals and configure IP address-based authentication-free rules for users who use dumb terminals.
- In MAC address + Portal authentication, the device performs MAC address authentication first for an access terminal. If MAC address authentication fails, the device performs Portal authentication. This is MAC address-prioritized Portal authentication. MAC address-prioritized Portal authentication is mainly used in wireless user access scenarios to ensure that Portal online users can automatically go online after they go offline because their terminals enter or leave a wireless signal coverage area. This authentication mode prevents users from entering their user names and passwords frequently, improving the access experience.
 - In wireless access scenarios, either 802.1X + MAC address authentication or 802.1X + Portal authentication is supported.

211. Which of the following IEEE 802.11 standards support operation in the 5GHz frequency band?

- A. 802.11b
- B. 802.11g
- C. 802.11n
- D. 802.11a

Answer: BC C,D

Justification

Source: Slide#16, Page number 7

Frequency Band	2.4GHz	2.4GHz	2.4GHz、 5GHz	2.4GHz & 5GHz	5GHz	5GHz	2.4GHz & 5GHz
Throughput	2Mbit/s	11Mbit/s	54Mbit/s	300Mbit/s	1300Mbit/s	6.9Gbit/s	9.6Gbit/s
Standard	802.11	802.11b	802.11a、 802.11g	802.11n	802.11ac wave1	802.11ac wave2	802.11ax

212. In order to join the wireless network, STA needs to obtain wireless network information first. Which of the following messages is used by the STA to actively obtain the SSID information?

- A. Probe
- B. Join
- C. Discovery
- D. Beacon

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100112363/75acc8a8/sta-access>

STA Access

STA access includes three phases: scanning, link authentication, and association.

Scanning

A STA can actively or passively scan wireless networks.

Active Scanning

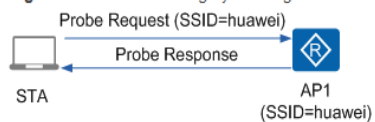
In active scanning, a STA periodically searches for surrounding wireless networks. The STA can send two types of Probe Request frames: containing SSID and not containing SSID.

- The STA sends a Probe Request frame containing an SSID in each channel to search for the AP with the same SSID. Only the AP with the same SSID will respond to the STA. For example, in Figure 2-6, the STA sends a Probe Request frame containing SSID huawei to search for an AP with SSID huawei.

This method applies to the scenario where a STA actively scans wireless networks to access a specified wireless network.

 Feedback

Figure 2-6 Active scanning by sending a Probe Request frame containing an SSID



213. Broadly speaking, which of the following does the transmission medium that constitutes the wireless local area network WLAN not include?

- A. Ray
- B. Radio waves
- C. Infrared
- D. Laser

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100320943/957921b6>

Overview of WLANs

Definition

A wireless local area network (WLAN) is a network that uses wireless channels such as radio waves, laser, and infrared rays to replace the transmission media used on a wired LAN. The WLAN technology described in this document is implemented based on 802.11 standards. That is, a WLAN is a network that uses high-frequency signals (for example, 2.4 GHz or 5 GHz signals) as transmission media.

802.11 was originally a wireless communications standard defined by the Institute of Electrical and Electronics Engineers (IEEE) in 1997. The IEEE then made amendments to the standard, forming the 802.11 family, including 802.11, 802.11a, 802.11b, 802.11e, 802.11g, 802.11i, 802.11n and 802.11ac.

214. Which field of the IPv6 packet header can be used for QoS (Quality of Service)?

- A. Traffic Class
- B. Payload Length
- C. Version
- D. Next Header

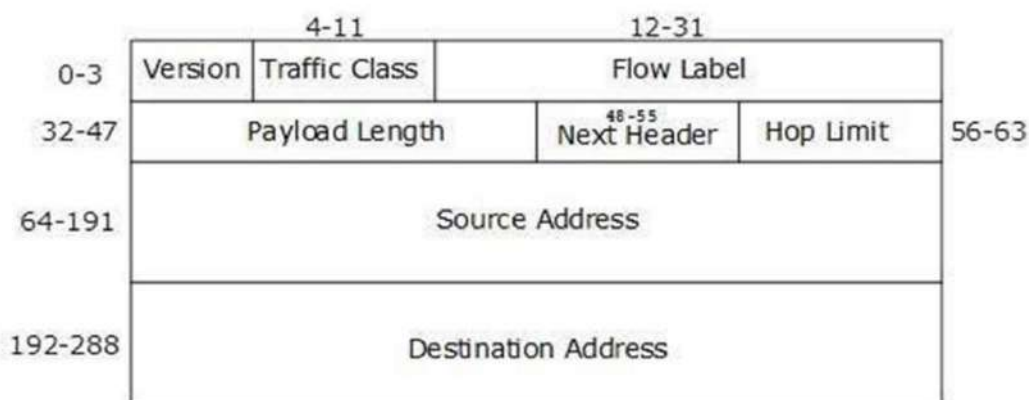
Answer: A

Justification

Source:

<https://www.microsoftpressstore.com/articles/article.aspx?p=2225063&seqNum=3#:~:text=Traffic%20Class%20The%20Traffic%20Class,this%20field%20is%208%20bits.>

Fixed Header



[Image: IPv6 Fixed Header]

- **Traffic Class** The Traffic Class field indicates the IPv6 packet's class or priority. The size of this field is 8 bits. This field provides functionality similar to the IPv4 Type of Service field. Like the Type of Service field in the IPv4 header, the first 6 bits of the Traffic Class field represent the DSCP field as defined in RFC 2474, and the last 2 bits are used for ECN as defined in RFC 3168.

215. Connection between switches and hosts, which port link type is often used by the switch?

- A. Access link
- B. Trunk link
- C. Trunk link
- D. Hybrid link

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000178168/8ebf93dc/link-and-interfacetypes>

<https://jumpcloud.com/it-index/what-is-an-access-port>

Access ports play an important role in network switch configurations, especially for managing VLANs (Virtual Local Area Networks). They connect devices to specific VLANs, helping to separate traffic and improve security in both enterprise and small office networks. Access ports keep communication organized and efficient.

This guide will explain everything you need to know about access ports, including how they work and their uses in modern networks.

Definition and Core Concepts

An access port is a type of switch port that lets untagged traffic move between end devices and the network. It's assigned to one VLAN, making sure the traffic from the connected device stays within that specific VLAN. Access ports work at Layer 2 of the OSI model, which handles data link layer functions.

216. When using the FTP protocol to upgrade the router software, which of the following transmission mode should be selected?

- A. Binary mode
- B. Byte mode
- C. Text mode
- D. Stream byte mode

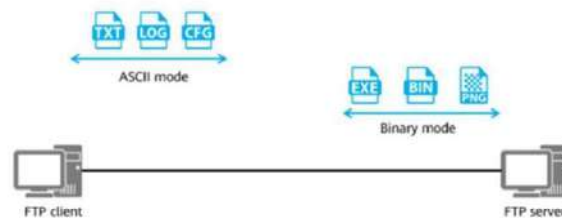
Answer: A

Justification

Source: Slide#15, Page Number 6

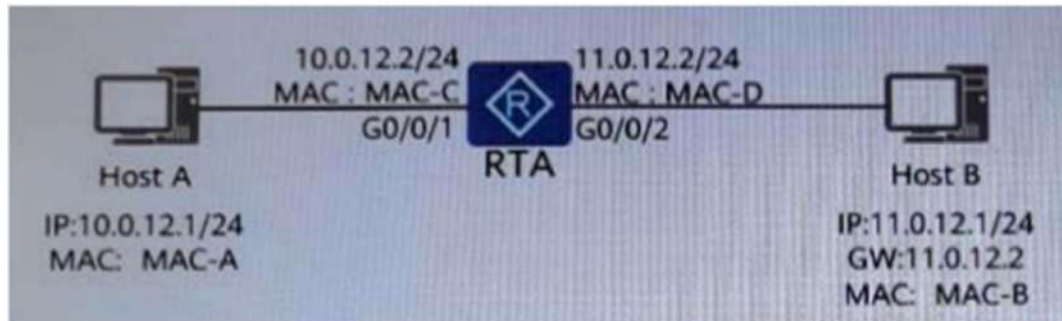


Basic Concepts of FTP



- FTP adopts the typical client/server (C/S) architecture. After an FTP client establishes a TCP connection with an FTP server, files can be uploaded and downloaded.
- FTP uses different transfer modes based on the file type:
 - **ASCII mode:** When a text file (in TXT, LOG, or CFG format) is transferred, the encoding mode of the text content is converted to improve the transfer efficiency. This mode is recommended for transferring configuration files and log files of network devices.
 - **Binary mode:** Non-text files (in CC, BIN, EXE, or PNG format), such as images and executable programs, are transferred in binary mode. This mode is recommended for transferring version files of network devices.

217. In the network shown below, Host A does not configure the gateway; Host B exists a gateway ARP cache. Which of the following statements is correct?



- A. Enable ARP proxy on the G0/0/1 port of the router, then host A can communicate with host B
- B. Host A and Host B cannot communicate in both directions
- C. The data packet sent by host B with the destination IP address of 10.0.12.1 can be forwarded to host A
- D. The data packet sent by host A with the destination IP address of 11.0.12.1 can be forwarded to host B

Answer: B

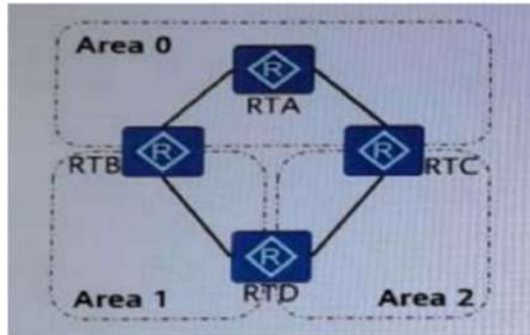
Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100279002/64996475/vlanconfiguration>

Host A does not have a gateway configured, hence it cannot route packets to Host B's network (11.0.12.0/24).

218. The network structure and OSPF partition are shown in the following figure; in addition to RTA, routers RTB, RTC, and RTD are ABR routers.



- A. True
- B. False

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100034072/32a400ec/ospf-fundamentals>

None of the routers (RTA, RTB, RTC, RTD) connect to both a non-backbone area and the backbone area, they cannot be considered ABRs under the provided network structure.

219. The network administrator hopes to effectively use the IP address of the 192.168.176.0/25 network segment. Now that the company's marketing department has 20 hosts, which of the following is the best address segments allocated to the marketing department?

- A. 192.168.176.160/27
- B. 192.168.176.0/25
- C. 192.168.176.96/27
- D. 192.168.176.48/29

Answer: C

Justification

The correct subnet that fits the requirement of at least 20 hosts within the 192.168.176.0/25 network is Option C: 192.168.176.96/27. This subnet provides 30 usable IP addresses, which is more than enough for the 20 hosts in the marketing department.

192.168.176.0/25

For 20 Hosts = $2^5 = 32$

The host will be given 5 bits

11111111.11111111.11111111.11100000 (27)

$160 + x = 255 \Rightarrow x = 255 - 160 = 95$

$96 + x = 255 \Rightarrow x = 255 - 96 = 159$

A)

192	168	176	160
11000000	10101000	10110000	10100000
AND 11111111.11111111.11111111.11100000			
11000000.10101000.10110000.10100000			
192	168	176	160

C)

192	168	176	96
11000000	10101000	10110000	01100000
AND 11111111.11111111.11111111.11100000			
11000000.10101000.10110000.01100000			
192	168	176	96

Scanned with CamScanner

220. What is the main function of the dynamic routing protocol?

- A. Manage routers
- B. Generate IP address
- C. Dynamically generate routing entries
- D. Control router interface status

Answer: C

Justification

Source: Slide#06, Page Number 18

OSPF Packet Types

- There are five types of OSPF protocol packets and implement different functions in interaction between OSPF routers.

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.

221. The device port running STP protocol is in the forwarding state. Which of the following statements is correct?

- A. The port not only forwards user traffic but also processes BPDU packets.
- B. The port only receives and processes BPDUs and does not forward user traffic.
- C. The port will build a MAC address table based on the received user traffic, but it does not forward user traffic.
- D. The port not only does not process BPDU packets, but also does not forward user traffic.

Answer: A

Justification

Source: Huawei slide#09, page 33

STP Port States	
Port State	Description
Disabled	The port cannot send or receive BPDUs or service data frames. That is, the port is Down.
Blocking	The port is blocked by STP. A blocked port cannot send BPDUs but listens to BPDUs. In addition, the blocked port cannot send or receive service data frames or learn MAC addresses.
Listening	STP considers the port in Listening state as the root port or designated port, but the port is still in the STP calculation process. In this case, the port can send and receive BPDUs but cannot send or receive service data frames or learn MAC addresses.
Learning	A port in Learning state listens to service data frames but cannot forward them. After receiving service data frames, the port learns MAC addresses.
Forwarding	A port in Forwarding state can send and receive service data frames and process BPDUs. Only the root port or designated port can enter the Forwarding state.

222. Which of the following descriptions about spanning tree designated ports is correct?

- A. The designated port can forward configuration BPDU messages to the network segment connected to it.
- B. The port on the root switch must not be a designated port.
- C. Each switch has only one designated port.
- D. The port on the root switch must be a designated port.

Answer: A

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100096325/99f4e7eb/understanding-stp-rstp>

Table 8-31 Description of the designated bridge and designated port

Object	Designated Bridge	Designated Port
Device	Device that forwards configuration BPDUs to a directly connected device	Designated bridge port that forwards configuration BPDUs to a device
LAN	Device that forwards configuration BPDUs to a network segment	Designated bridge port that forwards configuration BPDUs to a network segment.

223. Which of the following statements about link aggregation is correct?

- A. GE electrical interfaces and GE optical interfaces cannot be added to the same Eth-Trunk interface.
- B. Eth-Trunk interfaces cannot be nested.
- C. When two devices are connected, it is necessary to ensure that the link aggregation mode on both devices is the same.
- D. GE interface and FE interface cannot be added to the same Eth-Trunk interface.

Answer: BCD

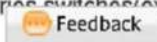
Justification

Source: <https://support.huawei.com/enterprise/en/knowledge/KB0000482703>

<https://support.huawei.com/enterprise/en/doc/EDOC1000060766/d8898e92/which-types-of-interfaces-can-join-the-same-eth-trunk>

Configuration Notes

- Member interfaces of an Eth-Trunk must use the same Ethernet type and rate.
- Both devices of the Eth-Trunk must use the same number of physical interfaces, interface rate, duplex.
- If an interface of the local device is added to an Eth-Trunk, an interface of the remote device directly connects to the Eth-Trunk.
- **Both devices of an Eth-Trunk must use the same link aggregation mode.**
- This example applies to all versions of all S series switches (except the S5731-L and S5731S-L).



, the FE interface and the GE interface cannot be added to the same Eth-trunk.

When adding an interface to an Eth-Trunk, ensure that the interface is a hybrid interface, which is the default interface type. An Eth-Trunk cannot be nested, that is, **its member interfaces cannot be Eth-Trunk**. An Ethernet interface can be added to only one Eth-trunk interface.

224. What is the total length of the VLAN frame format defined by IEEE802.1Q?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100088104>

VLAN Frame Format

VLAN Frame Format

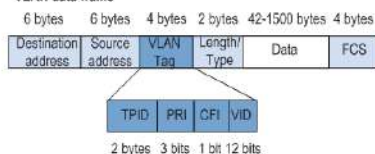
A switch identifies packets from different VLANs according to the information contained in its VLAN tags. IEEE 802.1Q adds a 4-byte VLAN tag between the Source/Destination MAC address and Length/Type fields of an Ethernet frame to identify the VLAN to which the frame belongs. Figure 1-1 shows the position of a VLAN tag in a VLAN data frame.

Figure 1-1 IEEE 802.1Q frame format

Traditional Ethernet data frame

6 bytes	6 bytes	2 bytes	46-1500 bytes	4 bytes
Destination address	Source address	Length/Type	Data	FCS

VLAN data frame



225. CSS refers to the combination of two switch devices that support cluster features and logically combine them into one switch device.

- A. True
- B. False

Answer: A

Justification

Source: <https://info.support.huawei.com/infofinder/encyclopedia/en/CSS.html>

What Is Cluster Switch System (CSS)?

A CSS combines two clustering-capable switches into a single logical switch. CSS is a widely used horizontal virtualization technology that simplifies configuration and management, expands bandwidth, and implements inter-chassis link redundancy.

226. The host uses the IP address 192.168.1.2 to access the Internet, which must go through NAT.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086645>

The IP address 192.168.1.2 is a private IP address. Private IP addresses are not routable on the public internet. In order for a host with a private IP address to access the internet, it must go through NAT (Network Address Translation). NAT translates the private IP address of the host to a public IP address, which can then be routed on the public internet.

So, if a host with the IP address 192.168.1.2 wants to access the internet, it must go through NAT.

227. Before the STA associates with the AP, it needs to exchange some messages to obtain the SSID. What are these messages?

- A. Discovery
- B. Beacon
- C. Probe
- D. Response Probe Request

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100112363/75acc8a8/staaccess>

STA Access

STA access includes three phases: scanning, link authentication, and association.

Scanning

A STA can actively or passively scan wireless networks.

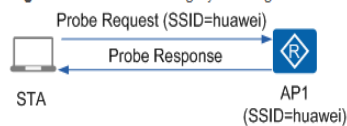
Active Scanning

In active scanning, a STA periodically searches for surrounding wireless networks. The STA can send two types of Probe Request frames: containing SSID and not containing SSID.

- The STA sends a Probe Request frame containing an SSID in each channel to search for the AP with the same SSID. Only the AP with the same SSID will respond to the STA. For example, in **Figure 2-6**, the STA sends a Probe Request frame containing SSID huawei to search for an AP with SSID huawei.

This method applies to the scenario where a STA actively scans wireless networks to access a specified wireless network.

Figure 2-6 Active scanning by sending a Probe Request frame containing an SSID



228. Which of the following statements about the MPLS Label statement is wrong?

- A. The label is encapsulated between the network layer and the transport layer.
- B. The tags are carried by the head of the packet, do not contain topology Information.
- C. The label is used to uniquely identify the forwarding equivalence class FEC to which a packet belongs.
- D. The label is a short identifier with a fixed length. It has only a local meaning.

Answer: A encapsulated b/w data-link and network layer

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100118961>

In MPLS, the label is inserted between the Layer 2 (Data Link layer) and Layer 3 (Network layer) headers. This encapsulation method is often referred to as a "shim" header.

Definition of MPLS

MPLS is implemented between the data link layer and network layer in the TCP/IP protocol stack. MPLS replaces IP forwarding with label switching. A label is a short and fixed-length connection identifier with local significance. It is similar to the virtual path identifier (VPI)/virtual channel identifier (VCI) of ATM and the data link connection identifier (DLCI) of Frame Relay.

MPLS can use any Layer 2 media to transfer packets. It is not limited by any specific protocol at the data link layer. MPLS was derived from the Internet Protocol version 4 (IPv4), and the core MPLS technology can be extended to multiple network protocols, such as the Internet Protocol version 6 (IPv6), Internet Packet Exchange (IPX), Appletalk, DECnet, and Connectionless Network Protocol (CLNP). As its name suggests, MPLS supports label switching between multiple network protocols.

229. Which of the following refers to network communication? (Multiple choice)

- A. Use instant messaging software (such as QQ, WeChat) to chat with friends
- B. Use a computer to visit the company's official website
- C. Use a computer to watch a video online
- D. Download emails from the company mailbox to your computer

Answer: ABC

Justification

Source: <https://www.huawei.com/en/giv/communications-network-2030>

- A. Use instant messaging software (such as, QQ, WeChat) to chat with friends
- B. Use a computer to visit the company's official website
- C. Use computer to watch video online

Network communication is the process of exchanging information between two or more devices over a network. The devices can be computers, smartphones, tablets, or other devices that are connected to the network.

while, Option D, downloading emails from the company mailbox to your computer, does not involve network communication. The emails are already stored on the company's server, and you are simply downloading them to your computer.

230. In the VRP interface, use the command **startup saved-configuration backup.cfg** to configure the **startup.cfg** file for the next startup.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000041693/f6e1ef7f/startup-savedconfigura%C6%9Fon>

Function

The **startup saved-configuration** command specifies the system configuration file for next startup.

Format

startup saved-configuration configuration-file [**slot** slot-id]

Parameters

Parameter	Description	Value
configuration-file	Specifies the name of a configuration file. Make sure that the file exists.	The value is a string of 5 to 64 case-sensitive characters without spaces. The file name extension can be .zip, .dat, or .cfg.
slot slot-id	Specifies a member device in a stack.	The value is an integer. The range of the integer is dependent on the specific device.

231. Which of the following fields does the ARP packet format include? (Multiple choice)

- A. Hardware Type
- B. Protocol Address of sender
- C. Operation Code
- D. Protocol Type Verified

Answer: ABC

Justification

Source: <https://www.geeksforgeeks.org/arp-protocol-packet-format/>

The ARP packet format is used for ARP requests and replies and consists of multiple fields including hardware type, protocol type, hardware and protocol size, operation, sender and target hardware, and IP addresses.

232. Both the `pwd` and `dir` commands in the VRP operating platform can view the file information under the current directory.

- A. True
- B. False

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064353/35605cd/pwd-user-view>

In the VRP (Versatile Routing Platform) operating platform, the `pwd` command is used to display the current working directory (folder), while the `dir` command is used to list the contents of the current directory.

233. The OSI reference model from high-level to bottom-level is ().

- A. Application layer, transport layer, network layer, data link layer, physical layer
- B. Application layer, session layer, presentation layer, transport layer, network layer, data link layer, physical layer
- C. Application layer, presentation layer, session layer, network layer, transport layer, data link layer, physical layer
- D. Application layer, presentation layer, session layer, transport layer, network layer, data link layer, physical layer

Answer: D

Justification

Source: Slide #02, Page Number: 08

7. Application Layer	Provides interfaces for applications.
6. Presentation Layer	Translates data formats to ensure that the application-layer data of one system can be identified by the application layer of another system.
5. Session Layer	Establishes, manages, and terminates sessions between communicating parties.
4. Transport Layer	Establishes, maintains, and cancels an end-to-end data transmission process; controls transmission speeds and adjusts data sequences.
3. Network Layer	Defines logical addresses and transfers data from sources to destinations.
2. Data Link Layer	Encapsulates packets into frames, transmits frames in P2P or P2MP mode, and implements error checking.
1. Physical Layer	Transmits bitstreams over transmission media and defines electrical and physical specifications.

234. Which of the following attributes cannot be used as a parameter to measure Cost? (Multiple choice)

- A. Bandwidth
- B. sysname
- C. Delay Don't know but is a metric in EIGRP
- D. Hop count

Answer: BC

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086956>

Route Metric

A route metric specifies the cost of a route to a specified destination address. The following factors often affect the route metric:

- Path length

Path length is the most common factor that affects the route metric. Link-state routing protocols allow you to assign a link cost for each link to identify the path length of a link. In this case, the path length is the sum of the link costs of all the links that packets pass through. Distance-vector routing protocols use the hop count to identify the path length. The hop count is the number of devices that packets pass through from the source to the destination. For example, the hop count from a router to its directly connected network is 0, and the hop count from a router to a network that can be reached through just one other router is 1. Other lengths can be deduced in the same manner.

- Network bandwidth

Network bandwidth is the transmission capability of a link. For example, a 10-Gigabit link has a higher transmission capability than a 1-Gigabit link. Although bandwidth defines the maximum transmission rate of a link, routes over high-bandwidth links are not necessarily better than routes over low-bandwidth links. For example, when a high-bandwidth link is congested, forwarding packets over this link will require more time.

- Load

The load is the degree to which a network resource is busy. You can calculate the load by calculating the CPU usage and packets processed per second. Continually monitoring the CPU usage and packets processed per second helps you learn more about network usage.

- Communication cost

The communication cost is the operating cost of a route over a link. The communication cost is another important indicator, especially if you do not care about network performance but are concerned about the operating expenditure.

235. A company applies for a Class C address segment, which needs to be evenly distributed to 8 subsidiaries. The largest subsidiary has 14 computers. Different subsidiaries must be in different network segments. The subnet mask should be designed as ().

- A. 255.255.255.240
- B. 255.255.255.128
- C. 255.255.255.192
- D. 255.255.255.0

Answer: A

Justification

Source: Slide 04 Page Number: 21

1. **Class C address segment** has the default subnet mask of 255.255.255.0, which provides 256 IP addresses (including network and broadcast addresses).
2. To divide the network into 8 segments, we need 3 bits (since $2^3=8$) for subnetting.
3. The subnet mask for 8 subnets in a Class C network is 255.255.255.240. This subnet mask gives:
 - o 16 IP addresses per subnet (14 usable, since 2 addresses are reserved for network and broadcast).

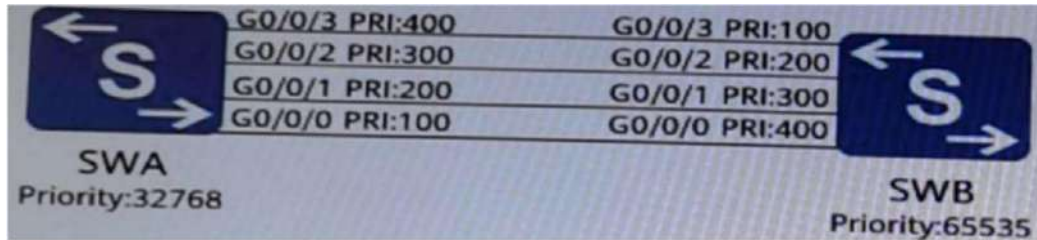
each subnet will have 16 IP addresses, which can support up to 14 hosts (computers) as required.

Calculation of Subnet Mask:

- The binary form of the default Class C subnet mask is:
11111111.11111111.11111111.00000000 (255.255.255.0).
- To create 8 subnets, we need to borrow 3 bits from the host part:
11111111.11111111.11111111.11100000 (255.255.255.240).

Therefore, the correct answer is: A. 255.255.255.240

236. The priority of the LACP protocol is shown in the figure. Switch A and Switch B adopt LACP mode link aggregation, and all interfaces join the link aggregation group. At the same time, set the maximum number of active ports to 3. Which interface of switch A is not an active port?



- A. G0/0/1
- B. G0/0/3
- C. G0/0/0
- D. G0/0/2

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000178168/b1d8b5a3/link-aggrega%C6%9Fon-inlACP-mode>

Concepts

- LACP system priority

LACP system priorities are set on devices at both ends of an Eth-Trunk. In LACP mode, active member interfaces selected by both devices must be consistent; otherwise, an LAG cannot be established. To keep active member interfaces consistent at both ends, set a higher priority for one end so that the other end selects active member interfaces based on the selection of the end with a higher priority. The smaller the LACP system priority value, the higher the LACP system priority.

- LACP interface priority

Interface LACP priorities are set to prioritize interfaces of an Eth-Trunk. Interfaces with higher priorities are selected as active interfaces. The smaller the LACP interface priority value, the higher the LACP interface priority.

237. If the Ethernet data frame Length/Type=0X0806, which of the following statements are correct? (Multiple choice)

- A. The source MAC address of the data frame must not be FFFF-FFFF-FFFF
- B. The data frame is an IEEE 802.3 frame
- C. The data frame is an Ethernet II frame
- D. The destination MAC address of the data frame may be FFFF-FFFF-FFFF broadcast-frame ?

Answer: AC

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100198444/7404084d/mac-address-tableconfiguration-commands>

Source: <https://notes.networklessons.com/ethernet-frame-types>

input digits. For example, if you enter e0, the system changes e0 to 00e0. The MAC address cannot be FFFF-FFFF-FFFF, or a multicast MAC address.

The **EtherType** field is used to identify the upper layer protocol (e.g. IP) that encapsulates the frame. For example, a value of 0x0800 indicates an IPv4 packet while a value of 0x0806 indicates an ARP frame. These values share the same numbering scheme as those found in the PTYPE field in the header of an ARP message.

238. VLAN 4095 cannot be created on Huawei switches, and VLAN 1 cannot be created.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100278261/55c108dd/creating-and-deleting-a-vlan>

Create a VLAN and enter the VLAN view. If the VLAN already exists, this command will directly display the VLAN view. The VLAN ID ranges from 1 to 4094 (0 and 4095 are reserved).

As VLAN 1 is the default VLAN, it cannot be deleted and does not need to be created.

As VLAN 1 is the default VLAN, it cannot be deleted and does not need to be created.

Procedure

- **Creating a VLAN**

Enter the system view.

```
system-view
```

Create a VLAN and enter the VLAN view. If the VLAN already exists, this command will directly display the VLAN view. The VLAN ID ranges from 1 to 4094 (0 and 4095 are reserved).

239. Which of the following port states is not included in the RSTP protocol?

- A. Learning
- B. Discarding
- C. Forwarding
- D. Blocking

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100247750/2bd0e592/rstp-portroles-and-port-states>

Table 13-8 Comparison between port states defined in STP and RSTP

STP Port State	RSTP Port State	Port Role
Forwarding	Forwarding	Root port or designated port
Learning	Learning	Root port or designated port
Listening	Discarding	Root port or designated port
Blocking	Discarding	Alternate port or backup port
Disabled	Discarding	Disabled port

240. Which statement about the Hybrid port is correct?

- A. When the Hybrid port sends data frames, it must carry a VLAN TAG
- B. Hybrid port only receives data frames with a VLAN TAG
- C. Hybrid port can strip the TAG of the VLAN in the direction of the interface
- D. Hybrid port does not require PVID

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064351/923935bf/port-hybrid-untagged-vlan>

Usage Guidelines

Usage Scenario

A hybrid interface can connect to either a user device or a wireless access controller. When a hybrid interface is connected to a user device, it must be added to VLANs in untagged mode so that it sends untagged frames.

After a hybrid interface is added to VLANs in untagged mode, the interface removes VLAN tags of frames before sending frames.

241. What elements can NAT convert?

- A. MAC address + port number
- B. Only IP address
- C. IP address + port number
- D. Only MAC address

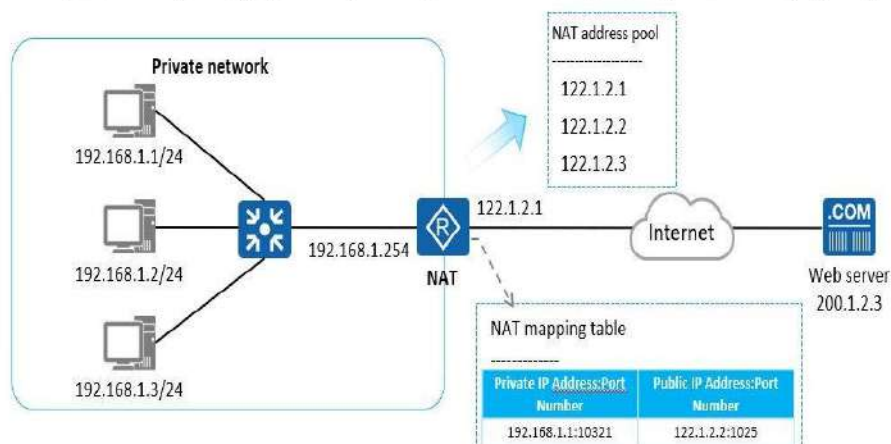
Answer: C

Justification

Source: Huawei slide 14, page number 20

NAPT Implementation

- Dynamic NAT does not translate port numbers. It belongs to No-Port Address Translation (No-PAT). In this mode, the mapping between public and private addresses is still 1:1, which cannot improve public address utilization.
- Network Address and Port Translation (NAPT): translates both IP addresses and port numbers from multiple internal hosts to one public IP address in an address pool. In this way, 1:n mapping between public and private addresses is implemented, which effectively improves public address utilization.



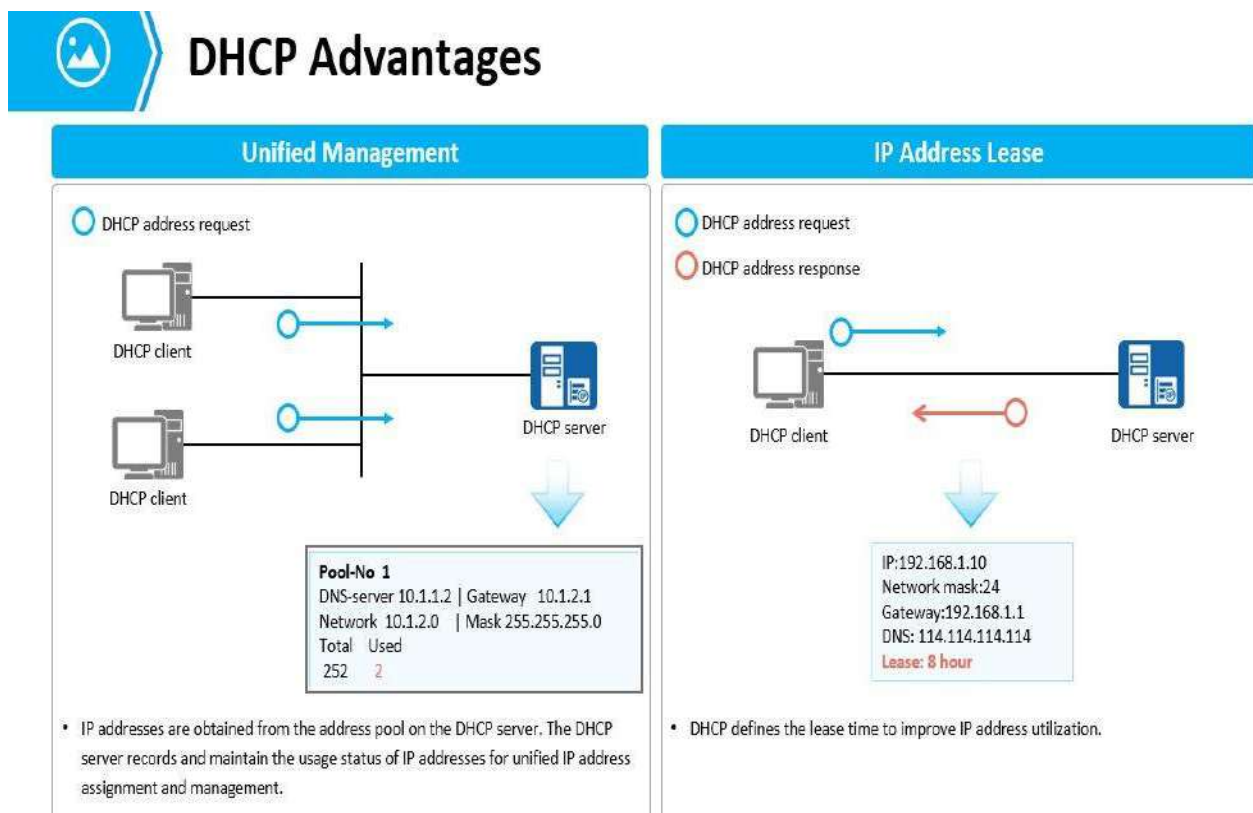
242. What are the advantages of using the Dynamic Host Configuration Protocol DHCP to assign IP addresses? (Multiple choice)

- A. Heavy workload and difficult to manage
- B. Reuse of IP address can be realized
- C. Avoid IP address conflicts
- D. When the configuration information changes (such as DNS), only the administrator needs to modify it on the DHCP server to facilitate unified management

Answer: BCD

Justification

Source: Huawei slide 15, page number 27



243. The role of the collaboration layer in the SDN architecture is to complete job deployment based on user intentions. OpenStack belongs to the business collaboration layer.

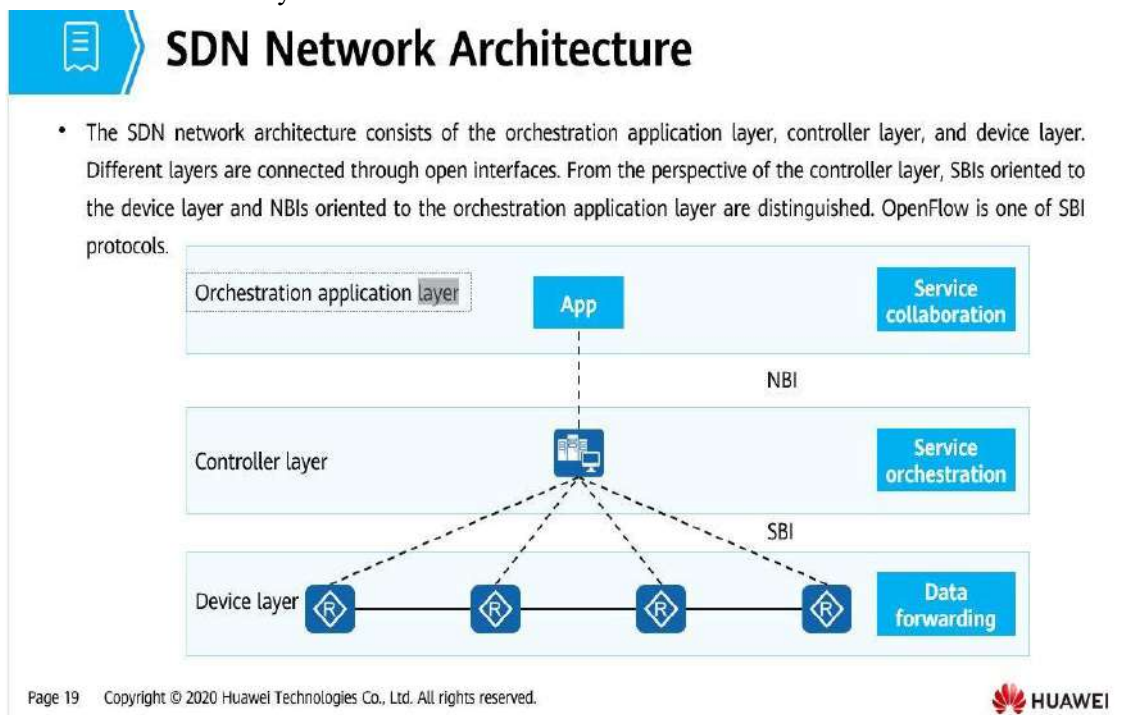
- A. True
- B. False

Answer: B

Justification

Source: Huawei slide 20, page number 19

There is no collaboration layer in SDN.



244. Which of the following statement about Python is incorrect?

- A. Is a fully open-source high-level programming language
- B. Has a rich third-party library
- C. Have a clear grammatical structure, simple and easy to learn, colleagues run efficiently
- D. can be used for the operation and maintenance of automation scripts, Artificial Intelligence, Data science, and many other fields

Answer: A

Justification

Explanation: Although Python is considered an open-source language, the term "**fully open-source**" can be misleading. While much of Python's source code is freely available and can be modified and distributed under an open-source license, not every aspect of its development and distribution is entirely free from restrictions. Therefore, calling it fully open-source is not entirely accurate. It is open-source to a large extent, but not in an absolutely unrestricted way

245. The administrator has configured the AR2200 as shown in the figure. Then, which of the following description of the configuration information is correct?

```
<Huawei>reset saved-configuration
Warning: The action will delete the saved configuration in the device.
The configuration will be erased to reconfigure. Continue? [Y/N]:
```

- A. The saved configuration file will be replaced by the running configuration file
- B. If the user wants to clear the next startup configuration file of the device configuration, then should select "N"
- C. If the user wants to clear the next startup configuration file of the device configuration, then should select "Y"
- D. The next startup file of the device configuration will be retained

Answer: C

Justification

- ☑ *The command reset saved-configuration deletes the current saved configuration file.*
- ☑ *If the user confirms by selecting "Y", the saved configuration file will be deleted, meaning the next startup configuration will be cleared.*
- ☑ *If the user selects "N", the action will be canceled and the saved configuration file will not be deleted.*

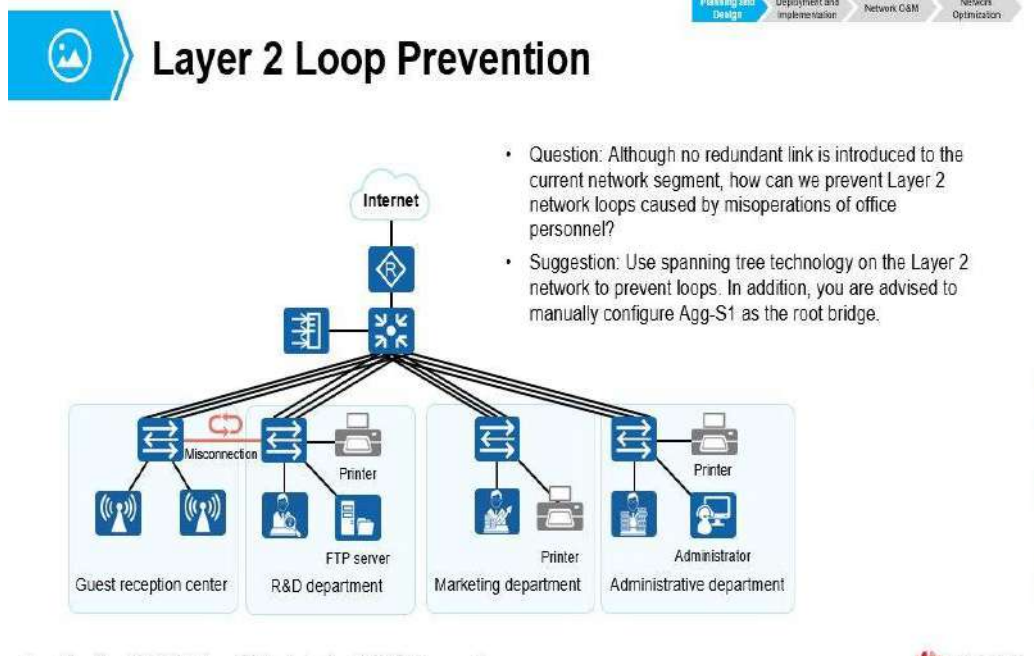
246. When building a campus network, which of the following protocols can be used to avoid layer 2 loops?

- A. SNMP
- B. RSTP
- C. NAT
- D. OSPF

Answer: B

Justification

Source: Huawei slide 22, page number 26



247. The tree network topology is actually a hierarchical star structure, which makes it easy to expand the network scale, but the higher the level, the more serious the network problem caused by the node failure.

- A. True
- B. False

Answer: A

Justification

Source: Huawei slide 01, page number 21

In the tree network topology the tree structure is actually a hierarchical star structure.

- Advantages: Multiple star networks can be quickly combined, which facilitates network expansion.
- Disadvantages: A fault on a node at a higher layer is more severe.

248. The routing table is shown in the figure. Which of the following statements is correct?
(Multiple choice)

```
<R1>display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public
Destinations : 14 Routes : 15

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.2.2/32	Static	60	0	RD	10.0.12.2	Ethernet0/0/0
10.0.3.3/32	Static	60	0	RD	10.0.2.2	Ethernet0/0/0
10.0.12.0/24	Direct	0	0	D	10.0.12.1	Ethernet0/0/0
10.0.12.1/32	Direct	0	0	D	127.0.0.1	Ethernet0/0/0

- A. The router forwards data packets with the destination IP address 10.0.3.3 from Ethernet0/0/0
- B. Router from Ethernet0/0/0 forwarding destination IP address 10.0.2.2 packet
- C. The router forwards data packets with the destination IP address 10.0.12.1 from Ethernet0/0/0
- D. The NextHop of the destination network 10.0.3.3/32 is not directly connected, so the router will not forward packets with the destination IP address of 10.0.3.3

Answer: ABC

Justification

Source: Huawei slide 05, page number 26

- When a router receives a packet, it searches its IP routing table for the outbound interface and next hop based on the destination IP address of the packet. If it finds a matching routing entry, it forwards the packet according to the outbound interface and next hop specified by this entry. Otherwise, it discards the packet.

249. The network administrator uses Ping to test the connectivity of the network. Which of the following protocols may be used in this process? (Multiple choice)

- A. ICMP
- B. UDP
- C. TCP
- D. ARP

Answer: AD

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100064365/f5ddba7b/understanding-arp>

IP Ping

IP ping uses encapsulated Internet Control Message Protocol (ICMP) packets to check whether a specific interface in the same routing domain is reachable. If the packets fail to reach the destination interface, a fault occurs on the fabric network. During IP ping, MAC, ARP, and route forwarding entries of switches are checked. Currently, the IP ping operation is supported between a VM and a physical server or between two VMs. Figure 2-23 shows the IP ping process.

250. The router needs to modify the destination IP address in the data packet when forwarding data packets.

- A. True
- B. False

Answer: B

Justification

Source: Huawei slide 05, page number 26

- When a router receives a packet, it searches its IP routing table for the outbound interface and next hop based on the destination IP address of the packet. If it finds a matching routing entry, it forwards the packet according to the outbound interface and next hop specified by this entry. Otherwise, it discards the packet.

**251. Which of the following descriptions about DD packets and LSA of OSPF is correct?
(Multiple choice)**

- A. The header of an LSA can uniquely identify an LSA
- B. The LSA header is only a small part of LSA
- C. DD packet contains detailed information about LSA
- D. The DD packet contains only LSA header information

Answer: ABD

Justification#1:

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100174721/fd1dc73c/ospf-dd>

2) A DD packet contains LSA header information, including LS Type, LS ID, Advertising Router, LS Sequence Number, and LS Checksum.

Justification#2:

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100174721/fd1dc73c/ospf-dd>

Protocol Stack Structure

OSPF DD packets are encapsulated into IP packets. The protocol number is 89.

An OSPF DD packet contains the header of each LSA in the LSDB. An LSA header uniquely identifies an LSA. The LSA header is only a small part of the entire LSA and the peer router can judge whether it has the LSA according to the LSA header. This reduces the amount of traffic transmitted between two routers. When two routers exchange DD packets, one acts as the master and the other one acts as the slave. The master defines a start sequence number and increments the sequence number by 1 each time it sends a DD packet. After the slave receives a DD packet, it uses the sequence number carried in the DD packet for acknowledgement.

252. Which of the following statements about the main functions of the router is wrong?
(Multiple choice)

- A. Realize the mutual communication between the same network disconnection equipment
- B. Forwarding according to the source IP address of the received data packet
- C. Establish routing tables through multiple protocols
- D. Guide data forwarding according to the routing table

Answer: AB

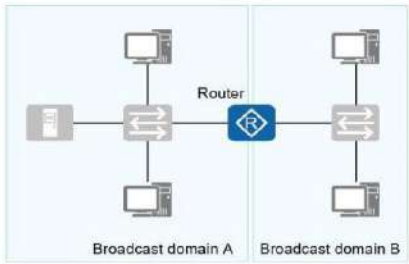
Justification

Source: Huawei Slide 1, Page number 12

Network communication → Data communication network

Routers

- Router: a network-layer device that forwards data packets on the Internet. Based on the destination address in a received packet, a router selects a path to send the packet to the next router or destination. The last router on the path is responsible for sending the packet to the destination host.
 - Implementing communication between networks of the same type or different types
 - Isolating broadcast domains
 - Maintaining the routing table and running routing protocols
 - Selecting routes and forwarding IP packets
 - Implementing WAN access and network address translation
 - Connecting Layer 2 networks established through switches



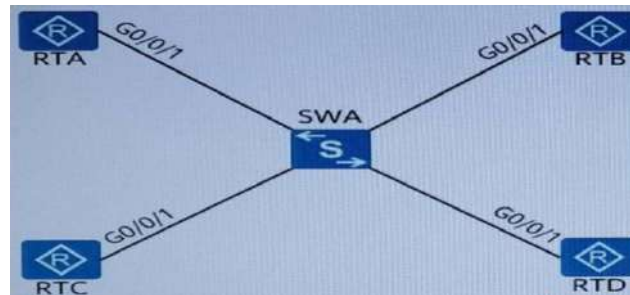
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Explanation:

- Option A: Routers primarily facilitate communication between different networks. Devices within the same network can typically communicate directly without a router.
- Option B: Routers forward data packets based on the destination IP address. This address determines the packet's intended recipient and guides the router's forwarding decision.

253. In the broadcast network as shown in the figure, OSPF runs on four routers in the same area and on the same network segment. OSPF will automatically elect one DR and multiple BDRs to achieve better backup results.



- A. True
- B. False

Answer: B

Justification

Source: verified from eNSP

```
OSPF Process 1 with Router ID 10.0.0.1
  Neighbors

Area 0.0.0.0 interface 10.0.0.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.0.2      Address: 10.0.0.2
  State: Full Mode:Nbr is Master Priority: 1
  DR: 10.0.0.1 BDR: 10.0.0.4 MTU: 0
  Dead timer due in 32 sec
  Retrans timer interval: 5
  Neighbor is up for 00:00:30
  Authentication Sequence: [ 0 ]

Router ID: 10.0.0.3      Address: 10.0.0.3
  State: Full Mode:Nbr is Master Priority: 1
  DR: 10.0.0.1 BDR: 10.0.0.4 MTU: 0
  Dead timer due in 40 sec
  Retrans timer interval: 5
  Neighbor is up for 00:00:04
  Authentication Sequence: [ 0 ]

Router ID: 10.0.0.4      Address: 10.0.0.4
  State: Full Mode:Nbr is Master Priority: 1
  DR: 10.0.0.1 BDR: 10.0.0.4 MTU: 0
  Dead timer due in 38 sec
  Retrans timer interval: 5
  Neighbor is up for 00:01:07
  Authentication Sequence: [ 0 ]
```

254. The Router ID of the OSPF process takes effect immediately after the modification.

- A. True
- B. False

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100082074>

After the router has been running OSPF and selected its router ID, it still uses this router ID if the interface whose IP address is used as the router ID is Down or disappears (for example, the undo interface loopback loopback-number command is run) or a larger interface IP address exists. The router can obtain a new router ID only after a router ID is reconfigured for the router or an OSPF router ID is reconfigured and the OSPF process restarts.

255. How many types of OSPF messages are there?

- A. 2
- B. 4
- C. 5
- D. 3

Answer: C

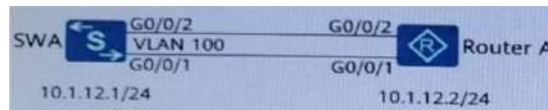
Justification

Source: Huawei slide 06, page number 18

- There are five types of OSPF protocol packets and implement different functions in interaction between OSPF routers.

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.

256. In the network shown in the figure below, switch A and router A are connected through two links, and the two links are aggregated in manual load-sharing mode. The aggregated port number is 1, and the aggregated link performs data when forwarding, you need to carry VLAN TAG 100. Switch A uses the Trunk link. Which of the following configurations should be used for Switch A?



- A.

```
interface Eth-Trunk1
mode lacp-static
port link-type trunk
port trunk allow-pass vlan 100
interface GigabitEthernet0/0/1
eth-trunk 1
interface GigabitEthernet0/0/2
eth-trunk 1
```
- B.

```
interface Eth-Trunk1
port link-type trunk
port trunk allow-pass vlan 100
interface GigabitEthernet0/0/1
eth-trunk 1
interface GigabitEthernet0/0/2
eth-trunk 1
```
- C.

```
interface Vlanif100
ip address 10.0.12.1 255.255.255.0
```
- D.

```
interface Eth-Trunk1
ip address 10.0.12.1 255.255.255.0
```

Answer: B

Justification

Source: Huawei slide 11, page number 37

Example for Configuring an Eth-Trunk in Manual Mode



SW1 configuration:

```
[SW1] interface eth-trunk 1
[SW1-Eth-Trunk1] trunkport gigabitethernet 0/0/1 to 0/0/2
[SW1-Eth-Trunk1] port link-type trunk
[SW1-Eth-Trunk1] port trunk allow-pass vlan 10 20
```

• Requirement description:

- SW1 and SW2 are connected to the networks of VLAN 10 and VLAN 20.
- SW1 and SW2 are connected through two Ethernet links. To provide link redundancy and enhance transmission reliability, configure an Eth-Trunk in manual mode between SW1 and SW2.

SW2 configuration:

```
[SW2] interface eth-trunk 1
[SW2-Eth-Trunk1] trunkport gigabitethernet 0/0/1 to 0/0/2
[SW2-Eth-Trunk1] port link-type trunk
[SW2-Eth-Trunk1] port trunk allow-pass vlan 10 20
```

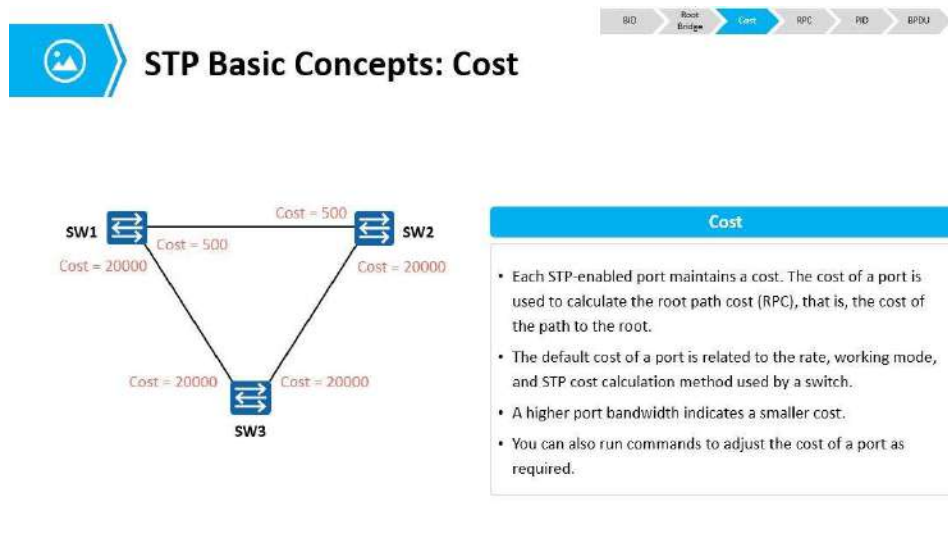
257. By default, the port cost calculated by STP has a certain relationship with the port bandwidth; that is, the greater the bandwidth, the () the cost.

- A. Greater
- B. Consistent
- C. Not necessarily
- D. Smaller

Answer: D

Justification

Source: Huawei slide 9, page number 16



258. Which of the following port sales does the RSTP protocol include? (Multiple choice)

- A. Learning
- B. Discarding
- C. Listening
- D. Forwarding

Answer: ABD

Justification

Source: Huawei slide 9, page number 51

Port States in RSTP

- RSTP deletes two port states defined in STP, reducing the number of port states to three.
 - If the port does not forward user traffic or learn MAC addresses, it is in Discarding state.
 - If the port does not forward user traffic but learns MAC addresses, it is in Learning state.
 - If the port forwards user traffic and learns MAC addresses, it is in Forwarding state.

STP Port State	RSTP Port State	Port Role
Forwarding	Forwarding	Root port or designated port
Learning	Learning	Root port or designated port
Listening	Discarding	Root port or designated port
Blocking	Discarding	Alternate port or backup port
Disabled	Discarding	Disabled port

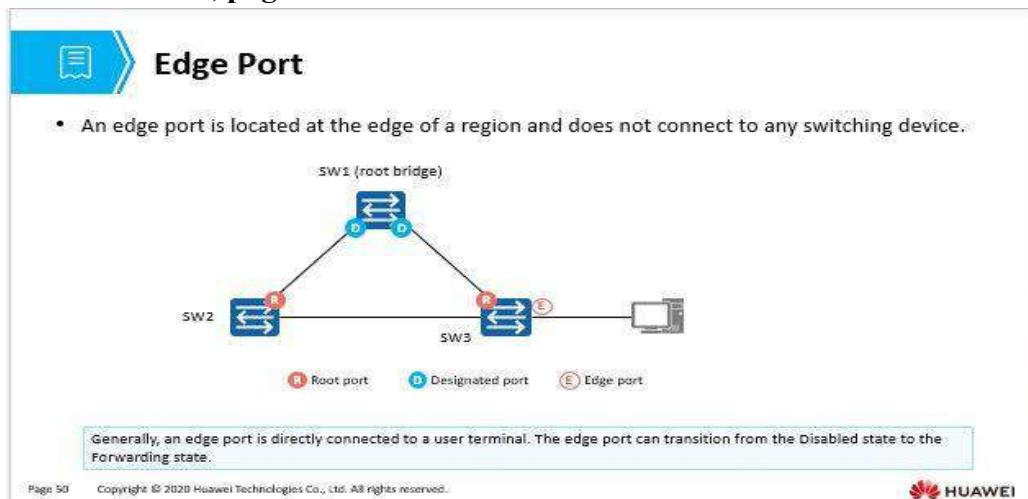
259. In the RSTP standard, in order to improve the convergence speed, the port switch is directly connected to the terminal can be defined as ().

- A. Edge port
- B. Root port
- C. Fast port
- D. Backup port

Answer: A

Justification

Source: Huawei slide 9, page number 50



260. The configuration information of a certain port of the switch is shown in the figure, then the port will strip off the VLAN TAG when sending data frames carrying which VLAN?

```
#
interface GigabitEthernet0/0/1
 port hybrid tagged vlan 2 to 3 100
 port hybrid untagged vlan 4 6
#
```

- A. 4.6
- B. 1.4.5.6
- C. 4.5.6
- D. 1.4.6

Answer: D

Justification

Source: Huawei slide 8, page number 22

A hybrid interface requires the configuration of a PVID (Port VLAN ID) as well as two VLAN ID lists: one for untagged VLANs and another for tagged VLANs. By default, VLAN 1 is included in the untagged VLAN ID list. Only frames with VLAN IDs specified in these two lists are allowed to pass through the hybrid interface. This interface type supports the transmission of tagged frames from multiple VLANs and can send frames either tagged or untagged, depending on the VLAN settings. Unlike trunk interfaces, hybrid interfaces are capable of handling untagged frames from multiple VLANs, offering more flexible VLAN traffic management.

261. The Backup ports in RSTP may alternatively fail the root port.

- A. True
- B. False

Answer: B

Justification

Source: Huawei slide 09, page number 47

- RSTP introduces new port roles. When the root port fails, the switch can enable the alternate port to obtain an alternate path from the designated bridge to the root bridge. RSTP defines three states for a port based on whether the port forwards user

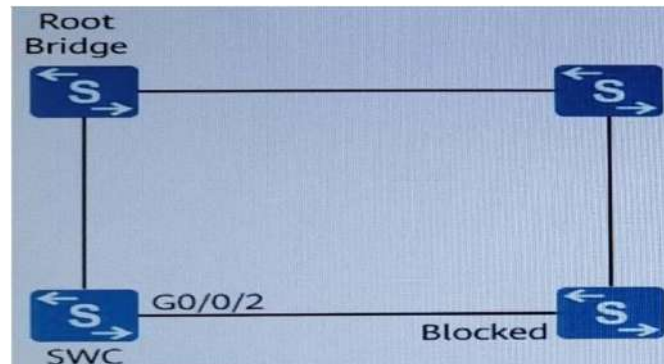
Description: If the root port fails, the alternate will be the alternate port, not the backup port.

Huawei slide 09, page number 49 notes:

From the perspective of user traffic:

- An alternate port acts as a backup of the root port and provides an alternate path from the designated bridge to the root bridge.
- A backup port backs up a designated port and provides a backup path from the root bridge to the related network segment.

262. As shown in the figure, all four switches run STP, and various parameters adopt default values. If the G0/0/2 port of switch C is blocked and the configuration BPDU cannot be sent through this port, how long will it take for the blocked port in the network to enter the forwarding state?

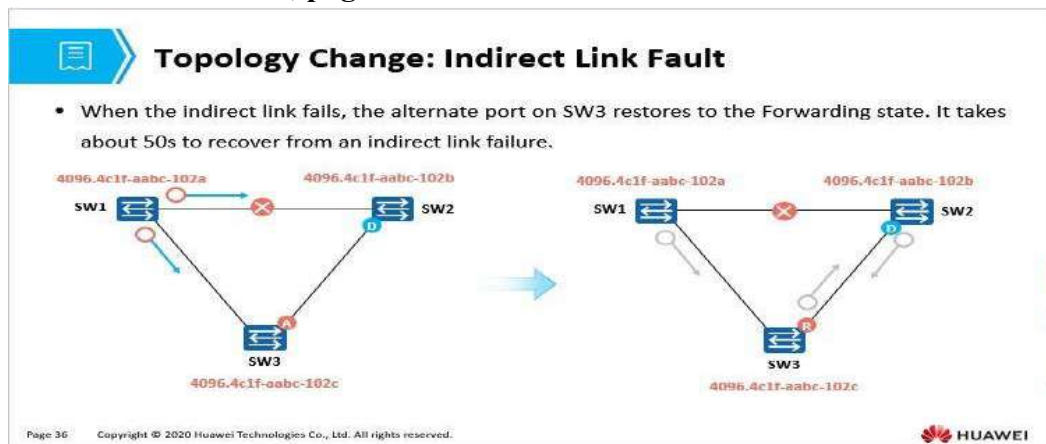


- A. about 3 seconds
- B. about 50 seconds
- C. about 30 seconds
- D. about 15 seconds

Answer: B

Justification

Source: Huawei Slide 09, page number 36



Notes:

Convergence time: It takes about 50s to recover from an indirect link failure, which is equal to the value of the max age timer plus twice the value of the Forward Delay timer.

263. If the largest number in an ACL rule is 12, by default, the user does not specify a number when configuring a new rule, then what is the number assigned by the system for the new rule?

- A. 13
- B. 15
- C. 14
- D. 16

Answer: B

Justification

Source: Huawei slide 12, page number 9

ACL Composition

ACL Classification

ACL Matching Rules



Rule ID

acl number 2000

	Rule ID			
rule	5	deny	source	10.1.1.1 0
rule	10	deny	source	10.1.1.2 0
rule	15	permit	source	10.1.1.0 0.0.0.255

Step = 5

How do I add a rule?

rule 11 deny source 10.1.1.3 0

acl number 2000

rule	5	deny	source	10.1.1.1 0
rule	10	deny	source	10.1.1.2 0
rule	11	deny	source	10.1.1.3 0
rule	15	permit	source	10.1.1.0 0.0.0.255

Rule ID and Step

- Rule ID**
Each rule in an ACL has an ID.
- Step**
A step is an increment between neighboring rule IDs automatically allocated by the system. The default step is 5. Setting a step facilitates rule insertion between existing rules of an ACL.
- Rule ID allocation**
If a rule is added to an empty ACL but no ID is manually specified for the rule, the system allocates a step value (5 for example) as the ID of the rule. If an ACL contains rules with manually specified IDs and a rule with no manually specified ID is added, the system allocates to this rule an ID that is greater than the largest rule ID in the ACL and is the smallest integer multiple of the step value.

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NOTES:

Rule ID and step:

Rule ID: Each ACL rule has an ID, which identifies the rule. Rule IDs can be manually defined or automatically allocated by the system.

Step: When the system automatically allocates IDs to ACL rules, the increment between neighbouring rule IDs is called a step. The default step is 5. Therefore, rule IDs are 5, 10, 15, and so on.

264. Complete the ACL configuration shown in the figure on the router RTA, which of the following description is correct?

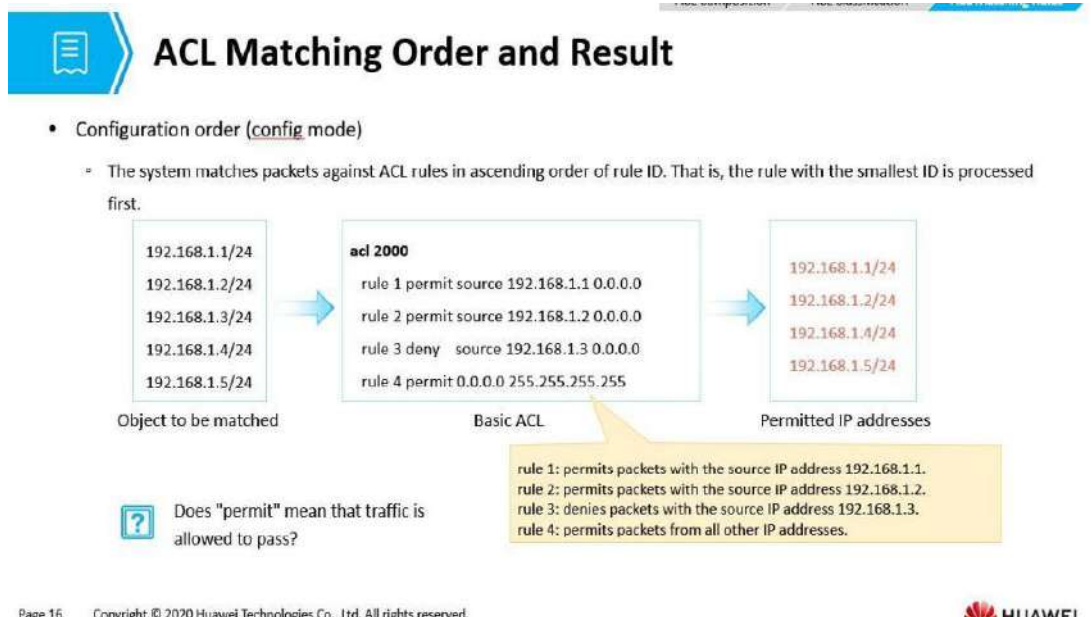
```
[RTA]acl 2001
[RTA-acl-basic-2001]rule 20 permit source 20.1.1.0 0.0.0.255
[RTA-acl-basic-2001]rule 10 deny source 20.1.1.0 0.0.0.255
```

- A. The VRP system will not adjust the sequence number, but will first match the rule permit source 20.1.1.0 0.0.0.255
- B. Configuration error, the sequence number of the rules must be configured from small to large
- C. The VRP system will adjust the sequence number of the first rule to 5 according to the configuration sequence
- D. The VRP system will first match the rule by numbering in sequence deny source 20.1.1.0 0.0.0.255

Answer: D

Justification

Source: Huawei slide 12, page number 16



265. Using the ACL shown in the figure on the router RTA to match routing entries, which of the following entries will be matched? (Multiple choice)

```
[RTA]acl 2002
[RTA-acl-basic-2002]rule deny source 172.16.1.1 0.0.0.0
[RTA-acl-basic-2002]rule deny source 172.16.0.0 0.255.0.0
```

- A. 172.16.1.0/24
B. 192.17.0.0/24
C. 172.18.0.0/16
D. 172.16.1.1/32

Answer: CD

Justification

Source: Huawei slide 12, page number 10

ACL Composition
ACL Classification
ACL Matching Rules



Wildcard (1)

acl number 2000

rule	action	type	source	destination	wildcard
rule 5	deny	source	10.1.1.1	0	
rule 10	deny	source	10.1.1.2	0	
rule 15	permit	source	10.1.1.0	0.0.0.255	

Wildcard

Wildcard

- A wildcard is a 32-bit number that indicates which bits in an IP address need to be strictly matched and which bits do not need to be matched.
- A wildcard is usually expressed in dotted decimal notation, as a network mask is expressed. However, their meanings are different.

- Matching rule
 - 0: matching; 1: random allocation

?

How do I match the network segment address corresponding to 192.168.1.1/24?

192.168.1.1	1 1 0 0 0 0 0 0	1 0 1 0 1 0 0 0	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	192.168.1.0/24 network segment
0.0.0.255	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	1 1 1 1 1 1 1 1	

Strict matching

Random allocation

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Notes:

When matching an IP address in a network rule, a 32-bit mask called a wildcard is used to determine which bits must match exactly and which can vary. This wildcard, expressed in dotted decimal notation, uses the value 0 to indicate that a corresponding bit must match and 1 to indicate that it does not matter. For example, Rule 5 denies packets from the source IP address 10.1.1.1, using a wildcard of all 0s to ensure an exact match at every bit level. In contrast, Rule 15 permits packets from the 10.1.1.0/24 network segment by using a wildcard of 0.0.0.255 (with the last eight bits set to 1), allowing any value in the last byte. To match the network address 192.168.1.1/24, the network bits must be fixed while the host bits can vary, resulting in a wildcard of 0.0.0.255.

266. The configuration information of a router is shown in the figure. Which of the following statements are correct? (Multiple choice)

```
#
aaa
 authentication-scheme default
 authentication-scheme huawei
   authentication-mode radius
 authorization-scheme default
 authorization-scheme huawei
 accounting-scheme default
 domain default
 domain default_admin
 domain huawei
   authentication-scheme huawei
   authorization-scheme huawei
 local-user huawei password cipher 123456
 local-user huawei@huawei password cipher 654321
#
```

- A. The authentication method adopted by the domain with the domain name huawei is local ID
- B. The authorization method for the domain with the domain name huawei is local authorization
- C. Use the username huawei for authentication; the password needs to be 654321
- D. The domain with the domain name huawei does not use the billing plan

Answer: BCD

Justification

Source:

For option B: <https://support.huawei.com/enterprise/en/doc/EDOC1100156624/9713e4a4/configuring-local-authentication-and-authorization>

Procedure	Command	Description
Apply an authentication scheme to the domain.	authentication-scheme authentication-scheme-name	By default, the authentication scheme named radius is applied to the default domain, the authentication scheme named default is applied to the default_admin domain, and the authentication scheme named default is applied to other domains.
Apply an authorization scheme to the domain.	authorization-scheme authorization-scheme-name	By default, no authorization scheme is applied to a domain.

For option C: Huawei, Slide 13, page number 15

For example: huawei@huawei password 654321.

Running this command will allow to authenticate the user by username “huawei” & password “654321”.

4. Create a user.

```
[Huawei-aaa] local-user user-name password cipher password
```

Create a local user and configure a password for the local user.

- If the username contains a delimiter "@", the character before "@" is the username and the character after "@" is the domain name.
- If the value does not contain "@", the entire character string represents the username and the domain name is the default one.

For option D: This statement is correct. There is no mention of a billing plan in the provided configuration for the domain with the domain name "huawei".

267. What is the main function of the DNS protocol?

- A. Domain name resolution
- B. File Transfer
- C. Remote access
- D. Mail transmission

Answer: A

Justification#01:

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100278532/612469cc/overview-of-dns>

Overview of DNS

Definition

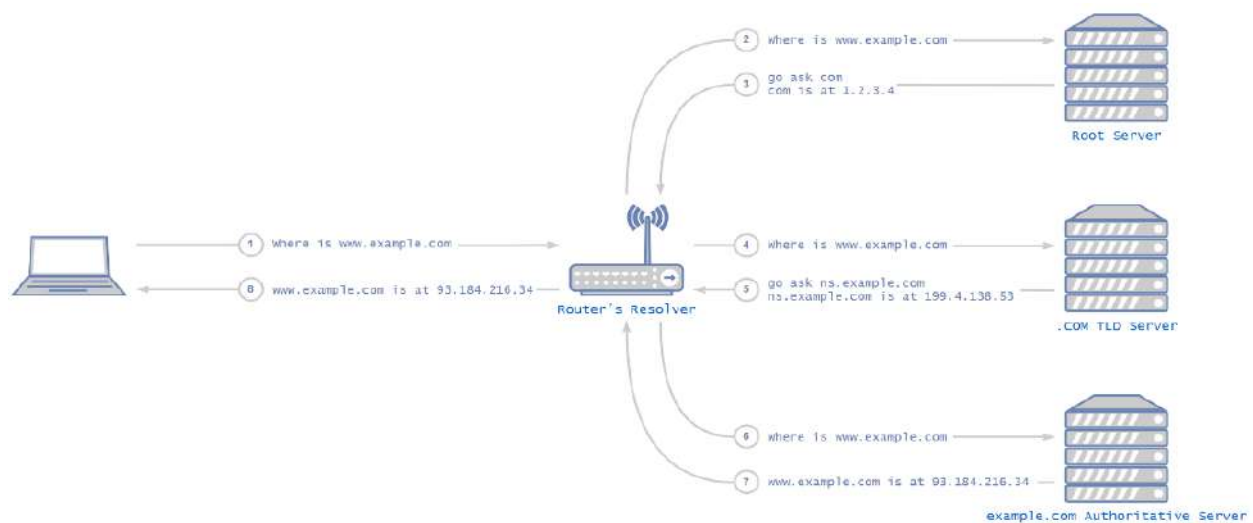
Domain Name System (DNS) is a distributed database for TCP/IP applications that provides conversion between domain names and IP addresses.

Purpose

DNS uses a hierarchical naming method to specify a meaningful name for each device on the network and uses a resolver to establish mappings between IP addresses and domain names. DNS allows users to use meaningful and easy-to-memorize domain names instead of IP addresses to identify devices.

Justification#02

Source: <https://unit42.paloaltonetworks.com/dns-vulnerabilities/>



268. What is the data link layer MAC address of each VAP on the AP?

- A. SSID
- B. BSS
- C. ESS
- D. BSSID

Answer: D

Justification

Source: Huawei slide 16, page number 22



The diagram illustrates a Basic Service Set (BSS) represented by a large light blue circle. Inside the circle, there is a central Access Point (AP) icon. Three laptop icons, each labeled 'Discover guest', are positioned around the AP. A yellow box next to the AP contains the text: 'SSID: guest' and 'BSSID: 00e0.fc45.24a0'. The label 'BSS' is written in red in the top right corner of the circle.

- Basic service set (BSS):
 - An area covered by an AP.
 - STAs in a BSS can communicate with each other.
- Basic service set identifier (BSSID):
 - An identifier of a WLAN, which is represented by the AP's MAC address.
- Service set identifier (SSID):
 - An identifier of a WLAN, which is represented by a string of characters.
 - SSIDs can replace BSSIDs to help users identify different WLANs.

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BSSID:

AP's MAC address on the data link layer.

STAs can discover and find an AP based on the BSSID.

Each BSS must have a unique BSSID. Therefore, the AP's MAC address is used to ensure the uniqueness of the BSSID

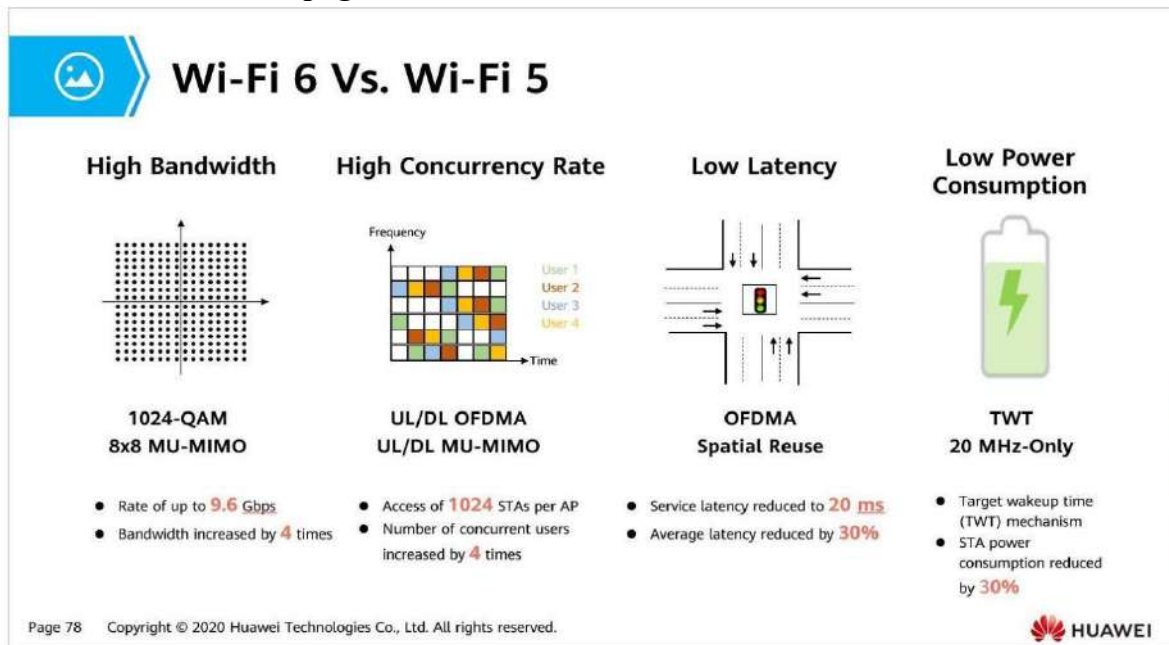
269. Which of the following does not include the advantages of Wi-Fi 6 compared to Wi-Fi 5?

- A. Higher bandwidth
- B. Lower transmission delay
- C. Higher power consumption
- D. Higher number of access terminals per AP

Answer: C

Justification

Source: Huawei slide 16, page number 78



270. PPPoE session can only use CHAP authentication

- A. True
- B. False

Answer: B

Justification

Source: Huawei slide 17, page 15

The course also covers the implementation of the Point-to-Point Protocol (PPP), focusing on its parameter negotiation during link establishment, including authentication and network layer negotiations. It provides a detailed analysis of the two commonly used PPP authentication protocols: PAP (Password Authentication Protocol) and CHAP (Challenge Handshake Authentication Protocol), explaining their operational procedures and differences. While both ends of a PPP link can use different authentication protocols to verify each other, the device being authenticated must support the authentication method specified by the authenticating device and must be correctly configured with valid credentials, such as a username and password.

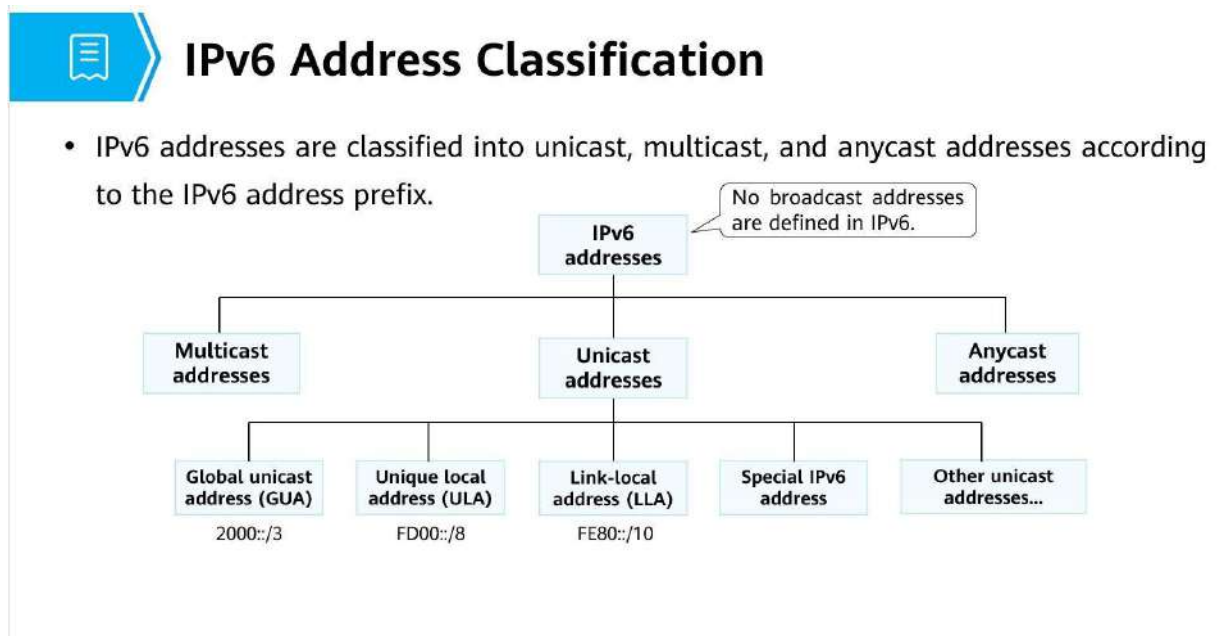
271. Which of the following types does an IPv6 address contain? (Multiple choice)

- A. Unicast address
- B. Broadcast address
- C. Multicast address
- D. Anycast address

Answer: ACD

Justification

Source: Huawei Lecture # 17, Page # 15



272. Regarding the IPv6 address 2031:0000:720C:0000:0000:09E0:839A:130B, which of the following abbreviations is correct?

- A. 2031:0:720C:0:0:9E:839A:130B
- B. 2031:0:720C::9E0:839A:130B
- C. 2031:0:720C:0:0:9E0:839A:130B
- D. 2031::720C::9E0:839A:130B

Answer: B

Justification

Source: Huawei Lecture # 19, Page # 14

IPv6 Address Abbreviation Specifications

- For convenience, IPv6 can be abbreviated according to the following rules.

Abbreviation Specifications

2001 : 0DB8 : 0000 : 0000 : 0008 : 0800 : 200C : 417A

The leading 0s in each 16-bit segment can be omitted. However, if all bits in a 16-bit segment are 0s, at least one 0 must be reserved. The trailing 0s cannot be omitted.

2001 : DB8 : 0 : 0 : 8 : 800 : 200C : 417A

If one or more consecutive 16-bit segments contain only 0s, a double colon (::) can be used to represent them, but only one :: is allowed in an entire IPv6 address.

2001 : DB8 : : 8 : 800 : 200C : 417A

If an abbreviated IPv6 address contains two double colons (::), the IPv6 address cannot be restored to the original one.

Abbreviation Examples

Before 0000:0000:0000:0000:0000:0000:0000:0001

After ::1

Before 2001:0DB8:0000:0000:FB00:1400:5000:45FF

After 2001:DB8:FB00:1400:5000:45FF

Before 2001:0DB8:0000:0000:0000:2A2A:0000:0001

After 2001:DB8::2A2A:0:1

Before 2001:0DB8:0000:1234:FB00:0000:5000:45FF

After 2001:DB8::1234:FB00:0:5000:45FF

or 2001:DB8:0:1234:FB00::5000:45FF

273. Which of the following protocols does the RA message used by IPv6 stateless auto configuration belong to

- A. IGMPv6
- B. ICMPv6
- C. UDPv6
- D. TCPv6

Answer: B

Justification #1

ICMPv6 Router Advertisement (RA) Message: Routers send out Router Advertisement message periodically, or in response to a Router Solicitation message. The ICMPv6 Router Advertisement messages include information about prefixes and some flag bits.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100174721/ff83a724/icmpv6-packet-format>

Justification #2

Source: Huawei Lecture # 19, Page # 28

Notes:

IPv6 supports stateful and stateless address autoconfiguration. The managed address configuration flag (M flag) and other stateful configuration flag (O flag) in ICMPv6 RA messages are used to control the mode in which terminals automatically obtain addresses.

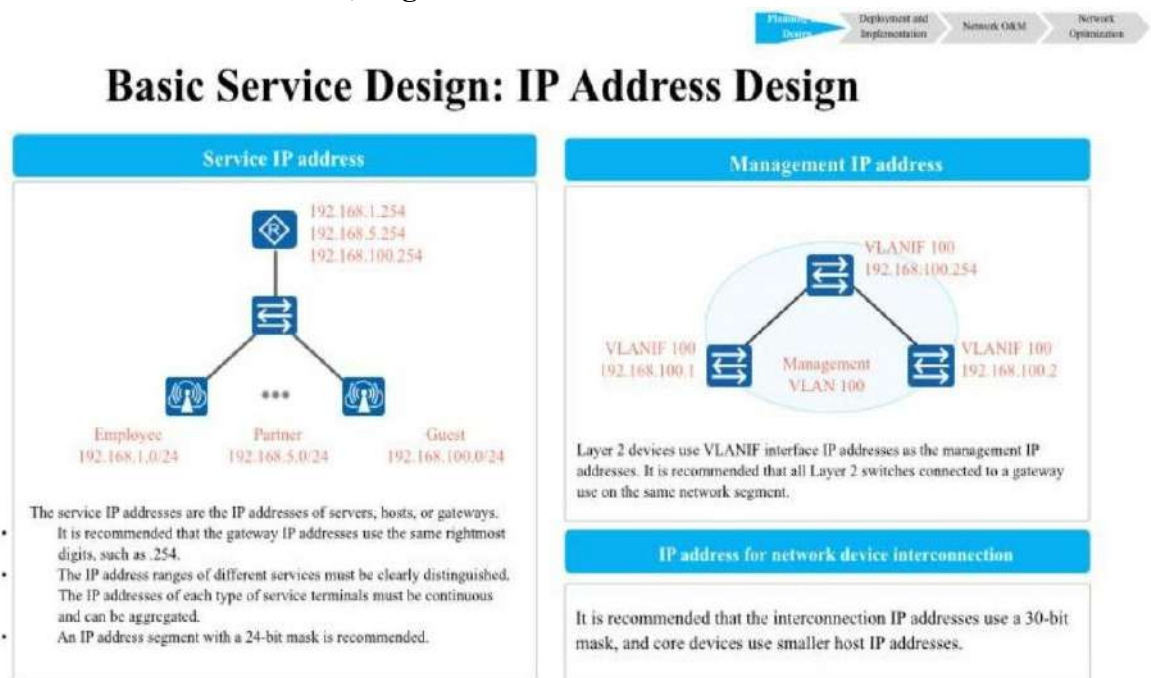
274. When planning the campus network, which of the following mask lengths is recommended for device interconnection IP addresses?

- A. 16
- B. 32
- C. 24
- D. 30

Answer: D

Justification

Source: Huawei Lecture # 22, Page # 18



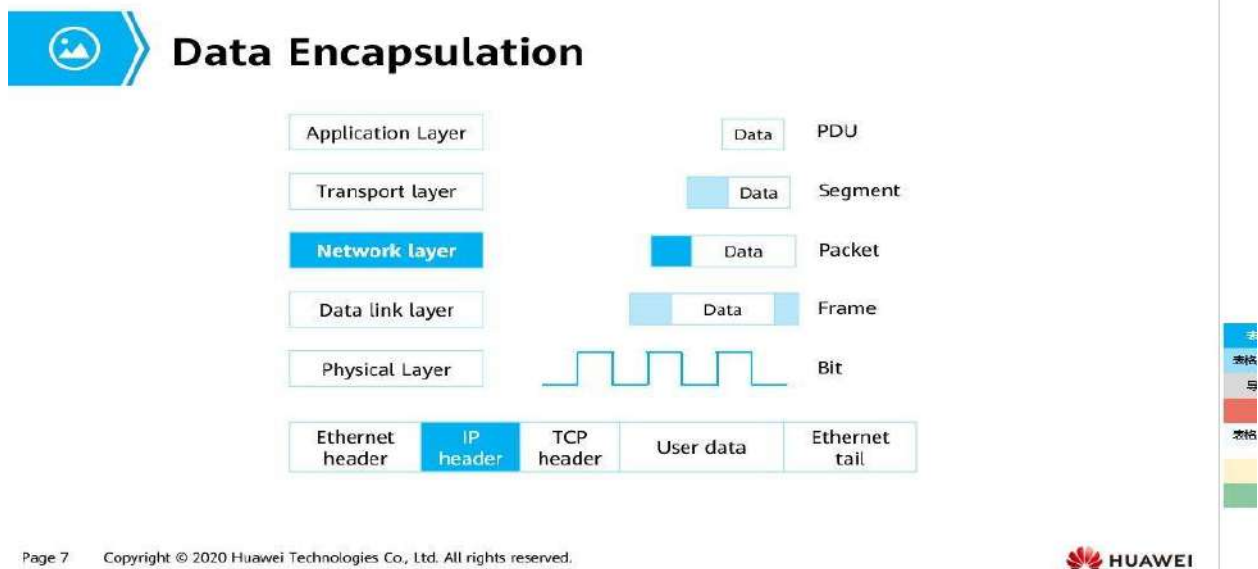
275. Which of the following are related to the purpose of encapsulation and decapsulation? (Multiple choice)

- A. Encapsulation and decapsulation are important steps to realize different protocol functions.
- B. Interworking between different networks.
- C. Shorten the length of the message.
- D. Encapsulation and decapsulation can complete the network fault location

Answer: AB

Justification

Source: Huawei Lecture # 4, Page # 7



Notes:

- For example, after a TCP header is added to the upper-layer data in a PDU at the transport layer, the PDU is called a segment. The data segment is transmitted to the network layer. After an IP header is added to the PDU at the network layer, the PDU is called a packet. The data packet is transmitted to the data link layer. After the data link layer header and trailer are encapsulated into the PDU, the PDU becomes a frame. Ultimately, the frame is converted into bits and transmitted through network media.
- The process in which data is delivered following the protocol suite from top to bottom and is added with headers and tails is called encapsulation.

**276. Which of the following statements about the transport layer protocol is correct?
(Multiple choice)**

- A. UDP is suitable for the transmission of delay-sensitive traffic, and can be reorganized according to the sequence number field in the message header
- B. The range of well-known port numbers is 0–1023
- C. UDP uses the STNOACK flag to request and confirm the establishment of a connection
- D. TCP establishes a connection is a three-way handshake process; TCP terminates the connection will have to go through a four-way handshake

Answer: BD

Justification#1

B: Well-known ports: also called system ports. The port number ranges from 0 to 1023.

D: TCP connection (three-way handshake) and disconnection (four-way wavehand)

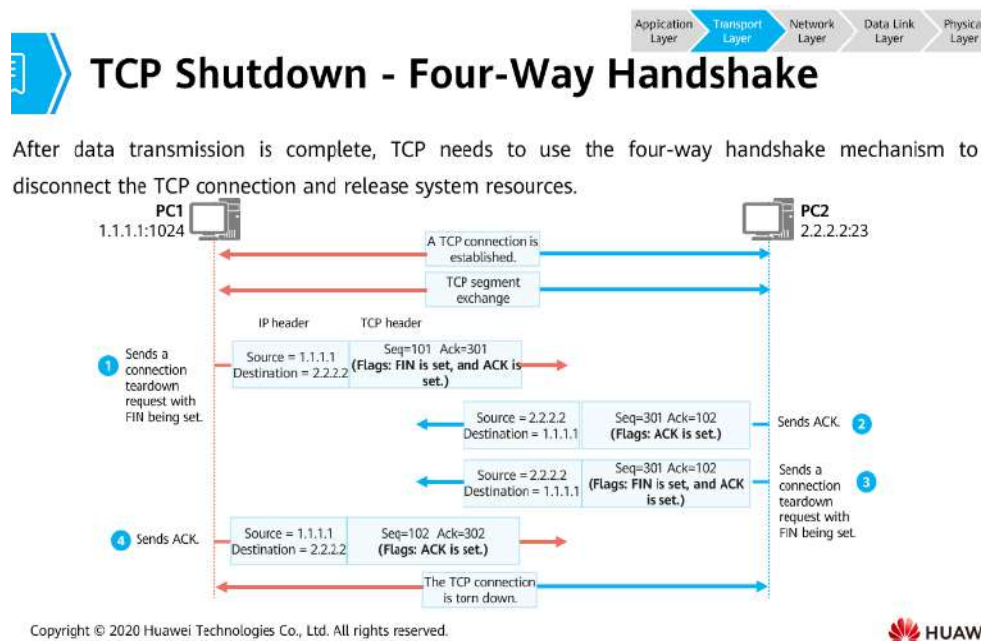
Source:

B: <https://support.huawei.com/enterprise/en/doc/EDOC1100297670>

D: <https://forum.huawei.com/enterprise/intl/en/thread/transmission-control-protocol-tcp/667261370510491648?blogId=667261370510491648>

Justification#2

Source: Huawei Lecture # 2, Slide # 24



277. Which of the following descriptions about wireless devices is correct?

- A. The wireless device has no wired interface
- B. FIT AP-like independent user access, Authentication, Forwarding services and other functions
- C. Wireless Controller AC-like at the aggregation layer of the entire network, providing high-speed, Safe, and Reliable VLAN business
- D. FAT AP generally works with a wireless controller

Answer: C

Justification

Source: Huawei Lecture # 16, Page # 22

- An AC is usually deployed at the aggregation layer of a network to provide high-speed, secure, and reliable WLAN services.

278. What is the full name of the general routing platform VRP?

- A. Versatile Redundancy Platform
- B. Versatile Routing Protocol
- C. Versatile Routing Platform
- D. Virtual Routing Platform

Answer: C

Justification#1

VRP (Versatile Routing Platform) is the Operating System that many Huawei devices operate.

Source:

https://forum.huawei.com/enterprise/intl/en/thread/what-is-vrp-versatile-routing-platform/667281128492515329?utm_source=chatgpt.com&blogId=667281128492515329

Justification#2

Source: Huawei Lecture # 2, Slide # 2

 **Foreword**

- The Versatile Routing Platform (VRP) is a universal operating system (OS) platform for Huawei datacom products. It is based on IP and adopts a component-based architecture. It provides rich features and functions, including application-based tailorable and extensible functions, greatly improving the running efficiency of the devices that use this OS. To efficiently manage such devices, you must be familiar with VRP and VRP-based configuration.
- This course describes the basic concepts, common commands, and command line interface (CLI) of VRP.

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279. VRP supports OSPF multi-process. If the process number is not specified when OSPF is enabled, the process number used by default is ().

- A. 10
- B. 100
- C. 0
- D. 1

Answer: D

Justification

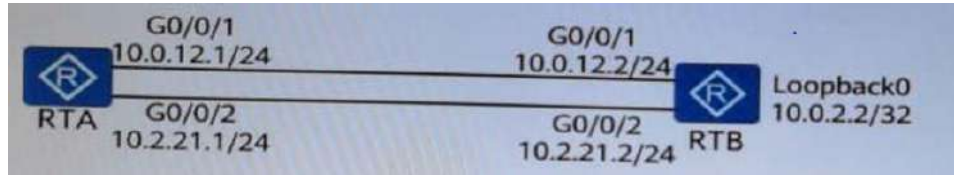
An OSPF process is created, and the OSPF view is displayed.

The parameter process-id specifies the ID of an OSPF process. The default value is 1.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100034072/270867b9/creating-an-ospf-process>

280. The network is shown in the figure below. The administrator hopes that RTA uses static routing, and preferentially sends data packets to RTB's Loopback0 interface through G0/0/1 port, while G0/0/2 serves as a backup. So, how to configure on RTA to achieve this requirement?



- A. `ip route-static 10.0.2.2 255.255.255.255 10.0.12.2`
`ip route-static 10.0.2.2 255.255.255.255 10.0.21.2`
`preference 40`
- B. `ip route-static 10.0.2.2 255.255.255.255 10.0.12.2`
`preference 70`
`ip route-static 10.0.2.2 255.255.255.255 10.0.21.2`
- C. `ip route-static 10.0.2.2 255.255.255.255 10.0.12.2`
`ip route-static 10.0.2.2 255.255.255.255 10.0.21.2`
- D. `ip route-static 10.0.2.2 255.255.255.255 10.0.12.2`
`ip route-static 10.0.2.2 255.255.255.255 10.0.21.2`
`preference 70`

Answer: D

Justification

Source: Huawei Lecture # 5, Slide # 19

Protocol	Route Type	Default Preference
Direct	Direct route	0
Static	Static route	60

Explanation:

The default priority of a static route is 60. So, RTA will prefer to send the data packet through G0/0/1 first. For the second option, it will then prefer G0/0/2 since the preference is set to 70. (Less has more preference).

281. If the broadcast address of a network is 172.16.1.255, then what may its network address be?

- A. 172.16.1.128
- B. 172.16.2.0
- C. 172.16.1.1
- D. 172.86.1.253

Answer: A

Justification

- **Options B & D** cannot be chosen as they are completely different from the IP given, as they come after the given IP.
- **Option C** can't be the correct answer because if the subnet is 24, then the network IP would be 172.16.1.0, and if the subnet is 25, then the network IP would be 172.16.1.128, which is **option A**.

282. The administrator wants to configure a hundred IP addresses on the G0/0/0 interface of the router. Which of the following addresses is correct?

- A. 192.168.10.112/30
- B. 237.6.1.2/24
- C. 127.3.1.4/28
- D. 145.4.2.55/26

Answer: B

Justification

Explanation:

A. 192.168.10.112/30

- /30 subnet mask gives 2 usable IP addresses.-Not suitable since it only provides 2 usable addresses.

B. 237.6.1.2/24

- /24 subnet mask gives $256 - 2 = 254$ usable IP addresses-Suitable as it provides more than 100 usable IP addresses.

C. 127.3.1.4/28

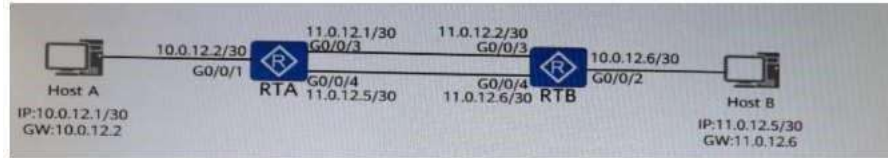
- /28 subnet mask gives $16 - 2 = 14$ usable IP addresses-Not suitable since it provides fewer than 100 usable addresses.

D. 145.4.2.55/26

- /26 subnet mask gives $64 - 2 = 62$ usable IP addresses-Not suitable since it provides fewer than 100 usable addresses.

Therefore, the correct answer is B. 237.6.1.2/24, which provides a sufficient number of usable IP addresses (254) to accommodate a hundred IP addresses on the G0/0/0 interface of the router.

283. The network shown below, it is desirable host traffic network administrator during planning A destined for host B data packets through the router between the G0/0/3 interfaces, the host B to the host PC A data packet through the router Between the G0/0/4 interface, which of the following commands can achieve this requirement? (Multiple choice)



- A. RTA : ip route-static 10.0.12.5 255.255.255.252 11.0.12.2
RTB : ip route-static 10.0.12.1 255.255.255.252 11.0.12.5
- B. RTA : ip route-static 10.0.12.5 255.255.255.252 11.0.12.6
RTB : ip route-static 10.0.12.1 255.255.255.252 11.0.12.1
- C. RTA : ip route-static 0.0.0.0 0.0.0.0 11.0.12.2
RTB : ip route-static 0.0.0.0 0.0.0.0 11.0.12.5
- D. RTA : ip route-static 0.0.0.0 0.0.0.0 11.0.12.6
RTB : ip route-static 0.0.0.0 0.0.0.0 11.0.12.1

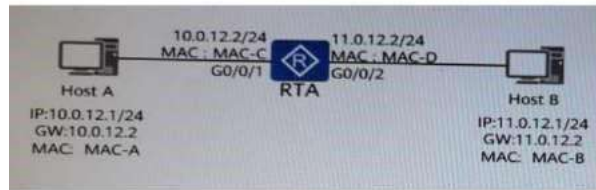
Answer: AC

Justification

Explanation:

- In **option A**, the data packet received by router A with destination address 10.0.12.5 should be sent to the G0/0/3 interface (11.0.12.2) of router B (there might be a mistake in Host B's IP address as it should be of network 10.0.12.4/30, otherwise communication between router B and Host B cannot be possible).
- In **option C**, the data packet received by router A with any destination address will be sent to the G0/0/3 interface (11.0.12.2) of router B. And the data packet received by router B with any destination address will be sent to the G0/0/4 interface (11.0.12.5) of router A.

284. In the network as shown in the figure below, the host has an ARP cache. Which of the following statements is correct? (Multiple Choice)



- A. The following entries may exist in the ARP cache of host A. 10.0.12.2 MAC-C
- B. The router needs to be configured with static routing, otherwise host A and host B cannot communicate in both directions
- C. The following entries exist in the ARP cache of host A 11.0.12.1 MAC-B
- D. Host A and Host B can communicate in both directions

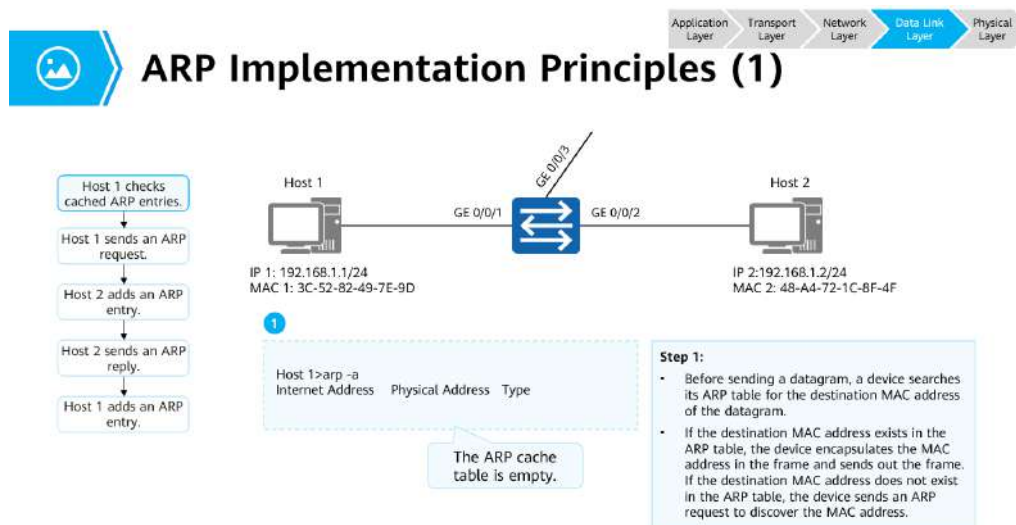
Answer: AD

Justification#1

- In **option A**, Host A needs to communicate with its **default gateway** (RTA at 10.0.12.2) to reach outside networks. So it **ARP-resolves 10.0.12.2 to MAC-C** and saves it in its ARP table.
- For **option D**, The router (RTA) connects both networks (10.0.12.0/24 and 11.0.12.0/24). Since their default gateways are set properly and interfaces exist, **they can communicate through RTA**

Justification#2

Source: Huawei Lecture # 2, Slide # 30



**285. Compared with STP, which of the following port roles are newly defined in RSTP?
(Multiple choice)**

- A. Designated port
- B. Root port
- C. Alternate port
- D. Backup port

Answer: CD

Justification#1

RSTP defines four port roles: root port, designated port, alternate port, and backup port.

Source :

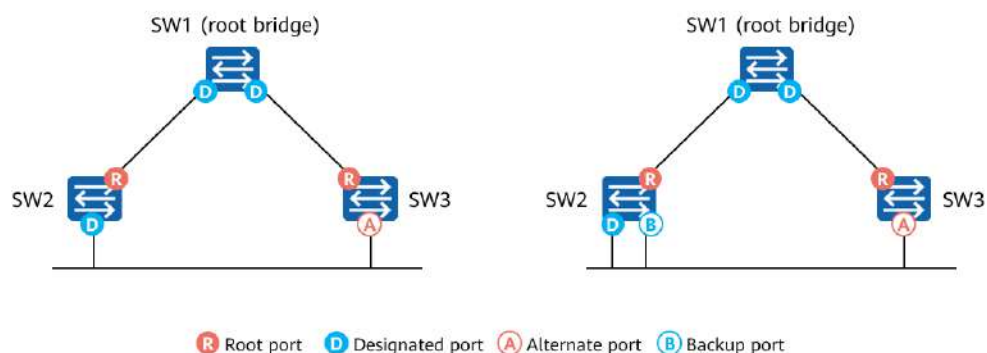
https://info.support.huawei.com/hedex/api/pages/EDOC1100363264/AEN0403J/05/resources/vrp/feature_stp_rstp_mstp_0010.html

Justification#2

Source: Huawei Lecture # 9 Slide # 49

Port Roles in RSTP

- RSTP adds port roles to help understand RSTP and simplify RSTP deployment.



RSTP defines four port roles: root port, designated port, alternate port, and backup port.

286. The Layer 2 switch is a data link layer device, which can identify the MAC address information in the data frame, forward data according to the MAC address, and record these MAC addresses and corresponding port information in its own MAC address table.

- A. True
- B. False

Answer: A

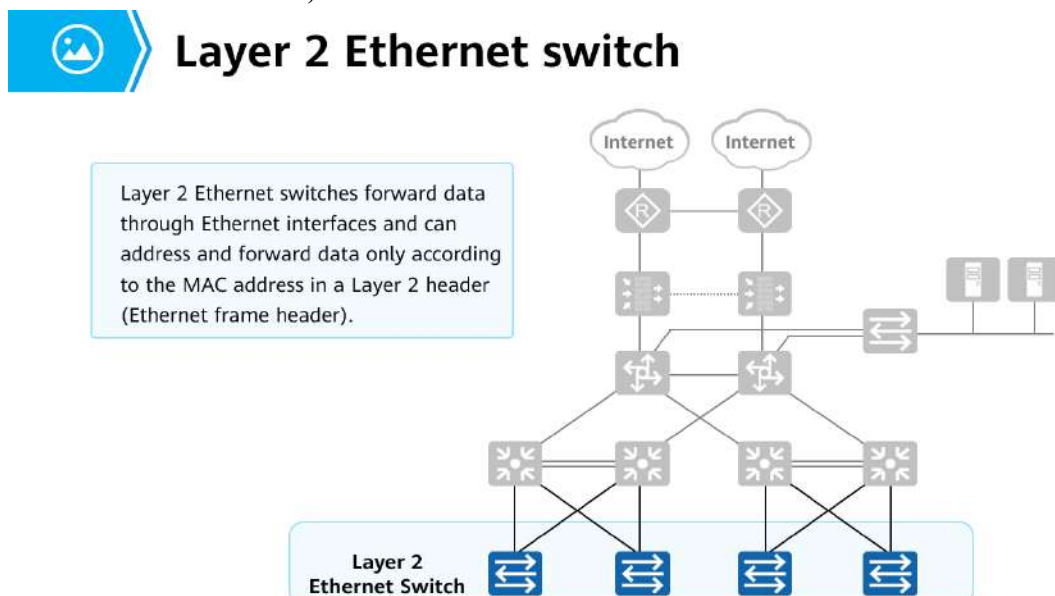
Justification#1

The device dynamically builds the address table by using the MAC source address of the frames received. When the device receives a frame for a MAC destination address not listed in its address table, it floods the frame to all LAN ports of the same VLAN except the port that received the frame. When the destination station replies, the device adds its relevant MAC source address and port ID to the address table.

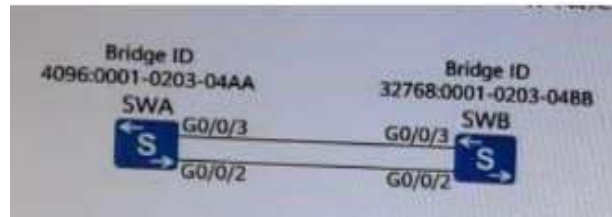
Source: https://www.cisco.com/c/en/us/td/docs/switches/datacenter/sw/4_2/nx-os/layer2/configuration/guide/Cisco_Nexus_7000_Series_NX-OS_Layer_2_Switching_Configuration_Guide_Release_4-2_chapter3.html

Justification#2

Source: Huawei Lecture # 7, Slide # 22



287. The two switches shown in the figure below have both enabled the STP protocol. Which port will eventually be in the Blocking state?



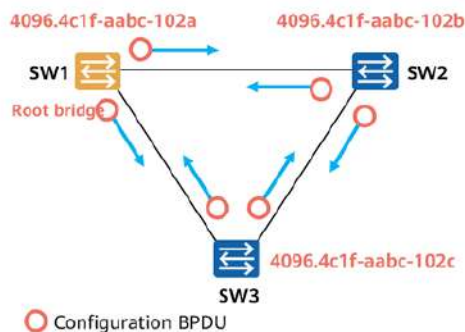
- A. G0/0/2 port of SWB
- B. G0/0/3 port of SWB
- C. G0/0/2 port of SWA
- D. G0/0/3 port of SWA

Answer: B

Justification

Source: Huawei Lecture # 9 Slide # 24

STP Calculation (1)



Select a Root Bridge on the Switching Network

- After STP starts to work on a switching network, each switch sends configuration BPDUs to the network. The configuration BPDU contains the BID of a switch.
- The switch with the smallest bridge ID becomes the root bridge.
- Only one root bridge exists on a contiguous STP switching network.
- The role of the root bridge can be preempted.
- To ensure the stability of the switching network, you are advised to plan the STP network in advance and set the bridge priority of the switch that is planned as the root bridge to the minimum value 0.

Explanation:

1. Identify the Root Bridge: SWA is the Root Bridge because it has the lower Bridge ID (4096 vs. 32768).
 2. Determine the Root Port on SWB: SWB's Root Port will be G0/0/2, as it connects to the Root Bridge (SWA) with the lowest cost.
 3. Designated Ports: Both ports on SWA (G0/0/2 and G0/0/3) will be Designated Ports since SWA is the Root Bridge.
 4. Blocking Port: Any port that is neither a Root Port nor a Designated Port will be in the Blocking state. Therefore, SWB's G0/0/3 will be in the Blocking state.
- So, the port that will eventually be in the Blocking state is: G0/0/3 port of SWB.

288. On Huawei switches, you can use the command `vlan batch` to create multiple VLANs in batches, simplifying the configuration process.

- A. True
- B. False

Answer: A

Justification#1

```
[HUAWEI] vlan batch 11 to 20
```

Create 10 contiguous VLANs: VLAN 11 to VLAN 20.

```
<HUAWEI> system-view
```

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000178168/a197c0b1/how-do-i-create-vlans-in-a-batch>

Justification#2

Source: Huawei Lecture # 8 Slide # 31



Basic VLAN Configuration Commands

Interface-based VLAN Assignment

MAC Address-based VLAN Assignment

1. Create one or more VLANs.

```
[Huawei] vlan vlan-id
```

This command creates a VLAN and displays the VLAN view. If the VLAN to be created already exists, this command directly displays the VLAN view.

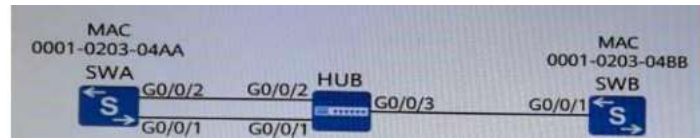
 - The value of *vlan-id* is an integer ranging from 1 to 4094.

```
[Huawei] vlan batch { vlan-id1 [ to vlan-id2 ] }
```

This command creates VLANs in a batch. In this command:

 - **batch**: creates VLANs in a batch.
 - *vlan-id1*: specifies a start VLAN ID.
 - *vlan-id2*: specifies an end VLAN ID.

289. As shown in the figure, the two switches use the default parameters to run STP, the configuration command STP root primary is used on switch A, and the configuration command STP priority 0 is used on switch B. Then which port will be blocked?



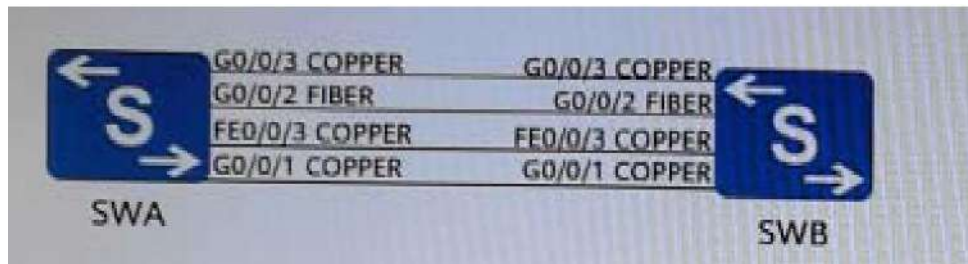
- A. G0/0/1 of switch B
- B. G0/0/1 of switch A
- C. HUB's G0/0/3
- D. G0/0/2 of switch A

Answer: D

Justification

Explanation: Because in switch A, two links are connected to the hub, so one is the root port and the other is blocked, which has the highest port ID.

290. In the network shown in the figure below, two switches are connected by four links. CUPPER refers to the electrical interface, and FIBER refers to the optical interface. Which of the following two interfaces can achieve link aggregation?



- A. G0/0/2 and G0/0/1
- B. G0/0/3 and FE0/0/3
- C. G0/0/3 and G0/0/1
- D. G0/0/2 and FE0/0/3

Answer: C

Justification

Explanation: Link aggregation groups (LAGs) combine multiple physical ports into a single logical link, increasing bandwidth and providing redundancy. For LAGs to function, the ports must be of the same type (copper or fiber) and speed. In the image, both G0/0/3 and G0/0/1 are copper Gigabit Ethernet interfaces, making them compatible for LAGs.

- A:G0/0/2 and G0/0/1: While both are copper interfaces, G0/0/2 is Fast Ethernet while G0/0/1 is Gigabit Ethernet.
- B:G0/0/3 and FE0/0/3: Though both are Gigabit Ethernet interfaces, G0/0/3 is copper while FE0/0/3 is fiber.

291. Trunk ports can allow multiple VLANs to pass, including VLAN4096.

- A. True
- B. False

Answer: B

Justification

The **port trunk allow-pass vlan** command adds a trunk interface to the specified VLANs.

The **undo port trunk allow-pass vlan** command deletes a trunk interface from the specified VLANs.

By default, a trunk interface is in VLAN 1.

Format

port trunk allow-pass vlan { { *vlan-id1* [**to** *vlan-id2*] } <1-10> | **all** }

undo port trunk allow-pass vlan { { *vlan-id1* [**to** *vlan-id2*] } <1-10> | **all** }

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064352/a0f4f7e4/port-trunk-allow-pass-vlan>

292. The following statements are about the election of active ports in the link aggregation LACP mode. Which is correct?

- A. Compare the priority of the interface first, and it is impossible to determine the better one to continue comparing the interface number, the smaller the better.
- B. Only compare interface priority
- C. Compare device priority
- D. Only compare interface numbers

Answer: A

Justification#1

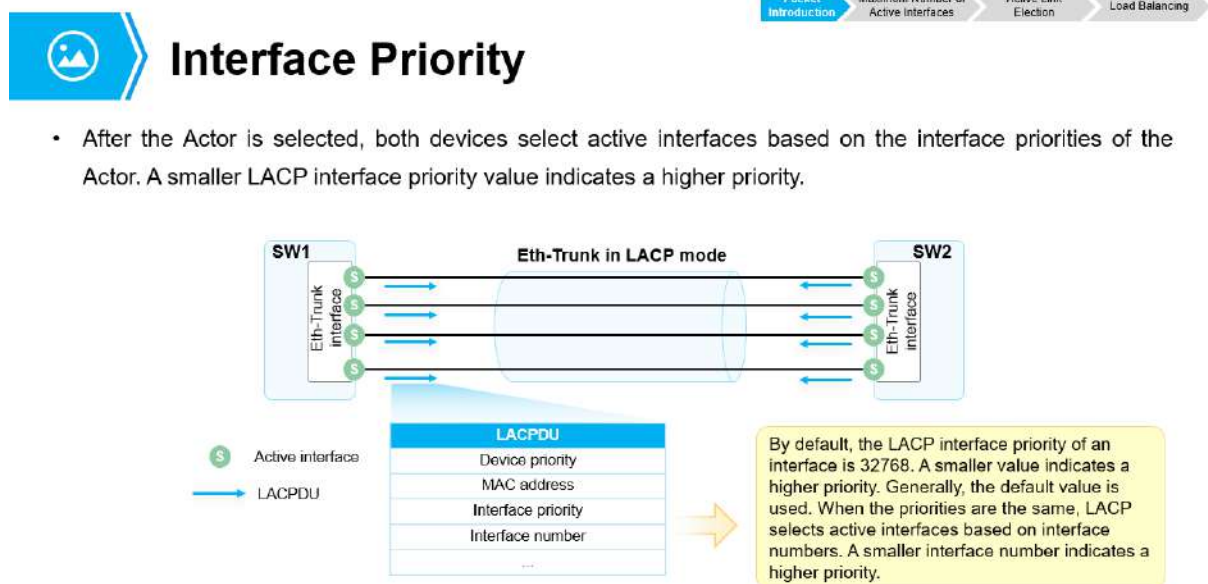
LACP uses the port priority to decide which ports should be put in standby mode when there is a hardware limitation that prevents all compatible ports from aggregating. LACP also uses the port priority with the port number to form the port identifier.

Source:

https://www.cisco.com/c/en/us/td/docs/switches/metro/me1200/config/guide/b_nid_config_book/b_nid_config_book_chapter_011000.html

Justification#2

Source: Huawei Lecture # 11, Slide # 20



293. Which of the following parameters is not included in the configuration BPDU of the STP protocol?

- A. Port ID
- B. VLAN ID
- C. Bridge ID
- D. Root ID

Answer: B

Justification#1

Field	Bytes	Description
Root Identifier	8	Indicates the BID of the current root bridge.
Bridge Identifier	8	Indicates the BID of the bridge that sends the BPDU.
Port Identifier	2	Indicates the ID of the port that sends the BPDU.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100198844/e99e1364/stp-bpdu-format>

Justification#2

Source: Huawei Lecture # 9, Slide # 21



Format of Configuration BPDUs

BID

Root Bridge

Cost

RPC

PID

BPDU

PID	PVI	BPDU Type	Flags	Root ID	RPC	Bridge ID	Port ID	Message Age	Max Age	Hello Time	Forward Delay
-----	-----	-----------	-------	---------	-----	-----------	---------	-------------	---------	------------	---------------

Byte	Field	Description
2	PID	For STP, the value of this field is always 0.
1	PVI	For STP, the value of this field is always 0.
1	BPDU Type	Type of BPDUs. The value 0x00 indicates a configuration BPDU and the value 0x80 indicates a TCN BPDU.
1	Flags	STP uses only the leftmost two bits and the rightmost two bits: Topology Change Acknowledgment (TCA) and Topology Change (TC).
8	Root ID	BID of the root bridge.
4	RPC	STP cost of the path from the current port to the root bridge.
8	Bridge ID	BID of the sender.
2	Port ID	ID of the port that sends this BPDU, which consists of the port priority and port number.
2	Message Age	Number of seconds after a BPDU is sent from the root bridge. The value increases by 1 each time the BPDU passes through a network bridge. It refers to the number of hops to the root bridge.
2	Max Age	If the bridge does not receive any BPDU for a period of time and the lifetime of the network bridge reaches the maximum, the network bridge considers that the link connected to the port is faulty. The default value is 20s.
2	Hello Time	Interval at which the root bridge sends configuration BPDUs. The default value is 2s.
2	Forward Delay	Time that is spent in Listening or Learning state. The default value is 15s.

294. Which of the following options are configured in a Layer 2 ACL? (Multiple choices)

- A. rule 10 permit 12-protocol arp
- B. rule 25 permit source 192.168.1.1 0.0.0.0
- C. rule 20 permit source-mac 0203-0405-0607
- D. rule 15 permit vlan-id 100

Answer: CD

Justification#1

A Layer 2 ACL defines rules to filter IPv4 and IPv6 packets based on Ethernet frame information, such as source MAC addresses, destination MAC addresses, VLANs, and Layer 2 protocol types.

This rule specifies a source MAC address and VLAN ID, which is a Layer 2 attribute.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178177/c18db3d9/configuring-a-layer-2-acl>

Justification#2

Source: Huawei Lecture # 12, Slide # 12

ACL Composition > **ACL Classification** > ACL Matching Rules

 **ACL Classification and Identification**

- ACL classification based on ACL rule definition methods

Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
Advanced ACL	3000 to 3999	Defines rules based on source and destination IPv4 addresses, IPv4 protocol types, ICMP types, TCP source/destination port numbers, UDP source/destination port numbers, and effective time ranges.
Layer 2 ACL	4000 to 4999	Defines rules based on information in Ethernet frame headers of packets, such as source and destination MAC addresses and Layer 2 protocol types.
User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

- ACL classification based on ACL identification methods

Category	Description
Numbered ACL	Traditional ACL identification method. A numbered ACL is identified by a number.
Named ACL	A named ACL is identified by a name.

295. A company has 50 private IP addresses. The administrator uses NAT technology to connect the company's network to the public network, but the company has only one public network address, which is not fixed. Which of the following NAT conversion methods meets the needs?

- A. easy-ip (Correct)
- B. NATP NATP requires the administrator to configure a fixed public IP address pool.
- C. Static NAT
- D. Basic NAT

Answer: B A

Justification#1

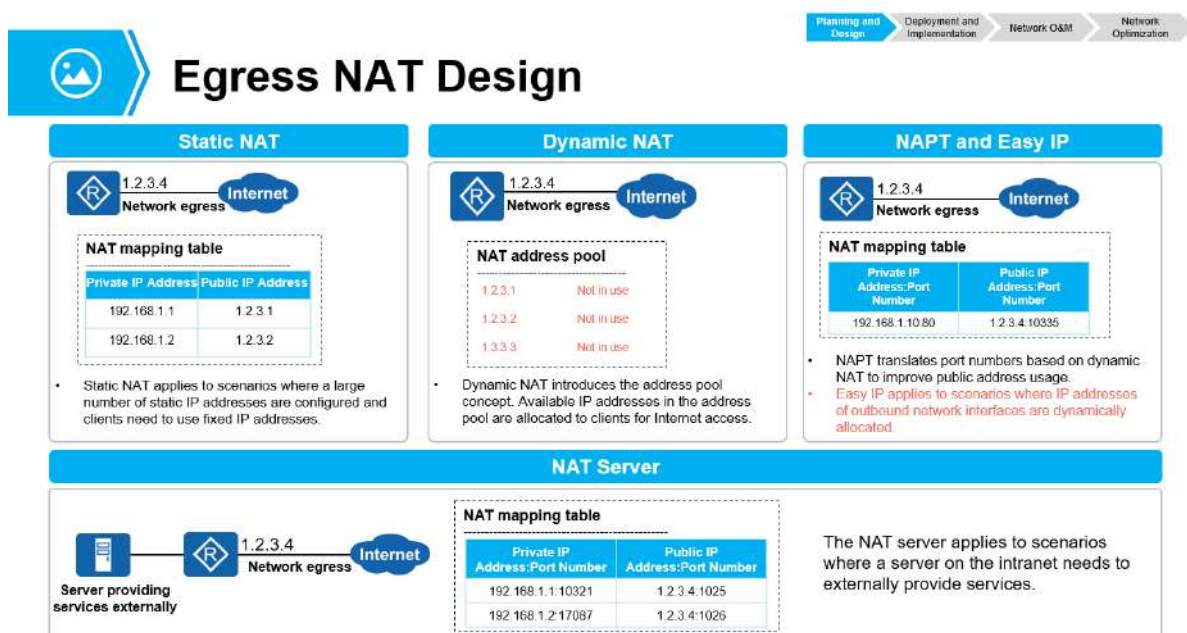
NAPT translates both source IP addresses and ports (default post-NAT port range: 2048–65535) to allow multiple private addresses to be translated into the same public address.

Source:

https://support.huawei.com/enterprise/search/#/index?keyword=napt&lang=en&searchModelType=30&searchType=SUPE_DOC&site=0&sortType=Relevance&pageSize=20&start=1&ebgSearchFlag=true

Justification#2

Source: Huawei Lecture # 22, Slide # 27



296. In the network shown in the figure, the administrator hopes that all hosts cannot access the web service (the port number is 80), and other services are normally accessed, so in the interface configuration of G0/0/1, the traffic filter outbound needs to define which ACL rule?

- A. `acl number 3000 rule 5 deny tcp destination-port eq www rule 10 permit ip`
- B. `acl number 3001 rule 5 deny udp destination-port eq www rule 10 permit ip`
- C. `acl number 3003 rule 5 permit ip rule 10 deny udp destination-port eq www`
- D. `acl number 3002 rule 5 permit ip rule 10 deny tcp destination-port eq www`

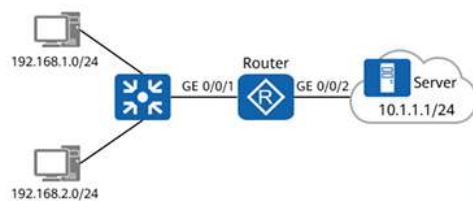
Answer: A

Justification

Source: Huawei Lecture # 12, Slide # 23



Case: Use a Basic ACL to Filter Data Traffic



• Requirements:

To prevent the user host on the network segment 192.168.1.0/24 from accessing the network where the server resides, configure a basic ACL on the router. After the configuration is complete, the ACL filters out the data packets whose source IP addresses are on the network segment 192.168.1.0/24 and permits other data packets.

1. Configure IP addresses and routes on the router.

2. Create a basic ACL on the router to prevent the network segment 192.168.1.0/24 from accessing the network where the server resides.

```
[Router] acl 2000
[Router-acl-basic-2000] rule deny source 192.168.1.0 0.0.0.255
[Router-acl-basic-2000] rule permit source any
```

3. Configure traffic filtering in the inbound direction of GE 0/0/1.

```
[Router] interface GigabitEthernet 0/0/1
[Router-GigabitEthernet0/0/1] traffic-filter inbound acl 2000
[Router-GigabitEthernet0/0/1] quit
```

Explanation:

To block all web traffic (port 80) while allowing all other services, you need an ACL that:

- 1. Denies TCP traffic to port 80.
- 2. Allows all other traffic.

Option A correctly does this:

`rule 5 deny tcp destination-port eq www`

- `deny tcp`: This specifies that the rule will block TCP traffic.
- `destination-port eq www`: This specifies that the rule applies to traffic destined for port 80. "www" is a common alias for port 80, which is used for HTTP traffic.

`rule 10 permit ip`

- `permit ip`: This specifies that the rule will allow all IP traffic. This rule allows all other traffic that does not match the previous rule (i.e., any traffic not destined for TCP port 80).

297. At present, the company has a network administrator, and the AR2200 in the company network can be remotely managed by directly entering the password through Telnet. After the two new network administrators arrived, the company hopes to assign their own user names and passwords to all administrators, as well as different levels of authority. So, how should it be done? (Multiple choices)

- A. Each administrator uses a different public IP address of the device when running the Telnet command
- B. The user authentication mode configured by Telnet must be selected AAA mode
- C. When configuring each administrator's account, different permission levels need to be configured
- D. Configure three user names and their corresponding passwords in the AAA view

Answer: BCD

Justification

B) The authentication mode must be specified on the device; otherwise, users cannot log in to the device through Telnet. The device supports non-authentication, password authentication, and AAA authentication, in which AAA authentication has the highest security.

C)

Level	Description
Administrator	An administrator has partial administrative permissions on the storage device but cannot manage users, upgrade the storage device, modify the system time, perform batch configuration, restart the device, or power off the device.

D) It is recommended to define a username and password on the access server before you start the AAA configuration

Source:

B)

<https://support.huawei.com/enterprise/en/doc/EDOC1000057410?section=j03w&topicName=configuring-authentication-for-telnet-login-users-aaa-local-authentication>

C) <https://support.huawei.com/enterprise/en/doc/EDOC1100112625/b1a6458f/user-levels-roles-and-permissions>

D) <https://www.cisco.com/c/en/us/support/docs/security-vpn/terminal-access-controller-access-control-system-tacacs-/10384-security.html>

Explanation:

- B (AAA mode) ensures that user authentication is managed centrally, enhancing security by leveraging a dedicated authentication server.
- C (Different permission levels) ensures that administrators have appropriate access based on their roles, reducing the risk of unauthorized access or unintended changes.
- D (Configuring usernames and passwords in AAA) is necessary to enforce individual authentication credentials for each administrator, ensuring accountability and facilitating easier management of access.

It is a broadcast message

298. What is the main function of the DHCP DISCOVER message?

- A. The server's confirmation response to the REQUEST message
- B. The client is used to find the DHCP server
- C. The client requests configuration confirmation, or renews the lease
- D. The DHCP server is used to respond to the DHCP DISCOVER message, which carries various configuration information

Answer: B

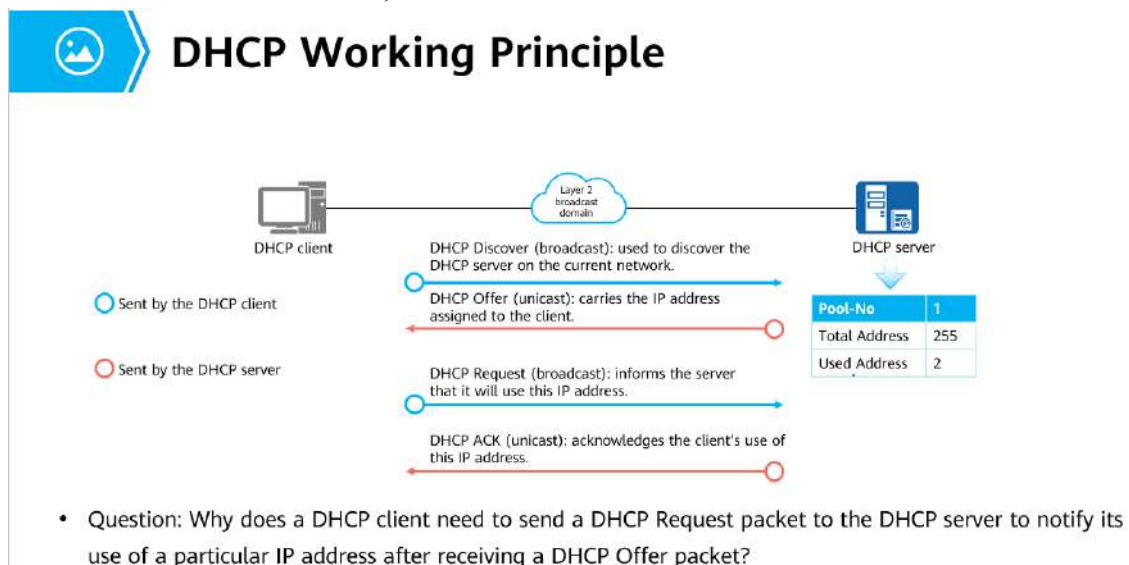
Justification

DHCP Message	Description
DHCP Discover	A DHCP client broadcasts this message to locate a DHCP server when the client attempts to connect to a network for the first time.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100276747/225eec10/dhcp-messages>

Justification

Source: Huawei Lecture # 15, Slide # 28



299. Which of the following network parameters can be assigned by the Dynamic Host Configuration Protocol DHCP? (Multiple choices)

- A. IP address
- B. Gateway address
- C. Operating system name
- D. DNS address

Answer: ABD

Justification#1

A & B) A DHCP client sends a message to a DHCP server to request configuration parameters, such as the IP address, subnet mask, and default gateway address.

D) The DHCP server can be configured to assign additional parameters such as the IP address of the Domain Name System (DNS) server

Source:

A & B)

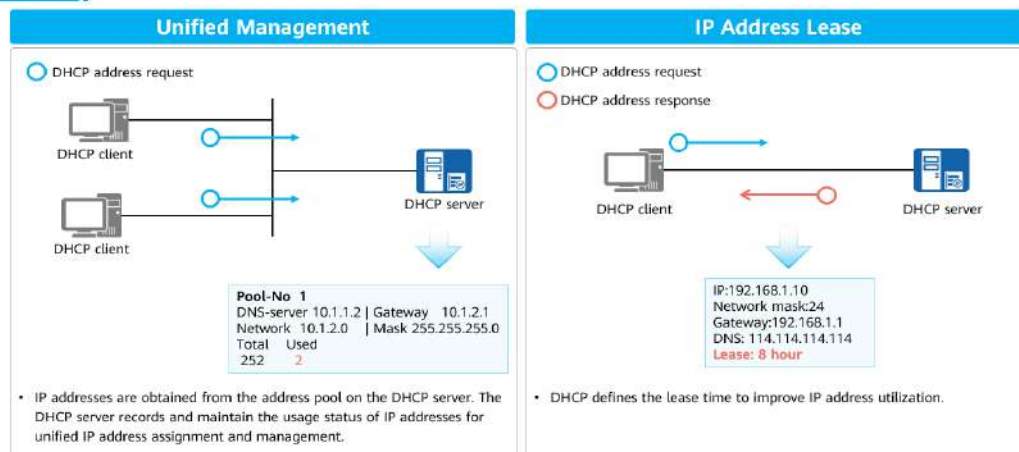
<https://support.huawei.com/enterprise/en/doc/EDOC1100279014/4b28c08d/understanding-dhcp>

D) https://www.cisco.com/c/en/us/td/docs/ios-xml/ios/ipaddr_dhcp/configuration/12-4/dhcp-12-4-book/config-dhcp-server.html

Justification#2

Source: Huawei Lecture # 15, Slide # 27

DHCP Advantages



300. In the WLAN architecture, the FIT AP cannot work independently and needs to be managed by the AC. Which of the following protocols are used to communicate between FIT AP and AC?

- A. IPsec
- B. WEP
- C. CAPWAP
- D. WAP

Answer: C

Justification#1

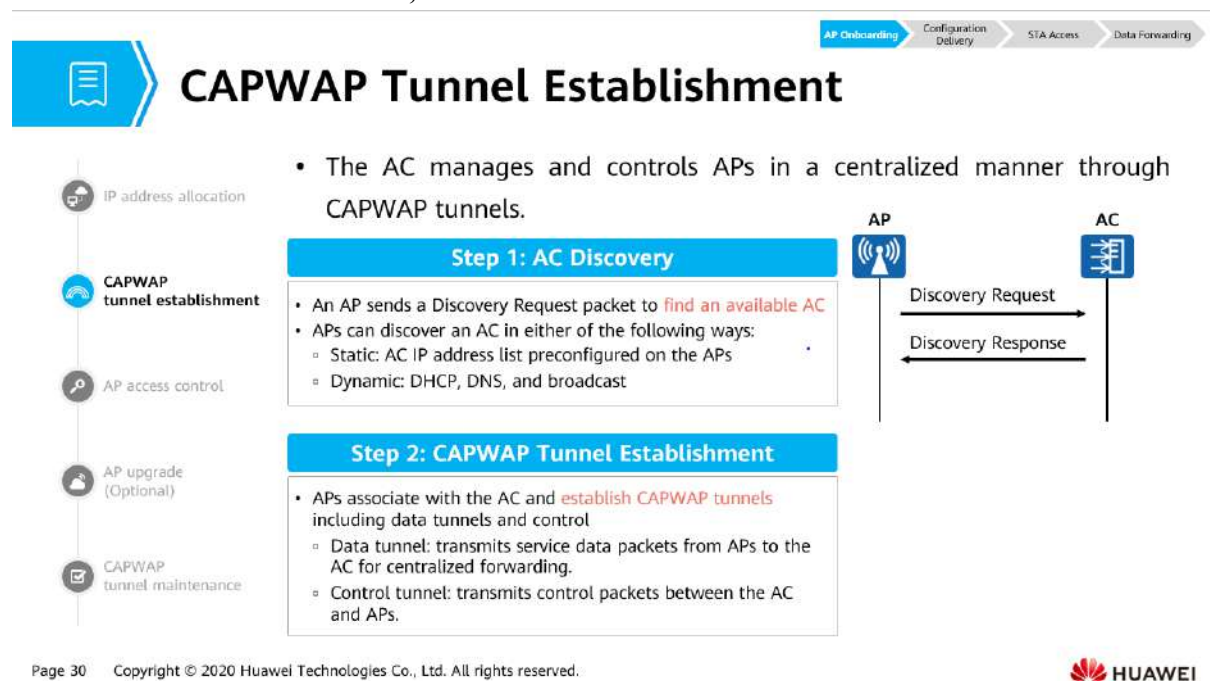
Control and Provisioning of Wireless Access Points (CAPWAP) provides a universal encapsulation and transmission mechanism widely applied to wireless local area networks (WLANs), implementing communication between access points (APs) and WLAN access controllers (WACs).

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100334297/2471df23/understanding-capwap>

Justification#2

Source: Huawei Lecture # 16, Slide # 30



301. What is the first operation performed by the FIT AP after obtaining the AC's IP address?

- A. Upgrade software version
- B. Download the configuration file
- C. Establish CAPWAP tunnel
- D. Request configuration file

Answer: C

Justification#1

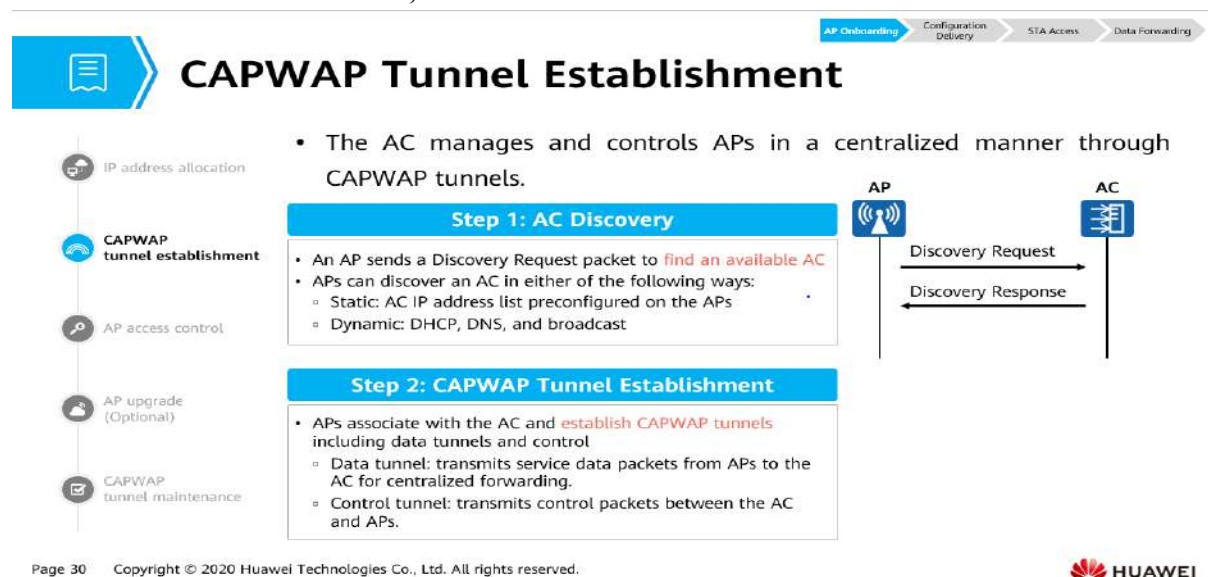
An AC IP address list is preconfigured on the AP. When the AP goes online, it unicasts a Discovery Request packet to each AC whose IP address is specified in the preconfigured AC IP address list. After receiving the Discovery Request packet, the ACs return Discovery Response packets to the AP. The AP then selects an AC to establish a CAPWAP tunnel according to the received Discovery Response packets.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100364427/78331d51/ap-online-process>

Justification#2

Source: Huawei Lecture # 16, Slide # 30



302. How many types of data flows with different service levels can be marked based on MPLS labels?

- A. 4
- B. 2
- C. 16
- D. 8

Answer: D

Justification#1

- **MPLS DiffServ**

On an MPLS network, EXP values are used to identify a maximum of eight service priorities.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100335695/b0d08e70/class-based-qos-configuration>

Justification#2

- Selection based on number of service types: Up to eight PHBs are supported if only E-LSPs are used. To support more than eight PHBs, you must use L-LSPs or use both E-LSPs and L-LSPs.

Figure 1-1444 Structure of the MPLS packet header

20 bits	3 bits	1 bit	8 bits
Label	Exp	S	TTL

The Exp value can be copied from the DSCP or IP Precedence in an IP packet or be set by MPLS network carriers. The LSP whose PHB is inferred from the Exp field is called an E-LSP, which is short for Exp-Inferred-PHB Scheduling Class (PSC) LSP.

The E-LSP scheme is applicable to networks that support not more than eight PHBs. The IP Precedence field in an IP header also has three bits, same as the Exp field length, which allows one-to-one mapping. However, the DSCP field in an IP header has six bits, different from the Exp length, which does not allow complete mapping between an E-LSP and an IP QoS. As the IEEE standard defines, the three left-most bits in the DSCP field (the CSCP value) correspond to the Exp value, regardless of what the three right-most bits are.

In the E-LSP scheme, traffic classification is performed when an MPLS packet enters a device. The Exp value carried in the packet is mapped to the service class and drop precedence of the device. After traffic classification, QoS operations on an MPLS network, such as traffic shaping, traffic policing, and congestion avoidance, are implemented in the same manner as those on an IP network.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100279023/f46b5cc0/mpls-diffservmodeconfiguration>

**303. Which of the following functions does the LCP protocol in the PPP protocol support?
(Multiple Choice)**

- A. Negotiate the network layer address
- B. Negotiate an authentication agreement
- C. Detect link loops
- D. Negotiate the maximum receiving unit MRU

Answer: BCD

Justification#1

B & C) It can provide loop detection, authentication, transmission encryption,[2] and data compression.

D) The ppp mru command sets the maximum receive unit (MRU) in PPP negotiation.

Source:

B & C) https://en.wikipedia.org/wiki/Point-to-Point_Protocol

D)

https://info.support.huawei.com/hedex/api/pages/EDOC1100277644/AEM10221/03/resources/command/yunshan/PPP_MRU.html

Justification#2

2. In the Establish phase, the two devices perform an LCP negotiation to negotiate the following items: working mode (SP or MP), MRU (Maximum Receive Unit), authentication mode, and magic number (SP is short for single-link PPP and MP is short for MultiLink PPP). If the LCP negotiation succeeds, LCP turns Opened, which indicates that a lower-layer link has been established.

Source:

Huawei website: <https://support.huawei.com/enterprise/en/doc/EDOC1100112361/935e3094/ppp-link-establishment-process>

Notes:

- **"Authentication mode"** refers to LCP's role in agreeing on PAP or CHAP for secure link setup.
- **"Magic number"** is a unique identifier used by LCP to detect looped-back links (if a device receives its own magic number, the link is looped).
- **"MRU (Maximum Receive Unit)"** is the maximum packet size LCP agrees on for the link, preventing fragmentation issues.

304. The network management system can only view the operating status of the device through the SNMP protocol, but cannot issue configurations.

- A. True
- B. False

Answer: B

Justification

Source: Huawei Lecture # 18, Slide # 6



Basic Network Management Functions



OSI defines five functional models for network management:

- Configuration management: monitors network configuration information so that network administrators can generate, query, and modify hardware and software running parameters and conditions, and configure services.
- Performance management: manages network performance so that the network can provide reliable, continuous, and low-latency communication capabilities with as few network resources as possible.
- Fault management: ensures that the network is always available and rectifies faults as soon as possible.
- Security management: protects networks and systems from unauthorized access and attacks.
- Accounting management: records the network resource usage of users, charges users, and collects statistics on network resource usage.

305. The network devices running the SNMP protocol are running locally-an agent process.

- A. True
- B. False

Answer: A

Justification#1

A router that runs the SNMP server program is called an agent,

Source: <https://www.geeksforgeeks.org/simple-network-management-protocol-snmp/>

Justification#2

Source: Huawei Lecture # 18 Slide # 11

- On a network where SNMP is used for network management, a network management system (NMS) functions as a network management center and runs management processes. Each managed device needs to run an agent process. The management process and agent process communicate with each other through SNMP messages.

306. Which extension headers are supported by IPv6 packets? (Multiple choice)

- A. Destination option extension header
- B. Fragmentation extension header
- C. Hop-by-hop option extension header
- D. VLAN extension header

Answer: ABC

Justification#1

Order	Header Type	Next Header Code
2	Hop-by-Hop Options	0
3	Destination Options (with Routing Options)	60
5	Fragment Header	44

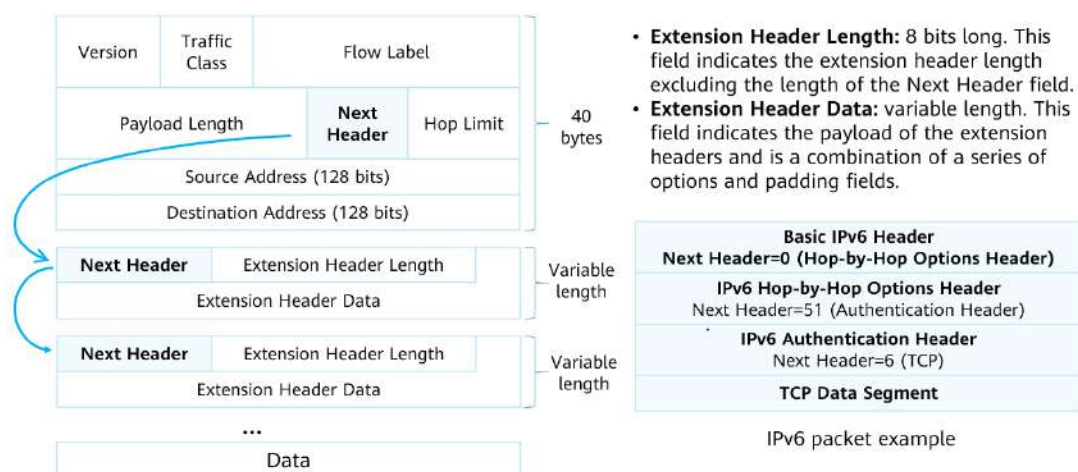
Source:

https://www.cisco.com/en/US/technologies/tk648/tk872/technologies_white_paper0900aecd8054d37d.html

Justification#2

Source: Huawei Lecture # 19 Slide # 10

IPv6 Extension Header



**307. Which of the following are not open interfaces of Huawei network equipment?
(Multiple choice)**

- A. NETCONF
- B. RESTCONF
- C. JSON
- D. XML

Answer: CD

Justification#1

D) NETCONF uses Extensible Markup Language (XML)-based data encoding for the configuration data and protocol messages

C) The RESTCONF client obtains both configuration and status data from a running RESTCONF server.

After processing the user request, the server replies with a response to the client. The request and response messages are coded in either XML or JavaScript Object Notation (JSON) format.

Source:

D) <https://info.support.huawei.com/info-finder/encyclopedia/en/NETCONF.html>

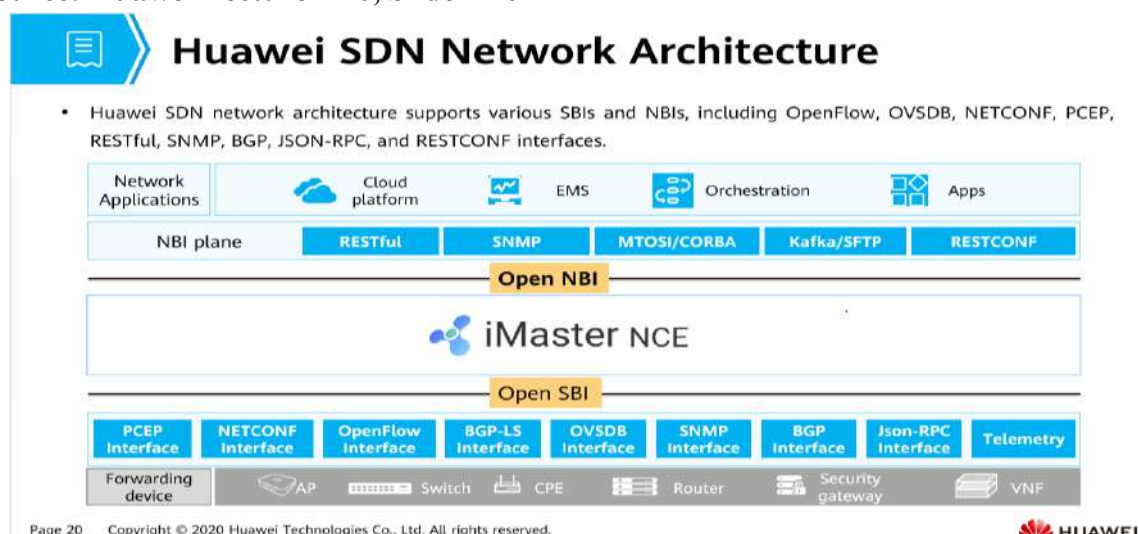
C) <https://support.huawei.com/enterprise/en/doc/EDOC1100278266/3e5858fb/restconf-fundamentals>

Explanation:

Therefore, in the context of Huawei network equipment, **NETCONF** and **RESTCONF** are considered open interfaces, while **JSON** and **XML** serve as data formats utilized by these interfaces.

Justification#2

Source: Huawei Lecture # 20, Slide # 20



308. Which of the following are not the characteristics of a small-sized campus network?

- A. Simple network level
- B. Fewer users
- C. Simple network requirements
- D. Wide coverage

Answer: D

Justification#1

The LAN is the networking infrastructure that provides access to network communication services and resources for end users and devices spread over a single floor or building. You create a campus network by interconnecting a group of LANs that are spread over a local geographic area. Campus network design concepts include small networks that use a single LAN switch, up to very large networks with thousands of connections.

Source:

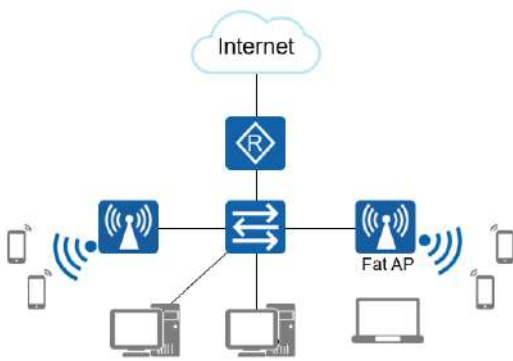
<https://www.cisco.com/c/en/us/td/docs/solutions/CVD/Campus/cisco-campus-lan-wlan-design-guide.html>

Justification#2

Source: Huawei Lecture # 22, Slide # 7



Typical Architecture of Small Campus Networks



Network topology of a chain cafe

- Small campus networks are typically deployed in scenarios where the number of access users is small (several or dozens of users). A small campus network can cover only one location, has a simple architecture, and is constructed to enable mutual access between internal resources.
- Characteristics of small campus networks:
 - Small number of users
 - Only one location
 - Simple network architecture
 - Simple network requirements

Number of terminals	< 200
Number of NEs	< 25

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 HUAWEI

309. Which statements are correct about the shortcut keys of the VRP operating platform? (Multiple choice)

- A. TAB prompts the most recently entered command
- B. CTRL+C to stop the current command
- C. The left cursor key (<) moves the cursor to the left position.
- D. CTRL+Z back to user view

Answer: BCD

Justification#1

Table 2-2 *System-defined shortcut keys*

Key	Function
Ctrl+C	Stops performing current functions.
Ctrl+Z	Returns to the user view
Ctrl+B	Moves the cursor back one character.

Source:

B, C & D) <https://support.huawei.com/enterprise/en/doc/EDOC1000178166/36ae01a3/using-command-line-shortcut-keys>

Justification#2

C) Source: Huawei Lecture # 3 Slide # 22, 27

1. Command editing through function keys

- Backspace: deletes the character before the cursor and moves the cursor to the left. When the cursor reaches the beginning of the command, an alarm is generated.
- Left cursor key ← or Ctrl+B: moves the cursor one character to the left. When the cursor reaches the beginning of the command, an alarm is generated.

System Shortcut Keys

- CTRL_A: moves the cursor to the beginning of the current line.
- CTRL_B: moves the cursor one character to the left.
- CTRL_C: stops the running of the current command.
- CTRL_E: moves the cursor to the end of the current line.
- CTRL_X: deletes all characters on the left of the cursor.
- CTRL_Y: deletes the character at the cursor and all characters on the right of the cursor.
- CTRL_Z: returns to the user view.
- CTRL+]: terminates the current connection or switches to another connection.

310. Which of the following information prompts will the VRP operating platform use for incomplete input commands?

- A. Error: Ambiguous command found at ' position
- B. Error: Incomplete command found at ' position
- C. Error: Too many parameters found at ' position
- D. Error: Wrong parameter found at ' position

Answer: B

Justification#1

Table 2-3 *Common error messages of the command line*

Error Message	Cause of the Error
Error: Incomplete command found at '^' position.	The entered command is incomplete.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100096311/941a7a88/interpreting-command-line-error-messages>

Justification#2

Source: Huawei Lecture # 3, Slide # 25

Interpreting Command Line Error Messages

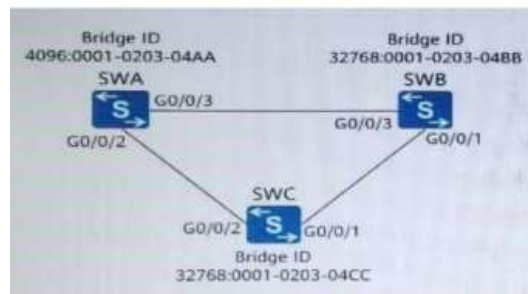
- If a command passes the syntax check, the system executes it. Otherwise, the system reports an error message.

```
[Huawei] sysname
      ^
Error:Incomplete command found at '^' position.  #A supplement needs to be made at the position pointed by the arrow.
```

Explanation:

- **Scenario:** Entering a command without required parameters (e.g., sysname without specifying a name).
- **Error B:** Indicates the command is **missing necessary input** at the marked position.

311. As shown in the figure below, which port of the following switches will be blocked?

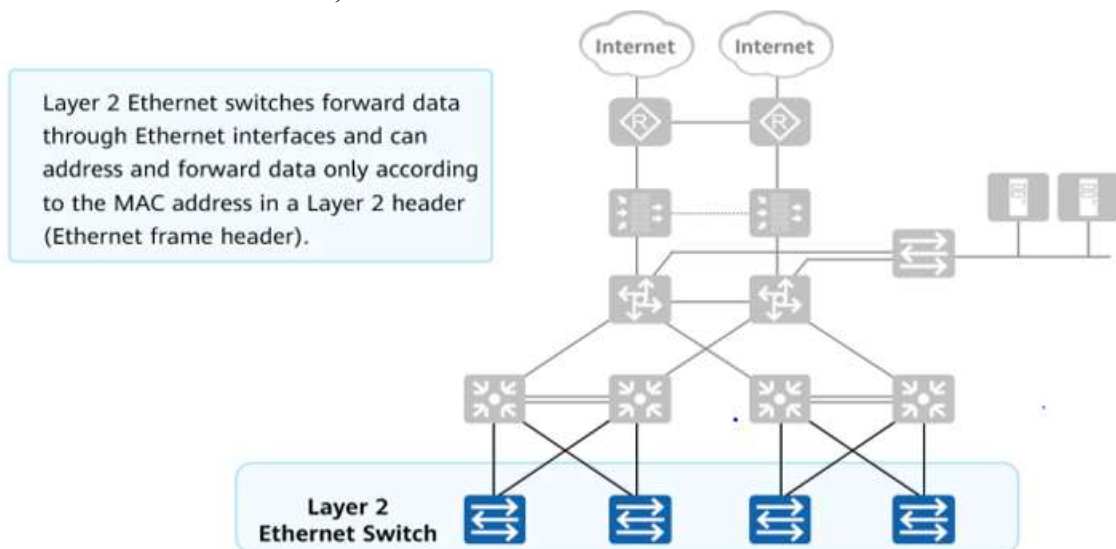


- A. SWC G0/0/1
- B. SWC G0/0/2
- C. SWB G0/0/3
- D. SWA G0/0/3

Answer: A

Justification

Source: Huawei Lecture # 7, Slide # 22



Explanation:

1. Identify the Root Bridge: SWA has the lowest Bridge ID (4096-0001-0203-0AAA), so it becomes the root bridge.
2. Root Ports: SWB and SWC determine their root ports based on the lowest path cost to SWA:
 - SWB: G0/0/1 (connected to SWA's G0/0/2)
 - SWC: G0/0/1 (connected to SWA's G0/0/1)
3. Designated Ports: SWA's ports G0/0/1 and G0/0/2 are designated ports since SWA is the root bridge.
4. Blocked Ports: A port that is neither a root port nor a designated port will be blocked to prevent loops.
 - On SWB, G0/0/3 (connected to SWC's G0/0/2) is not a root or designated port.
- On SWC, G0/0/2 (connected to SWB's G0/0/3) is not a root or designated port.

Therefore, the port that will be blocked in this network setup is SWC's G0/0/1, as it is not selected as either a root port or a designated port. So, the correct answer is: A. SWC G0/0/1

312. An existing switch is connected to a designated port through a port, but the port does not forward any packets, and can monitor network changes by receiving BPDUs. then what is the role of the port?

- A. Designated port
- B. Alternate port
- C. Disable port
- D. Root port

Answer: B

Justification#1

An alternate port receives BPDUs from another switch but remains in a blocked state. For example, let's say a switch has two paths to the root bridge. It will elect one of the two ports as a root port and the other will become an alternate port.

Source:

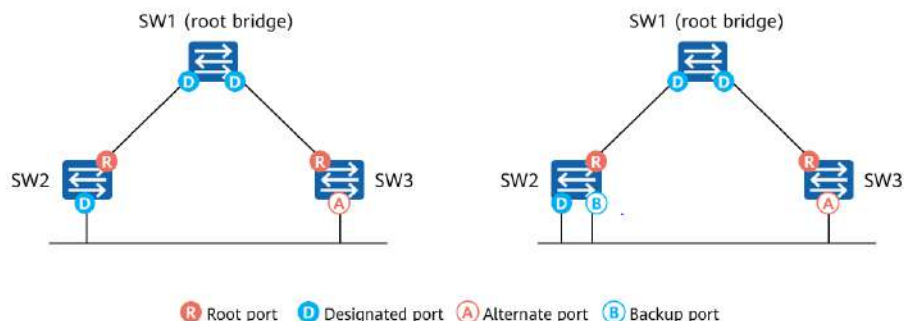
[https://documentation.meraki.com/MS/Port_and_VLAN_Configuration/Spanning_Tree_Protocol_\(STP\)_Overview#:~:text=An%20alternate%20port%20receives%20BPDUs,will%20be%20an%20alternate%20port](https://documentation.meraki.com/MS/Port_and_VLAN_Configuration/Spanning_Tree_Protocol_(STP)_Overview#:~:text=An%20alternate%20port%20receives%20BPDUs,will%20be%20an%20alternate%20port)

Justification#2

Source: Huawei Lecture # 9, Slide # 49

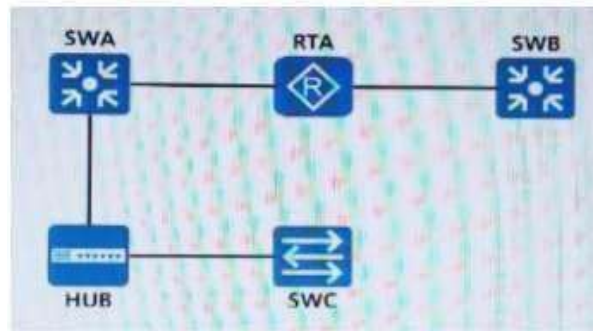
Port Roles in RSTP

- RSTP adds port roles to help understand RSTP and simplify RSTP deployment.



RSTP defines four port roles: root port, designated port, alternate port, and backup port.

313. In the network shown in the figure, which of the following descriptions is correct?



- A. The network between RTA and SWC is the same conflict domain
- B. The network between SWA and SWC is the same conflict domain
- C. The network between SWA and SWB is the same broadcast domain
- D. The network between SWA and SWC is the same broadcast domain

Answer: B

Justification

Explanation:

- Because of the SWA and SEC are connected to the hub that's why both belong to the same conflict domain.
- **Conflict Domain:** A hub (HUB) does not segment collision domains, so devices connected through a hub share the same collision domain.
- **SWA ↔ HUB ↔ SWC:** Since SWA and SWC are connected via a hub (HUB), they are in the **same collision domain**.

314. The output information of a switch is shown in the figure. Which of the following statements is correct? (Multiple choice)

```
[SWA]display vlan
The total number of vlans is : 5
```

U: Up;	D: Down;	TG: Tagged;	UT: Untagged;
MP: Vlan-mapping;		ST: Vlan-stacking;	
#: ProtocolTransparent-vlan;		*: Management-vlan;	

VID	Type	Ports
1	common	UT:GEO/0/3(D) GEO/0/4(D) GEO/0/5(D) GEO/0/6(D) GEO/0/7(D) GEO/0/8(D) GEO/0/9(D) GEO/0/10(D) GEO/0/11(D) GEO/0/12(D) GEO/0/13(D) GEO/0/14(D) GEO/0/15(D) GEO/0/16(D) GEO/0/17(D) GEO/0/18(D) GEO/0/19(D) GEO/0/20(D) GEO/0/21(D) GEO/0/22(D) GEO/0/23(D) GEO/0/24(D) TG:GEO/0/1(U)
10	common	UT:GEO/0/1(U) GEO/0/2(U)
20	common	TG:GEO/0/1(U)
30	common	TG:GEO/0/1(U)
40	common	TG:GEO/0/1(U)

- A. The GE0/0/2 port of the switch carries VLAN TAG when sending VLAN20 data frames
- B. The user manually created 4 VLANs
- C. The GE0/0/1 port of the switch does not carry VLAN TAG when sending VLAN20 data frames
- D. The GE0/0/1 port of the switch does not carry VLAN TAG when sending VLAN10 data frames

UT = untagged

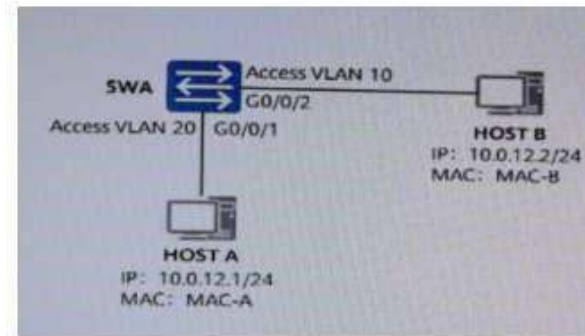
Answer: BD

Justification

In an untagged vlan interface, the vlan id would be stripped while sending the data frame hence data packets sent from the interface 01 02 would not carry vlan tag

- For **option B**, the user manually created 4 VLANs (B). VLANs 10, 20, 30, and 40 are all listed as "common" which means they were manually created by the user.
- In **option D**, this is because under the "VID" (VLAN ID) column, "10" has "UT" (untagged) under the "Ports" column with G 0/0/1

315. As shown in the figure, if host A has the ARP cache of host B, host A can ping host B.



- A. True
- B. False

Answer: A

Justification

HOSTA searches the local ARP table for the ARP entry corresponding to HOSTB. If the corresponding ARP entry is found, HOSTA encapsulates the IP packets into Ethernet frames and forwards them to HOSTB based on its MAC address.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100169990/f5ddba7b/understanding-arp>

Explanation:

- By default, VLANs isolate broadcast traffic, so ARP requests from Host A (VLAN 20) will not reach Host B (VLAN 10), and vice versa. This means Host A cannot dynamically learn Host B's MAC address via ARP.
- Even if Host A has Host B's MAC address statically configured in its ARP cache, the switch will not forward traffic between VLANs without a router or Layer 3 device to route between the VLANs.

Since there is no indication of inter-VLAN routing (e.g., a router or Layer 3 switch), Host A cannot ping Host B, even with Host B's MAC in its ARP cache. The switch will drop the traffic due to VLAN segregation.

316. By default, what is the STP Forward Delay time in seconds?

- A. 5
- B. 10
- C. 15
- D. 20

Answer: C

Justification#1

By default, the Forward Delay timer is 1500 centiseconds (15 seconds).

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178168/4dfa81e4/setting-the-stp-timers>

Justification#2

Huawei Slides: Lecture 9 Slide 21

2	Forward Delay	Time that is spent in Listening or Learning state. The default value is 15s.
---	---------------	--

317. The G0/0/1 interface configuration information of router A is shown in the figure.
Which of the following statements is correct? (Multiple choice)

```
#
acl number 3000
 rule 5 deny 17
 rule 10 deny 89
 rule 15 deny 6
#
interface GigabitEthernet0/0/1
 traffic-filter inbound acl 3000
#
```

89 protocol = OSPF

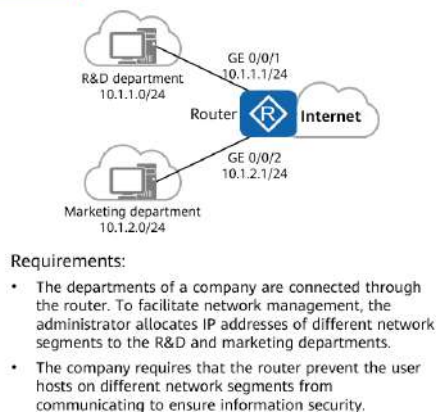
- A. This interface will not forward received FTP packets
- B. This interface will not forward ICMP packets
- C. This interface will not forward received SNMP packets SNMP uses UDP
- D. This interface can establish OSPF neighbor relationships with other routers

Answer: AC

Justification

Source: Huawei Lecture # 12 Slide # 28

Case: Use Advanced ACLs to Prevent User Hosts on Different Network Segments from Communicating (2)



4. Configure traffic filtering in the inbound direction of GE 0/0/1 and GE 0/0/2.

```
[Router] interface GigabitEthernet 0/0/1
[Router-GigabitEthernet0/0/1] traffic-filter inbound acl 3001
[Router-GigabitEthernet0/0/1] quit

[Router] interface GigabitEthernet 0/0/2
[Router-GigabitEthernet0/0/2] traffic-filter inbound acl 3002
[Router-GigabitEthernet0/0/2] quit
```

Source: Huawei Website (<https://support.huawei.com/enterprise/en/doc/EDOC1100086647>)

Explanation

- Rule 5 denies protocol 17 (UDP).
 - Rule 10 denies protocol 89 (OSPF).
 - Rule 15 denies protocol 6 (TCP).
- A. This interface will not forward received FTP packets: FTP uses TCP (protocol 6). Since TCP packets are denied by rule 15, this statement is correct.
- B. This interface will not forward ICMP packets: ICMP uses protocol 1. There is no rule denying ICMP packets in the ACL, so this statement is incorrect.
- C. This interface will not forward received SNMP packets: SNMP uses UDP (protocol 17). Since UDP packets are denied by rule 5, this statement is correct.
- D. This interface can establish OSPF neighbour relationships with other routers: OSPF uses protocol 89. Since OSPF packets are denied by rule 10, this statement is incorrect.

318. To apply the ACL configured in the figure in Telnet, which of the following statements is correct?

```
#
acl number 2000
 rule 5 deny source 172.16.105.3 0
 rule 10 deny source 172.16.105.4 0
 rule 15 deny source 172.16.105.5 0
 rule 20 permit
#
```

- A. The device with IP address 172.16.105.4 can use Telnet service
- B. The device with IP address 172.16.105.3 can use Telnet service
- C. The device with IP address 172.16.105.5 can use Telnet service
- D. The device with IP address 172.16.105.6 can use Telnet service

Answer: D

Justification

Because we use ACL rule to not use Telnet service in IP addresses 176.16.105.3 & 176.16.105.4 & 176.16.105.5, deny means to stop. Only the option D is correct because we cannot apply any rule on this IP address.

These rules are applied in order, and if a rule matches, no further rules are evaluated. Here's the evaluation of each option:

- **Option A:** The device with IP address 172.16.105.4 can use Telnet service
 - This device is denied by rule 10.
- **Option B:** The device with IP address 172.16.105.3 can use Telnet service
 - This device is denied by rule 5.
- **Option C:** The device with IP address 172.16.105.5 can use Telnet service
 - This device is denied by rule 15.
- **Option D:** The device with IP address 172.16.105.6 can use Telnet service
 - This device is not explicitly denied by any of the deny rules and thus is permitted by rule 20.

Therefore, the correct answer is: The device with IP address **172.16.105.6** can use Telnet service.

319. Which of the following fields is not included in the HELLO packet of the OSPF protocol?

- A. Neighbor
- B. Network Mask
- C. HelloInterval
- D. Sysname

Answer: D

Justification#1

Table 2 *Hello packet fields*

Field	Length	Description
Network Mask	32 bits	Mask of the network on which the interface that sends the Hello packet resides.
HelloInterval	16 bits	Interval at which Hello packets are sent.
Neighbor	32 bits	Router ID of the neighbor.

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100277644/AEM10221/04/resources/vrp/feature_0003998103.html

Justification#2

Source: Huawei Lecture # 6, Slide # 18

OSPF Packet Types

- There are five types of OSPF protocol packets and implement different functions in interaction between OSPF routers.

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.

320. All interfaces of the router belong to the same broadcast domain.

- A. True
- B. False

Answer: B

Justification#1

Routers do not forward broadcast traffic; they break broadcast domains.

Source:

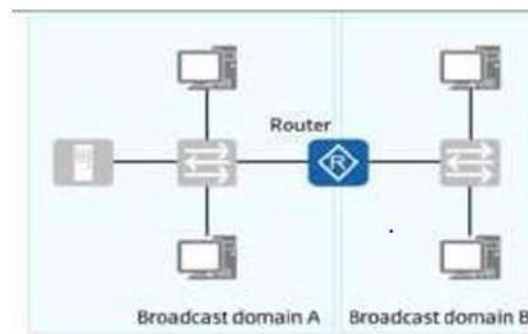
<https://networklessons.com/switching/broadcast-domain>

Justification#2

Because on the router both interfaces belong to different networks so the broadcast domain is not same if two or more networks are running on the router.

Source:

Huawei Website: <https://forum.huawei.com/enterprise/intl/en/thread/datacom-basics-network-deviceandtheir-functions/667246568694235136?blogId=667246568694235136>



321. The host's IPv4 address is 200.200.200.201/30. Which of the following hosts has an IPv4 address and its communication does not need to be forwarded by a router?

- A. 200.200.200.202 direct communication without forwarding
- B. 200.200.200.200
- C. 200.200.200.203
- D. 200.200.200.1

Answer: A

Justification

Source: Huawei Lecture # 4 Slide # 20

IP Address Types

- A network range defined by a network ID is called a network segment.

- **Network address:** identifies a network.

Example: 192.168.10.0/24

192.	168.	10.	00000000
------	------	-----	----------

- **Broadcast address:** a special address used to send data to all hosts on a network.

Example: 192.168.10.255/24

192.	168.	10.	11111111
------	------	-----	----------

- **Available addresses:** IP addresses that can be allocated to device interfaces on a network.

Example: 192.168.10.1/24

192.	168.	10.	00000001
------	------	-----	----------

Note

- Network and broadcast addresses cannot be directly used by devices or their interfaces.
- Number of available addresses on a network segment is $2^n - 2$ (n is the number of bits in the host part).

Explanation:

To determine which host's communication does not need to be forwarded by a router, we need to understand the subnetting of the given network address 200.200.200.201/30.

A /30 subnet has a subnet mask of 255.255.255.252, which provides 2 usable IP addresses. The addresses in this subnet range from 200.200.200.200 to 200.200.200.203.

The subnet includes the following addresses: 200.200.200.200 (network address)

- 200.200.200.201 (host address)
- 200.200.200.202 (host address)
- 200.200.200.203 (broadcast address)

Among these addresses, only the address 200.200.200.202 falls within the same subnet as 200.200.200.201, so communication between these two hosts does not require forwarding by a router. Therefore, the **correct answer is: A. 200.200.200.202**

Default Route ---> `ip route-static 0.0.0.0 0.0.0.0 <next-hop>`

322. Among the following commands to configure the default route, which one is correct?

- A. `[Huawei-Serial10] ip route-static 0.0.0.0 0.0.0.0 0.0.0.0`
- B. `[Huawei] ip route-static 0.0.0.0 0.0.0.0 192.168.1.1`
- C. `[Huawei] ip route-static 0.0.0.0 0.0.0.0 0.0.0.0`
- D. `[Huawei] ip route-static 0.0.0.0 255.255.255.255 192.168.1.1`

Answer: B

Justification#1

Set default gateway addresses of PC1, PC2, and PC3 to 192.168.1.1, 192.168.2.1, and 192.168.3.1 respectively.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100034072/14e80e16/example-for-configuring-ipv4-static-routes>

Justification#2

Source: Huawei Lecture # 5, Slide # 29

Static Route Configuration

- Specify a next-hop IP address for a static route.

```
[Huawei] ip route-static ip-address { mask | mask-length } nexthop-address
```

- Specify an outbound interface for a static route.

```
[Huawei] ip route-static ip-address { mask | mask-length } interface-type interface-number
```

- Specify both the outbound interface and next hop for a static route.

```
[Huawei] ip route-static ip-address { mask | mask-length } interface-type interface-number [ nexthop-address ]
```

When creating a static route, you can specify both the outbound interface and next hop. Alternatively, you can specify either the outbound interface or next hop, depending on the interface type:

For a point-to-point interface (such as a serial interface), you must specify the outbound interface.

For a broadcast interface (for example, an Ethernet interface) or a virtual template (VT) interface, you must specify the next hop.

Explanation:

(Option B is correct because it properly specifies the default route with the next hop IP address (192.168.1.1), allowing any traffic with a destination address not matching any other route to be forwarded to that next hop. Options A, C, and D are incorrect because they either lack the specification of the next hop IP address (A and C) or use an inappropriate subnet mask for a default route (D).)

323. If the network address of a network is 192.168.1.0, then its broadcast address must be 192.168.1.255.

- A. True
- B. False

Answer: B

Justification

Source: Huawei Lecture # 4, Slide # 27

IP Address Calculation

- Example: What are the network address, broadcast address, and number of available addresses of class B address 172.16.10.1/16?

	172.	16.	00001010.	00000001	
IP address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 1 0	0 0 0 0 0 0 0 1	
Network mask	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	
Network address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0	172.16.0.0/16 The network address is obtained, with all host bits set to 0s.
Broadcast address	1 0 1 0 1 1 0 0	0 0 0 1 0 0 0 0	1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1	172.16.255.255/16 The broadcast address is obtained, with all host bits set to 1s.
Number of IP addresses	$2^{16} = 65536$				Quiz Example: What are the network address, broadcast address, and number of available addresses of class A address 10.128.20.10/8?
Number of available addresses	$2^{16} - 2 = 65534$				
Range of available addresses	172.16.0.1/16–172.16.255.254/16				

Explanation:

The statement "If the network address of a network is 192.168.1.0, then its broadcast address must be 192.168.1.255" is only true if the subnet mask is 255.255.255.0 (/24). If a different subnet mask is used, the broadcast address will be different.

For different subnet:

- Subnet Mask: 255.255.255.128 (/25)
- Network Address: 192.168.1.0
- Broadcast Address: 192.168.1.127

324. On Huawei AR G3 routers, what is the -i parameter of the Ping command in VRP used to set?

- A. The destination IP address of the Echo Reply message
- B. The source IP address of the Echo Request message
- C. Interface for receiving Echo Reply messages
- D. Interface for sending Echo Request messages

Answer: D

Justification#1

- i: specifies an interface for sending ICMP Echo Request packets.

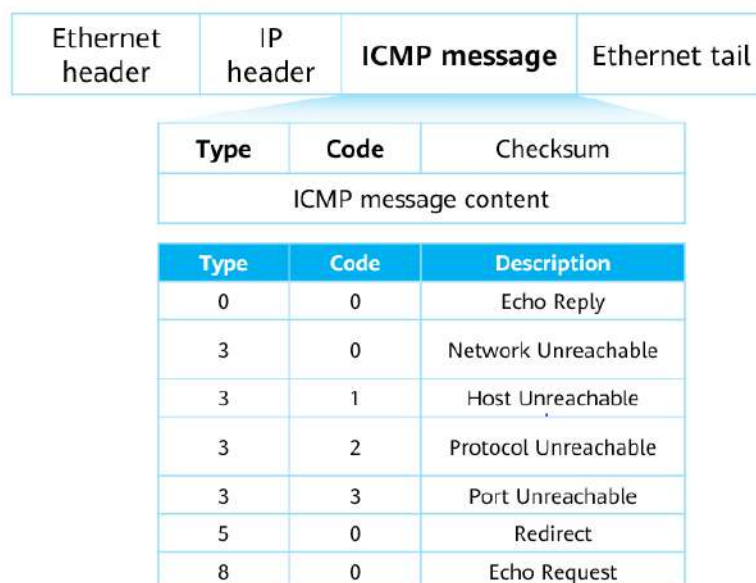
Source:

<https://forum.huawei.com/enterprise/intl/en/thread/ping-command-options/667235016977039360?blogId=667235016977039360>

Justification#2

Huawei Slides: Lecture 4 Slide 34

On Huawei AR G3 routers running VRP (Versatile Routing Platform), the -i parameter of the Ping command is used to set the interface for sending Echo Request messages.



325. If the Proto of a route information in the routing table is OSPF, the priority of this route must be 10.

- A. True
- B. False

Answer: B

Justification#1

Parameter	Description	Value
priority	Specifies the priority of an interface that participates in the DR or BDR election. A larger value indicates a higher priority.	The value is an integer ranging from 0 to 255.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100333403/15a37ba9/ospf-dr-priority>

Justification#2

Source: Huawei Lecture # 6, Slide # 40

Basic OSPF Configuration Commands (2)

4. (Interface view) Set an OSPF interface cost.

```
[Huawei-GE1/0/1] ospf cost cost
```

The **ospf cost** command sets a cost for an OSPF interface. By default, OSPF automatically calculates the cost of an interface based on the interface bandwidth. The cost value is an integer ranging from 1 to 65535.

5. (OSPF view) Set an OSPF bandwidth reference value.

```
[Huawei-ospf-1] bandwidth-reference value
```

The **bandwidth-reference** command sets a bandwidth reference value that is used to calculate interface costs. The value ranges from 1 to 2147483648, in Mbit/s. The default value is 100 Mbit/s.

6. (Interface view) Set the priority of an interface for DR election.

```
[Huawei-GigabitEthernet0/0/0] ospf dr-priority priority
```

The **ospf dr-priority** command sets a priority for an interface that participates in DR election. A larger value indicates a higher priority. The value ranges from 0 to 255.

326. In the switch MAC address table, which of the following entries will not age?

- A. Dynamic MAC address table entry
- B. Static MAC address table entry
- C. Port MAC address table entry
- D. Device MAC address table entry

Answer: B

Justification

A static MAC address entry will not be aged out. After being saved, a static MAC address entry will not be lost after a system restart, and can only be deleted manually.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000128406/7fe5507/configuring-a-mac-address-table>

327. What is the transmission rate of an Eth-Trunk port in manual link aggregation mode related to? (Multiple choice)

- A. Number of member ports
- B. Bandwidth of member ports
- C. If an IP address is configured on the member port
- D. If the member port is on the public or private network

Answer: AB

Justification#1

The number of member links in the Up state affects the status and bandwidth of a trunk interface. The bandwidth of an Eth-Trunk interface equals the total bandwidth of all member interfaces in the Up state.

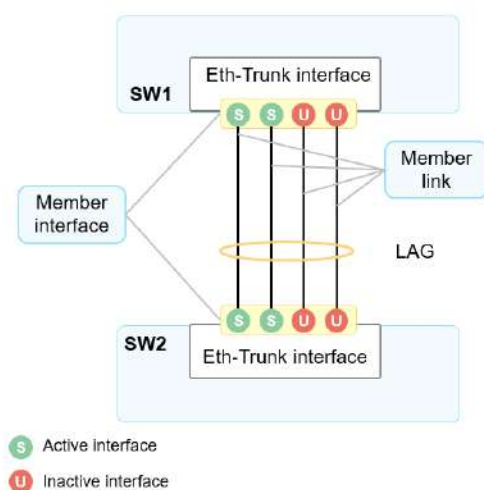
Source:

https://support.huawei.com/enterprise/en/doc/EDOC1100279274/5a9f722c/eth-trunk-interface-configuration?utm_source=chatgpt.com

Justification#2

Source: Huawei Lecture # 11, Slide # 13

Basic Concepts of Eth-Trunk



- A link aggregation group (LAG) is a logical link formed by bundling several links. Each LAG has one logical interface, known as an LAG interface or Eth-Trunk interface.
- Member interface and member link: Physical interfaces that constitute an Eth-Trunk interface are called member interfaces, and the link corresponding to a member interface is known as a member link.
- Active interface and active link: An active interface is also called a selected interface and is a member interface that participates in data forwarding. The link corresponding to an active interface is called an active link.
- Inactive interface and inactive link: An inactive interface is also called an unselected interface and is a member interface that does not participate in data forwarding. A link corresponding to an inactive interface is referred to as an inactive link.
- Link aggregation mode: Based on whether the Link Aggregation Control Protocol (LACP) is enabled, link aggregation can be classified into manual mode and LACP mode.
- Other concepts: upper and lower thresholds for the number of active interfaces

328. What is the role of the command port trunk allow pass vlan all?

- A. If the port default vlan 3 command is configured for the connected remote device, VLAN 3 cannot communicate between the two devices
- B. All VLAN data frames are allowed to pass through this port
- C. The connected peer device can dynamically determine which VLAN IDs are allowed to pass
- D. The opposite port connected to the port must be configured with port trunk permit VLAN all

Answer: B

Justification#1

To allow all packets from one or multiple VLANs to pass through a trunk interface, the trunk interface must be added to the VLANs using the port trunk allow-pass vlan command.

Source:

https://support.huawei.com/enterprise/en/doc/EDOC1100064351/a0f4f7e4/port-trunk-allow-pass-vlan?utm_source=chatgpt.com

Justification#2

Source: Huawei Lecture # 8 Slide # 23

Basic Trunk Interface Configuration Commands

1. Set the link type of an interface.

```
[Huawei-GigabitEthernet0/0/1] port link-type trunk
```

In the interface view, set the link type of the interface to trunk.

2. Add the trunk interface to specified VLANs.

```
[Huawei-GigabitEthernet0/0/1] port trunk allow-pass vlan { [ vlan-id1 [ to vlan-id2 ] ] | all }
```

In the interface view, add the trunk interface to specified VLANs.

3. (Optional) Configure a default VLAN for the trunk interface.

```
[Huawei-GigabitEthernet0/0/1] port trunk pvid vlan vlan-id
```

In the interface view, configure a default VLAN for the trunk interface.

329. Hybrid port can be connected to user host and other switches.

- A. True
- B. False

Answer: A

Justification#1

Hybrid ports are switch ports that connect a local switch to hosts and to other switches.

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100149308/AEJ0713J/17/resources/admin/sec_admin_network_vlan_0002.html

Justification#2

Source: Huawei Lecture # 8 Slide # 17

- **Hybrid interface**

Similar to a trunk interface, a hybrid interface also allows frames that belong to multiple VLANs to pass through and differentiates the frames using the 802.1Q tag. You can determine whether to allow a hybrid interface to carry VLAN tags when sending the frames of one or more VLANs.

330. In which working state of the STP protocol can the port be changed to the Forwarding state without going through other states?

- A. Blocking
- B. Learning
- C. Listening
- D. Disabled (Correct) for edge-port in RSTP

Answer: B

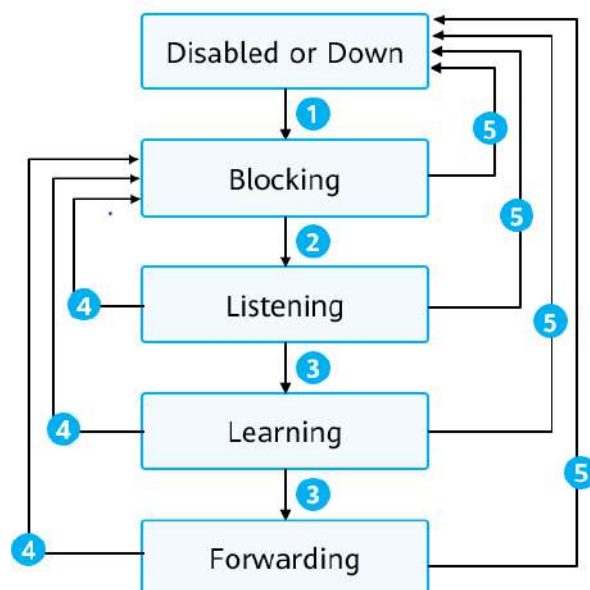
Justification#1

Port State	Description
Learning	This is a transitional state. In Learning state, a loop-free MAC address forwarding table is established. The port does not forward user traffic.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100410551/e99e1364/stp-port-roles-and-port-states>

Justification#2

Source: Huawei Lecture # 9 Slide # 33



331. Switches running STP protocol, the port needs to wait for the forwarding delay in the Learning state before it can be transformed into the Forwarding state.

- A. True
- B. False

Answer: A

Justification#1

The newly selected root port and designated port must wait for two Forward Delay intervals before transitioning to the Forwarding state.

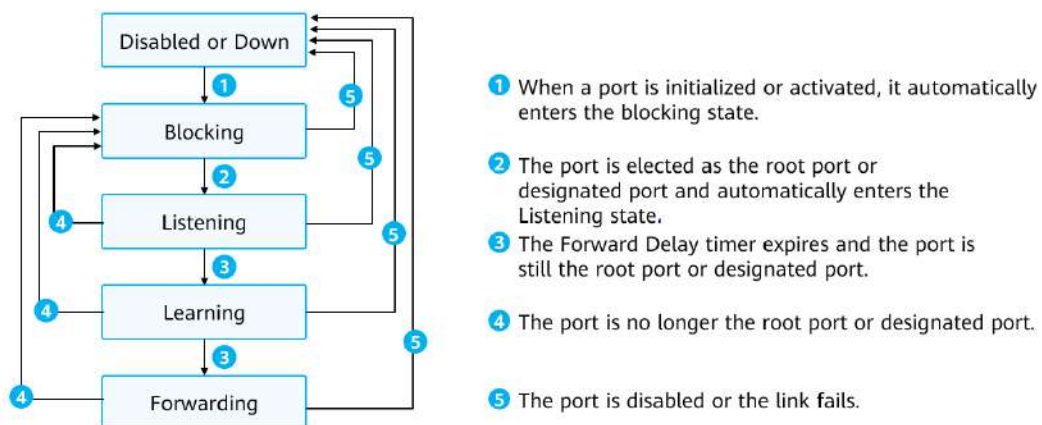
Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100092143>

Justification#2

Source: Huawei Lecture # 9 Slide # 33

STP Port State Transition



332. In order to realize the network layer function on the VLANIF interface, what need to be configured on the VLANIF interface? (Multiple choice)

- A. MAC address
- B. Subnet mask
- C. IP prefix
- D. IP address

Answer: BD

Justification#1

Run `ip address ip-address { mask | mask-length } [ sub ]`

An IP address is assigned to the VLANIF interface for communication at the network layer.

Source:

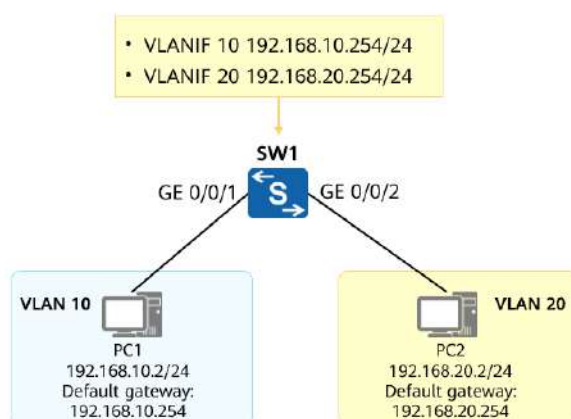
https://info.support.huawei.com/hedex/api/pages/EDOC1100277644/AEM10221/04/resources/vrp/dc_vrp_vlan_cfg_0013.html

Justification#2

Source: Huawei Lecture # 10, Slide # 14



Example for Configuring VLANIF Interfaces



- **Configuration Requirements**
Configure VLANs 10 and 20 for the interfaces connecting to PC1 and PC2, respectively. Configure the Layer 3 switch to allow the two PCs to communicate with each other.

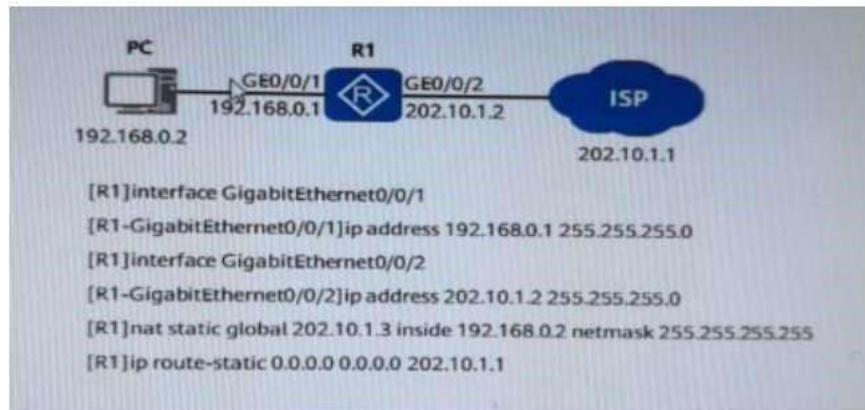
Basic configurations:

```
[SW1]vlan batch 10 20
[SW1] interface GigabitEthernet 0/0/1
[SW1-GigabitEthernet0/0/1] port link-type access
[SW1-GigabitEthernet0/0/1] port default vlan 10
[SW1] interface GigabitEthernet 0/0/2
[SW1-GigabitEthernet0/0/2] port link-type access
[SW1-GigabitEthernet0/0/2] port default vlan 20
```

Configure VLANIF interfaces:

```
[SW1]interface Vlanif 10
[SW1-Vlanif10]ip address 192.168.10.254 24
[SW1]interface Vlanif 20
[SW1-Vlanif20]ip address 192.168.20.254 24
```

333. As shown in the figure, a static NAT command is deployed on router R1. When the PC accesses the Internet, the destination address in the data packet will not change.



- A. True
- B. False

Answer: A

Justification

Source: Huawei Lecture # 14, Slide # 11

Configuring Static NAT

1. Method 1: Configure static NAT in the interface view.

```
[Huawei-GigabitEthernet0/0/0] nat static global { global-address } inside { host-address }
```

global { global-address } is used to configure an external public IP address, and **inside { host-address }** is used to configure an internal private IP address.

2. Method 2: Configure static NAT in the system view.

```
[Huawei] nat static global { global-address } inside { host-address }
```

The command format in the system view is the same as that in the interface view. After this configuration, enable static NAT on a specific interface.

```
[Huawei-GigabitEthernet0/0/0] nat static enable
```

This command enables static NAT on the interface.

Explanation:

- NAT Translation: The NAT (Network Address Translation) configured on R1 is translating the private IP address of the PC (192.168.0.2) to a public IP address (202.10.1.3). The relevant NAT command is `nat static global 202.10.1.3 inside 192.168.0.2 netmask 255.255.255.255`, which specifies that traffic from 192.168.0.2 will appear to come from 202.10.1.3 when going to the ISP.
- Destination Address: While the source address of packets from the PC (192.168.0.2) will be translated to the global address (202.10.1.3), the destination address will remain the same as it was set by the PC when initiating the connection. The NAT process does not modify the destination address; it only changes the source address for outgoing packets and the destination address for incoming packets back to the PC.

334. A DHCP server is deployed in the network, but the administrator found that some hosts did not correctly obtain the address specified by the DHCP server. What are the possible reasons? (Multiple choice)

- A. The address pool of the DHCP server has all been allocated
- B. Some hosts cannot communicate with the DHCP server normally, and the client systems of these hosts automatically generate addresses in the range of 169.254 0.0
- C. There is another DHCP server in the network that works more efficiently
- D. Some hosts cannot communicate with the DHCP server normally, and the client systems of these hosts automatically generate addresses in the range of 127.254.0.0

Answer: ABC

Justification

- **A:** Address pool exhaustion is a documented cause ("IP addresses in the pool have been used up").
- **B:** Confirmed—hosts generate 169.254.x.x (APIPA) if DHCP fails ("client auto-configures an address in 169.254.0.0/16").
- **C:** Competing DHCP servers are flagged as a conflict source ("another server responds faster").
- **D:** Invalid—Huawei does not mention 127.254.0.0 (APIPA uses 169.254.0.0/16).

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100034071/e0653f04/a-client-fails-to-obtain-an-ip-address>

335. On the AC, you can manually specify the source address or source interface of the CAPWAP tunnel.

- A. True
- B. False

Answer: A

Justification#1

1. Run the capwap source interface vlanif vlan-id command to specify a VLANIF interface as the source interface or run the capwap source ip-address ip-address command to specify an IP address of the VLANIF interface as the source IP address.

If a source interface or a source IP address has been configured on the device, you must run the undo capwap source interface or undo capwap source ip-address command first before configuring a new source interface or a source IP address.

After the undo capwap source interface or undo capwap source ip-address command is executed, all APs get offline on the AC. Therefore, exercise caution before running the command.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100064365/89b2f9cc/configuring-a-source-interface-or-source-address>

Justification#2

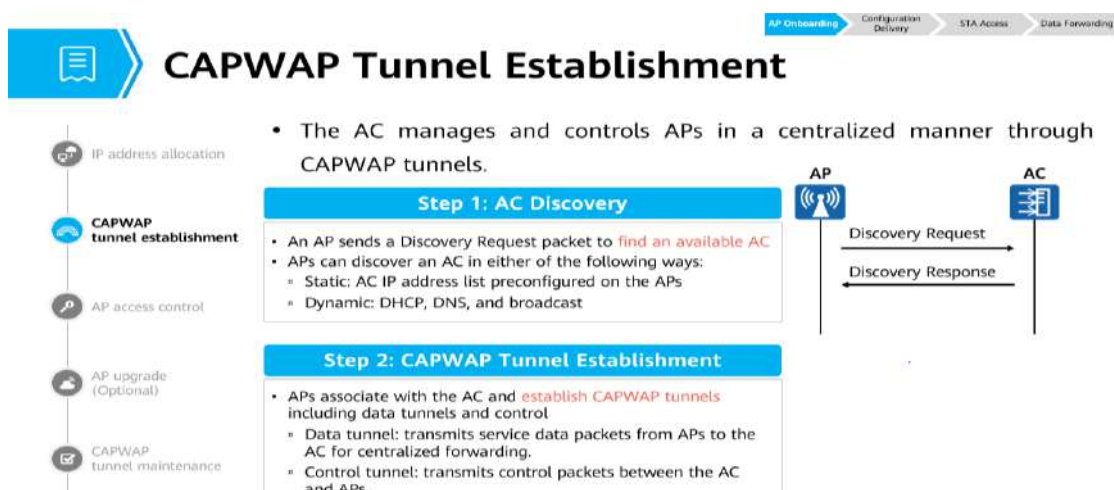
A maximum of one IPv4 address and one IPv6 address can be configured for the CAPWAP source interface or as the source address.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100096312/21011940/capwap-source-interface>

Justification#3

Source: Huawei Lecture # 16 Slide # 30



336. AC uses CAPWAP control tunnel to transmit management messages to the FIT AP.

- A. True
- B. False

Answer: A

Justification#1

The AC sends management packets over the control tunnel to manage APs in a centralized manner. Data packets of users are all forwarded to the AC for centralized processing through the data tunnel. To improve link reliability and prevent CAPWAP control tunnels from being terminated when the service traffic volume is high, configure a high priority for CAPWAP management packets.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100212081/f9d80d43/configuring-capwap-tunnel-parameters>

Justification#2

Source: Huawei Lecture # 16 Slide # 16 (notes section)

- To meet the requirements of large-scale networking and unified management of APs on the network, the Internet Engineering Task Force (IETF) sets up a CAPWAP Working Group and formulates the CAPWAP protocol. This protocol defines how an AC manages and configures APs. That is, CAPWAP tunnels are established between the AC and APs, through which the AC manages and controls the APs.
- CAPWAP is an application-layer protocol based on UDP transmission.
 - CAPWAP functions in the transmission of two types of messages:
 - Data messages, which encapsulate wireless data frames through the CAPWAP data tunnel.
 - Control messages, which are exchanged for AP management through the CAPWAP control tunnel.
 - CAPWAP data and control packets are transmitted on different UDP ports:
 - UDP port 5246 for transmitting control packets
 - UDP port 5247 for transmitting data packets

337. What is the coverage area of an AP in WLAN?

- A. BSS
- B. SSID
- C. BSSID
- D. ESS

Answer: A

Justification#1

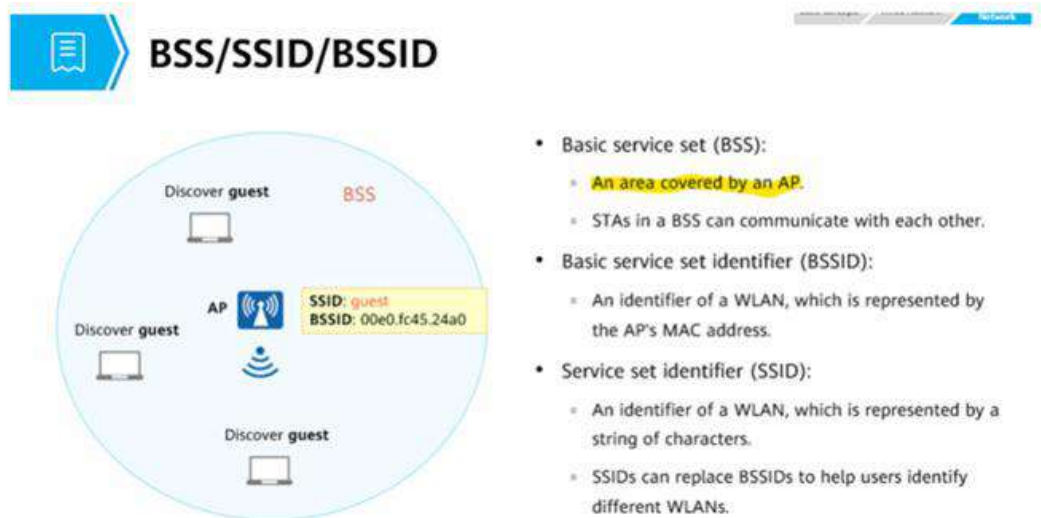
- Basic service set (BSS): an area covered by an AP. STAs in a BSS can communicate with each other.

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100331202/FEM1012J/08/resources/en-us_concept_0000001172291501.html

Justification#2

Source: Huawei Lecture # 16, Slide # 22



338. FIT AP can work independently without relying on AC.

- A. True
- B. False

Answer: B

Justification#1

Fat APs can implement such functions independently, whereas Fit APs must interwork with the WLAN Access Controller (AC).

Source: <https://e.huawei.com/ph/blogs/enterprise-networking/wifi6/201903011058>

Justification#2

Source: Huawei Lecture # 16 Slide # 11 (notes section)

- Fit AP: applies to medium- and large-sized enterprises. Fit APs are managed and configured by the AC in a unified manner, provide various functions, and have high requirements on network maintenance personnel's skills. Fit APs must work with an AC for user access, AP going-online, authentication, routing, AP management, security, and QoS.

339. In the CAPWAP tunnel establishment phase, after the AC receives the Join Request message sent by the AP, the AC will check the validity of the AP. Which of the following is the authentication method supported by the AC? (Multiple choice)

- A. SN certification
- B. Password authentication
- C. MAC authentication
- D. Not certified

Answer: ACD

Justification#1

AP authentication includes non-authentication, MAC address authentication, and SN authentication

Source:

https://info.support.huawei.com/network/ptmngsys/Web/tsrev_wlan/en/content/wlan/11_edesk_ap_online_fail/edesk_ap_online_fail_edesk004.html

Justification#2

Source: Huawei Lecture # 16 Slide # 34

AP Access Control

- After discovering and AC, the AP sends a Join Request packet to the AC. The AC then determines **whether to allow the AP access** sends a Join Response packet to the AP.
- The AC supports three AP authentication modes: MAC address authentication, SN authentication, and non-authentication.

Notes:

"Non-authentication" (or "not certified") is a valid operational mode where no validation is performed, it is not considered an authentication method—it simply bypasses security checks. The question specifically asks for supported authentication methods, so only A (SN) and C (MAC) qualify.

340. Which of the following statements about the S field in the MPLS packet header are correct? (Multiple choice)

- A. s bit has only 1 bit in frame mode and 2 bit in cell mode
- B. s bit exists in every MPLS packet header
- C. It is used to mark whether there are other labels behind this label, 1 means yes, 0 means no
- D. It is used to mark whether there are other labels behind this label, 0 means yes, 1 means no

Answer: BD

Justification



An MPLS label header contains the following fields:

- Label: a 20-bit field that identifies a label value.
- Exp: a 3-bit field used for extension. This field is used by the class of service (CoS) function, which is similar to Ethernet 802.1p.
- BoS: a 1-bit field that identifies the bottom of a label stack. MPLS supports multiple labels, which are stacked. If the BoS field of a label is set to 1, the label is at the bottom of the label stack.
- TTL: an 8-bit field indicating a time to live (TTL) value. This field implements the same function as the TTL field in IP packets.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100118961>

Explanation:

- The S-bit (1 bit) is present in every MPLS label header.
- S=1 indicates the bottom of the stack (no more labels); S=0 means additional labels follow.
- Option A falsely claims the S-bit size varies (e.g., 2 bits in cell mode), it is always 1 bit.
- Option D inverts the standard logic (S=1 = last label). Huawei's text strictly adheres to RFC 3032's definition.

341. The network management workstation manages network equipment through the SNMP protocol. When an abnormality occurs in the managed device, what kind of SHMP message will the network management workstation receive?

- A. trap message
- B. get-response message
- C. set-request message
- D. get-request message

Answer: A

Justification#1

Unsolicited (asynchronous) notifications can be generated as traps or inform requests. Traps are messages alerting the SNMP manager to a condition on the network.

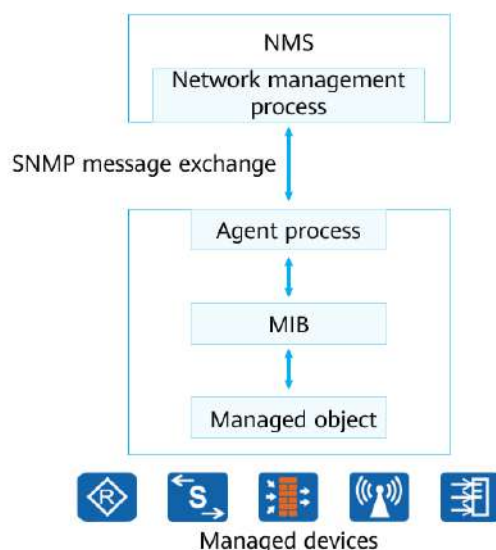
Source:

https://learningnetwork.cisco.com/s/article/Understanding-Simple-Network-Management-Protocol--SNMP?utm_source=chatgpt.com

Justification#2

Source: Huawei Lecture # 18, Slide # 15

SNMP Management Model



- **Query/Modify operation:**
 - The NMS sends an SNMP request message to an agent process.
 - The agent process searches the MIB on the device for information to be queried or modified and sends an SNMP response message to the NMS.
- **Trap operation:**
 - If the trap triggering conditions defined for a module are met, the agent process sends a message to notify the NMS that an event or trap has occurred on a managed object. This helps network administrators promptly process network faults.

342. What type of IPv6 address FE80: 2E0: FCFF: FE6F: 4F36 belongs to?

- A. Multicast address
- B. Link local address
- C. Global semicast address
- D. Anycast address

Answer: B

Justification#1

A Link-Local address is an IPv6 unicast address that can be automatically configured on any interface that uses the Link-Local prefix FE80::/10 (1111 1110 10)

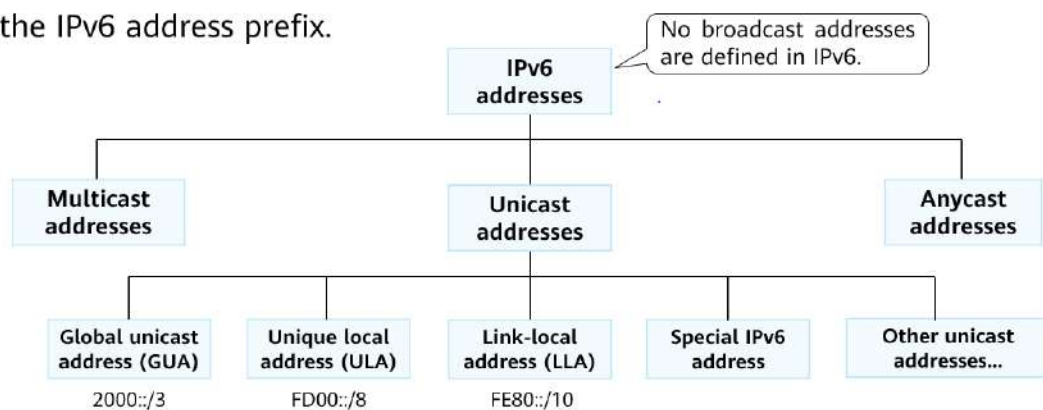
Source: https://www.cisco.com/c/en/us/support/docs/ip/ip-version-6-ipv6/113328-ipv6-lla.html?utm_source=chatgpt.com

Justification#2

Source: Huawei Lecture # 19, Slide # 15

IPv6 Address Classification

- IPv6 addresses are classified into unicast, multicast, and anycast addresses according to the IPv6 address prefix.



343. The origin of SDN is the separation of transfer and control. The separation of transfer and control is a way to realize SDN, not the essence.

- A. True
- B. False

Answer: A

Justification

Source: Huawei Lecture # 20 Slide # 18



Essential Requirements of SDN

- The essence of SDN is to make networks more open, flexible, and simple. It builds a centralized brain for a network and implements fast service deployment, traffic optimization, or network service openness through centralized control in the global view.
-

344. Which of the following IEEE802.11 standards only supports 2.4GHz frequency band for communication?

- A. 802.11a
- B. 802.11g
- C. 802.11ax
- D. 802.11n

Answer: B

Justification

Table 8-3 802.11 protocols

Protocol	Frequency Band (GHz)
802.11b	2.4
802.11a	5
802.11g	2.4
802.11n	2.4 and 5
802.11ac	5
802.11ax	2.4, 5, and 6
802.11be	2.4, 5, and 6

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000113315/bd9557ad/80211-protocols>

345. Through SR (Segment Routing), an explicit path can be easily defined. The nodes in the network only need to maintain the SegmentRouting information to cope with the real-time rapid development of the business. What are the following characteristics of Segment Routing? (Multiple choice)

- A. By extending the existing protocol (e.g, IGP), the existing network can better and smoothly evolve
- B. SR (Segment Routing) adopts IP forwarding, and no additional maintenance is required for another label forwarding table
- C. Adopting source routing technology to provide the ability to quickly exchange text between the network and upper-layer applications
- D. Support the centralized control mode of the controller and the distributed control mode of the transponder at the same time, and provide a balance between centralized control and distributed control

Answer: ACD

Justification

- Existing protocols are extended, enabling the network to smoothly evolve.
- The ingress controls and adjusts service paths through the source routing technology, enabling the network to quickly respond to the requirements of upper-layer applications.
- Balancing centralized control and distributed control/forwarding prevents controller performance from becoming a service bottleneck.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100092117>

346. The MPLS label header is encapsulated between the data link layer header and the network layer header of the message.

- A. True
- B. False

Answer: A

Justification

For MPLS to work, a label needs to be added to the packet. The label is added as a shim header between the Layer 2 Frame Header and the Layer 3 Packet Header.

Source:

<https://community.cisco.com/t5/networking-knowledge-base/understanding-multi-protocol-label-switching-mpls/ta-p/5053726>

347. If the MAC address of an interface is 00EO-FCEF-OFEC, what is its corresponding EUI-64 address?

- A. 00EO-FCEF-FFFE-OFEC
- B. 02EO-FCFF-FEEF-0FEC
- C. 00EO-FCFF-FEEF-0FEC
- D. 02EO-FCEF-FFFE-0FEC

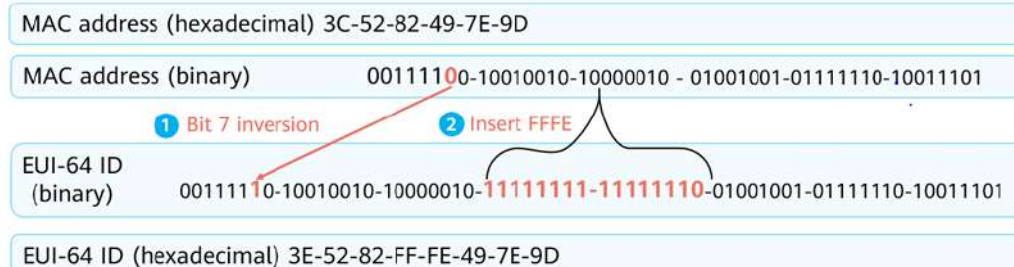
Answer: B

Justification

Source: Huawei Lecture # 19, Slide # 17

Interface ID of an IPv6 Unicast Address

- 3 methods to generate an interface ID:
 - Manual configuration
 - Automatic generation by the system
 - Using the IEEE 64-bit extended unique identifier (EUI-64) standard
- EUI-64 is most commonly used. It converts the MAC address of an interface into an IPv6 interface ID.



348. Which of the following descriptions of static and dynamic routing is wrong?

- A. Dynamic routing protocols take up more system resources than static routing protocols.
- B. After the link fails, the static route can automatically complete the network convergence.
- C. After the administrator deploys the dynamic routing protocol in the corporate network, later maintenance and expansion can be more convenient
- D. Static routing is simple to configure and easy to manage when applied in the enterprise

Answer: B

Justification#1

The disadvantage of static routes is that they cannot automatically adapt to network topology changes. Therefore, static routes require subsequent maintenance.

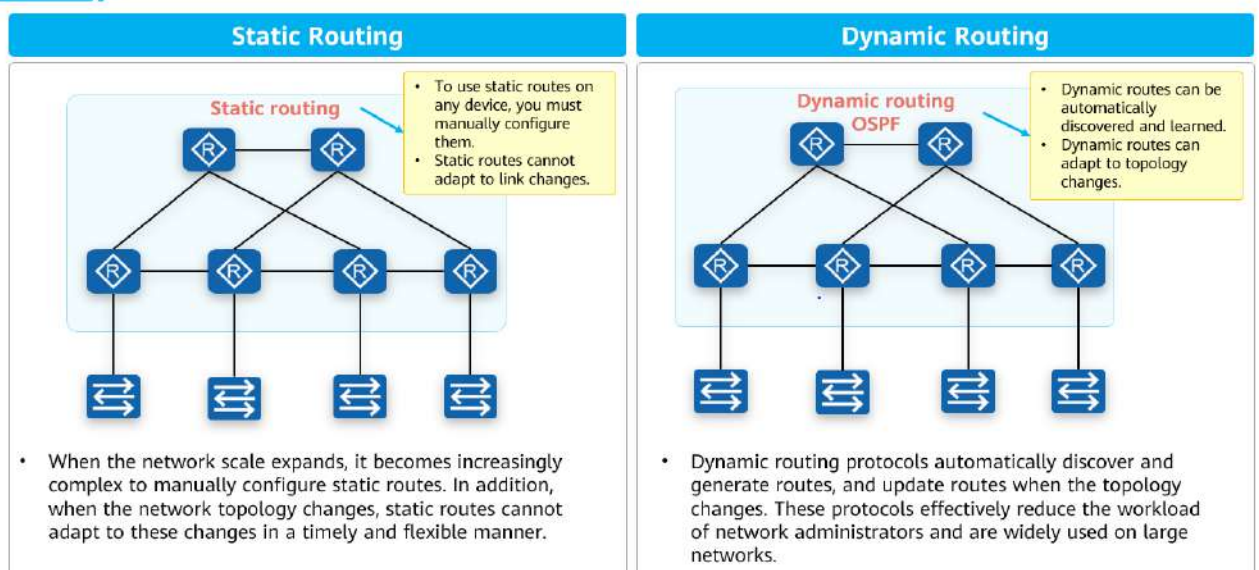
Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100149308/AEJ0713J/18/resources/admin/sec_admin_router_basic_0003.html

Justification#2

Source: Huawei Lecture # 5, Slide # 34

Overview of Dynamic Routing



349. Which of the following descriptions of OSPF areas are correct? (Multiple choice)

- A. All networks should be declared in zone 0
- B. The number of the backbone area cannot be 2
- C. Before configuring the OSPF area, an IP address must be configured for the loopback interface of the router
- D. The number range of the area is from 0.0.0.0 to 255.255.255.255

Answer: BD

Justification#1

B) The backbone area (Area 0) is the core of an OSPF network

D) Entering an area ID of 0 or 0.0.0.0 automatically joins the routing switch to the backbone area. The maximum area ID value is 255.255.255.254 (4,294,967,294.)

Source:

B) <https://docs.paloaltonetworks.com/pan-os/11-1/pan-os-networking-admin/ospf/ospf-concepts/ospf-areas>

D) <https://arubanetworking.hpe.com/techdocs/AOS-S/16.11/MRG/KB/content/kb/cnf-osp-bac-nor-are.htm>

Justification#2

Source: Huawei Lecture # 6 Slide # 35 (notes section), 39

- Types of areas: Areas can be classified into backbone areas and non-backbone areas. Area 0 is a backbone area. All areas except area 0 are called non-backbone areas.

[Huawei] **area** *area-id*

The **area** command creates an OSPF area and displays the OSPF area view.

The *area-id* value can be a decimal integer or in dotted decimal notation. If the value is an integer, it ranges from 0 to 4294967295.

350. The Layer 2 Ethernet switch generates entries in the MAC address table according to the () of the Ethernet frame received by the port.

- A. Source IP address
- B. Source MAC address
- C. Destination MAC address
- D. Destination IP address

Answer: B

Justification#1

When receiving a data frame, the switch records the source MAC address of the data frame and the interface that receives the data frame in a MAC address entry.

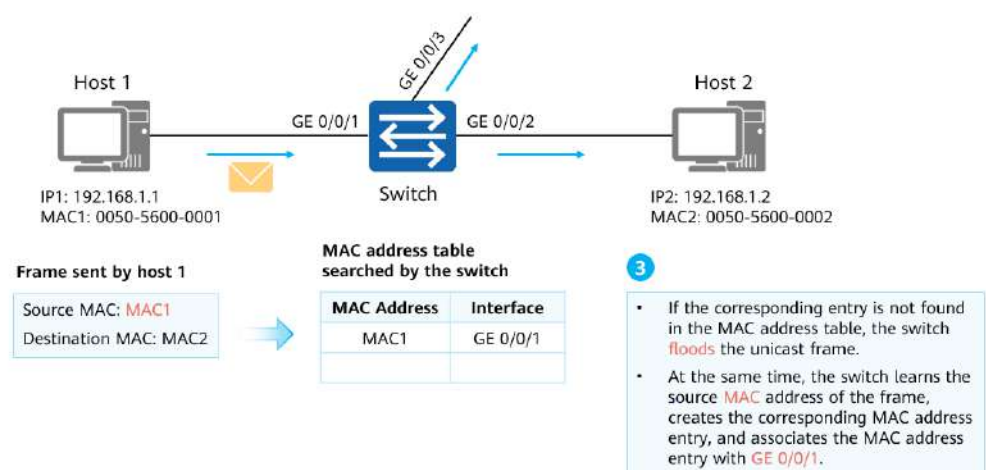
Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178168/e48192c/disabling-mac-address-learning>

Justification#2

Source: Huawei Lecture # 7 Slide # 31

MAC Address Learning on a Switch (3)



351. The MAC address table of existing switches is shown in the figure. Which of the following statement is correct?

```
<Huawei>display mac-address
MAC address table of slot 0:
```

MAC Address	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR-ID MAC-Tunnel
5489-9811-0b49	1	-	-	Eth0/0/3	static	-

Total matching items on slot 0 displayed = 1

```
MAC address table of slot 0:
```

MAC Address	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR-ID MAC-Tunnel
5489-989d-1d30	1	-	-	Eth0/0/1	dynamic	0/-
5489-9885-18a8	1	-	-	Eth0/0/2	dynamic	0/-

Total matching items on slot 0 displayed = 2

- A. The MAC address 5489-9885-18a8 is manually configured by the administrator.
- B. The MAC address 5489-9811-0b49 is manually configured by the administrator.
- C. The MAC address 5489-989d-1d30 is manually configured by the administrator.
- D. After the switch restarts, all MAC addresses need to be relearned.

Answer: B

Justification

This is because 5489-9811-0b49 has "static" listed under the "Type" column.

The other MAC addresses in the table, 5489-989d-1d30 and 5489-9885-18a8, have "dynamic" listed under the "Type" column, which means they were learned dynamically by the switch.

. Verify the configuration.

Run the `display mac-address static vlan 2` command in any view to check whether static MAC address entries were su

```
[Switch] display mac-address static vlan 2
```

MAC Address	VLAN/VSI	Learned-From	Type
xxxx-xxxx-xxx2	2/-	GE0/0/1	static
xxxx-xxxx-xxx4	2/-	GE0/0/2	static

Total items displayed = 2

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000141870/f355890a/example-for-configuring-static-mac-address-entries>

352. When the switch port sends a data frame carrying the same VLAN TAG and PVID, it must strip the VLAN TAG and forward it.

- A. True
- B. False

Answer: A

Justification#1

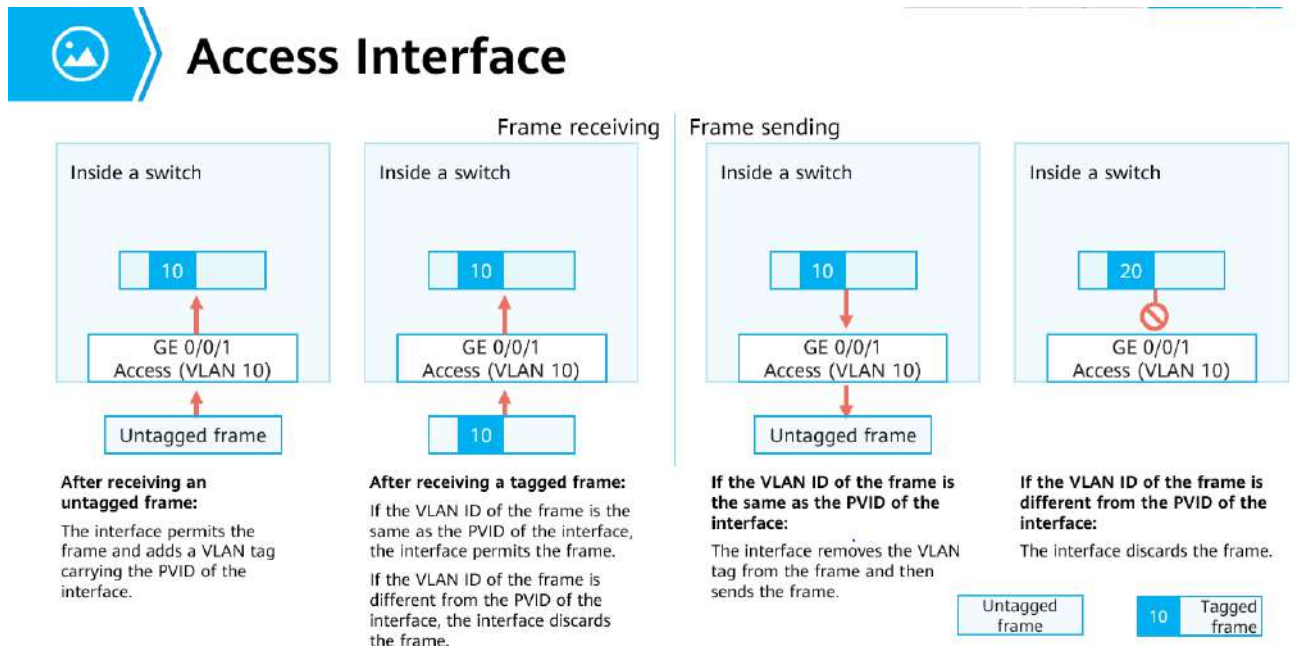
An access interface directly removes VLAN tags from frames before sending the frames.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178168/f9b8a34a/adding-and-removing-vlan-tags>

Justification#2

Source: Huawei Lecture # 8 Slide # 18



353. Which of the following levels is not a common network level in a mid-sized campus network architecture?

- A. Access layer
- B. Core layer
- C. Convergence layer
- D. Network layer

Answer: D

Justification#1

A medium-sized campus network can use a three-layer intranet networking architecture that consists of the core layer, aggregation layer, and access layer.

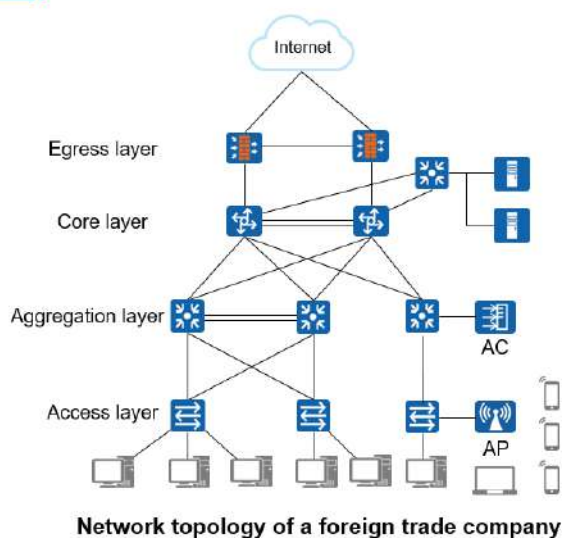
Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100331202/FEM1012J/08/resources/en-us_topic_0000001489814749.html

Justification#2

Source: Huawei Lecture # 22 Slide # 8

Typical Architecture of Midsize Campus Networks



- A midsize campus network supports access of hundreds to thousands of users.
- The modular design is introduced to midsize campus networks, that is, the networks can be partitioned by function. However, the number of function modules is small. In most cases, a midsize campus network is flexibly partitioned based on service requirements.

- Characteristics of midsize campus networks:

- Midsize network scale
- Most commonly used
- Function partition

Number of terminals	200 to 2000
Number of NEs	25 to 100

- Typical three-layer network architecture: core, aggregation, and access

354. UDP cannot guarantee the reliability of data transmission, and does not provide the functions of message sequencing and flow control. It is suitable for flows with low transmission reliability requirements but high transmission speed and delay requirements.

- A. True
- B. False

Answer: A

Justification

Advantages of UDP:

Speed: UDP is faster than TCP because it does not have the overhead of establishing a connection and ensuring reliable data delivery.

Disadvantages of UDP:

No reliability: UDP does not guarantee delivery of packets or order of delivery, which can lead to missing or duplicate data.

Source:

Geeksforgeeks website: <https://www.geeksforgeeks.org/user-datagram-protocol-udp/>

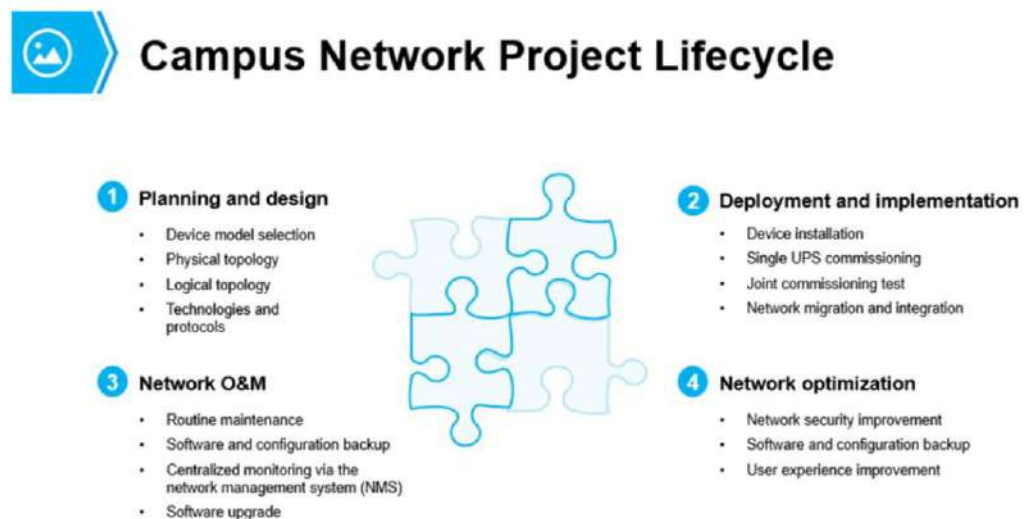
355. Which stage of the life cycle of campus network construction is the beginning of a project?

- A. Network optimization
- B. Network implementation
- C. Planning and design
- D. Network operation and maintenance

Answer: C

Justification

Source: Huawei Lecture # 22 Slide # 13



Source:

<https://forum.huawei.com/enterprise/intl/en/thread/campus-network-building-cases-and-practices/667246826295803905?blogId=667246826295803905>

356. When the router is powered on, it will read the configuration file from the default storage path to initialize the router. If there is no configuration file in the default storage path, what will the router use for initialization?

- A. Initial configuration
- B. Current configuration
- C. New configuration
- D. Default parameters

Answer: D

Justification

When a device is powered on, it reads the preset default configuration file from the default storage path to initialize the system. The configurations in this configuration file are called preset configurations. If no preset default configuration file exists in the default storage path, the device uses the factory configurations for initialization.

Source:

Huawei Website

<https://support.huawei.com/enterprise/en/doc/EDOC1100335705/c80a8cf/configuration-filemanagement-configuration>

357. In the OSPF protocol, which of the following descriptions of DR is correct? (Multiple choice)

- A. If the priority values of the two routers are equal, the router with the larger Router ID is selected as the DR
- B. By default, all routers in this broadcast network will participate in the DR election
- C. If the priority values of the two routers are different, select the router with the lower priority as the DR
- D. DR and BDR should also establish an adjacency relationship

Answer: ABD

Justification#1

- A. During DR/BDR election, if two routers use the same DR priority, they need to compare their router IDs, and the router with a larger router ID will become the DR or BDR.
- B. A priority of zero removes a router from the election.
- D. and an adjacency relationship is established between the DR and the BDR.

Source:

- A. <https://support.huawei.com/enterprise/en/doc/EDOC1100301642/32a400ec/ospf-fundamentals>
- B. <https://community.cisco.com/t5/switching/dr-and-bdr-in-ospf/td-p/2716502>
- D. <https://forum.huawei.com/enterprise/intl/en/thread/application-scenarios-for-routing-policies-1/667223541210300416?blogId=667223541210300416>

Justification#2

Only the DR and BDR can establish adjacencies with other OSPF routers; the others don't do this, and they are in the 2-way state.

Source: Huawei Lecture # 6 Slide # 32, 33 (notes section)

- On an MA network, if each OSPF router establishes OSPF adjacencies with all the other routers, excessive OSPF adjacencies exist on the network, which increases the load on the devices and the number of OSPF packets flooded on the network.

- Election rule: The interface with a higher OSPF DR priority becomes the DR of the MA. If the priorities (default value of 1) are the same, the router (interface) with a higher OSPF router ID is elected as the DR, and the DR is non-preemption.

358. Which of the following fields is not included in the HELLO packet of the OSPF protocol?

- A. Neighbor
- B. Network Mask
- C. HelloInterval
- D. Sysname

Answer: D

Justification#1

Table 2 *Hello packet fields*

Field	Length	Description
Network Mask	32 bits	Mask of the network on which the interface that sends the Hello packet resides.
HelloInterval	16 bits	Interval at which Hello packets are sent.
Neighbor	32 bits	Router ID of the neighbor.

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100277644/AEM10221/04/resources/vrp/feature_0003998103.html

Justification#2

Source: Huawei Lecture # 6 Slide # 18

OSPF Packet Types

- There are five types of OSPF protocol packets and implement different functions in interaction between OSPF routers.

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.

359. If the network address of a network is 10.1.1.0/30, what is its broadcast address?

- A. 10.1.1.2
- B. 10.1.1.3
- C. 10.1.1.4
- D. 10.1.1.1

Answer: B

Justification

Source: Huawei Lecture # 4 Slide # 20

IP Address Types

- A network range defined by a network ID is called a network segment.

- **Network address:** identifies a network.

Example: 192.168.10.0/24

192.	168.	10.	00000000
------	------	-----	----------

- **Broadcast address:** a special address used to send data to all hosts on a network.

Example: 192.168.10.255/24

192.	168.	10.	11111111
------	------	-----	----------

- **Available addresses:** IP addresses that can be allocated to device interfaces on a network.

Example: 192.168.10.1/24

192.	168.	10.	00000001
------	------	-----	----------

Note

- Network and broadcast addresses cannot be directly used by devices or their interfaces.
- Number of available addresses on a network segment is $2^n - 2$ (n is the number of bits in the host part).

Explanation:

The subnet mask /30 corresponds to a subnet mask of 255.255.255.252 in dotted decimal notation. This means:

- The network address is 10.1.1.0
- The subnet allows for 4 IP addresses in total ($2^{(32-30)} = 4$), but 2 of them are reserved for network and broadcast addresses.

In a /30 subnet:

- 10.1.1.0: Network address
- 10.1.1.3: Broadcast address

360. If the broadcast address of a network is 172.16.1.255, what might its network address be?

- A. 172.16.2.0
- B. 172.16.1.128
- C. 172.16.1.1
- D. 172.16.1.253

Answer: B

Justification

Source: Huawei Lecture # 4 Slide # 20



IP Address Types

- A network range defined by a network ID is called a network segment.
- Network address:** identifies a network.
Example: 192.168.10.0/24

192.	168.	10.	00000000
------	------	-----	----------
- Broadcast address:** a special address used to send data to all hosts on a network.
Example: 192.168.10.255/24

192.	168.	10.	11111111
------	------	-----	----------
- Available addresses:** IP addresses that can be allocated to device interfaces on a network.
Example: 192.168.10.1/24

192.	168.	10.	00000001
------	------	-----	----------

Note

- Network and broadcast addresses cannot be directly used by devices or their interfaces.
- Number of available addresses on a network segment is $2^n - 2$ (n is the number of bits in the host part).

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Explanation:

If the broadcast address is 172.16.1.255, the subnet is likely /25 (255.255.255.128). In this case, the network range is 172.16.1.128 – 172.16.1.255, making the network address 172.16.1.128.

361. When the IPv4 protocol is used as the network layer protocol, which field is not included in the network layer header?

- A. Source IPv4 address
- B. Sequence number
- C. Destination IPv4 address
- D. TTL

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100334399/114a4ceb/i/%20pv4-basic-configuration>

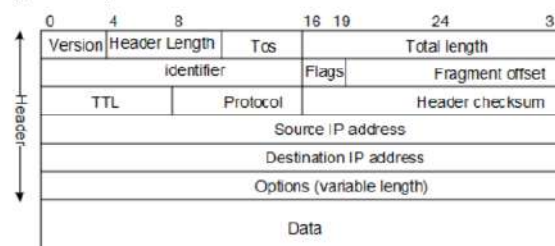
Table 2-4 Meaning of each field in an IPv4 datagram

Field	Length	Meaning
Version	4 bits	IP version, which can be IPv4 or IPv6.
Header Length	4 bits	Length of the IPv4 header.
Type of Service	8 bits	This field takes effect only in the differentiated service model.
Total Length	16 bits	Total length of the header and data.
Identification	16 bits	A device maintains a counter on a storage device to record the number of IPv4 datagrams. The counter value increases by 1 each time an IPv4 datagram is sent.
Flags	3 bits	This field has three bits. The most significant bit is reserved and must be set to 0. If the middle bit is 0, the datagram can be fragmented. If the middle bit is 1, the datagram cannot be fragmented. If the least significant bit is 0, the fragment is the last one. If the least significant bit is 1, there are more fragments.
Fragment Offset	13 bits	Location of a fragment in a datagram.
TTL	8 bits	Life span of a datagram on a network. TTL is measured by the number of hops.
Protocol	8 bits	Type of the protocol carried in a datagram.
Header Checksum	16 bits	A device calculates the header checksum for each datagram received. If the checksum is 0, the device knows that the header remains unchanged and retains the datagram. This field checks only the header but not the data.
Source IP Address	32 bits	IPv4 address of a sender.
Destination IP Address	32 bits	IPv4 address of a receiver.
Options	0 to 40 bytes	This field supports various options, such as error correction, measurement, and security. Pad bytes with a value of 0 are added if necessary.
Data	Variable	Pads an IP datagram.

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100334321/AEM1020X/05/resources/dc/dc_fd_IPv4_0006.html

Figure 1 IPv4 packet format



362. When the router forwards a data packet, if it does not match the corresponding detailed route and there is no default route, it will directly discard the data packet.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086956>

Default Routes

Default routes are special routes used only when packets to be forwarded do not match any routing entry in a routing table. If the destination address of a packet does not match any entry in the routing table, the packet is sent through a default route. **If no default route exists and the destination address of the packet does not match any entry in the routing table, the packet is discarded.** An Internet Control Message Protocol (ICMP) packet is then sent, informing the originating host that the destination host or network is unreachable.

In a routing table, a default route is the route to network 0.0.0.0 (with the mask 0.0.0.0). You can run the **display ip routing-table** command to check whether a default route is configured. Generally, administrators can manually configure default static routes. Default routes can also be generated through dynamic routing protocols such as OSPF and IS-IS.

363. What kind of message does the OSPF protocol use to send LSA to neighbors?

- A. LSU
- B. LSACK
- C. LSR
- D. LSA

Answer: A

Justification

Source: Huawei Slide # 06, Page # 18

Packet Name	Function
Hello	Is periodically sent to discover and maintain OSPF neighbor relationships.
Database Description	Describes the summary of the local LSDB, which is used to synchronize the LSDBs of two devices.
Link State Request	Requests a needed LSA from a neighbor. LSRs are sent only after DD packets have been successfully exchanged.
Link State Update	Is sent to advertise a requested LSA to a neighbor.
Link State ACK	Is used to acknowledge the receipt of an LSA.

364. Apply the ACL configured as shown in the figure in Telnet. Only the device with the IP address of 172.16.105.2 is allowed to log in remotely:

```
#
acl number 2000
rule 5 permit source 172.16.105.2 0
#
```

- A. True
- B. False

Answer: A

Justification

Source: Huawei Slide # 12, Page # 13

Here, the Wildcard is 0, and the IP to permit is 172.16.105.2. So true

- Basic ACL

Number range:
2000 to 2999

Source IP address				
IP Header		TCP/UDP Header		Data
acl number 2000				
rule	5	deny	source 10.1.1.1	0
rule	10	deny	source 10.1.1.2	0
rule	15	permit	source 10.1.1.0	0.0.0.255

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086647>

- Example 3: Apply an ACL to Telnet, to allow only the administrator's host (172.16.105.2) to Telnet to the device and reject other users.
The default ACL action of the Telnet module is **deny**, and a few packets are permitted and most packets are denied. Therefore, you only need to configure **rule permit xxx**.

```
#
acl number 2000
rule 5 permit source 172.16.105.2 0
#
```

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178177/d8968298/configuring-a-basic-acl>

Configuring rules for a basic ACL

- **Configuring a packet filtering rule based on the source IP address (host address)**
To allow the packets from a host to pass, add a rule to an ACL. For example, to allow packets from host 192.168.1.3 to pass, create the following rule in ACL 2001.

```
<HUAWEI> system-view
[HUAWEI] acl 2001
[HUAWEI-acl-basic-2001] rule permit source 192.168.1.3 0
```

365. Regarding the corresponding relationship between the access control list number and type, which of the following descriptions is correct?

- A. The basic access control list number range is 1000-2999
- B. The access control list number range of the second layer is 4000-4999
- C. The number range of interface-based access control list is 1000-2000
- D. The range of advanced access control list numbers is 3000-4000

Answer: B

Justification

Source: Huawei Slide # 12, Page # 12

- ACL classification based on ACL rule definition methods

Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
Advanced ACL	3000 to 3999	Defines rules based on source and destination IPv4 addresses, IPv4 protocol types, ICMP types, TCP source/destination port numbers, UDP source/destination port numbers, and effective time ranges.
Layer 2 ACL	4000 to 4999	Defines rules based on information in Ethernet frame headers of packets, such as source and destination MAC addresses and Layer 2 protocol types.
User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

366. What is the number range of the Layer 2 ACL?

- A. 2000 — 2999
- B. 0 — 3999
- C. 4000 — 4999
- D. 6000 — 6031

Answer: C

Justification

Source: Huawei Slide # 12, Page # 12

- ACL classification based on ACL rule definition methods

Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
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User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

367. What is the destination IP address of the DHCP DISCOVER message?

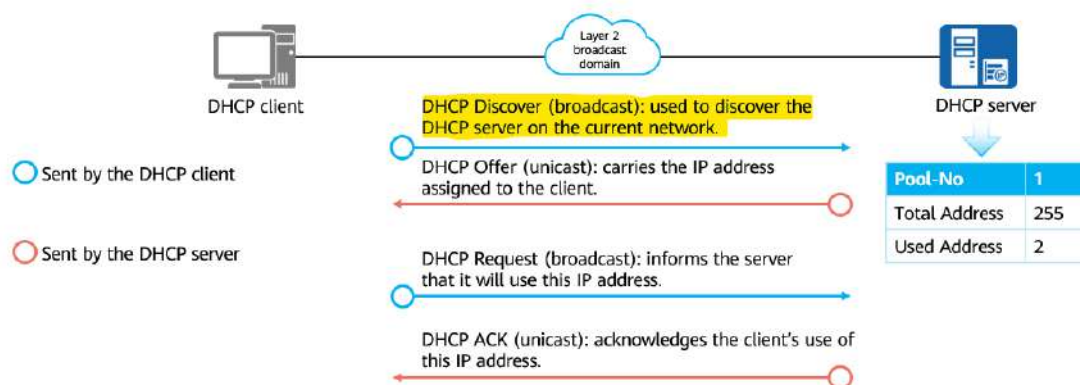
- A. 127.0.0.1
- B. 224.0.0.1
- C. 224.0.0.2
- D. 255.255.255.255

Answer: D

Justification

Source: Huawei Slide # 15, Page # 28

DHCP Working Principle



Source: <https://info.support.huawei.com/info-finder/encyclopedia/en/DHCP.html>

Stage 1: Discovery stage

The newly connected DHCP client does not know the [IP address](#) of the DHCP server. To learn the IP address of the DHCP server, the DHCP client broadcasts a DHCPDISCOVER message with the destination IP address of 255.255.255.255 to all devices on the same network segment. Information carried in a DHCPDISCOVER message includes the client's MAC address ([chaddr field](#)), parameter request list ([Option 55](#)), and broadcast flag ([flags field](#)).

368. In the history of WLAN development, which of the following is not included in the main wireless protocol standards in the primary mobile office era?

- A. 802.11g
- B. 802.11a
- C. 802.11b
- D. 802.11ac

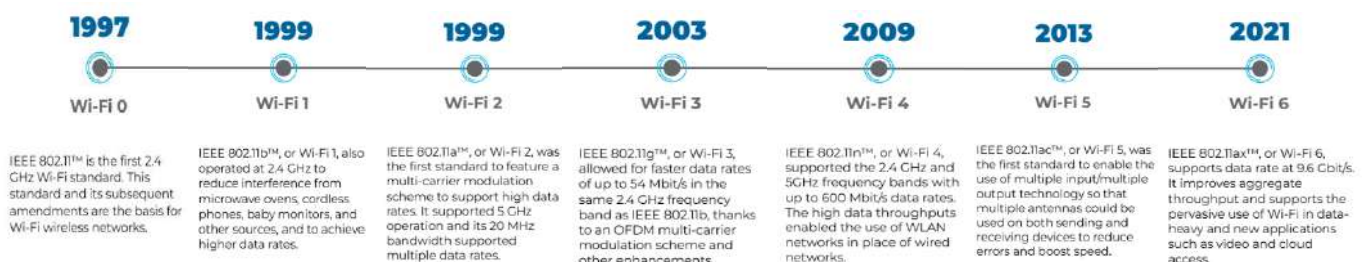
Answer: D

Justification

Source: <https://www.ciscopress.com/articles/article.asp?p=2999384&seqNum=3>

IEEE WLAN Standard	Radio Frequency	Description
802.11	2.4 GHz	Speeds of up to 2 Mbps
802.11a	5 GHz	Speeds of up to 54 Mbps Small coverage area Less effective at penetrating building structures Not interoperable with 802.11b and 802.11g
802.11b	2.4 GHz	Speeds of up to 11 Mbps Longer range than 802.11a Better able to penetrate building structures
802.11g	2.4 GHz	Speeds of up to 54 Mbps Backward compatible with 802.11b with reduced bandwidth capacity
802.11n	2.4 GHz 5 GHz	Data rates ranging from 150 Mbps to 600 Mbps with a distance range of up to 70 m (230 feet) APs and wireless clients require multiple antennas using MIMO technology Backward compatible with 802.11a/b/g devices with limiting data rates
802.11ac	5 GHz	Provides data rates ranging from 450 Mbps to 1.3 Gbps (1300 Mbps) using MIMO technology Up to eight antennas can be supported Backward compatible with 802.11a/n devices with limiting data rates
802.11ax	2.4 GHz 5 GHz	Released in 2019 (latest standard) Also known as High-Efficiency Wireless (HEW) Higher data rates and increased capacity Handles many connected devices Improved power efficiency 1 GHz and 7 GHz capable when those frequencies become available

Source: <https://standards.ieee.org/wp-content/uploads/interactive/web/wi-fi-timeline/index.html>



369. The router running the OSPF protocol can reach the FULL state after completing the LSDB synchronization.

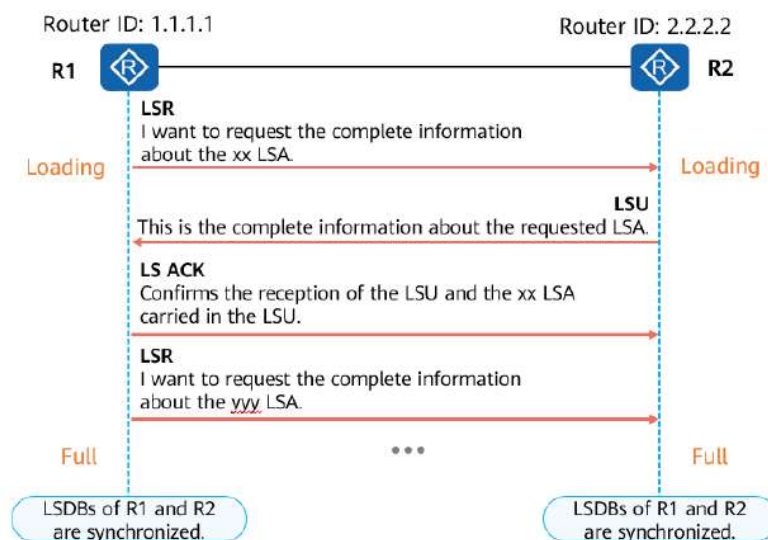
- A. True
- B. False

Answer: A

Justification

Source: Huawei Slide # 06, Page # 2

Process of Establishing an OSPF Adjacency - Step 4



Source: <https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/13685-13.html>

Full

In this state, routers are fully adjacent with each other. All the router and network LSAs are exchanged, and the routers' databases are fully synchronized.

Full is the normal state for an OSPF router. If a router is stuck in another state, it is an indication that there are problems when the adjacencies are formed. The only exception to this is the 2-way state, which is normal in a broadcast network. Routers achieve the FULL state with their DR and BDR in NBMA/broadcast media and the FULL state with every neighbor in the residual media, such as point-to-point and point-to-multipoint.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100301642/32a400ec/ospf-fundamentals>

The OSPF neighbor state machine has the following states:

- **Full:** This state occurs when two neighbors complete their LSDB synchronization and establish an adjacency.

370. In order to distinguish different users and provide different network services on the AP. Which of the following can be configured?

- A. VT
- B. VAP
- C. VAC
- D. WAP

Answer: B

Justification

Source: Huawei Slide # 16, Page # 23

- VAP:
 - A physical AP can be virtualized into multiple VAPs, each of which provides the same functions as the physical AP.
 - Each VAP corresponds to one BSS. In this way, one AP may provide multiple BSSs that can have different SSIDs specified.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100034241/2cc5a91d/configuring-a-virtual-access-point-vap>

Procedure

When a WLAN is available for a user and the user starts to connect to the WLAN, the user actually connects to a virtual access point (VAP). A VAP is a functional entity on an AP. You can create different VAPs on an AP to provide wireless access services for different users so that these users can obtain different network resources. A VAP is also a binding between an AP radio and a service set. A VAP is generated when a service set is bound to an AP radio.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100034241/2cc5a91d/configuring-a-virtual-access-point-vap>

Configuring a VAP

Context

A VAP is a functional entity on an AP. You can create a VAP on a radio by binding a service set to the radio.

When a VAP is delivered to an AP, the service set parameters in the VAP are delivered to the AP. The AP then provides services for users based on the service set parameters.

371. Network equipment running the SNMP protocol can take the initiative to report alarm information so that the network administrator can discover the fault in time.

- A. True
- B. False

Answer: A

Justification

Source: Huawei Slide # 18, Page # 12

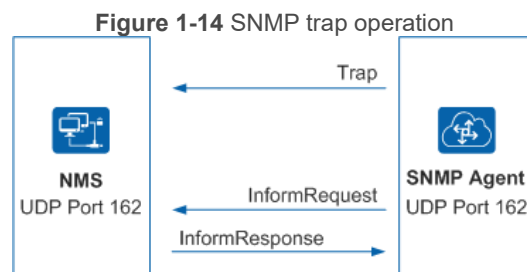
- The NMS and managed devices exchange messages in the following modes:
 - The NMS sends a request for modifying or querying configuration information to a managed device through SNMP. The agent process running on the managed device responds to the request from the NMS.
 - The managed device can proactively report traps to the NMS so that the network administrator can detect faults in a timely manner.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086963>

SNMP Traps

SNMP traps are notification messages sent by an SNMP agent to inform the NMS of alarms or events generated by the device. In this way, the administrator can learn the running status of the device in a timely manner.

There are two types of SNMP traps: trap and inform. SNMPv1 does not support inform. The difference between trap and inform is that, after an SNMP agent sends an alarm or event to the NMS through an InformRequest message, the NMS needs to reply with an InformResponse message, as shown in [Figure 1-14](#).



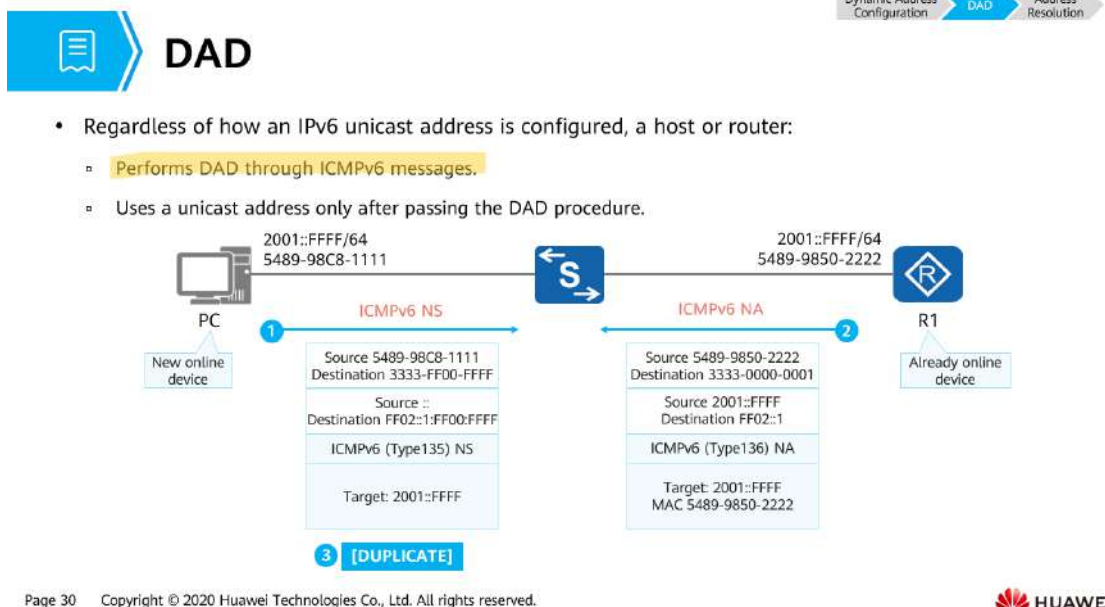
372. The IPv6 protocol uses NS and NA packets for duplicate address detection (DAD).

- A. True
- B. False

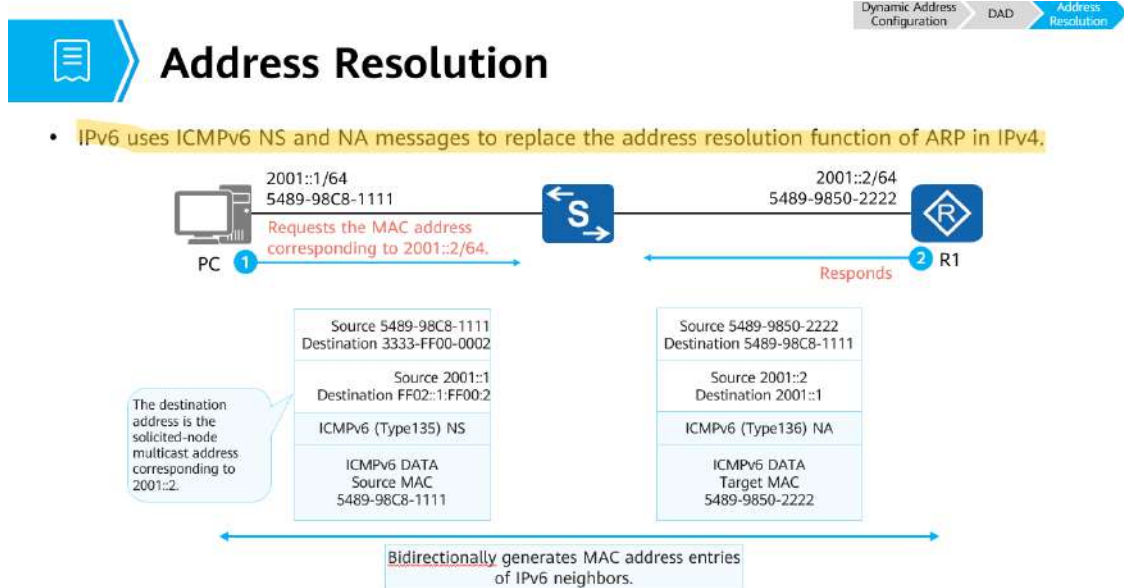
Answer: A

Justification

Source: Huawei Slide # 19, Page # 30



Source: Huawei Slide # 19, Page # 30



373. Which of the following descriptions are correct about the concept of IEV architecture? (Multiple choice)

- A. MANO is composed of NFVO, VNFM and VIM.
- B. VNFM. Virtual network function management, responsible for the life cycle management of VNF.
- C. VIM (Virtualised Infrastructure Manager) virtual resource management, responsible for the virtualization of infrastructure.
- D. NFVO (NFV Orchestrator) is responsible for business orchestration.

Answer: AB

Justification

Source: Huawei Slide # 20, Page # 45

Functional Modules of the NFV Architecture

- Main functional modules defined in the standard NFV architecture:

OSS or BSS	Management system for a service provider. It is not a functional component in the NFV architecture, but the MANO must provide an interface for interoperation with the OSS or BSS.
MANO	NFV management and orchestration. The MANO includes the VIM, VNFM, and NFVO, and provides unified management and orchestration for VNFs and the NFVI. • VIM: NFVI management module that runs on an infrastructure site. The VIM provides functions such as resource discovery, virtual resource management and allocation, and fault handling. • VNFM: It controls the VNF lifecycle (including instantiation, configuration, and shutdown). • NFVO: It orchestrates and manages all the software resources and network services on an NFV network.
VNF	VNFs refer to VMs as well as service NEs and network function software deployed on the VMs.
NFVI	NFV infrastructure, including required hardware and software. The NFVI provides a running environment for VNFs. • Hardware layer: includes hardware devices that provide compute, network, and storage resources. • Virtualization layer: abstracts hardware resources to form virtual resources, such as virtual compute, storage, and network resources. The virtualization function is implemented by Hypervisor^[1].

374. Which of the following methods can read data non-blocking in Python's telnetlib?

- A. telnet_Read_all
- B. telnet_read_very_lazy
- C. telnet.read_eager
- D. telnet_read_very_eager

Answer: D

Justification

Source: Huawei Slide # 21, Page # 24

Method	Function
Telnet.read_until (expected, timeout=None)	Read data until a given byte string, <i>expected</i> , is encountered or until <i>timeout</i> seconds have passed.
Telnet.read_all ()	Read all data until EOF as bytes; block until connection closed.
Telnet.read_very_eager()	Read everything that can be without blocking in I/O (eager). Raise EOFError if connection closed and no cooked data available. Return b" if no cooked data available otherwise. Do not block unless in the midst of an IAC sequence.
Telnet.write(buffer)	Write a byte string to the socket, doubling any IAC characters.
Telnet.close()	Close the connection.

375. Regarding the control plane of the device, which of the following statements is correct? (Multiple choice)

- A. The control plane is responsible for the interaction of routing protocols between devices
- B. The control plane provides equipment monitoring capabilities
- C. The control plane is responsible for directing data forwarding
- D. The control plane is responsible for generating the IP routing table

Answer: ACD

Justification

Source: Huawei Slide # 21, Page # 24 (Notes section)

- **Control plane:** provides functions such as protocol processing, service processing, route calculation, forwarding control, service scheduling, traffic statistics collection, and system security. The control plane of a switch is used to control and manage the running of all network protocols. The control plane provides various network information and forwarding query entries required for data processing and forwarding on the data plane.

Source: <https://www.ibm.com/think/topics/control-plane>

Control plane benefits

Control planes have many benefits when it comes to network management. Here are some of the most common:

- **Efficiency:** The control plane provides a single point from which every device or node on a network can be managed. For example, network administrators use the control plane to configure security settings, such as access control, and automate the delivery of critical software updates.
- **Adaptability:** The control plane allows devices on a network to react dynamically to changes in network functionality, such as a link failure or power outage. For example, when it comes to routing data across a network, the control plane can adjust when a node fails and reroute data accordingly.
- **Scalability:** Control planes are considered highly scalable because additional resources can be added easily without increasing the complexity of the network. Some control planes are even configured to be automatically scalable, a feature known as [autoscaling](#). When user traffic on a network reaches a certain threshold, autoscaling triggers the [provisioning](#) of additional compute resources.
- **Resiliency:** Control planes are highly resilient because of certain aspects of their architecture. First, they are usually kept separate from the data plane, meaning that a failure on the control plane—for example, a load balancer malfunctioning—won't affect the data plane.
- **Low latency:** Control planes monitor [latency](#) and other performance metrics for connected systems and devices on a network (such as computers, mobile devices and graphics cards) so they can be kept below certain levels. In particular, [CPU](#) latency is important to monitor as it reflects the amount of time it takes data to move through a system.
- **Security:** Control planes with additional security features—known as endpoint control planes—give network administrators the ability to monitor devices connected to a network, identify threats and enforce additional security policies.

376. Which of the following commands can modify the device name to Huawei?

- A. Hostname Huawei
- B. Sysname Huawei
- C. Rename Huawei
- D. Do name Huawei

Answer: B

Justification

Source: Huawei Slide # 03, Page # 32

1. Configure a system name.

[Huawei] **sysname** *name*

377. Which statement about the ARP message is wrong?

- A. ARP reply packets are sent unicast
- B. Any link layer protocol requires ARP messages to assist in obtaining data link layer identification
- C. ARP packets cannot be forwarded to other broadcast domains
- D. ARP request message is sent by broadcast

Answer: B

Justification

Source: Huawei Slide # 02, Page # 29 (Notes section)

- The Address Resolution Protocol (ARP) is a TCP/IP protocol that discovers the data link layer address associated with a given IP address.
- ARP is an indispensable protocol in IPv4. It provides the following functions:
 - Discovers the MAC address associated with a given IP address.
 - Maintains and caches the mapping between IP addresses and MAC addresses through ARP entries.
 - Detects duplicate IP addresses on a network segment.

Explanation: Not all link-layer protocols use ARP. ARP is specifically used for resolving IP addresses to MAC addresses in IPv4 networks. Protocols such as IPv6 use the Neighbor Discovery Protocol (NDP) instead of ARP.

378. Which of the following networks belongs to a local area network? (Multiple choice)

- A. home network
- B. Office network of a company
- C. Internet
- D. Broadband Metropolitan Area Network

Answer: AB

Justification

Source: <https://www.cisco.com/c/en/us/products/switches/what-is-a-lan-local-area-network.html>

What Is a LAN?

A local area network (LAN) is a collection of devices connected together in one physical location, such as a building, office, or home. A LAN can be small or large, ranging from a home network with one user to an enterprise network with thousands of users and devices in an office or school.

379. Which statement about the OSPF protocol DR is correct?

- A. DR must be the highest priority device in the network based on interface priority, not the whole device.
- B. Router ID, the larger the value, the more preferred to be elected as DR
- C. DR election is preemptive
- D. An interface priority is 0; then the interface cannot be a DR

It is non-preemptive—once elected, a higher-priority router joining later does not replace the existing DR.

Answer: D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100262536/aa8766d8/ospfpacket-format>

DR priority. The default value is 1.

NOTE:

If the DR priority of a device interface is set to 0, the interface cannot participate in a DR or BDR election.

380. Which of the following elements are included in the routing table? (Multiple choice)

- A. Interface
- B. Destination/mask
- C. Nexthop
- D. Protocol
- E. Cost

Answer: ABCDE

Justification

Source: https://support.huawei.com/enterprise/en/doc/EDOC1100086956#EN-US_TOPIC_0291822184

Routing Table Contents

You can run the **display ip routing-table** command on a router to view basic information about the routing table of the router. The command output is as follows:

```
<HUAWEI> display ip routing-table
Proto: Protocol          Pre: Preference
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table: _public_
      Destinations : 14          Routes : 14

Destination/Mask    Proto    Pre    Cost    Flags NextHop          Interface
-----
0.0.0.0/0           Static   60     0        RD    10.137.216.1       Vlanif20
10.10.10.0/24        Direct   0       0         D    10.10.10.10        Vlanif20
10.10.10.10/32       Direct   0       0         D    127.0.0.1           InLoopBack0
10.10.10.255/32      Direct   0       0         D    127.0.0.1           InLoopBack0
10.10.11.0/24        Direct   0       0         D    10.10.11.1          LoopBack0
10.10.11.1/32        Direct   0       0         D    127.0.0.1           InLoopBack0
10.10.11.255/32      Direct   0       0         D    127.0.0.1           InLoopBack0
10.137.216.0/23      Direct   0       0         D    10.137.217.208      Vlanif20
10.137.217.208/32    Direct   0       0         D    127.0.0.1           InLoopBack0
10.137.217.255/32    Direct   0       0         D    127.0.0.1           InLoopBack0
127.0.0.0/8          Direct   0       0         D    127.0.0.1           InLoopBack0
127.0.0.1/32         Direct   0       0         D    127.0.0.1           InLoopBack0
127.255.255.255/32   Direct   0       0         D    127.0.0.1           InLoopBack0
255.255.255.255/32   Direct   0       0         D    127.0.0.1           InLoopBack0
```

381. The following rules exist in a certain ACL of the router. rule deny tcp source 192.168.1.0 0.0.0.255 destination 172.16.10.1 0.0.0.0 destination-port eq 21 Which statement is correct?

- A. The source IP is 192.168.1.1, the destination IP is 172.16.10.1, and all TCP packets whose destination port number is 21 match this rule
- B. The source IP is 192.168.1.1, the destination IP is 172.16.10.2, and all TCP packets whose destination port number is 21 match this one
- C. The source IP of the rule is 192.168.1.1, and the destination IP is 172.16.10.3. All TCP packets with the destination port number of 21 match this rule
- D. The type of ACL is a basic ACL

Answer: A

Justification

Explanation: The Rule says to deny any source in the range 192.168.1.0 with wildcard 0.0.0.255, meaning 255 IPs from 192.168.1.0 with destination 172.16.10.1 0.0.0.0, meaning only the exact IP of a destination with port 21. Only option (A) matches this. Option D is incorrect because basic ACLs do not specify destination IPs, protocols, or ports.

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100149308/AEJ0713J/17/resources/cli/rule_acl_view.html

- For TCP:

```
rule [ rule-id] { permit | deny } { 6 | tcp } [ source { source-ip-address { 0 | source-wildcard }  
| address-set address-set-name | any } | destination { destination-ip-address { 0 | destination-  
wildcard } | address-set address-set-name | any } | source-port operator port [ port2 ]  
| destination-port operator port [ port2 ] | precedence precedence | tos tos | time-range time-  
name | logging | dscp dscp-value | tcp-flag { tcp-flag [ mask mask-value ] | established |  
{ ack | fin | psh | rst | syn | urg } * } ] * [ description description ]
```

Example

Add a rule to deny the hosts at 172.16.0.0 to establish a connection to the WWW port (80) of the hosts at 172.16.160.0 and log violation events.

```
[sysname-acl-adv-3101] rule deny tcp source 172.16.0.0 0.0.255.255 destination 172.16.160.0 0.0.0.255 destination-port eq www logging
```

382. Which of the following parameters cannot be used in the advanced visit control list?

- A. Protocol number
- B. Destination port number
- C. Physical interface
- D. Time range

Answer: C

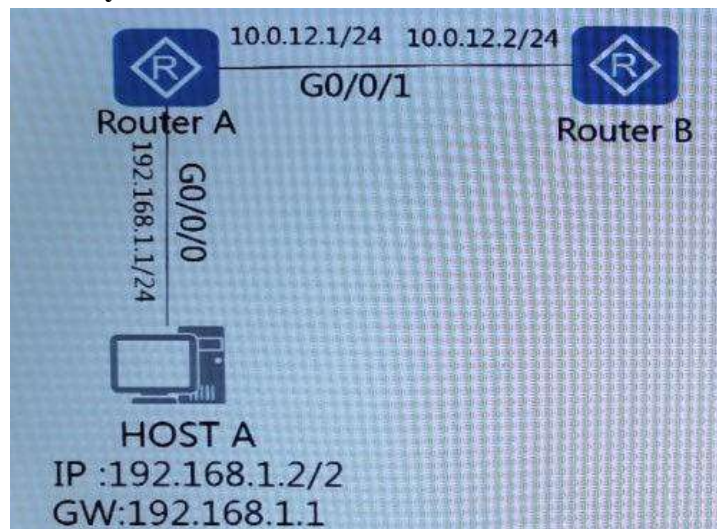
Justification

Source: Huawei Slide # 12, Page # 12

- ACL classification based on ACL rule definition methods

Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
Advanced ACL	3000 to 3999	Defines rules based on source and destination IPv4 addresses, IPv4 protocol types, ICMP types, TCP source/destination port numbers, UDP source/destination port numbers, and effective time ranges.
Layer 2 ACL	4000 to 4999	Defines rules based on information in Ethernet frame headers of packets, such as source and destination MAC addresses and Layer 2 protocol types.
User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

383. In the network shown in the figure below, host A logs in to router A through Telnet, and then obtains the configuration file of router B through FTP on the remote interface. How many TCP connections exist on router A at this time?



- A. 1
- B. 2
- C. 3
- D. 4

Answer: C

Justification

Explanation:

How many TCP connections does Router A handle?

1. Telnet Session (Host A → Router A)

- Telnet uses TCP (port 23)
- One TCP connection established between Host A and Router A

2. FTP Control Connection (Host A → Router B)

- FTP uses two TCP connections:
 - Control connection (usually on port 21)
 - Data connection (for file transfer)

These both pass through Router A, so:

3. FTP Data Connection (Host A → Router B)

- A second TCP connection for data transfer is established

Total TCP connections that Router A handles:

- 1 TCP connection terminated at Router A (Telnet)
- 2 TCP connections passing through Router A (FTP control + FTP data)

384. Which type of message is used by network devices running the SNMPv1 protocol to actively send alarm information?

- A. Trap
- B. Response
- C. Get Request
- D. Get-Next Request

Answer: A

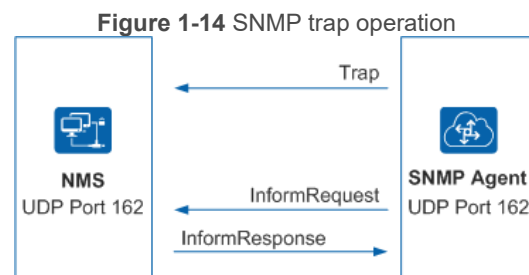
Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086963>

SNMP Traps

SNMP traps are notification messages sent by an SNMP agent to inform the NMS of alarms or events generated by the device. In this way, the administrator can learn the running status of the device in a timely manner.

There are two types of SNMP traps: trap and inform. SNMPv1 does not support inform. The difference between trap and inform is that, after an SNMP agent sends an alarm or event to the NMS through an InformRequest message, the NMS needs to reply with an InformResponse message, as shown in [Figure 1-14](#).



385. By default, network devices running the SNMPv2c protocol use which of the following port numbers to send trap messages to the network management system?

- A. 6
- B. 161
- C. 17
- D. 162

Answer: D

Justification

Source: Huawei Questions Library, Page # 135

2. (Single Choice) Which of the following statements regarding traps in SNMP is correct?

- A. Traps are transmitted using UDP to destination port number 162.
- B. Traps are transmitted using UDP to destination port number 161.
- C. Traps are transmitted using TCP to destination port number 162.
- D. Traps are transmitted using TCP to destination port number 161.

2.A

Source:

https://support.huawei.com/enterprise/en/doc/EDOC1100086963#ENUS_TOPIC_017288170%203

SNMP Port Numbers

SNMP packets are common UDP packets. SNMP defines two default port numbers:

- Port 161: This port number is used when the NMS sends Get, GetNext, GetBulk, and Set requests and the SNMP agent responds to these requests.
This port number is configurable. Ensure that the port number used by the NMS to send request packets is the same as that used by the SNMP agent to respond to request packets.
- Port 162: This port number is used by the SNMP agent to send traps or inform messages to the NMS.
This port number is configurable. Ensure that the port number used by the SNMP agent to send traps or inform messages is the same as that used by the NMS to listen to traps or inform messages.

386. Which of the following types of addresses is not included in IPv6 addresses?

- A. Anycast address
- B. Broadcast address
- C. Unicast address
- D. Multicast address

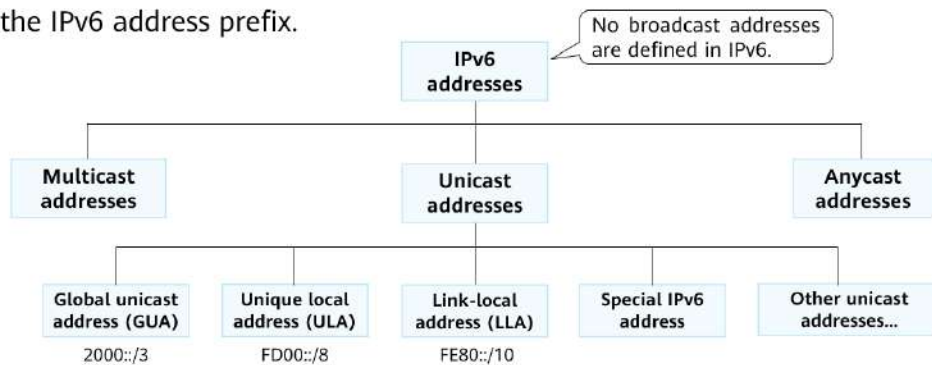
Answer: B

Justification

Source: Huawei Slide # 19, Page # 15

IPv6 Address Classification

- IPv6 addresses are classified into unicast, multicast, and anycast addresses according to the IPv6 address prefix.



387. In the VRP interface, use the command `startup saved-configuration backup.cfg` to configure the `backup.cfg` file for the next startup.

- A. True
- B. False

Answer: A

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100083037/a43ef6dd/backingup-the-%20configuration-file>

NOTICE

The functions of the **save** and **save config-filename** commands are different. Note the following when using them:

- The **save** command saves the current configurations to the next-startup configuration file on the storage device. You can run the **display startup** command to check the name of the next-startup configuration file. By default, the next-startup configuration file is the **vrpcfg.dat** file saved in `home:/`.
- The **save config-filename** command backs up the current configurations to the file specified by *config-filename* on the storage device. If the storage path and name of the configuration file specified using *configuration-file* are the same as those of the next-startup configuration file, the **save configuration-file** command functions the same as the **save** command.
- If you have run the **save config-filename** command to back up the current configurations and still want to deliver a new configuration, you need to run the **save config-filename** command again to back up the new configuration to the configuration file. This ensures that the new configuration is restored after the router restarts.

388. Which of the following statements about Trunk port and Access port are correct?

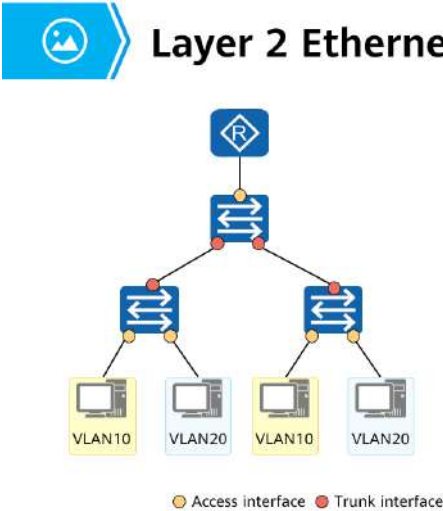
- A. Trunk ports can only send untagged frames
- B. Access port can only send tagged frames
- C. Trunk ports can only send tagged frames
- D. Access port can only send untagged frames

Answer: D

Justification

Source: Huawei Slide # 08, Page # 17

VLAN Identification
VLAN Assignment
VLAN Frame Processing



Layer 2 Ethernet Interface Types

Interface Types

- Access interface**
 An access interface is used to connect a switch to a terminal, such as a PC or server. In general, the NICs on such a terminal receive and **send only untagged frames**. An access interface can be added to only one VLAN.
- Trunk interface**
 A trunk interface allows frames that belong to multiple VLANs to pass through and differentiates the frames using the 802.1Q tag. This type of interface is used to connect a switch to another switch or a sub-interface on a device, such as a router or firewall.
- Hybrid interface**
 Similar to a trunk interface, a hybrid interface also allows frames that belong to multiple VLANs to pass through and differentiates the frames using the 802.1Q tag. You can determine whether to allow a hybrid interface to carry VLAN tags when sending the frames of one or more VLANs.

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Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100096323/819ade70/understanding-vlans>

Table 8-12 Frame processing based on the port type

Port Type	Untagged Frame Processing	Tagged Frame Processing	Frame Transmission
Access port	Accepts an untagged frame and adds a tag with the default VLAN ID to the frame.	- Accepts the tagged frame if the frame's VLAN ID matches the default VLAN ID. - Discards the tagged frame if the frame's VLAN ID differs from the default VLAN ID.	After the PVID tag is stripped, the frame is transmitted.
Trunk port	- Adds a tag with the default VLAN ID to the untagged frame and then transmits it if the default VLAN ID is permitted by the port - Adds a tag with the default VLAN ID to the untagged frame and then discards it if the default VLAN ID is denied by the port.	- Accepts the tagged frame if the frame's VLAN ID is permitted by the port. - Discards the tagged frame if the frame's VLAN ID is denied by the port.	- If the frame's VLAN ID matches the default VLAN ID and the VLAN ID is permitted by the port, the switch removes the tag and transmits the frame. - If the frame's VLAN ID differs from the default VLAN ID, but the VLAN ID is still permitted by the port, the switch will directly transmit the frame.

389. If the Length/Type of an Ethernet data frame is 0X8100, which of the following is correct?

- A. This data frame must carry a VLAN TAG
- B. There must be a UDP header in the upper layer of this data frame
- C. There must be an IP header in the upper layer of this data frame
- D. There must be a TCP header in the upper layer of this data frame

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100088104>

VLAN Frame Format

A switch identifies packets from different VLANs according to the information contained in its VLAN tags. IEEE 802.1Q adds a 4-byte VLAN tag between the Source/Destination MAC address and Length/Type fields of an Ethernet frame to identify the VLAN to which the frame belongs. [Figure 1-1](#) shows the position of a VLAN tag in a VLAN data frame.

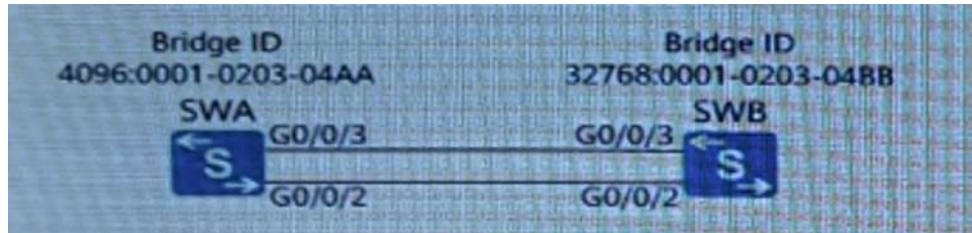
Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000178168/3e01a4f2/tpid>

TPID

The Tag Protocol Identifier (TPID) specifies the protocol type of a VLAN tag. The TPID value defined in IEEE 802.1Q is 0x8100.

[Figure 10-6](#) shows the Ethernet packet format defined in IEEE 802.1Q. An IEEE 802.1Q tag, containing the TPID, lies between the Source Address field and the Length/Type field. A device checks the TPID value in a received packet to determine whether the VLAN tag is an S-VLAN tag or C-VLAN tag. The device compares the configured TPID value with the TPID value in the packet. For example, if a frame carries the VLAN tag with TPID 0x8100 but the TPID configured for a customer network on a device is 0x8200, the device considers the frame untagged.

390. The two switches shown below have both enabled the STP protocol. An engineer made the following conclusions on this network. Which conclusions do you think are correct? (Multiple choice)



- A. The G0/0/3 port of SWA is stable in the Forwarding state
- B. The G0/0/2 port of SWA is stable in the Forwarding state
- C. Both ports of SWA are designated ports
- D. G0/0/2 port of SWB is stable in the Forwarding state
- E. Both ports of SWB are designated ports

Answer: ABCD

Justification

Explanation: The bridge priority of SWA is small, so it will be the root bridge. So, all its ports will be designated ports. G0/0/2 of SWB will be the root port, and G0/0/3 of SWB will be the blocked port. So only option (E) is wrong

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086964>

Root Bridge, Root Port, and Designated Port

Three elements are involved in trimming a ring network into a tree network: root bridge, root port, and designated port.

- Root bridge: There is only one root bridge on the entire STP network. The root bridge is the logical center, but not necessarily the physical center, of the network. Typically, the root bridge should have high performance and be located at a higher layer of the network. When the network topology changes, the root bridge will be re-elected accordingly. It is recommended that you manually configure the root bridge and secondary root bridge. Configure an optimal core switch as the root bridge to ensure stability of the STP Layer 2 network. Otherwise, new access devices may trigger STP root bridge switching, causing short service interruptions.
- Root port: The root port on an STP device has the smallest path cost to the root bridge and is responsible for forwarding data to the root bridge. Among all STP-capable ports on a device, the port with the smallest root path cost is a root port. An STP device has only one root port, and there is no root port on the root bridge.
- Designated port: The designated port is located on a designated bridge and forwards configuration BPDUs to the device or LAN, as illustrated in [Figure 1](#). AP1 and AP2 are ports of S1; BP1 and BP2 are ports of S2; CP1 and CP2 are ports of S3.
 - S1 sends configuration BPDUs to S2 through AP1, so S1 is the designated bridge for S2, and AP1 is the designated port on S1.
 - S2 and S3 are connected to the LAN. If S2 forwards configuration BPDUs to the LAN, S2 is the designated bridge for the LAN, and BP2 is the designated port on S2.

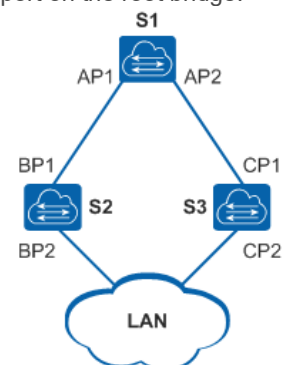


Figure 1-1 Designated bridge and designated port

391. What is the length of the Ethernet II frame containing the Ethernet header?

- A. 60-1560B
- B. 46-1500B
- C. 64-1500B
- D. 64-1518B

Answer: D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100247749/e1be8fdc/minimum-frame-length-and-maximum-transmission-distance>

Minimum Frame Length and Maximum Transmission Distance

Due to the limitation of the CSMA/CD algorithm, an Ethernet frame must be longer than or equal to a specified length. On the Ethernet, the minimum frame length is 64 bytes, which is determined jointly by the maximum transmission distance and the collision detection mechanism.

The use of minimum frame length can prevent the following situation: station A finishes sending the last bit, but the first bit does not arrive at station B, which is far from station A. Station B considers that the line is idle and begins to send data, leading to a collision.

Figure 1-2 Ethernet II frame format

6bytes	6bytes	2bytes	46~1500bytes	4bytes
DMAC	SMAC	Type	Data	CRC

The upper layer protocol must ensure that the Data field of a packet contains at least 46 bytes, so that the total length of the Data field, the 14-byte Ethernet frame header, and the 4-byte check code at the frame tail can reach the minimum frame length, as shown in [Figure 1-2](#). If the Data field is less than 46 bytes, the upper layer must pad the field to 46 bytes.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100174721/ea0a043c/ethernet-ii-frame>

Table 1-1 Link layer fields in an Ethernet II frame

Field	Length	Description
DMAC	6 bytes	Destination MAC address. Six bytes for IPv4. This field identifies the receiver of a frame.
SMAC	6 bytes	Source MAC address. Six bytes for IPv4. This field identifies the sender of a frame.
Type	2 bytes	Protocol type. Table 1-3 lists the types of protocols that are directly encapsulated on a link.
Data	46~1500 bytes	Data field, which identifies the payload of a frame (which may contain padding bits). The minimum length of the Data field must be 46 bytes to ensure that the frame length is at least 64 bytes. This means that the 46-byte Data field is required even if you want to transmit only 1 byte of information. If the payload of the Data field is less than 46 bytes, the Data field must be padded to 46 bytes. The maximum length of the Data field is 1500 bytes. The frame length must be integral bytes. If the frame length is not integral bytes, the Data field must be padded to make the frame length be integral bytes.
FCS	4 bytes	This field provides a method for the receiver to determine whether a transmission error occurs. If errors are detected, involved frames are discarded. FCS is a general name for multiple verification methods. In an Ethernet frame, CRC is usually used as the FCS.

Explanation: For minimum frame length, add the above lengths: $6 + 6 + 2 + 46 + 4 = 64$

For minimum frame length, add the above lengths: $6 + 6 + 2 + 1500 + 4 = 1518$

392. By default, the bridge priority of Huawei switches is 32768.

- A. True
- B. False

Answer: A

Justification

Source: Huawei Slide # 09, Page # 41



Basic STP Configuration Commands (2)

1. (Optional) Configure the STP priority of a switch.

```
[Huawei] stp priority priority
```

By default, the priority value of a switch is 32768.

2. (Optional) Configure a path cost for a port.

```
[Huawei] stp pathcost-standard { dot1d-1998 | dot1t | legacy }
```

Configure a path cost calculation method. By default, the IEEE 802.1t standard (**dot1t**) is used to calculate path costs.

All switches on a network must use the same path cost calculation method.

```
[Huawei-GigabitEthernet0/0/1] stp cost cost
```

Set the path cost of the port.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086964>

Configuring Basic STP

1. Run the **system-view** command to enter the system view.
2. Run the **stp mode stp** command to set the working mode of the device to STP.
By default, the working mode of a device is MSTP. MSTP is compatible with STP and RSTP.
3. (Optional) Configure the root bridge and secondary root bridge.
 - Run the **stp root primary** command on a device to configure the device as the root bridge.
 - Run the **stp root secondary** command on another device to configure the device as the secondary root bridge.
4. (Optional) Run the **stp priority** priority command to set a priority for the device.
The default priority value of a device is 32768. The value is an integer multiple of 4096, such as 0, 4096 and 8192. A smaller value indicates a higher priority of the device. A device with a higher priority is more likely to be elected as the root bridge. Low-performance devices at lower network layers are not suitable for root bridges, so you need to set low priorities for these devices.

393. On the switch, which VLANs can be deleted by using the undo command? (Multiple choice)

- A. vlan 4094
- B. vlan 2
- C. vlan 1
- D. vlan 1024

Answer: ABD

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100041734/834147df/vlan-configuration-commands>

vlan

Function: The **vlan** command creates a VLAN and displays the VLAN view. If the VLAN exists, the VLAN view is displayed.

The undo vlan command deletes a VLAN.

By default, all interfaces belong to the default VLAN, named VLAN 1.

Format:

```
vlan vlan-id
vlan batch { vlan-id1 [ to vlan-id2 ] } <1-10>
undo vlan vlan-id
undo vlan batch { vlan-id1 [ to vlan-id2 ] } <1-10>
```

Parameters

Parameter	Description	Value
vlan-id	Specifies the VLAN ID.	The value is an integer ranging from 1 to 4094.
batch	Configures VLANs in batches.	-
vlan-id1 to vlan-id2	Specifies range of VLANs to be configured in batches: - vlan-id1 specifies the start VLAN ID. - vlan-id2 specifies the end VLAN ID. vlan-id2 must be greater than or equal to vlan-id1. vlan-id1 and vlan-id2 define a range together. - If the parameter to vlan-id2 is not specified, only the VLAN specified by vlan-id1 is created.	The vlan-id1 and vlan-id2 are integers ranging from 1 to 4094.

Views

System view

Default Level

2: Configuration level

Usage Guidelines

Usage Scenario: To reduce broadcast domains and enhance security on a complex network, VLANs can be created on the network to isolate the devices that do not need to communicate each other. The **vlan batch** command creates multiple VLANs at one time, simplifying VLAN configuration.

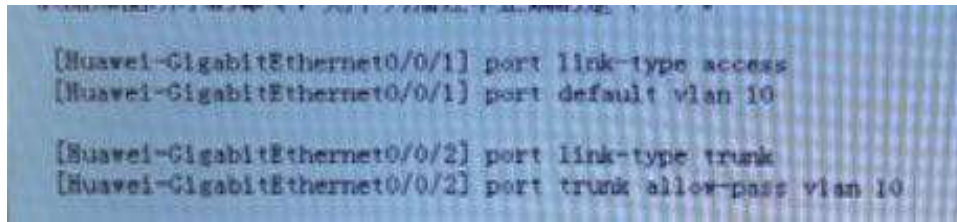
Follow-up Procedure: Assign VLANs according to network requirements.

Precautions: **VLAN 1 is the default VLAN, which cannot be deleted and does not need to be created.**

The **vlan** command can be used to create a VLAN and enter the VLAN view. If a VLAN has been created, the VLAN view is displayed after this command is used. The **vlan** command multiple times creates multiple VLANs. If a VLAN has been created, using this command cannot be used to create the same VLAN or modify the configurations of the VLAN.

Explanation: **All Vlan types can be deleted, but not 1 because it is the default pre-made Vlan.**

394. According to the command shown in the figure, which of the following descriptions is correct? (Multiple choice)



```
[Huawei-GigabitEthernet0/0/1] port link-type access
[Huawei-GigabitEthernet0/0/1] port default vlan 10

[Huawei-GigabitEthernet0/0/2] port link-type trunk
[Huawei-GigabitEthernet0/0/2] port trunk allow-pass vlan 10
```

- A. The PVID of the GigabitEthernet0/0/2 port is 1
- B. The PVID of the GigabitEthernet0/0/1 port is 10
- C. The PVID of the GigabitEthernet0/0/2 port is 10
- D. The PVID of the GigabitEthernet0/0/1 port is 11

Answer: AB

Justification

Explanation: PVID of trunk port is 1 as it is the default ID every port has. GE0/0/1 is configured with VLAN ID 10. If PVID (default VLAN) is not explicitly configured, it defaults to VLAN 1. Only VLAN 10 is allowed to pass, but that does not change the PVID unless configured.

Source: https://support.huawei.com/enterprise/en/doc/EDOC1100064351/299a5aeb/port-trunk-pvid-vlan?utm_source=chatgpt.com

port trunk pvid vlan

Function

The **port trunk pvid vlan** command specifies the default VLAN for a trunk interface. The **undo port trunk pvid vlan** command restores the default VLAN of a trunk interface to the default setting. By default, VLAN 1 is the default VLAN of trunk interfaces.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064351/a25e7c6d/port-default-vlan>

Prerequisites

The VLAN to be configured as the default VLAN has been created.
The interface type is access. If not, run the **port link-type** command to change the interface type.

Precautions

You can also run the **port** command in the VLAN view to configure the default VLAN of an interface. The two commands have the same function.

A super VLAN cannot be configured as the default VLAN of interfaces.

This command is invalid on a member interface of an Eth-Trunk.

If you run the **port default vlan** command multiple times in the same interface view, only the latest configuration takes effect.

Example

Configure VLAN 3 (an existing VLAN) as the default VLAN of GE0/0/1 (an access interface).

```
<AC6605> system-view
[AC6605] interface gigabitethernet 0/0/1
[AC6605-GigabitEthernet0/0/1] port link-type access
[AC6605-GigabitEthernet0/0/1] port default vlan 3
```


395. Which of the following advantages does the router-on-a-stick approach achieve routing intercommunication between VLANs?

- A. Reduce the number of equipment
- B. Reduce routing table entries
- C. Reduce the use of IP addresses
- D. Reduce the number of chain connections

Answer: D

Justification

Source: Huawei Slide # 10, Page # 09

Using a Router's Sub-interfaces

Using Physical Interfaces

Using Sub-Interfaces

Physical Connection

```
graph TD
    R1[R1] --- SW1[SW1]
    R1 --- S10[GE 0/0/1.10 192.168.10.254]
    R1 --- S20[GE 0/0/1.20 192.168.20.254]
    SW1 --- T24[G 0/0/24 Trunk VLANs 10 20]
    SW1 --- A10[GE 0/0/1 Access VLAN 10]
    SW1 --- A20[GE 0/0/2 Access VLAN 20]
    PC1[PC1 192.168.10.2/24 gateway 192.168.10.254] --- A10
    PC2[PC2 192.168.20.2/24 gateway 192.168.20.254] --- A20
```

- A sub-interface is a logical interface created on a router's Ethernet interface and is identified by a physical interface number and a sub-interface number. Similar to a physical interface, a sub-interface can perform Layer 3 forwarding.
- Different from a physical interface, a sub-interface can terminate data frames with VLAN tags.
- You can create multiple sub-interfaces on one physical interface. After connecting the physical interface to the trunk interface of the switch, the physical interface can provide Layer 3 forwarding services for multiple VLANs.

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Explanation: The term "chain connections" in this context refers to the multiple physical links that would otherwise be necessary to connect the router to each VLAN individually. By implementing the router-on-a-stick method, these multiple connections are consolidated into a single trunk link between the router and the switch. This effectively reduces the number of physical connections, aligning with the advantage described in choice D.

396. When the host frequently moves, which VLAN division method is most appropriate?

- A. Based on the MAC address division
- B. Based on the strategy division
- C. Based on IP subnet division
- D. Based on port division

Answer: C

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000017271/a6bce928/dividing-a-lan-into-vlans-based-on-ip-subnets>

Dividing a LAN into VLANs Based on IP Subnets

Context

IP subnet-based and protocol-based VLAN division are called network layer-based VLAN division, which reduces manual VLAN configuration workload and allows users to easily join a VLAN, transfer from one VLAN to another, and exit from a VLAN. IP subnet-based VLAN division is applicable to networks that have traveling users and require simple management.

VLANs configured based on IP subnets process only untagged frames. and treat tagged frames in the same manner as VLANs configured based on ports.

After receiving untagged frames, a device determines the VLANs to which the frames belong based on their source IP addresses before sending them to the corresponding VLANs.

Explanation: Based on the MAC address division, as the MAC address does not change and the condition is host moves, so it is not suitable.

Based on port division: if the device itself moves, then no matter whether the VLAN is done on the port.

Based on IP subnet division: this method includes the IP being assigned to the device to if the host moves then also it will not create the problem. As IP will be assigned.

397. Which of the following descriptions are wrong about the Forwarding state in the spanning tree protocol? (Multiple choice)

- A. The port in the Forwarding state can forward data packets
- B. The port in the Forwarding state does not learn the source MAC address of the packet
- C. Ports in Forwarding state can receive BPDU packets
- D. The port in the Forwarding state will definitely send BPDU packets
depends on whether STP is enabled and the port role.

Answer: BD

Justification

Source: Huawei Slide # 09, Page # 32

 STP Port States

Port State	Description
Disabled	The port cannot send or receive BPDUs or service data frames. That is, the port is Down.
Blocking	The port is blocked by STP. A blocked port cannot send BPDUs but listens to BPDUs. In addition, the blocked port cannot send or receive service data frames or learn MAC addresses.
Listening	STP considers the port in Listening state as the root port or designated port, but the port is still in the STP calculation process. In this case, the port can send and receive BPDUs but cannot send or receive service data frames or learn MAC addresses.
Learning	A port in Learning state listens to service data frames but cannot forward them. After receiving service data frames, the port learns MAC addresses.
Forwarding	A port in Forwarding state can send and receive service data frames and process BPDUs. Only the root port or designated port can enter the Forwarding state.

Explanation:

- B.** The port in the Forwarding state does not learn the source MAC address of the packet.
- Incorrect. Ports in the Forwarding state learn the source MAC addresses of incoming packets to update the MAC address table.
- D.** The port in the Forwarding state will send BPDU packets.
- Incorrect. Only designated ports send BPDUs. Non-designated ports in the Forwarding state do not send BPDUs.

398. On the VRP platform, what is the role of the command "interface vlanif <vlanid>"?

- A. Create a VLAN
- B. No such command
- C. Create or enter the VLAN virtual interface view
- D. Configure VLAN for a port

Answer: C

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178172/433bc017/configuring-a-vlanif-interface>

Procedure

1. Run **system-view**
The system view is displayed.
2. Run **interface vlanif** vlan-id
A VLANIF interface is created, and the VLANIF interface view is displayed.

399. The second layer ACL can match the source MAC, destination MAC, source IP, destination IP, and other information. IP's require Layer-3 info

- A. True
- B. False

Answer: B

Justification

Source: Huawei Slide # 12, Page # 12

- ACL classification based on ACL rule definition methods

Category	Number Range	Description
Basic ACL	2000 to 2999	Defines rules based on source IPv4 addresses, fragmentation information, and effective time ranges.
Advanced ACL	3000 to 3999	Defines rules based on source and destination IPv4 addresses, IPv4 protocol types, ICMP types, TCP source/destination port numbers, UDP source/destination port numbers, and effective time ranges.
Layer 2 ACL	4000 to 4999	Defines rules based on information in Ethernet frame headers of packets, such as source and destination MAC addresses and Layer 2 protocol types.
User-defined ACL	5000 to 5999	Defines rules based on packet headers, offsets, character string masks, and user-defined character strings.
User ACL	6000 to 6999	Defines rules based on source IPv4 addresses or user control list (UCL) groups, destination IPv4 addresses or destination UCL groups, IPv4 protocol types, ICMP types, TCP source/destination port numbers, and UDP source/destination port numbers.

400. The configuration of the country code will affect the actual transmission frequency and maximum transmission power.

- A. True
- B. False

Answer: A

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100033726/82579c28/countrycode-%20channel-compliance-table-and-channel-and-power-restrictions>

Country Code & Channel Compliance Table, and Channel and Power Restrictions

Different countries or regions have specified the channels and maximum transmit power of radio signals that can be used locally. Radio signals working on different channels may have varying signal strength. The *Country Code & Channel Compliance Table* describes the country code and channel compliance, the maximum power allowed by each channel, and the mapping relationship between channels and frequencies. You can directly search for this table on the Huawei enterprise service technical support website (<https://support.huawei.com/enterprise>). N/A indicates that the country or region does not support the channel.

401. What equipment is generally involved in the WLAN deployment scheme in the enterprise scenario? (Multiple choice)

- A. AC (Access Controller)
- B. Poe switch
- C. CPE
- D. AP

Answer: ABD

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100064365/ed705f9/application-scenarios-for-poe>

APs are usually installed in the corridors or on the ceilings where power sockets are unavailable. In many WLAN applications, the administrator manages many APs that require unified power supply and management. In this case, power supply management is difficult. As shown in **Figure 5-24**, the AC that has the PoE function deployed directly provides power for each AP, without the need for additional external power supplies. This reduces cable connections, saves network construction costs, and facilitates AP management.

Explanation:

- **AP (Access Point):** Access Points are devices that allow wireless devices to connect to a wired network using Wi-Fi. They are essential components of the WLAN, providing wireless coverage to the users.
- **PoE Switch:** Power over Ethernet (PoE) switches provide both power and data connectivity to devices like Access Points over a single Ethernet cable.
- **AC (Access Controller):** The Access Controller manages and controls multiple Access Points (APs) in the network. It handles tasks such as configuration, firmware updates, security policies, and load balancing.

402. The role of `telnet.read_all()` in Telnetlib is to read all data until EOF. If the echo does not return EOF, it will always be blocked.

- A. True
- B. False

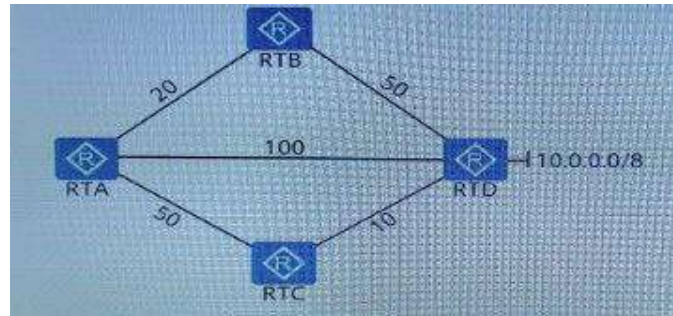
Answer: A

Justification

Source: Huawei Slide # 21, Page # 24

Method	Function
<code>Telnet.read_until</code> (<i>expected</i> , <i>timeout=None</i>)	Read data until a given byte string, <i>expected</i> , is encountered or until <i>timeout</i> seconds have passed.
<code>Telnet.read_all ()</code>	Read all data until EOF as bytes; block until connection closed.
<code>Telnet.read_very_eager()</code>	Read everything that can be without blocking in I/O (eager). Raise EOFError if connection closed and no cooked data available. Return b'' if no cooked data available otherwise. Do not block unless in the midst of an IAC sequence.
<code>Telnet.write(buffer)</code>	Write a byte string to the socket, doubling any IAC characters.
<code>Telnet.close()</code>	Close the connection.

403. In the network shown in the figure below, all routers run the OSPF protocol, and the link cost value is shown in the figure. What is the cost value of the route entry to the network 10.0.0.0/8 in the RTA routing table?



- A. 70
- B. 100
- C. 20
- D. 60

Answer: D

Justification

Source: Huawei slide # 06, Page # 17

Area
Router-ID
Cost Value

Basic OSPF Concepts: Cost Value

- OSPF uses costs as route metric values. Each OSPF-enabled interface maintains a cost value. Default cost value = $\frac{100 \text{ Mbit/s}}{\text{Interface bandwidth}}$, where, 100 Mbit/s is the default reference value specified by OSPF and is configurable.
- Generally, the cost of an OSPF route is the sum of costs of all inbound interfaces along a path from the destination network segment to the local router.

Cost Value of an OSPF Interface

* Each OSPF interface has a specific cost because of the particular bandwidth value.

Accumulated Costs on an OSPF Path

* In the routing table of R3, the cost of the OSPF route to 1.1.1.0/24 is 75 (10 + 1 + 64).

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Explanation: OSPF follows the shortest path, which means that the shortest path to router D was through RTC, which would cost 60.

Possible Paths and Cost Calculation:

1. **Path 1:** RTA → RTB → RID → RTA to RTB: 20, RTB to RID: 50, Total: 20 + 50 = 70
2. **Path 2:** RTA → RID (direct) → RTA to RID: 100
3. **Path 3:** RTA → RTC → RID → RTA to RTC: 50, RTC to RID: 10, Total: 50 + 10 = 60 (lowest cost)

404. The router output information is shown in the figure. Which of the following statements is wrong?

```
<RI>display ip interface Ethernet 0/0/0
Ethernet0/0/0 current state : UP
Line protocol current state : UP
The Maximum Transmit Unit : 1500 bytes
input packets : 3645, bytes : 287468, multicasts : 3590
output packets : 3715, bytes : 294616, multicasts : 3654
Directed-broadcast packets:
  received packets:      0, sent packets:      0
  forwarded packets:    0, dropped packets:    0
ARP packet input number: 29
  Request packet:      13
  Reply packet:        16
  Unknown packet:      0
Internet Address is 10.0.12.1/24
Broadcast address : 10.0.12.255
```

- A. The IP address of the Ethernet0/0/0 interface is 10.0.12.1/24
- B. The MTU value of the Ethernet0/0/0 interface is 1480
- C. The physical link of the Ethernet0/0/0 interface is normal
- D. The broadcast address corresponding to the Ethernet0/0/0 interface is 10.0.12.255

Answer: B

Justification

Explanation:

- A. *Correct* — The IP address is shown as 10.0.12.1/24, which matches.
- B. *Incorrect* — The MTU is clearly shown as 1500 bytes, not 1480. So this statement is wrong.
- C. *Correct* — Both the physical state (current state: UP) and line protocol state (UP) indicate the link is operational.
- D. *Correct* — With a subnet mask of /24, the broadcast address 10.0.12.255 is indeed correct.

405. As shown in the figure below, which of the following statement is correct?

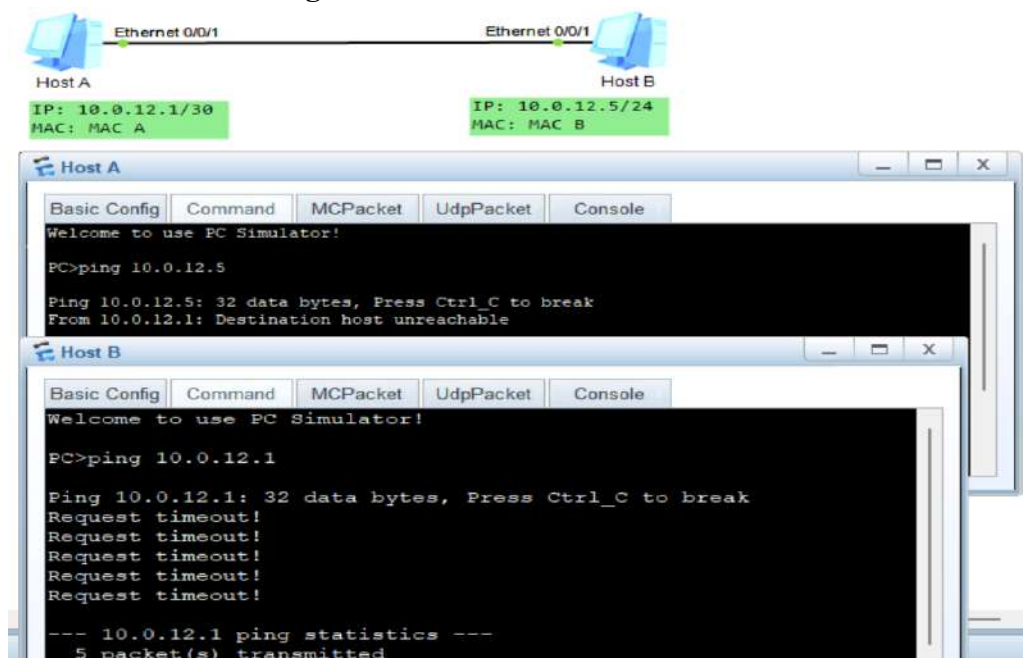


- A. Host A can ping host B
- B. Only when the masks of host A and host B are set to always, host A and host B can communicate
- C. The broadcast addresses of host A and host B are the same
- D. The IP address masks of host A and host B are different, so host A and host B cannot communicate

Answer: D

Justification

Source: Huawei eNSP Configuration



Explanation:

Host	IP Address	Subnet Mask	Subnet Range	Broadcast Address
A	10.0.12.1	/30	10.0.12.0 – 10.0.12.3	10.0.12.3
B	10.0.12.5	/24	10.0.12.0 – 10.0.12.255	10.0.12.255

- Host A's /30 subnet considers only .0 – .3 valid addresses.
- Host B's IP (.5) lies outside Host A's subnet.
- Host A assumes Host B is on a **different network** and tries to forward the packet via a default gateway, but **no gateway is configured**, so the **ping fails**. Even though Host B sees A within its /24 subnet, communication still fails because Host A never initiates ARP for Host B.

Why Other Options Are Incorrect

Option	Explanation
A. Host A can ping Host B	False – Host A sees B as out-of-subnet; ping fails
B. Only when the masks of host A and host B are set to always...	Incorrect grammar and logic
C. The broadcast addresses of Host A and Host B are the same	False – 10.0.12.3 ≠ 10.0.12.255

406. `ip route-static 10.0.12.0 255.255.255.0 192.168.1.1`, which the description about command is correct?

- A. This command configures a route to the 10.0.12.0 network
- B. The priority of this route is 100
- C. If the router learns a route with the same destination network as this route through other protocols, the router will prefer this route
- D. This command configures a route to the 192.168.1.1 network

Answer: A

Justification

Source: Huawei Slide # 05, Page # 29

- Specify both the outbound interface and next hop for a static route.

```
[Huawei] ip route-static ip-address { mask / mask-length } interface-type interface-number [ nexthop-address ]
```

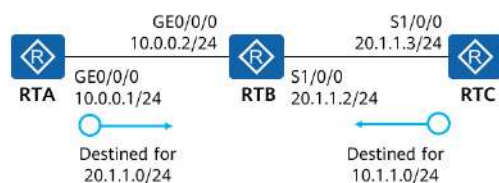
When creating a static route, you can specify both the outbound interface and next hop. Alternatively, you can specify either the outbound interface or next hop, depending on the interface type:

For a point-to-point interface (such as a serial interface), you must specify the outbound interface.

For a broadcast interface (for example, an Ethernet interface) or a virtual template (VT) interface, you must specify the next hop.

Source: Huawei Slide # 05, Page # 30

Configuration Example



- Configure static routes on RTA and RTC for communication between 10.0.0.0/24 and 20.1.1.0/24.

- Packets are forwarded hop by hop. Therefore, all the routers along the path from the source to the destination must have routes destined for the destination.
- Data communication is bidirectional. Therefore, both forward and backward routes must be available.

Configure RTA.

```
[RTA] ip route-static 20.1.1.0 255.255.255.0 10.0.0.2
```

Configure RTC.

```
[RTC] ip route-static 10.0.0.0 255.255.255.0 S1/0/0
```

Explanation: The default static route priority is 60 unless specified (Option B is wrong), and the router will only accept the defined route. The format for static route commands begins with the destination IP network, followed by its subnet mask, and then the next-hop IP.

This command does NOT configure a route to 192.168.1.1, but rather uses it as a next-hop for reaching 10.0.12.0 so Option D is also wrong

407. What address does 192.168.1.127/25 represent?

- A. Broadcast
- B. Multicast
- C. Network
- D. Host

Answer: A

Justification

Source: <https://jodies.de/ipcalc?host=192.168.1.1&mask1=24&mask2=25>

Subnets

Netmask:	255.255.255.128 = 25	11111111.11111111.11111111.1	0000000
Wildcard:	0.0.0.127	00000000.00000000.00000000.0	1111111
Network:	192.168.1.0/25	11000000.10101000.00000001.0	0000000
(Class C)			
Broadcast:	192.168.1.127	11000000.10101000.00000001.0	1111111
HostMin:	192.168.1.1	11000000.10101000.00000001.0	0000001
HostMax:	192.168.1.126	11000000.10101000.00000001.0	1111110
Hosts/Net:	126	(Private Internet)	

Explanation: A subnet mask of /25 means that the first 25 bits of the IP address are used for the network portion, leaving 7 bits for the host portion. The possible host can be 126.

Subnet mask /25 (or 255.255.255.128 in dotted-decimal notation) means: Network address: 192.168.1.0

Broadcast address: 192.168.1.127

192.168.1.127 does not fall within the range of addresses assignable to hosts. We can see that it is similar to broadcast IP.

408. What are the characteristics of VLAN based on the port?

- A. Add VLAN tags to the data frame according to the IP address carried in the message
- B. Assign a VLAN ID according to the protocol type and encapsulation format of the data frame
- C. No need to reconfigure the VLAN when the host moves
- D. The host mobile location needs to be reconfigured VLAN

Answer: D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100352667/526f31bd/example-for-assigning-vlans-based-on-ports>

Example for Assigning VLANs Based on Ports

Networking Requirements

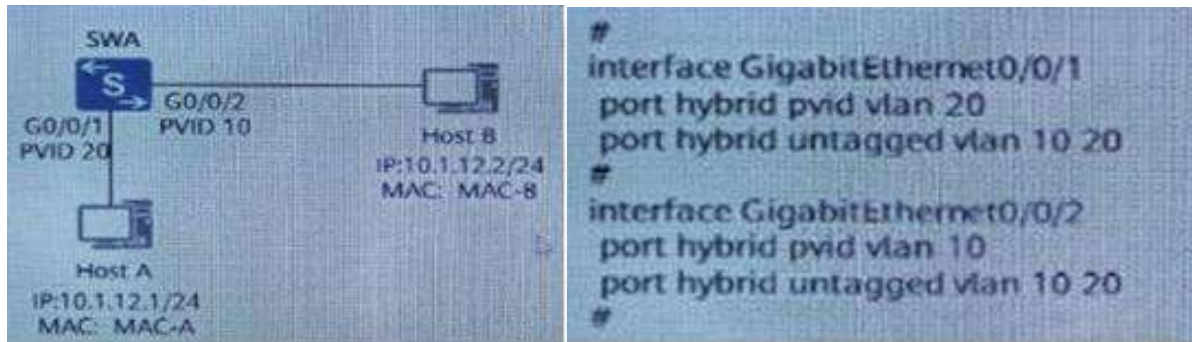
In [Figure 4-22](#), multiple user terminals are connected to switches in a data center. Users who use the same service all access the network using different devices.

To ensure communication security and avoid broadcast storms, the administrator wants to allow users who use the same service to communicate with each other but isolate users who use different services.

Configure the switch with port-based VLAN assignment and add ports connected to users who use the same service to the same VLAN. This way, users who use the same services can directly communicate, but users using different services cannot communicate over Layer 2.

Explanation: VLAN membership is tied to specific switch ports. If a host is moved to another port, it will not remain in the same VLAN unless the new port is also manually configured for the same VLAN.

409. As shown in the figure below, the network and switch configuration information are shown in the figure. Which of the following statements is correct? (Multiple choice)



- A. Data frames on the two links do not contain a VLAN TAG
- B. The PVID of the switch GigabitEthernet0/0/1 port is 20
- C. Host A and Host B can be pinged
- D. Host A and Host B cannot ping

Answer: ABD

Justification

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100278054/AEL11217/08/resources/dc/port_hybrid_pvid_vlan.html

Usage Scenario

An interface may receive untagged and tagged frames, but the AP processes only tagged frames. Therefore, the AP adds a tag to each untagged frame received by an interface. The default VLAN ID of the interface can be added to untagged frames so that these frames are forwarded in the default VLAN.

A hybrid interface processes Ethernet frames as follows:

- When the interface receives an untagged frame, it tags the frame with the default VLAN ID. If the default VLAN ID is allowed by the interface, the interface accepts the frame. Otherwise, the interface discards the frame.
- When the interface receives a tagged frame, it accepts the frame if the VLAN ID of the frame is in the list of allowed VLAN IDs. Otherwise, the interface discards the frame.
- If the VLAN ID of a frame is allowed by the interface, the interface forwards the frame.

Prerequisites

The VLAN to be specified as the default VLAN has been created.

Precautions

The **port hybrid pvid vlan** command only specifies the default VLAN for an interface but does not add the interface to the default VLAN.

If you run the **port hybrid pvid vlan** command multiple times in the same interface view, only the latest configuration takes effect.

Explanation: Picture 2 shows that the port is configured to have untagged VLAN 10 20, so it will not contain a VLAN tag. Moreover, both PCs are in different VLANs, so they cannot ping each other. Lastly, GigabitEthernet0/0/1 port is labelled as PVID20, so (B) is also correct.

410. In the STP protocol, which of the following factors will affect the election of the root switch? (Multiple choice)

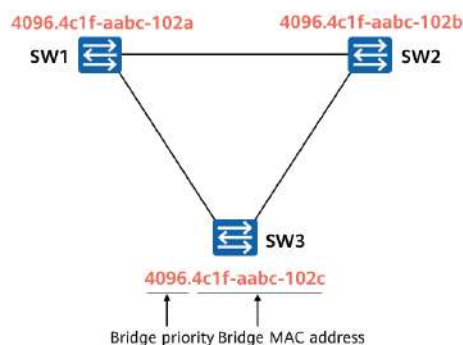
- A. The IP address of the switch
- B. Switch interface ID
- C. Switch interface bandwidth port
- D. Switch priority
- E. MAC address of the switch

Answer: DE

Justification

Source: Huawei Slide # 09, Page # 14

STP Basic Concepts: BID



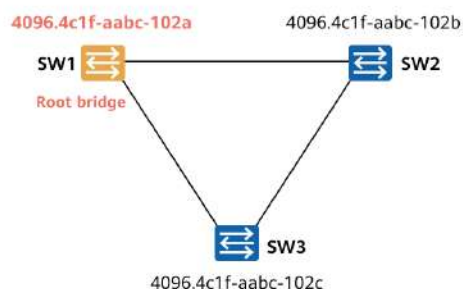
Bridge ID (BID)

- As defined in IEEE 802.1D, a BID consists of a 16-bit bridge priority and a bridge MAC address.
- Each switch running STP has a unique BID.
- The bridge priority occupies the leftmost 16 bits and the MAC address occupies the rightmost 48 bits.
- On an STP network, the device with the smallest BID acts as the root bridge.

Note: A bridge is a switch.

Source: Huawei Slide # 09, Page # 15

STP Basic Concepts: Root Bridge



Root Bridge

- One of the main functions of STP is to calculate a loop-free STP tree on the entire switching network.
- The root bridge is the root of an STP network.
- After STP starts to work, it elects a root bridge on the switching network. The root bridge is the key for topology calculation of the spanning tree and is the root of the loop-free topology calculated by STP.
- On an STP network, the device with the smallest BID acts as the root bridge.

During BID comparison, devices first compare bridge priorities. A smaller priority value indicates a higher priority of a device. The switch with the smallest priority value becomes the root bridge. If priority values are the same, the switch with the smallest MAC address becomes the root bridge.

411. The configuration information of an interface of the router is shown in the figure, then Which VLAN data packet can this port receive?

```
interface Gigabit Ethernet 0/0/2.30
dot1q termination vid 100
ip address 10.0.21.1 255.255.255.0
arp broadcast enable
```

- A. 1
- B. 100
- C. 30
- D. 20

Answer: B

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100112348/79dc6b8f/overview-of-vlan-termination>

Definition

VLAN termination is a VLAN tag processing mechanism. After VLAN termination is enabled on a device, the device identifies VLAN tags in received packets, removes single or double tags from the packets, and then forwards packets at Layer 3 or takes other actions. These VLAN tags are only useful before termination, and are not used in Layer 3 forwarding or other processing.

A device with VLAN termination enabled processes incoming and outgoing packets as follows:

- Removes VLAN tags from the packets received on interfaces, and then forwards the packets at Layer 3 or takes other actions.
- Adds VLAN tags to the packets that will be sent out through interfaces.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100276781/60c75ace/configuring-a-dot1q-termination-sub-interface-to-implement-inter-vlan-communication>

Run **dot1q termination vid** low-pe-vid [to high-pe-vid]

Dot1q termination is configured on the sub-interface.

Sub-interfaces of different main interfaces can be associated with the same VLAN, whereas sub-interfaces of the same main interface cannot be associated with the same VLAN.

Explanation:

- **Dot1q termination:** The device removes the outer VLAN tag from any received single-tagged or double tagged packets, and adds a VLAN tag to the packets to be sent by an interface.
- In Huawei routers and switches, when you configure dot1q termination vid 100 This means that the sub-interface (GigabitEthernet 0/0/2.30) is configured to **receive and process packets tagged with VLAN ID 100.**
- The .30 at the end of the interface name (i.e., 0/0/2.30) is just a sub-interface identifier and is not related to the VLAN ID.

Why the other options are wrong?

- **A. 1** – VLAN 1 is not specified anywhere in the configuration.
- **C. 30** – This is just the sub-interface number, not the VLAN.
- **D. 20** – Not mentioned in the config either.

412. In a Layer 2 network with redundant links, which of the following protocols can be used to avoid loops?

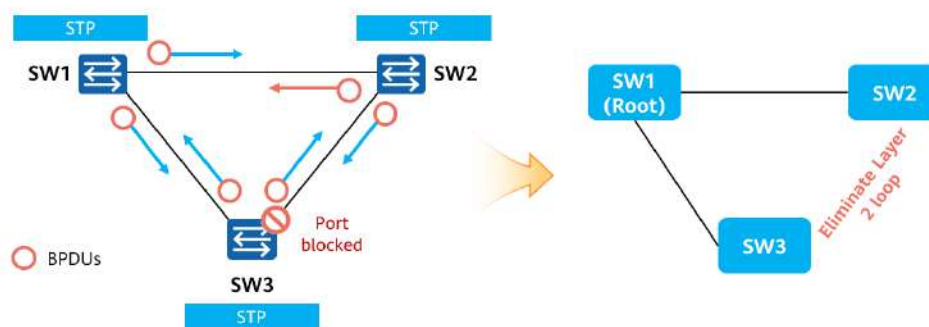
- A. VRRP
- B. ARP
- C. UDP
- D. STP

Answer: D

Justification

Source: Huawei Slide # 09, Page # 08

Introduction to STP



When STP is deployed on a network, switches exchange STP BPDUs and calculate a loop-free topology. Finally, one or more ports on the network are blocked to eliminate loops.

Source: Huawei Slide # 09, Page # 08 (notes section)

- On an Ethernet network, loops on a Layer 2 network may cause broadcast storms, MAC address flapping, and duplicate data frames. STP is used to prevent loops on a switching network.
- STP constructs a tree to eliminate loops on the switching network.
- The STP algorithm is used to detect loops on the network, block redundant links, and prune the loop network into a loop-free tree network. In this way, proliferation and infinite loops of data frames are avoided on the loop network.

413. When the switch port receives a data frame that does not carry a VLAN TAG, it must add PVID

- A. True
- B. False

Answer: A

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100096323/819ade70/understa%20nding-vlans>

VLAN Assignment

VLAN assignment can be based on interface numbers, and VLAN frames are processed depending on the interface type.

The network administrator configures a port default VLAN ID (PVID), that is, the default VLAN ID, for each port on the switching device. That is, a port belongs to a VLAN by default.

- When a data frame reaches a port, it is marked with the PVID if the data frame carries no VLAN tag and the port is configured with a PVID.
- If the data frame carries a VLAN tag, the switching device will not add a VLAN tag to the data frame even if the port is configured with a PVID.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100096323/819ade70/understa>

VLAN Communication

Implementation of VLAN Communication

To improve the efficiency in processing frames, frames within a switch all carry VLAN tags for uniform processing. When a data frame reaches a port of the switch, if the frame carries no VLAN tag and the port is configured with a PVID, the frame is marked with the port's PVID. If the frame has a VLAN tag, the switch will not mark a VLAN tag for the frame regardless of whether the port is configured with a PVID.

The switch processes frames differently according to the type of port receiving the frames. The following describes the frame processing according to the port type.

414. As shown in the figure below, the router uses a sub-interface as the gateway of the host. The IP address of the gateway is 10.0.12.2. Which of the following commands can achieve this requirement? (Multiple choice)



- A. interface GigabitEthernet0/0/0.20 dot1q termination vid 10
ip address 10.0.12.2 255.255.255.0
arp broadcast enable
#
- B. interface GigabitEthernet0/0/0.10 dot1q termination vid 10
ip address 10.0.12.2 255.255.255.0
arp broadcast enable
#
- C. interface GigabitEthernet0/0/0.10 dot1q termination vid 20
ip address 10.0.12.2 255.255.255.0
arp broadcast enable
#
- D. interface GigabitEthernet0/0/0.20 dot1q termination vid 20
ip address 10.0.12.2 255.255.255.0
arp broadcast enable
#

Answer: CD

Justification

Explanation: CD commands terminate VLAN 20, which is what needed to be done. AB terminates VLAN 10, which is wrong

Source: https://support.huawei.com/enterprise/en/doc/EDOC1100112348/cbf869f6/example-for-configuring-dot1q-termination-sub-interfaces-to-implement-inter-vlan-communication#EN-US_TASK_0177868189

1. Configure the interface on the Router connected to RouterB.
Create and configure Ethernet sub-interface GE2/0/0.1.
[Router] **interface gigabitEthernet 2/0/0.1**
[Router-GigabitEthernet2/0/0.1] **dot1q termination vid 20**
[Router-GigabitEthernet2/0/0.1] **ip address 10.10.20.1 24**
[Router-GigabitEthernet2/0/0.1] **quit**
2. Verify the configuration.
On PC1 in VLAN 10, set the IP address (10.10.10.1/24) of GE1/0/0.1 as the default gateway address.
On PC2 in VLAN 20, set the IP address (10.10.20.1/24) of GE2/0/0.1 as the default gateway address.

415. The output information of a switch is shown in the figure. Which of the following interfaces can forward data frames with VLAN ID 40 and not carry tags when forwarding?

```
<SWA>display vlan
The total number of vlans is : 5
```

VID	Type	Ports
10	common	UT:GE0/0/1(U) GE0/0/2(U)
20	common	TG:GE0/0/1(U) GE0/0/5(U)
30	common	TG:GE0/0/1(U)
40	common	UT:GE0/0/5(U) TG:GE0/0/1(U) GE0/0/3(U) GE0/0/4(U)

- A. GE0/0/2
- B. GE0/0/5
- C. GE0/0/3
- D. GE0/0/4

Answer: B

Justification

Explanation:

- Untagged port for VLAN 40:
 - GE0/0/5 is listed under UT (Untagged) for VLAN 40. This means:
 - VLAN 40 frames are sent and received without VLAN tags on this port.
- Tagged ports for VLAN 40:
 - GE0/0/1, GE0/0/3, and GE0/0/4 are listed under TG (Tagged), so they send VLAN 40 frames with tags.

416. Which of the following packet types is used by RADIUS to indicate authentication rejection?

- A. Access-Accept
- B. Access-Request
- C. Access-Challenge
- D. Access-Reject

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000177218/c58849f8/radius-packets>

Table 2-1 Authentication types

Packet Type	Description
Access-Request	This is the first packet transmitted in a RADIUS interaction process. This packet carries user authentication information, such as user name and password. The Access-Request packet is from the RADIUS client to the RADIUS server. The RADIUS server determines whether a user is allowed to access the network according to the user information carried in the received Access-Request packet.
Access-Accept	This packet is sent by the RADIUS server to respond to the Access-Request packet sent by the client. If all attributes in the Access-Request packet are acceptable, the server determines that the user passes the authentication and sends this packet. After receiving this packet, the client grants the network access rights to the user.
Access-Reject	This packet is sent by the RADIUS server to respond to the Access-Request packet sent by the client. If any attribute in the Access-Request packet is unacceptable, the RADIUS server determines that the user fails the authentication and sends this packet.
Access-Challenge	During an EAP authentication, when the RADIUS server receives an Access-Request packet carrying a user name, it generates a random MD5 challenge and sends the MD5 challenge to the client through an Access-Challenge packet. After the client encrypts the user password using the MD5 challenge, the client sends the encrypted password in an Access-Request packet to the RADIUS server. The RADIUS server compares the encrypted password received from the client with the locally encrypted password. If they are the same, the server determines the user is valid.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100306152/e1dc45f9/understanding-radius-based-aaa>

3. The RADIUS server verifies the user identity:

- If the user identity is valid, the RADIUS server returns an Access-Accept packet to the RADIUS client to permit further operations of the user. The Access-Accept packet contains authorization information because RADIUS provides both authentication and authorization functions.
- If the user identity is invalid, the RADIUS server returns an Access-Reject packet to the RADIUS client to reject access from the user.

417. NAPT distinguishes the IP addresses of different users through the protocol numbers in TCP, UDP, or IP packets.

- A. True
 B. False
- Source IP address
 Source port number (not TCP/UDP protocols)
 The key identifier is the port number, not the protocol number.

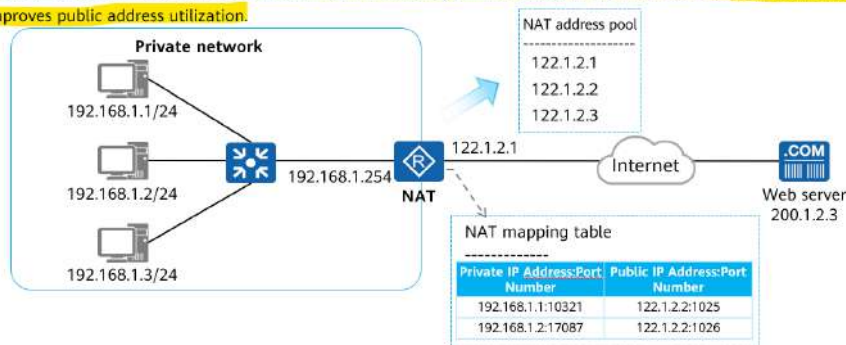
Answer: B

Justification

Source: Huawei Slide # 14, Page # 20

NAPT Implementation

- Dynamic NAT does not translate port numbers. It belongs to No-Port Address Translation (No-PAT). In this mode, the mapping between public and private addresses is still 1:1, which cannot improve public address utilization.
- Network Address and Port Translation (NAPT): translates both IP addresses and port numbers from multiple internal hosts to one public IP address in an address pool. In this way, 1:n mapping between public and private addresses is implemented, which effectively improves public address utilization.



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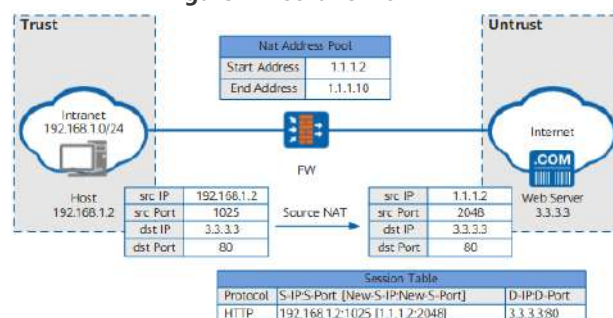
Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100149308/AEJ0713J/17/resources/admin/sec_admin_nat_0094_3.html#:~:text=NAPT%20translates%20both%20IP%20addresses,need%20to%20access%20the%20Internet

NAPT

NAPT translates both IP addresses and ports to enable multiple private addresses to share one or multiple public addresses. NAPT applies to scenarios with a few public addresses but many private users who need to access the Internet. [Figure 1](#) shows its mechanism.

Figure 1 Mechanism of NAPT



418. Which of the following authentication methods does not require the user name and password?

- A. authentication-mode local
- B. authentication-mode none
- C. authorization-mode hwtacacs
- D. Authentication-mode hwtacacs

Answer: B

Justification

Source:

https://support.huawei.com/enterprise/en/doc/EDOC1000178178/c57f8bee/aaascheme#dc_fd_aaa_0005_1

Authentication Scheme

An authentication scheme is used to define methods for user authentication and the order in which authentication methods take effect. An authentication scheme is applied to a domain. It is combined with the authorization scheme, accounting scheme, and server template in the domain for user authentication, authorization, and accounting.

Authentication Methods Supported by a Device

- RADIUS authentication: User information is configured on the RADIUS server through which user authentication is performed.
- HWTACACS authentication: User information is configured on the HWTACACS server through which user authentication is performed.
- Local authentication: The device functions as an authentication server and user information is configured on the device. This mode features fast processing and low operation costs. However, the information storage capacity is subject to the device hardware.
- Non-authentication: Users are completely trusted without validity check. This mode is rarely used.

419. Telnet is based on the TCP protocol.

- A. True
- B. False

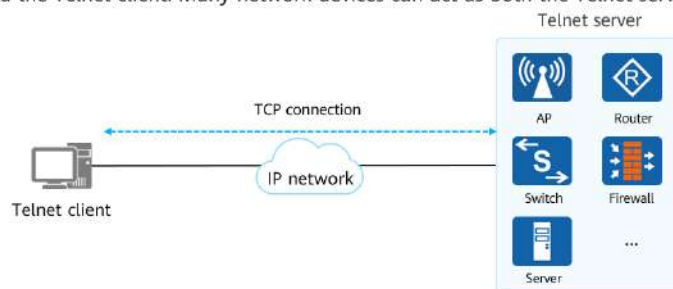
Answer: A

Justification

Source: Huawei Slide # 15, Page # 17

Application Scenario of Telnet

- To facilitate device management using commands, you can use Telnet to manage devices.
- Device management through Telnet is different from that using the console port. In Telnet-based device management mode, no dedicated cable is required to directly connect to the console port of the Telnet server, as long as the Telnet server's IP address is reachable and **Telnet clients can communicate with the Telnet server's TCP port 23.**
- The device that can be managed through Telnet is called the Telnet server, and the device connecting to the Telnet server is called the Telnet client. Many network devices can act as both the Telnet server and Telnet client.



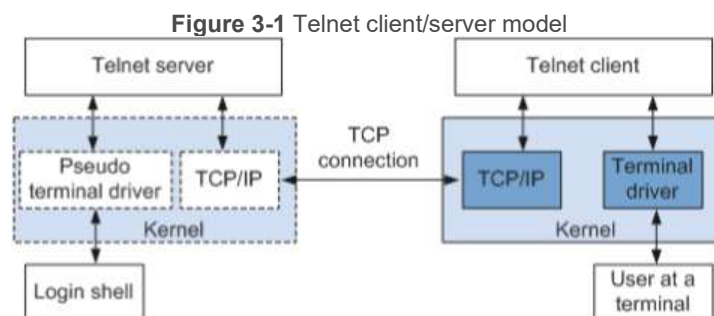
Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100366532/b5747c35/overview-of-telnet>

Overview of Telnet

The Telecommunication Network Protocol (Telnet) is derived from ARPANET released in 1969. It is the earliest Internet application.

A Telnet connection is a Transmission Control Protocol (TCP) connection used to transmit data with interspersed Telnet control information. Telnet uses the client/server model to present an interactive interface that enables a terminal to remotely log in to a server. A user can log in to one host and then use Telnet to remotely log in to and configure and manage multiple hosts without having to connect each one to a terminal. [Figure 3-1](#) shows the Telnet client/server model.



420. CAPWAP protocol stipulates the communication standard between AC and AP. Which of the following statements about the CAPWAP protocol is correct?

- A. CAPWAP is an application-layer protocol based on TCP transmission
- B. To establish a CAPWAP tunnel, FIT Ap can only discover the AC through broadcast messages
- C. CAPWAP In order to reduce the burden on the AP, a tunnel is used to transmit control messages and data messages at the same time
- D. AP can encapsulate the user's data message in CAPWAP and forward it by AC

Answer: D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100334297/2471df23/understa%20ndin g-%20capwap>

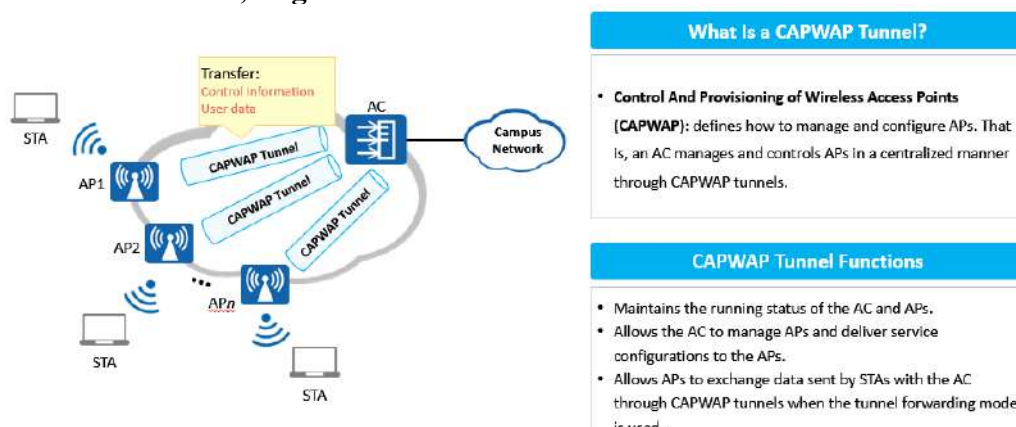
DTLS Encryption for CAPWAP Control Tunnels

Extending the protocol architecture of Transport Layer Security (TLS), DTLS ensures the security for **UDP packets, which are used by CAPWAP.**

Clients can discover servers in static or dynamic mode.

- **Static mode**
A static IP address list of servers is preconfigured on clients. When a client goes online, it unicasts a Discovery Request message to each server in the preconfigured IP address list. After receiving the Discovery Request message, the servers respond to the client with a Discovery Response message. The client then selects a server to establish a CAPWAP tunnel.
 - **Dynamic mode**
A client can dynamically discover servers in one of these modes: **Dynamic Host Configuration Protocol (DHCP), Domain Name Service (DNS), and broadcast.**
2. CAPWAP tunnel establishment phase:
CAPWAP tunnels fall into two categories: data tunnel and control tunnel. Whether both types need to be established or only the latter depends on the data forwarding mode.
- CAPWAP data tunnel: transmits data packets received by clients to the server for centralized forwarding.
 - CAPWAP control tunnel: transmits control packets between the server and clients. Datagram Transport Layer Security (DTLS) encryption can be enabled for CAPWAP control tunnels to encrypt and decrypt CAPWAP control packets.

Source: Huawei Slide 16, Page # 16



Explanation:

- CAPWAP communicates through UDP, so statement (A) is wrong. CAPWAP can be discovered through different methods, such as DHCP, DNS, and Static IP config. So, statement (B) is wrong.
- CAPWAP uses separate tunnels for control and data messages to optimize performance and security. Control messages are sent over a control tunnel, and data messages are sent over a data tunnel. So statement (C) is wrong.

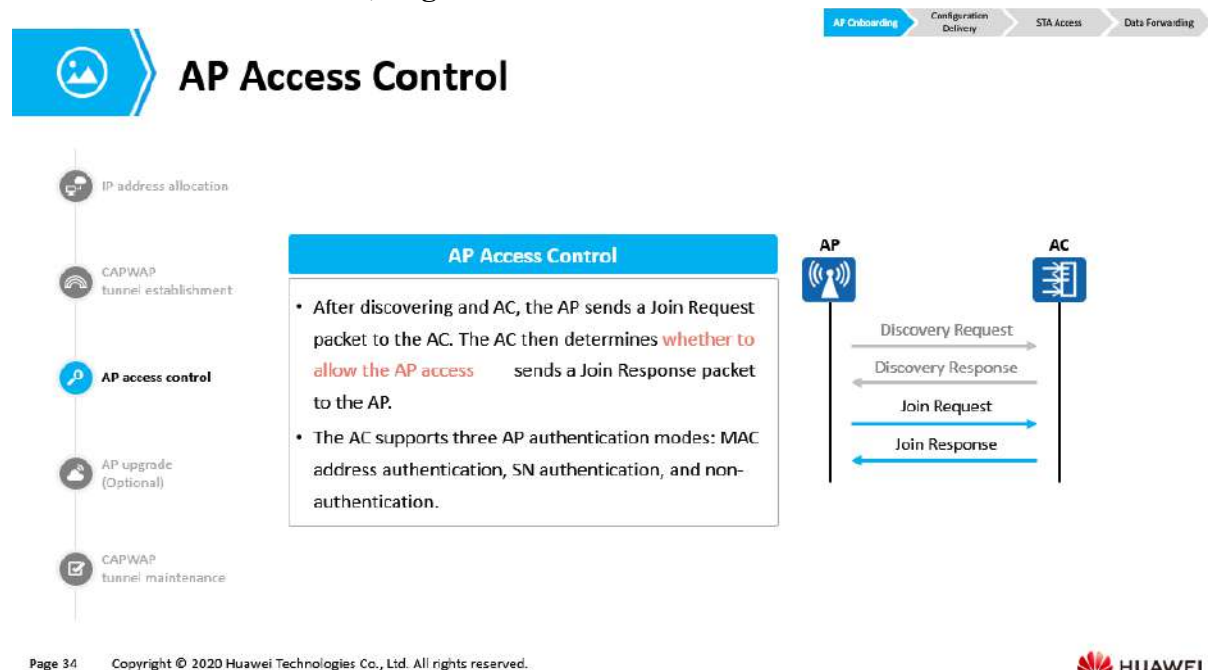
421. The wireless access controller (AC, Access Controller) is the unified management and control device in the FIT AP architecture. Which of the following description of the role of AC is wrong?

- A. No matter when the data is forwarded, the user's data message will be forwarded by AC
- B. User access control
- C. User access authentication
- D. AP configuration delivery

Answer: A

Justification

Source: Huawei Slide # 16, Page # 34



Explanation: No matter when the data is forwarded, the user's data message will be forwarded by AC. This statement is incorrect because, in FIT AP architectures, data forwarding can be performed locally by the APs in many cases, especially when local forwarding is configured to improve network performance and reduce the load on the AC.

422. How many more bytes are the packets using MPLS label double-layer nesting technology than the original IP packets?

- A. 32
- B. 8
- C. 4
- D. 16

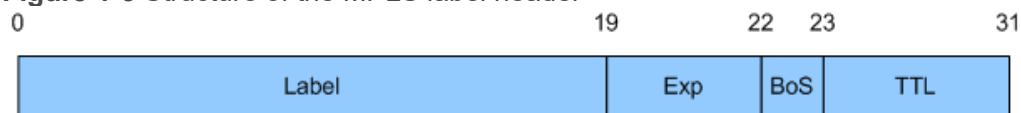
Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100118961>

Figure 1-5 illustrates the structure of a 4-byte MPLS label header.

Figure 1-5 Structure of the MPLS label header



An MPLS label header contains the following fields:

- Label: a 20-bit field that identifies a label value.
- Exp: a 3-bit field used for extension. This field is used by the class of service (CoS) function, which is similar to Ethernet 802.1p.
- BoS: a 1-bit field that identifies the bottom of a label stack. MPLS supports multiple labels, which are stacked. If the BoS field of a label is set to 1, the label is at the bottom of the label stack.
- TTL: an 8-bit field indicating a time to live (TTL) value. This field implements the same function as the TTL field in IP packets.

Calculation:

1. **MPLS Label Structure:** Each MPLS label is **32 bits (4 bytes)**. This includes fields like the label value, traffic class (QoS), bottom of stack, and TTL.
2. **Double-Layer Nesting:** This involves **two MPLS labels** (outer and inner).
 - **Total added bytes:** $2 \times 4 = 8$ bytes.

423. ::1/128 is the IPv6 loopback address.

- A. True
- B. False

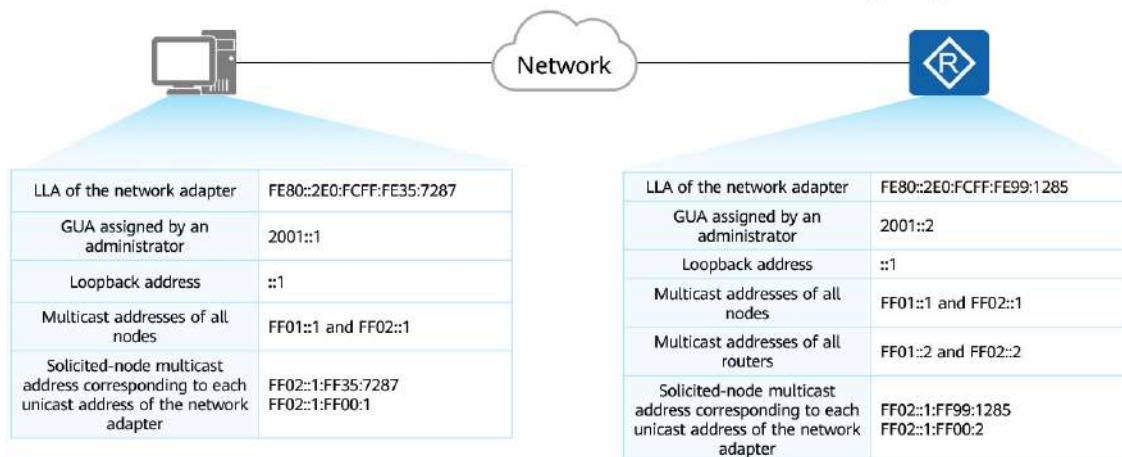
Answer: A

Justification

Source: Huawei Slide # 19, Page # 25

IPv6 Addresses of Hosts and Routers

- The unicast IPv6 addresses and multicast addresses of hosts and routers are typically as follows:



Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100334321/AEM1020X/05/resources/dc/dc_feature_ipv6_0004.html#:~:text=An%20IPv6%20loopback%20address%20is,a%20loopback%20interface%20for%20example

- Loopback address
An IPv6 loopback address is 0:0:0:0:0:0:0:1/128 or ::1/128. Similar to IPv4 loopback address 127.0.0.1, IPv6 loopback address is used when a node needs to send IPv6 packets to itself. This IPv6 loopback address is usually used as the IP address of a virtual interface (a loopback interface, for example). The loopback address cannot be used as the source or destination IP address of packets that need to be forwarded.

Explanation: The IPv6 loopback address is ::1/128, representing a single host (128-bit prefix) for internal communication. This is standardized in RFC 4291.

424. Which of the following IPv6 addresses is a global unicast address?

- A. 2001::12:1
- B. FF02::1
- C. FF::02::2E0:FCFF:FEEF:FEC
- D. FEE:FCFF:FEEF:FEC

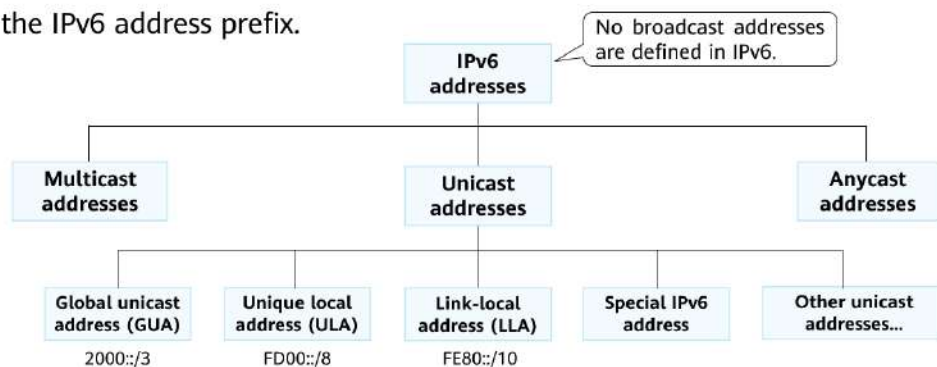
Answer: A

Justification

Source: Huawei Slide # 19, Page # 25

IPv6 Address Classification

- IPv6 addresses are classified into unicast, multicast, and anycast addresses according to the IPv6 address prefix.



Explanation: Option (A) 2001::12:1 This address falls within the range of 2000::/3 (which covers from 2000:: to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF), indicating that it is a global unicast address.

425. In the campus network planning, in which of the following ways can business VLAN be divided?

- A. Divide the VLAN by the access member structure
- B. Divide the VLAN by business
- C. Divide VLAN by logical area
- D. Divide the VLAN by geographical area

Answer: B

Justification

Source: Huawei Slide # 22, Page # 17



VLAN Planning

Design

Implementation

Optimization

- A management VLAN is reserved for Layer 2 devices.
- VLANs are classified into the guest VLAN, R&D department VLAN, marketing department VLAN, and administrative department VLAN.
- Layer 3 switches need to be connected to routers through VLANIF interfaces. Therefore, interconnection VLANs need to be reserved.
- A VLAN is established for CAPWAP tunnels between APs and ACs.

VLAN ID	VLAN Description
1	Guest VLAN or WLAN service VLAN
2	R&D department VLAN
3	Marketing department VLAN
4	Administrative department VLAN
100	Management VLAN of Layer 2 devices
101	Management VLAN of WLAN services
102	Interconnection VLAN between Agg-S1 and CORE-R1

Explanation:

In campus network planning, business VLANs are typically divided based on business functions or services (e.g., separating HR, finance, or IoT traffic) to ensure security, traffic prioritization, and efficient resource allocation.

426. When configuring the router through the Console, what is the correct setting of the terminal emulation program?

- A. 0bps, 8 data bits, 1 stop bit, no checksum, and no flow control
- B. 9600bps, 8 data bits, 1 stop bit, even parity, and hardware flow control
- C. 9600bps, 8 data bits, 1 stop bit, no checksum, and no flow control
- D. 4800bps, 8 data bits, 1 stop bit, odd checksum, and no flow control

Answer: C

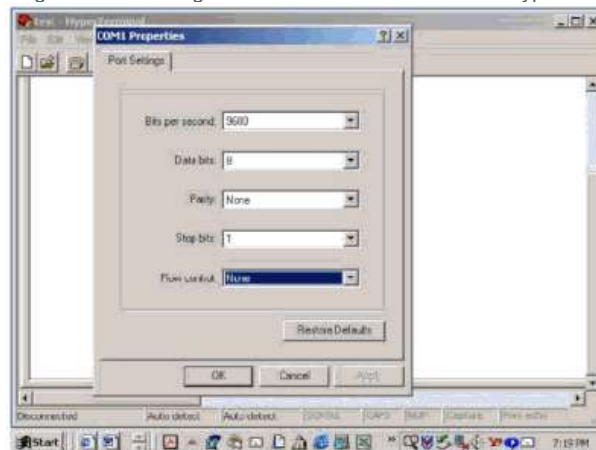
Justification

Source: <https://www.cisco.com/c/en/us/support/docs/storage-networking/management/217970-troubleshoot-and-apply-correct-terminal.html>

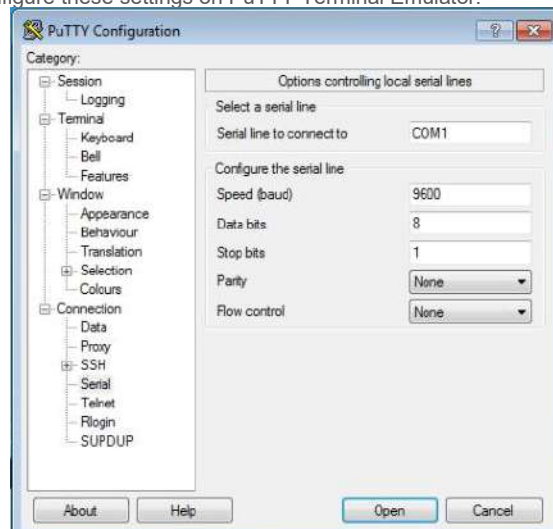
The settings in the Terminal Emulator software need to be set correctly; otherwise, unintelligible characters show up on the screen. When you set up the connection, use these settings:

```
Bits per sec    : 9600
Data bits       : 8
Parity          : none
Stop bits       : 1
Flow control    : none
```

Here is an example of how to configure these settings on a Windows-based PC with Hyper Terminal:



Here is an example of how to configure these settings on PuTTY Terminal Emulator:



427. In the VRP interface, use the command delete vrpcfg.zip to delete the file. You must empty the recycle bin to completely delete the file.

- A. True
- B. False

Answer: A

Justification

Source: Huawei Slide # 03, Page # 42

By default, configurations are saved in the vrpcfg.cfg file. You can also create a file for saving the configurations. VRPv5 and VRPv8 have the same command that is used to specify the configuration file for the next startup, but different directories for saving the file.

Source: Huawei Slide # 03, Page # 31



Common File System Operation Commands (3)

11. Restore a deleted file.

```
<Huawei>undelete
```

12. Permanently delete a file in the recycle bin.

```
<Huawei>reset recycle-bin
```

428. The router running the OSPF protocol reaches the FULL state first, and then synchronizes the LSDB.

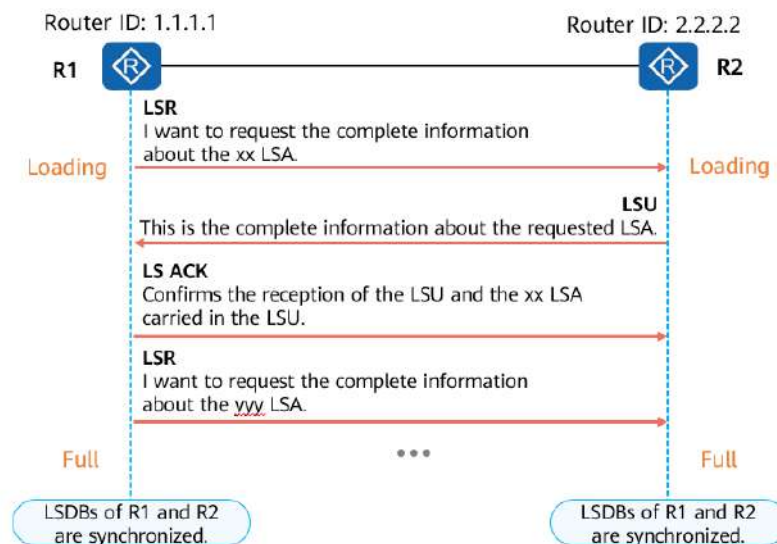
- A. True
- B. False

Answer: B

Justification

Source: Huawei Slide # 06, Page # 2

Process of Establishing an OSPF Adjacency - Step 4



Source: <https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/13685-13.html>

Full

In this state, routers are fully adjacent with each other. All the router and network LSAs are exchanged and the routers' databases are fully synchronized.

Full is the normal state for an OSPF router. If a router is stuck in another state, it is an indication that there are problems when the adjacencies are formed. The only exception to this is the 2-way state, which is normal in a broadcast network. Routers achieve the FULL state with their DR and BDR in NBMA/broadcast media and FULL state with every neighbor in the residual media such as point-to-point and point-to-multipoint.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100301642/32a400ec/ospf-fundamentals>

The OSPF neighbor state machine has the following states:

- **Full:** This state occurs when two neighbors complete their LSDB synchronization and establish an adjacency.

429. As shown in the following figure, router A and router B establish an OSPF neighbor relationship. The OSPF process number of router A is 1, and the area number is 0. Which of the following configurations on router A can enable router B to obtain the network segment of host A routing? (Multiple choice)



- A. `ospf 1`
`area 0.0.0.0`
`network 192.168.1.2 0.0.0.0`
- B. `ospf 1`
`import-route direct`
- C. `ospf 1`
`area 0.0.0.0`
`network 192.168.0.0 0.0.255.255`
- D. `ospf 1`
`area 0.0.0.0`
`network 192.168.1.0 0.0.0.255`

Answer: BD

Justification

Explanation:

- **Option B: import-route direct (Correct Answer)**
This command imports directly connected routes (like 192.168.1.0/24) into the OSPF routing process, making them available to OSPF neighbors even if the interfaces are not participating in OSPF. Since the interface connecting to Host A is not part of the OSPF adjacency, this method ensures the network is still advertised.
- **Option D: network 192.168.1.0 0.0.0.255 (Correct Answer)**
This OSPF configuration uses a wildcard mask of 0.0.0.255, which matches all IP addresses from 192.168.1.0 to 192.168.1.255 (i.e., the /24 subnet). It enables OSPF on the interface within this subnet and advertises the 192.168.1.0/24 network to OSPF neighbors like Router B.
- **Option A: network 192.168.1.2 0.0.0.0 (Incorrect Answer)**
This matches only a single host address (192.168.1.2) and does not enable OSPF on the entire subnet or ensure the connected network is advertised effectively.
- **Option C: network 192.168.0.0 0.0.255.255 (Incorrect Answer)**
Although this covers the correct IP range, it still does not ensure the advertisement of the connected subnet if the interface does not form an OSPF adjacency.

430. How to enter the view of OSPF area 0 in the VRP operating system? (Multiple choice)

- A. [Huawei-ospf-1] area 0 enable
- B. [Huawei-ospf-1] area 0
- C. [Huawei] ospf area 0
- D. [Huawei-ospf-1] area 0.0.0.0

Answer: BD

Justification

Source: Huawei Slide # 06, Page # 39



Basic OSPF Configuration Commands (1)

1. (System view) Create and run an OSPF process.

```
[Huawei] ospf [ process-id | router-id router-id ]
```

The *process-id* parameter specifies an OSPF process. The default process ID is 1. OSPF supports multiple processes. Multiple OSPF processes can separately run on the same device. The **router-id** command is used to manually specify the ID of a device. If no ID is specified, the system automatically selects the IP address of an interface as the device ID.

2. (OSPF view) Create an OSPF area and enter the OSPF area view.

```
[Huawei] area area-id
```

The **area** command creates an OSPF area and displays the OSPF area view.
The *area-id* value can be a decimal integer or in dotted decimal notation. If the value is an integer, it ranges from 0 to 4294967295.

3. (OSPF area view) Specify the interface that runs OSPF.

```
[Huawei-ospf-1-area-0.0.0.0] network network-address wildcard-mask
```

The **network** command specifies the interface that runs OSPF and the area to which the interface belongs. The *network-address* parameter specifies the network segment address of the interface. The *wildcard-mask* parameter is the wildcard of an IP address, which is equivalent to the reverse mask of the IP address (0 is converted to 1, and 1 to 0). For example, 0.0.0.255 indicates that the mask length is 24 bits.

431. What is the protocol number corresponding to the OSPF protocol at the network layer?

- A. 89
- B. 6
- C. 0
- D. 1

Answer: A

Justification

Source:

https://info.support.huawei.com/hedex/api/pages/EDOC1100277644/AEM10221/03/resources/vrp/feature_0003998103.html

OSPF Packet Format

OSPF packets are encapsulated into IP packets. The protocol number is 89. OSPF packets are classified into five types of packets: Hello packets, DD packets, LSR packets, LSU packets, and LSAck packets.

432. As shown in the figure below, all hosts can pass normally, so what is correct about the correspondence between the MAC address and the port in the MAC table of the SWB?



- A. MAC-A G0/0/2 MAC-B G0/0/2 MAC-C G0/0/3
- B. MAC-A G0/0/3 MAC-B G0/0/3 MAC-C G0/0/1
- C. MAC-A G0/0/1 MAC-B G0/0/1 MAC-C G0/0/3
- D. MAC-A G0/0/1 MAC-B G0/0/2 MAC-C G0/0/3

Answer: B

Justification

Explanation:

In a scenario where Switch SWB is connected to Host C on interface G0/0/1 and to another switch (SWA) on interface G0/0/3:

- When Host C sends a frame, SWB learns MAC-C on G0/0/1.
- When frames from Hosts A and B arrive via SWA, SWB learns MAC-A and MAC-B on G0/0/3.

433. Which of the following statements about the role of the STP protocol's Forward Delay is correct?

- A. Prevent temporary loops
- B. There is a delay when the blocking state and the Disabled state are converted
- C. Reduce the time interval of BPDU sending. Improve the convergence speed of STP
- D. Improve the survival time of BPDUs to ensure that the configured BPDUs can be forwarded to more switches

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100092143>

- Forward Delay

The Forward Delay timer specifies the delay in a port state transition. When a link fails, STP calculation is triggered and the spanning tree structure changes accordingly. However, new configuration BPDUs cannot be flooded immediately across the entire network. If the new root port and designated port forward data immediately, transient loops may occur. Therefore, STP defines a delay mechanism for port state transition. The newly selected root port and designated port must wait for two Forward Delay intervals before transitioning to the Forwarding state. During this period, the new configuration BPDUs can be transmitted over the network, preventing transient loops.

The Forward Delay timer defines the time that is spent in Listening or Learning state. The default Forward Delay value is 15 seconds. This means that the port stays in Listening state for 15 seconds and then stays in Learning state for another 15 seconds before transitioning to the Forwarding state. The port in Listening or Learning state does not forward user traffic, effectively preventing transient loops.

434. The configuration information of a certain port of the switch is shown in the figure.
Which of the following statements is wrong?

```
#
interface GigabitEthernet0/0/2
 port hybrid pvid vlan 100
 port hybrid tagged vlan 100
 port hybrid untagged vlan 200
#
```

- A. The port type is Hybrid
- B. If the port receives a data frame without a VLAN TAG, the switch needs to add a VLAN TAG 100
- C. If the VLAN TAG carried in the data frame is 200, the switch strips off the VLAN TAG
- D. If the VLAN TAG carried in the data frame is 100, the switch will strip off the VLAN TAG and send it out

Answer: D For a hybrid interface, the tagged vlan command explicitly instructs the switch to keep the VLAN tag when transmitting out of the port.

Justification

Source: <http://support.huawei.com/enterprise/en/doc/EDOC1000178168/f9b8a34a/adding-and-removing-vlan-tags>

Table 5-2 Frame processing based on the port type

Port Type	Untagged Frame Processing	Tagged Frame Processing	Frame Transmission
Hybrid port	▪ Adds a tag with the default VLAN ID to an untagged frame and accepts the frame if the port permits the default VLAN ID. ▪ Adds a tag with the default VLAN ID to an untagged frame and discards the frame if the port denies the default VLAN ID.	▪ Accepts a tagged frame if the VLAN ID carried in the frame is permitted by the port. ▪ Discards a tagged frame if the VLAN ID carried in the frame is denied by the port.	If the frame's VLAN ID is permitted by the port, the frame is transmitted. The port can be configured to transmit frames with tags.

Explanation of Each Line:

- port hybrid pvid vlan 100:
 - Sets the default VLAN ID (PVID) to 100 for untagged incoming frames.
 - Frames without a VLAN tag will be tagged with VLAN 100 on ingress.
- port hybrid tagged vlan 100:
 - Outgoing frames for VLAN 100 will retain the tag.
- port hybrid untagged vlan 200:
 - Outgoing frames for VLAN 200 will have their VLAN tags stripped (sent untagged).

435. Neither the Alternate interface nor the Backup port in RSTP can forward user traffic, nor can they receive, process, and forward BPDUs.

don't forward user-traffic but process BPDUs

- A. True
- B. False

- 1. Alternate Port
Backup path to Root Port
Used when RP fails
- 2. Backup Port (In RSTP not STP)
Backup for a Designated Port on the same switch
Appears when two ports connect to the same shared segment

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100276753/2bd0e592/rstpport-roles-and-%20port-states>

Alternate port and backup port

Category	Alternate Port	Backup Port
From the perspective of configuration BPDUs transmission	An alternate port is blocked after learning a configuration BPDU sent from another bridge.	An alternate port is blocked after learning a configuration BPDU sent from itself.
From the perspective of user traffic	An alternate port acts as a backup of the root port and provides an alternate path from the designated bridge to the root bridge.	A backup port acts as a backup of the designated port and provides a backup path from the root bridge to the related network segment.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100332771/b4e804a4/understanding-stp-rstp>

Port Role

Table 6-19 lists the port roles supported in STP and RSTP, among which the root port (RP) and designated port (DP) are the basic port roles.

Compared with STP, RSTP adds three port roles: alternate port, backup port, and edge port.

Table 6-19 Port roles defined in STP and RSTP

Spanning Tree Protocol	Root Port	Designated Port	Alternate Port	Backup Port	Edge Port
STP	√	√	x	x	x
RSTP	√	√	√	√	√

Alternate Port

An alternate port backs up the root port and provides an alternate path from the designated bridge to the root bridge. It is blocked after learning the configuration BPDUs sent by other bridges.

Backup Port

A backup port backs up a designated port and provides a backup path from the root bridge to the related network segment. It is blocked after learning the configuration BPDUs sent by itself.

Explanation: That's not entirely accurate. In Rapid Spanning Tree Protocol (RSTP), both the Alternate and Backup ports can indeed receive BPDUs to determine the best path to the root bridge. However, they do not forward user traffic or BPDUs. Their primary role is to provide redundancy; the Alternate port serves as a backup for the Root port, and the Backup port serves as a backup for another port on the same segment that forwards traffic.

436. Which of the following statements about VLANIF interfaces is correct?

- A. VLANIF interface does not need to learn the MAC address
- B. VLANIF interface has no MAC address virtual MAC address used for layer-3 ARP forward / replies
- C. Different VLANIF interfaces can use the same IP address IP is unique for each interface
- D. VLANIF interface is a three-layer interface

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1000060766/f3dc84c8/what-are-the-%20functions-of-vlan-and-vlanif-interfaces>

What Are the Functions of VLAN and VLANIF Interfaces?

The functions of VLAN assignment are as follows:

- Limits broadcast domains. Broadcast domains are limited to conserve bandwidth and improve network efficiency.
- Enhances LAN security. Packets from different VLANs are transmitted separately. Hosts in a VLAN cannot communicate directly with hosts in another VLAN.
- Improves network robustness. A fault in a VLAN does not affect hosts in other VLANs.
- Allows flexible construction of virtual groups. With VLAN technology, hosts in different geographical locations can be grouped together, simplifying network construction and maintenance.

The functions of configuring VLANIF interfaces are as follows: A VLANIF interface is a VLAN-based Layer 3 logical interface and can be configured with an IP address. After VLANs are assigned, users in the same VLAN can communicate with each other while users in different VLANs cannot. To implement communication between VLANs, you can configure a logical Layer 3 interface (VLANIF interface).

437. Which of the following statements about the edge port in the RSTP protocol is correct?

- A. The edge port discards the received configuration BPDU packet
- B. The edge port can be directly transferred from the Disabled to the Forwarding state
- C. Edge port participates in RSTP operation
- D. The interconnection ports between switches need to be set as edge ports

Answer: B

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100332771/9fc55767/configuring-edge-ports-and-bpdu-filtering-ports-for-rstp>

Configuring Edge Ports and BPDU Filtering Ports (for RSTP)

Context

RSTP defines a port that is located at the network edge and directly connected to a terminal as an edge port.

Edge ports do not receive or process configuration BPDUs or participate in RSTP calculation. They can change from the Disabled state to the Forwarding state without any delay, as if RSTP was disabled.

Edge ports send BPDUs. If the BPDUs are sent to other networks, flapping may occur on these networks. BPDU filtering can be configured on the edge ports to disable them from processing and sending BPDUs.

438. Static NAT can only achieve one-to-one mapping between private addresses and public addresses.

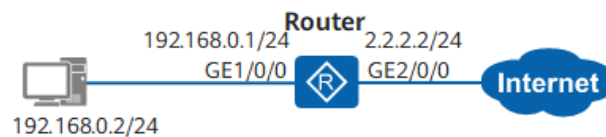
- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100034231/70655889/example-for-configuring-static-one-to-one-nat>

Figure 5-18 Networking diagram for configuring static one-to-one NAT



Configuration Roadmap

The configuration roadmap is as follows:

1. Configure the IP address of ports, default route, and static NAT on the WAN interface to implements **one-to-one translation between a private IP address and a public IP address.**

Source: https://support.huawei.com/enterprise/en/doc/EDOC1100086056#EN-US_CONCEPT_0233837380

Static NAT VS Dynamic NAT

Types of Destination NAT		Description
Dynamic NAT	Public addresses are randomly translated into addresses in the destination address pool.	Dynamic NAT applies when there are no fixed mappings between public and private addresses and public addresses are randomly translated into addresses in the destination address pool.
Static NAT	One-to-one mappings between public and private addresses.	This mode of static NAT applies when a public address is used to access a private address or multiple public addresses are used to access multiple private addresses.
	One-to-one mappings between public and private ports.	This mode of static NAT applies when multiple ports of a public address are used to access multiple ports of a private address.
	One-to-one mappings between multiple ports of a public address and multiple private addresses.	This mode of static NAT applies when multiple ports of a public address are used to access multiple private addresses.
	One-to-one mappings between multiple public addresses and multiple private ports.	This mode of static NAT applies when multiple public addresses are used to access multiple ports of a private address.
	NAT Server.	This mode of static NAT applies to scenarios where there are fixed mappings between private IP addresses and public IP addresses, between private IP addresses and public port numbers, between private port numbers and public IP addresses, and between private port numbers and public port numbers. NAT Server is implemented by running the nat server command.

439. A certain ACL of the router is in the following rules: rule deny source 192.168.2.0 0.0.0.255 destination 172.16.10.2 0.0.0.0. Which statement is correct?

- A. All TCP packets whose source IP is 172.16.10.2 and destination IP is 192.168.2.1 match this rule
- B. All TCP packets whose source IP is 192.168.2.1 and destination IP is 172.16.10.1 match this rule
- C. All TCP packets whose source IP is 172.16.10.2 and destination IP is 192.168.2.0 match this rule
- D. All TCP packets whose source IP is 192.168.2.1 and destination IP is 172.16.10.2 are configured with this rule.

Answer: D

Justification

Source: Huawei Slide # 12, Page # 13

- **Advanced ACL**
Number range: 3000-3999

Source IP address, destination IP address, and protocol type

Source and destination port numbers

IP Header	TCP/UDP Header	Data
<pre>acl number 3000 rule 5 permit ip source 10.1.1.0 0.0.0.255 destination 10.1.3.0 0.0.0.255 rule 10 permit tcp source 10.1.2.0 0.0.0.255 destination 10.1.3.0 0.0.0.255 destination-port eq 21</pre>		

Explanation:

The ACL rule is: rule deny source 192.168.2.0 0.0.0.255 destination 172.16.10.2 0.0.0.0

This means:

- **Source IP:** 192.168.2.0 with wildcard mask 0.0.0.255 ⇒ This matches 192.168.2.0 to 192.168.2.255
- **Destination IP:** 172.16.10.2 with wildcard mask 0.0.0.0 ⇒ This matches only 172.16.10.2
- This is a **deny** rule

So, it matches any packet (not just TCP unless specified) with:

- Source IP: 192.168.2.x
- Destination IP: 172.16.10.2

440. RADIUS is a common protocol to implement AAA.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100198635/896511fc/configuring-radius-aaa>

RADIUS Authentication, Authorization, and Accounting

RADIUS is often used to implement authentication, authorization, and accounting (AAA). It uses the client/server model and protects a network from unauthorized access. It is often used in network environments that require high security and control of remote user access.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100306152/e1dc45f9/understanding-radiusbased-aaa>

Overview of RADIUS

AAA can be implemented using multiple protocols, among which RADIUS is most frequently used in real-world applications.

RADIUS is a protocol that uses the client/server model in distributed mode and protects a network from unauthorized access. It is often used on networks that require high security and allow remote user access. It defines the UDP-based RADIUS packet format and transmission mechanism, and specifies UDP ports 1812 and 1813 as the default authentication and accounting ports, respectively.

441. What kind of message does the DHCP server use to confirm that the host can use the IP address?

- A. DHCP DISCOVER
- B. DHCP ACK
- C. DHCP OFFER
- D. DHCP REQUEST

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100276747/225eec10/dhcp-messages>

Table 3-1 DHCP message types

DHCP Message	Description
DHCP Discover	A DHCP client broadcasts this message to locate a DHCP server when the client attempts to connect to a network for the first time.
DHCP Offer	A DHCP server sends this message in response to a DHCP Discover message. A DHCP Offer message carries configuration information.
DHCP Request	A DHCP client sends this message in the following scenarios: • After the client starts, it broadcasts a DHCP Request message to respond to a DHCP Offer message sent by a DHCP server. • After the client restarts, it broadcasts a DHCP Request message to confirm the configuration (including the allocated IP address). • After the client obtains an IP address, it unicasts or broadcasts a DHCP Request message to renew the IP address lease.
DHCP ACK	A DHCP server sends this message to acknowledge a DHCP Request message sent from a DHCP client. After receiving a DHCP ACK message, the DHCP client obtains configuration parameters (including an IP address).
DHCP NAK	A DHCP server sends this message to reject a DHCP Request message from a DHCP client. For example, a DHCP server will send this message if it determines that there is no available IP address after receiving a DHCP Request message.
DHCP Decline	A DHCP client sends this message to notify the DHCP server that the allocated IP address conflicts with another IP address. The DHCP client then applies to the DHCP server for another IP address.
DHCP Release	A DHCP client sends this message to release its allocated IP address. After receiving a DHCP Release message, the DHCP server can allocate this IP address to another DHCP client.
DHCP Inform	A DHCP client sends this message to obtain network configuration parameters, such as the gateway address and DNS server address, after it has obtained an IP address.

442. The administrator has configured the router as shown in the figure. What IP address can a host connected to the G1/0/0 interface of the router obtain through DHCP?

```
[Router] ip pool pool1
[Router-ip-pool-pool1] network 10.10.10.0 mask 255.255.255.0
[Router-ip-pool-pool1] gateway-list 10.10.10.1
[Router-ip-pool-pool1] quit
[Router] ip pool pool2
[Router-ip-pool-pool2] network 10.20.20.0 mask 255.255.255.0
[Router-ip-pool-pool2] gateway-list 10.20.20.1
[Router-ip-pool-pool2] quit
[Router] interface gigabitethernet 1/0/0
[Router-GigabitEthernet1/0/0] ip address 10.10.10.1 24
[Router-GigabitEthernet1/0/0] dhcp select global
```

- A. The IP address data obtained by the host 10.10.10.0/24
- B. The IP address obtained by the host may belong to the 10.10.10.0/24 network, or it may belong to the 10.20.20.0/24 network
- C. The host cannot obtain an IP address
- D. The IP address obtained by the host belongs to the 10.20.20.0/24 network

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100306163/bece38d2/configuring-an-interface-address-pool>

A DHCPv4 server selects an address pool based on whether a DHCPv4 relay agent is deployed. If no DHCPv4 relay agent is deployed, the DHCPv4 server selects an address pool that is on the same network segment as the IPv4 address of the interface that receives DHCPv4 request messages. If a DHCPv4 relay agent is deployed, the DHCPv4 server selects an address pool that is on the same network segment as the IPv4 address specified in the giaddr field of received DHCPv4 request messages.

Explanation:

The router is configured with two DHCP address pools:

- pool1: for the **10.10.10.0/24** network
- pool2: for the **10.20.20.0/24** network

But the key to determining which pool will be used lies in the interface IP address:

```
[Router-GigabitEthernet1/0/0] ip address 10.10.10.1 24
```

The interface G1/0/0 is in the 10.10.10.0/24 network, and the router uses the command:

```
[Router-GigabitEthernet1/0/0] dhcp select global
```

This enables the global DHCP server. The router will match the interface IP address to the appropriate DHCP pool. Since the interface IP is 10.10.10.1/24, it matches pool1.

443. On the VRP operating platform, what is the command to display the current interface configuration in the interface view?

- A. display users
- B. display version
- C. display this
- D. display ip interface brief

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064353/cf86c2d8/display-this>

Function: The **display this** command displays the running configuration in the current view.

```
The device is running!

<R3>system-view
Enter system view, return user view with Ctrl+Z.
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]display this
#
interface GigabitEthernet0/0/1
 ip address 24.23.5.213 255.255.255.252
#
return
[R3-GigabitEthernet0/0/1]
```

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064353/b048882d/display-ip-interface>

Function: The **display ip interface brief** command displays brief information about interface IP addresses, including the IP address, subnet mask, physical status, link-layer protocol status, and number of interfaces in different states.

```
[R5]display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
The number of interface that is UP in Physical is 3
The number of interface that is DOWN in Physical is 8
The number of interface that is UP in Protocol is 3
The number of interface that is DOWN in Protocol is 8

Interface                IP Address/Mask      Physical  Protocol
Ethernet0/0/0             unassigned           down      down
Ethernet0/0/1             unassigned           down      down
GigabitEthernet0/0/0      unassigned           down      down
GigabitEthernet0/0/1      24.23.5.196/29       up        up
GigabitEthernet0/0/2      24.23.5.209/30       up        up
GigabitEthernet0/0/3      unassigned           down      down
NULL0                    unassigned           up        up(s)
Serial0/0/0               unassigned           down      down
Serial0/0/1               unassigned           down      down
Serial0/0/2               unassigned           down      down
Serial0/0/3               unassigned           down      down
```

444. What commands and views exist in the VRP operating platform? (Multiple choice)

- A. User view
- B. Interface view
- C. Agreement view
- D. System view

Answer: ABD

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000178166/c1bb51ce/enteringcommand-views>

Common Command Views

Name	How To Enter	Function
User view	When a user logs in to the device, the user enters the user view and the following prompt is displayed: <code><HUAWEI></code>	In the user view, you can view the running status and statistics of the device.
System view	Run the system-view command and press Enter in the user view. The system view is displayed. <code><HUAWEI> system-view</code> Enter system view, return user view with Ctrl+Z. <code>[HUAWEI]</code>	In the system view, you can set the system parameters of the device, and enter other function views from this view.
Interface view	Run the interface command and specify an interface type and number to enter the interface view. <code>[HUAWEI] interface gigabitethernet X/Y/Z</code> <code>[HUAWEI-GigabitEthernetX/Y/Z]</code> <i>X/Y/Z indicates the number of an interface that needs to be specified. It is in the format of stack ID/card number/interface sequence number. The interface GigabitEthernet is used as an example.</i>	In the interface view, you can configure interface parameters including physical attributes, link layer protocols, and IP addresses.

445. On the VRP platform, which of the following methods can be used to access the previous history commands?

- A. CTRL +U
- B. CTRL +P
- C. Left cursor
- D. Up cursor

Answer: B Both B,D if only 1 so then do D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000128405/6e85b486/displayhistory-command>

You can view historical commands using the following methods:

- To view the previous historical command, press the Up arrow key or **Ctrl+P**. If there is an earlier historical command, the earlier historical command is displayed.
- To view the next historical command, press the Down arrow key or **Ctrl+N**. If there is a new historical command, the new historical command is displayed.

446. If the subnet mask of a network segment 150.25.0.0 is 255.255.224.0, then what is a valid host address in the network segment?

- A. 150.25.2.24
- B. 150.25.0.0
- C. 150.25.1.255
- D. 150.15.3.30

Answer: AC

Justification

Explanation:

Network address: 150.25.0.0

Subnet mask: 255.255.224.0 → this is equivalent to /19

- Subnet mask /19 → gives 8192 IPs per subnet
- Range of IPs for this subnet:
 - First IP (Network Address): 150.25.0.0
 - Last IP (Broadcast Address): 150.25.31.255
 - Valid host addresses: From 150.25.0.1 to 150.25.31.254

447. On the broadcast network, both DR and BDR use the multicast address 224.0.0.6 to receive link-state update messages.

- A. True Correct
- B. False

Answer: B A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100034072/32a400ec/ospf-fundamentals>

Table 5-7 OSPF network types

Network Type	Description
Broadcast	A network with the link layer protocol of Ethernet or Fiber Distributed Data Interface (FDDI) is a broadcast network by default. On a broadcast network: - Hello packets, LSU packets, and LSAck packets are usually transmitted in multicast mode. 224.0.0.5 is an IP multicast address reserved for an OSPF device. 224.0.0.6 is an IP multicast address reserved for an OSPF DR or backup designated router (BDR). - DD and LSR packets are transmitted in unicast mode.
Non-Broadcast Multi-Access (NBMA)	A network with the link layer protocol of frame relay (FR) or X.25 is an NBMA network by default. On an NBMA network, protocol packets such as Hello packets, DD packets, LSR packets, LSU packets, and LSAck packets are sent in unicast mode.
Point-to-Multipoint (P2MP)	No network is a P2MP network by default, no matter what type of link layer protocol is used on the network. A network can be changed to a P2MP network. The common practice is to change a non-fully meshed NBMA network to a P2MP network. On a P2MP network: - Hello packets are transmitted in multicast mode using the multicast address 224.0.0.5. - Other types of protocol packets, such as DD packets, LSR packets, LSU packets, and LSAck packets are sent in unicast mode.
Point-to-point (P2P)	By default, a network where the link layer protocol is PPP, HDLC, or LAPB is a P2P network. On a P2P network, protocol packets such as Hello packets, DD packets, LSR packets, LSU packets, and LSAck packets are sent in multicast mode using the multicast address 224.0.0.5.

448. What are the roles of the OSPF protocol DR and BDR? (Multiple choice)

- A. Reduce the number of adjacencies
- B. Reduce the types of OSPF protocol packets
- C. Reduce the time to establish adjacency
- D. Reduce the number of exchanges of link-state information.

Answer: AD

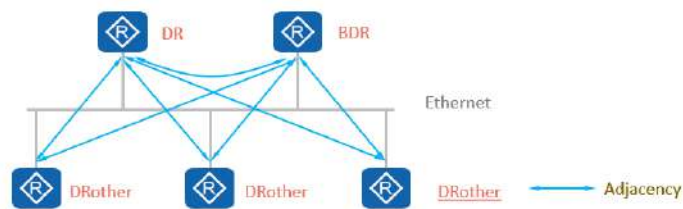
Justification

Source: Huawei Slide # 06, Page # 33



DR and BDR

- To optimize OSPF neighbor relationships on an MA network, the OSPF protocol specifies three types of OSPF routers: DR, BDR, and DRoother.
- Only the DR and BDR can establish adjacencies with other OSPF routers. DRoother do not establish OSPF adjacencies with one another, and their relationship is in the 2-way state.
- The BDR monitors the status of the DR and takes over the role of the DR if the existing DR fails.



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Explanation:

In OSPF on broadcast networks (like Ethernet), when many routers are on the same network segment, they would all try to form adjacencies with each other, leading to a large number of adjacencies and a flood of link-state advertisements (LSAs).

To solve this, Designated Router (DR) and Backup Designated Router (BDR) are elected:

Role of DR and BDR:

Role	Description
A. Reduce the number of adjacencies	✓ Only routers form full adjacencies with the DR and BDR, not with every other router.
B. Reduce the types of OSPF protocol packets	✗ Not true — all OSPF packet types are still used.
C. Reduce the time to establish adjacency	✗ Not necessarily; DR/BDR elections may delay initial adjacency.
D. Reduce the number of exchanges of link-state information	✓ Only DR forwards LSAs to all others, minimizing flooding.

449. Which of the following statements about the TTL field in the IP packet header is correct?

- A. TTL defines the time interval at which the source host can send data packets
- B. TTL defines the number of packets that the source host can send
- C. Every time an IP packet passes through a router, its TTL value will be reduced by 1
- D. Each time an IP packet passes through a router, its TTL value will be increased by 1

Answer: C

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100117312>

Table 1-1 Parameter description of the **tracert** command

Parameter	Description	Value
-a <i>source-ip-address</i>	Specifies the source address of trace packets. <i>source-ip-address</i> is the IP address of the local interface.	The value is in dotted decimal notation.
-f <i>first-TTL</i>	Specifies the initial time to live (TTL) of trace packets. Carried in the IP header, the TTL field indicates the lifetime of a trace packet and specifies the maximum number of devices that the packet can pass through. The TTL value is set on the source and reduced by 1 each time the packet passes through one device. When the TTL value is reduced to 0, the packet is discarded, and an ICMP Time Exceeded message is sent to the source. If first-TTL is specified and the number of hops that a trace packet passes through is less than the specified value, the TTL value will be greater than 0 after the packet passes through all the nodes. In this case, no ICMP Time Exceeded message is sent to the source. If max-TTL is also specified, the value of first-TTL must be smaller than the value of max-TTL.	The value is an integer ranging from 1 to 255. The default value is 1.
-m <i>max-TTL</i>	Specifies the maximum TTL. Generally, the value of <i>max-TTL</i> is set to the number of hops that the packet passes through. To change the TTL value, specify this parameter. If first-TTL is specified, the value of max-TTL must be greater than the value of first-TTL.	The value is an integer ranging from 1 to 255. The default value is 30.

450. In the network shown in the figure below, all routers run the OSPF protocol, which device is an ABR? (Multiple choice)



- A. Router A
- B. Router C
- C. Router D
- D. Router B

Answer: AD

Justification

Explanation:

An ABR (Area Border Router) is a router that connects two or more OSPF areas, with at least one interface in Area 0 (the backbone area).

Let's analyze each router:

Router	Areas Connected	Is ABR?	Reason
Router A	Area 0 & Area 2	Yes	Connects Area 0 to another area
Router B	Area 0 & Area 1	Yes	Connects Area 0 to another area
Router C	Area 2 & Area 4	No	Does not connect to Area 0 → Not ABR
Router D	Area 1 & Area 4	No	Does not connect to Area 0 → Not ABR

So, only **Router A** and **Router B** qualify as ABRs.

451. It is known that there are two entries in the routing table of a router. If the router wants to forward packets with the destination address of 9.1.4.5, which of the following statements is correct?

Destination/Mask	Protocol	Pre	Cost	NextHop	Interface
9.0.0.0/8	OSPF	10	50	1.1.1.1	Serial0
9.1.0.0/16	IS-IS	15	100	2.2.2.2	Ethernet0

- A. Choose the second item as the best matching item, because the cost of the RIP protocol is small
- B. Select the second item as the best matching item, because the route is a more accurate match relative to the destination 9.1.4.5
- C. Choose the second item as the best match, because Ethernet0 is faster than serial0
- D. Select the first item as the best matching item, because the priority value of the ospf protocol is higher

Answer: B

Justification

Source: Huawei Slide # 05, Page # 22

Longest Matching

- When a router receives an IP packet, it compares the destination IP address of the packet with all routing entries in the local routing table bit by bit until the longest matching entry is found. This is the longest matching mechanism.

		Bit-by-bit matching			
Destination IP address	172.16.2.1	172.	16.	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1
Routing entry 1	172.16.1.0 255.255.255.0	172.	16.	0 0 0 0 0 0 0 1	xxxxxx
Routing entry 2	172.16.2.0 255.255.255.0	172.	16.	0 0 0 0 0 0 1 0	xxxxxx
Routing entry 3	172.16.0.0 255.255.0.0	172.	16.	xxxxxxx	xxxx

452. For multiple paths to the same destination network, the router needs to select by comparing the magnitude of the Preference value. If the Preference is the same, then select according to the magnitude of the Cost value.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086956>

- Pre: indicates the routing protocol preference of a route. There may multiple routes to the same destination, which have different next hops and outbound interfaces. These routes may be discovered by different routing protocols or manually configured. A router selects the route with the highest preference (the smallest value) as the optimal route. For the routing protocol preference, see Routing Protocol Preference.
- Cost: indicates the route cost. When multiple routes to the same destination have the same preference, the route with the lowest cost is selected as the optimal route.

NOTE

The Preference value is used to compare the preferences of different routing protocols, while the Cost value is used to compare the preferences of different routes of the same routing protocol.

Explanation: There may be multiple routes to the same destination, which have different next hops and outbound interfaces. These routes may be discovered by different routing protocols or manually configured. A router selects the route with the highest preference (the smallest value) as the optimal route.

453. When the LACP mode is used for link aggregation, what is the default system interference level of Huawei switches?

- A. 36864
- B. 4096
- C. 24576
- D. 32768

Answer: D

Justification

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1000017271/63a1ce94/optional-setting-the-lacp-system-priority>

Procedure

1. Run:

```
system-view
```

The system view is displayed.

2. Run:

```
lacp priority priority
```

The LACP system priority is set.

A smaller LACP priority value indicates a higher priority. By default, the LACP system priority is 32768.

The end with a smaller priority value functions as the Actor. If the two ends have the same priority, the end with a smaller MAC address functions as the Actor.

454. After the network administrator uses the Traceroute function on the router device, in the data packet sent by the router, what is the value of the Protocol field in the IPv4 header?

- A. 17
- B. 2
- C. 6
- D. 1

Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100117312>

Table 1-1 Parameter description of the **tracert** command

Parameter	Description	Value
-a <i>source-ip-address</i>	Specifies the source address of trace packets. <i>source-ip-address</i> is the IP address of the local interface.	The value is in dotted decimal notation.
-f <i>first-TTL</i>	Specifies the initial time to live (TTL) of trace packets. Carried in the IP header, the TTL field indicates the lifetime of a trace packet and specifies the maximum number of devices that the packet can pass through. The TTL value is set on the source and reduced by 1 each time the packet passes through one device. When the TTL value is reduced to 0, the packet is discarded, and an ICMP Time Exceeded message is sent to the source. If first-TTL is specified and the number of hops that a trace packet passes through is less than the specified value, the TTL value will be greater than 0 after the packet passes through all the nodes. In this case, no ICMP Time Exceeded message is sent to the source. If max-TTL is also specified, the value of first-TTL must be smaller than the value of max-TTL.	The value is an integer ranging from 1 to 255. The default value is 1.

455. The information shown in the figure is the port status information displayed on a switch running STP. According to this information, which of the following descriptions is wrong?

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/2	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/13	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/21	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/22	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/23	DESI	FORWARDING	NONE

- A. The priority of this switch is 0
- B. This network may only contain this switch
- C. This switch may be connected to six other switches
- D. This switch is the root switch in the network

Answer: A

Justification

Explanation:

Correct Answer: A. The priority of this switch is 0

Wrong: The output only shows the MST instance ID (MSTID = 0), which refers to MST instance 0 and not the bridge priority value. The switch priority cannot be determined from this output alone. Therefore, stating that the priority is 0 is not supported by the given information.

Option B. This network may only contain this switch

Correct: All ports are in the Designated role and Forwarding state, which is consistent with a single-switch environment. In such a case, no STP election occurs, and the switch designates all ports.

Option C. This switch may be connected to six other switches

Correct: There are six ports shown as forwarding and in the Designated role. Each of these could potentially connect to a separate switch or device. Therefore, the statement is logically valid.

Option D. This switch is the root switch in the network

Correct: In STP, a switch with all ports in the Designated role and in the Forwarding state typically indicates that it is acting as the Root Bridge within the topology.

456. Two authentication domains, "Area 1" and "Area 2" are configured on a router acting as an authentication server. If the user authenticates with the correct user name "huawei" and password "hello", this user will be assigned to which authentication domain?

- A. Authentication domain "Area 1"
- B. Authentication domain "default domain"
- C. Authentication domain "default_admin domain"
- D. Authentication domain "Area 2"

Answer: B

Justification

Source: https://support.huawei.com/enterprise/en/doc/DOC1000141870/26361446/example-for-configuring-default-domain-based-user-management?utm_source=chatgpt.com

Domain and Default Domain Overview

The device manages access users based on domains. Each access user belongs to a domain.

The authentication, authorization, and accounting schemes can be bound to domain views. The device manages the access users in the same domain in the same manner, for example, using the same authentication, authorization, and accounting scheme.

As shown in [Figure 9-23](#), the users are authenticated in the specified domain when entered user names contain domain names or in the default domain when entered user names do not contain domain names.

If a user name contains a domain name, the user belongs to this domain; otherwise, the user belongs to the default domain. If most users on a network belong to the same domain, you can configure this domain as the default domain so that these users do not need to enter the domain name when logging in to the device.

Default domains fall into default administrative domain and default common domain.

- The administrator (logging in through Telnet, SSH, FTP, HTTP, or Terminal) is authenticated in the default administrative domain.
By default, the default administrative domain is **default_admin**.
- The common users (logging in through MAC, Portal, or 802.1X authentication) are authenticated in the default common domain.
By default, the default common domain is **default**.

Explanation:

In Huawei devices, when a user logs in without specifying an authentication domain (i.e., without appending @domain to the username), the system assigns the user to a default domain.

- **Default Domains:**
 - **default_admin:** Used for administrative access methods like Telnet, SSH, FTP, and HTTP.
 - **default:** Used for common user access methods like 802.1X, MAC, or Portal authentication.
- Therefore, if a user logs in with the username "huawei" and password "hello" without specifying a domain (e.g., "huawei@Area1"), the system assigns the user to the **default** domain.

457. By default, it takes at least 30 seconds for the port status in the STP protocol to change from Disabled to Forwarding.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100092143>

- **Forward Delay**

The Forward Delay timer specifies the delay in a port state transition. When a link fails, STP calculation is triggered and the spanning tree structure changes accordingly. However, new configuration BPDUs cannot be flooded immediately across the entire network. If the new root port and designated port forward data immediately, transient loops may occur. Therefore, STP defines a delay mechanism for port state transition. The newly selected root port and designated port must wait for two Forward Delay intervals before transitioning to the Forwarding state. During this period, the new configuration BPDUs can be transmitted over the network, preventing transient loops.

The Forward Delay timer defines the time that is spent in Listening or Learning state. The default Forward Delay value is 15 seconds. This means that the port stays in Listening state for 15 seconds and then stays in Learning state for another 15 seconds before transitioning to the Forwarding state. The port in Listening or Learning state does not forward user traffic, effectively preventing transient loops.

Explanation:

In the Spanning Tree Protocol (STP), when a port transitions from a disabled (or blocking) state to the forwarding state, it undergoes two intermediate states: Listening and Learning.

- **Listening State:** The port listens for Bridge Protocol Data Units (BPDUs) to ensure no loops occur. This state lasts for **15 seconds**.
- **Learning State:** The port learns MAC addresses to populate the MAC address table but still doesn't forward frames. This state also lasts for **15 seconds**.

Therefore, the total time for a port to move from a disabled state to the forwarding state is:

15 seconds (Listening) + 15 seconds (Learning) = 30 seconds

This default behavior ensures network stability by preventing loops during topology changes.

458. Which of the following parts compose SNMP? (Multiple choice)

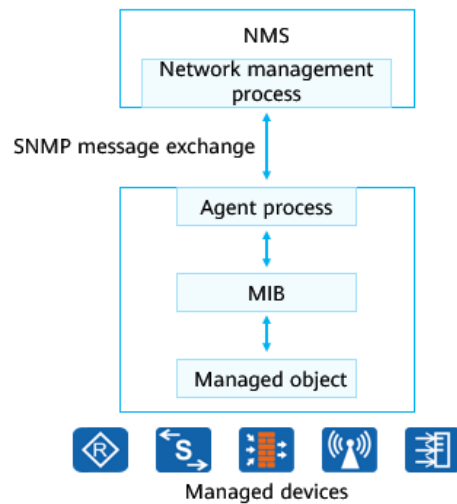
- A. NMS
- B. Agent process
- C. Managed object
- D. Management information base

Answer: ABCD

Justification

Source: Huawei Slide # 18, Page # 15

SNMP Management Model



- Query/Modify operation:
 - The NMS sends an SNMP request message to an agent process.
 - The agent process searches the MIB on the device for information to be queried or modified and sends an SNMP response message to the NMS.
- Trap operation:
 - If the trap triggering conditions defined for a module are met, the agent process sends a message to notify the NMS that an event or trap has occurred on a managed object. This helps network administrators promptly process network faults.

459. An AC is a unified management and control device in the AC + Fit AP architecture. Which of the following functions is not supported by the AC.

- A. User access control
- B. AP configuration delivery
- C. User access authentication
- D. User data packet forwarding regardless of data forwarding

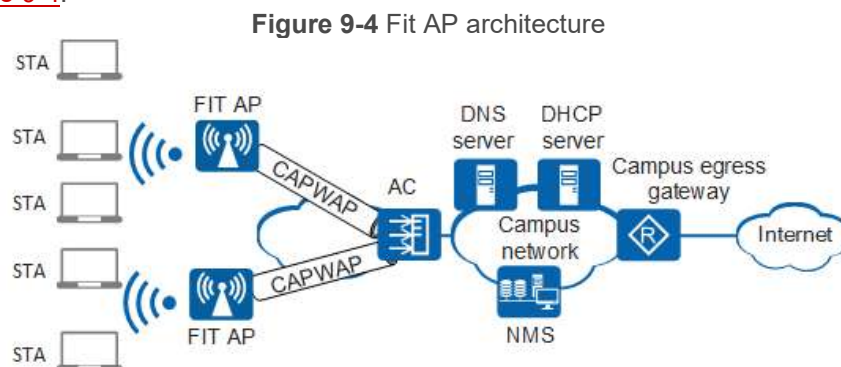
Answer: D

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100096325/f342dc7/wlan-architecture>

Fit AP Architecture

In Fit AP architecture, an AC manages and controls multiple APs (Fit APs) in a centralized manner, as shown in Figure 9-4.



In Fit AP architecture, APs work together with an AC to implement wireless access.

- The AC implements all security, control, and management functions. These functions include mobile user management, identity authentication, VLAN assignment, radio management, and data forwarding.
- Fit APs implement wireless radio access, including radio signal transmission and detection response, data encryption and decryption, and data transmission acknowledgment.
- The AC and APs communicate using Control and Provisioning of Wireless Access Points (CAPWAP). They can be connected across a Layer 2 or Layer 3 network.

460. What protocol does OSPF run over?

- A. IP
- B. TCP
- C. UDP
- D. HTTP

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100082074>

Definition

The Open Shortest Path First (OSPF) protocol, developed by the Internet Engineering Task Force (IETF), is a link-state Interior Gateway Protocol (IGP).

At present, OSPF Version 2, defined in RFC 2328, is intended for IPv4, and OSPF Version 3, defined in RFC 2740, is intended for IPv6. Unless otherwise stated, OSPF stated in this document refers to OSPF Version 2.

461. In RSTP, can a backup port replace a faulty root port?

- A. True
- B. False

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100276753/2bd0e592/rstp-port-roles-and-port-states>

Table 14-7 Alternate port and backup port

Category	Alternate Port	Backup Port
From the perspective of configuration BPDU transmission	An alternate port is blocked after learning a configuration BPDU sent from another bridge.	An alternate port is blocked after learning a configuration BPDU sent from itself.
From the perspective of user traffic	An alternate port acts as a backup of the root port and provides an alternate path from the designated bridge to the root bridge.	A backup port acts as a backup of the designated port and provides a backup path from the root bridge to the related network segment.

Explanation:

In RSTP (Rapid Spanning Tree Protocol), port roles and transitions are more dynamic than in traditional STP, but they still follow certain rules:

- A **Backup Port** is a redundant path to the same designated bridge as the designated port on the same switch. It's a backup to the **Designated Port**, not the Root Port.
- A **Root Port** is the port that receives the **best BPDU** from the **Root Bridge** and is used to reach the Root Bridge.

So, if a Root Port fails, a Backup Port cannot take over its role, because it is not connected to a better path toward the Root Bridge.

Instead, the port that can become the new Root Port would typically be an Alternate Port, which is a backup path to the Root Bridge.

462. The configuration order (config) is the default match order for ACL rules on Huawei devices.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100086647>

Matching Order

An ACL consists of multiple deny | permit clauses, each of which describes a rule. These rules may repeat or conflict. For example, an ACL contains two rules:

```
rule deny ip destination 10.1.0.0 0.0.255.255 //Reject the packets destined
for network segment 10.1.0.0/16.
rule permit ip destination 10.1.1.0 0.0.0.255 //Permit the packets destined
for network segment 10.1.1.0/24, which has a smaller range than 10.1.0.0/16.
```

The permit and deny rules conflict. If the system first matches a packet destined for 10.1.1.1 against the deny rule, the packet is discarded. However, if the system matches the packet against the permit rule first, the packet is forwarded.

Therefore, if ACL rules repeat or conflict, the matching order decides the matching result.

The device supports two matching orders: the configuration order (**config**) and the automatic order (**auto**). The default order is **config**.

463. The data link layer uses PPP encapsulation, and the IP addresses of the two ends of the link can be in different network segments.

- A. True
- B. False

Answer: A

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100278979/5e7a58c1/ppp-configuration>

Overview of PPP

This section describes the principles of the Point-to-Point Protocol (PPP) and related protocols. Point-to-point (P2P) connections are simple wide area network (WAN) connections. A P2P link uses PPP as its link-layer protocol.

PPP, which works at the second layer (data link layer) of the open systems interconnection (OSI) model, is mainly used on links that support full-duplex to transmit data. PPP is widely used because it provides user authentication, supports synchronous and asynchronous communication, and is easy to extend.

Context

Devices on both ends of a PPP link may have IP addresses at different network segments. When the two devices communicate with each other, one device automatically adds the host route of the other end to the local routing table of direct routes. If one end has an incorrect IP address configured, incorrect routing information will be advertised on the network. To prevent incorrect routing information, prohibit the local device from adding the peer host route to the local routing table of direct routes.

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100278760/5e7a58c1/ppp-configuration>

Configuring an Interface to Support PPP

Configure an interface to support PPP so that it can implement point-to-point (P2P) traffic transmission.

Usage Scenario

PPP is a link layer protocol that transmits network layer packets over P2P links. PPP defines a group of protocols, including LCP and NCP.

- During the LCP negotiation phase, an MRU, negotiation timeout period, and PPP LCP link dead duration are negotiated.
- During the NCP negotiation phase, network layer packet attributes and types are negotiated. For example, during the IPCP negotiation, the IP address of a DNS server will be negotiated.

To monitor link status in real time, configure link status monitoring parameters so that link faults can be identified in time. A PPP link does not require that the peer route and local route be on the same network segment. To avoid incorrect routing information, prohibit the addition of the peer host route to the local routing table of direct routes.

464. For VRP, the login timeout period can be set only on a VTY interface.

- A. True
- B. False

Answer: B

Justification

Source: Huawei Slide # 03, Page # 33



Basic Configuration Commands (2)

3. Configure a command level.

```
[Huawei] command-privilege level level view view-name command-key
```

This command configures a level for commands in a specified view. Command levels are classified into visit, monitoring, configuration, and management, which are identified by the numbers 0, 1, 2, and 3, respectively.

4. Configure the password-based login mode.

```
[Huawei] user-interface vty 0 4  
[Huawei-ui-vty0-4] set authentication password cipher information
```

This **user-interface vty** command displays the virtual type terminal (VTY) user interface view, and the **set authentication password** command configures the password authentication mode. The system supports the console user interface and VTY user interface. The console user interface is used for local login, and the VTY user interface is used for remote login. By default, a device supports a maximum of 15 concurrent VTY-based user accesses.

5. Configure user interface parameters.

```
[Huawei] idle-timeout minutes [seconds]
```

This command sets a timeout period to disconnect from the user interface. If no command is entered within the specified period, the system tears down the current connection. The default timeout period is 10 minutes.

465. On a layer 2 WLAN STAs uses an AC.

- A. True
- B. False

use AP to access the network
STA ↔ AP
AP ↔ AC

Answer: B

Justification

Source: <https://support.huawei.com/enterprise/en/doc/EDOC1100064365/e88c00c0/sta-%20access>

STA Access

STAs can access wireless networks after APs are logged in and CAPWAP tunnels are established. STA access involves the following steps:

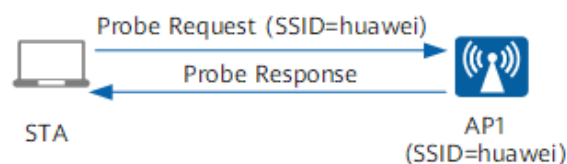
- Scanning
- Link authentication
- Association

A STA can access wireless networks in either IPv4 or IPv6 mode, with the IPv4 mode taking precedence.

STA access depends on the number of access users supported by the AC and a single AP.

- If the number of STAs associated with an AP reaches the maximum limit of the AP but not the maximum limit of the AC, a new STA cannot connect to the current AP. However, the STA can associate with another AP on the network.
- If the number of STAs associated with an AP reaches the maximum limit of the AC, a new STA cannot access the WLAN even though the maximum limit of the AP is not reached.
- If the number of STAs associated with an AP does not reach the maximum limit of the AP or AC, a new STA can access the WLAN.

Figure 8-10 Active scanning by sending a Probe Request frame containing an SSID



Explanation: This source tells us about the connections of STAs with APs. No connection with AC mentioned, thus the answer is false.

466. Using the commands "vlan batch 10 20" and "vlan batch 10 to 20", how many VLANs are created in each case, respectively?

- A. 11 and 11
- B. 11 and 2
- C. 2 and 2
- D. 2 and 11

Answer: D

Justification

Source: Huawei Slide # 08, Page # 31

Basic VLAN Configuration Commands

1. Create one or more VLANs.

```
[Huawei] vlan vlan-id
```

This command creates a VLAN and displays the VLAN view. If the VLAN to be created already exists, this command directly displays the VLAN view.

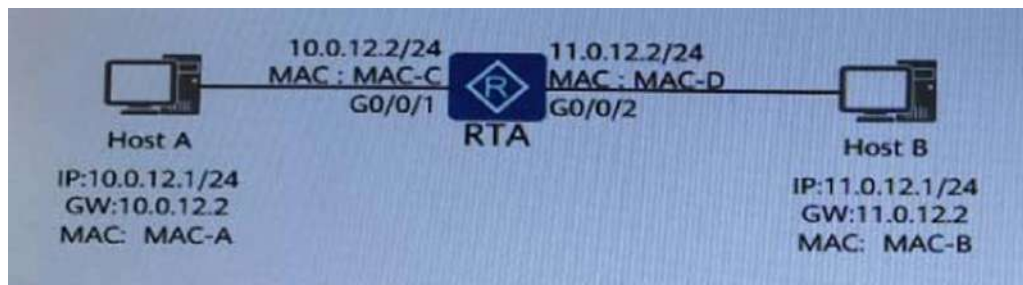
- The value of *vlan-id* is an integer ranging from 1 to 4094.

```
[Huawei] vlan batch { vlan-id1 [ to vlan-id2 ] }
```

This command creates VLANs in a batch. In this command:

- **batch**: creates VLANs in a batch.
- *vlan-id1*: specifies a start VLAN ID.
- *vlan-id2*: specifies an end VLAN ID.

467. In the network figure given below, a router receives a data packet from host A with the destination IP address 11.0.12.1. After the router forwards the packet, what will be the destination MAC and destination IP addresses



- A. MAC-C, 11.0.12.1
- B. MAC-D, 10.0.12.2
- C. MAC-D, 11.0.12.1
- D. MAC-B, 11.0.12.1

Answer: D

Justification

Explanation:

When Host A (10.0.12.1/24) sends a packet to Host B (11.0.12.1/24), it identifies that the destination IP address is outside its own subnet. Therefore, Host A forwards the packet to its configured gateway, which is Router RTA's G0/0/1 interface (IP: 10.0.12.2, MAC: MAC-C).

The router (RTA) receives the packet and checks its routing table to determine the appropriate outgoing interface for the destination IP 11.0.12.1. It determines that the packet should be forwarded out of interface G0/0/2 (MAC-D).

At this point:

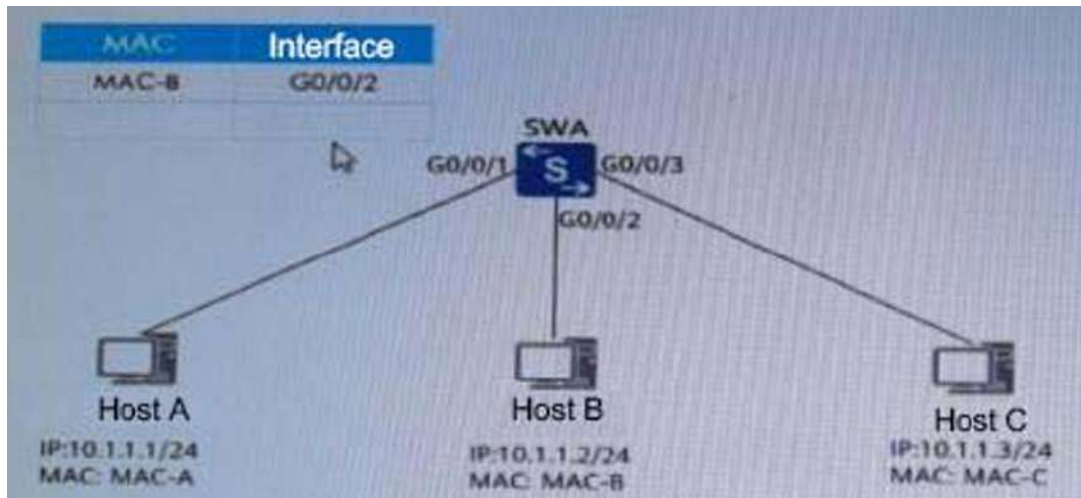
- **The destination IP remains unchanged:** 11.0.12.1
- **The destination MAC address is updated** to the MAC address of Host B (MAC-B)
- **The source MAC address is updated** to the MAC address of RTA's G0/0/2 interface (MAC-D)

Thus, the final packet leaving RTA has:

- **Source IP:** 10.0.12.1
- **Destination IP:** 11.0.12.1
- **Source MAC:** MAC-D
- **Destination MAC:** MAC-B

Therefore, **option C is incorrect**, and **option D (MAC-B, 11.0.12.1)** is the correct answer based on accurate Layer 2 forwarding behavior.

468. As shown in the figure below, consider the MAC address table of switch SWA. If host A sends a data frame with the destination MAC address MAC-B, which of the following statements is correct?



- A. Forward this data frame only from port G0/0/2
- B. Forward this data frame only from port G0/0/3
- C. STA discards the data frame
- D. Flood this data frame

Answer: A

Justification

Reason: When Host A sends a data frame to Host B, the switch (SWA) uses its MAC address table to determine the outgoing port. Since the MAC address table of SWA shows that the MAC address of Host B (MAC-B) is associated with port G0/0/2, the switch will specifically forward the data frame to this port.

Source:

<https://support.huawei.com/enterprise/en/doc/EDOC1100096325/406107ae/understanding-the-mac-address-table>

MAC Address Table

Each device maintains a MAC address table. A MAC address table records the MAC address, VLAN ID and outbound interfaces learned from other devices. When forwarding a data frame, the device searches the MAC table for the outbound interface according to the destination MAC address and VLAN ID in the frame. This helps the device reduce broadcasting.

Packet Forwarding Based on the MAC Address Table

The device forwards packets based on the MAC address table in either of the following modes:

- **Unicast mode:** If the destination MAC address of a packet can be found in the MAC address table, the device forwards the packet through the outbound interface specified in the matching entry.
- **Broadcast mode:** If a packet is a broadcast or multicast packet or its destination MAC address cannot be found in the MAC address table, the device broadcasts the packet to all the interfaces in the VLAN except the inbound interface.

469. The routing table output information of R1 is shown in the figure. Which of the following statements is correct? (Multiple choice)

```
<R1>display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public
Destinations : 10 Routes : 10

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.0.0/8	Static	60	0	RD	10.0.12.2	Ethernet0/0/0
10.0.2.2/32	Static	70	0	RD	10.0.21.2	Ethernet0/0/1

- A. The R1 to the NextHop of 10.0.2.2 is 10.0.21.2
- B. The R2 to the NextHop of 10.0.2.1 is 10.0.21.2
- C. The R3 to the NextHop of 10.0.2.2 is 10.0.12.2
- D. The R4 to the NextHop of 10.0.0.1 is 10.0.12.2

Answer: AD

Justification

Explanation:

- Option A:** Correct – This is an exact /32 match in the table. The next hop is indeed 10.0.21.2.
- Option B:** Incorrect – There is **no route** for 10.0.2.1 in the table, only for 10.0.2.2.
- Option C:** Incorrect – 10.0.2.2 is explicitly routed via 10.0.21.2, **not** 10.0.12.2.
- Option D:** Correct – 10.0.0.1 matches the 10.0.0.0/8 route, and the next hop is 10.0.12.2.